Second Master Plan
For
Chennai Metropolitan Area, 2026

Volume III
Sectoral Background

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Contents

Acronyms

A/R	Auto Rickshaw
ADB	Asian Development Bank
ATC	Area Traffic Control
BOO	Build, Own & Operate
BOOT	Build, Own, Operate & Transfer
BPL	Below Poverty Line
BSNL	Bharat Sanchar Nigam Ltd.
CAA	Constitutional Amendment Act
CBD	Central Business District
CBDRM	Community Based Disaster Risk Management
CBED	Community Based Environmental Development Programme
CBO	Community Based Organisation
CMA	Chennai Metropolitan Area
CMWSSB	Chennai Metropolitan Water Supply & Sewerage Board
CNG	Compressed Natural Gas
CO	Carbon Monoxide
CoC	Corporation of Chennai
CPHEEO	Central Public Health and Environmental Engineering Organisation
CPT	Chennai Port Trust
Crore	100 lakhs = 1 00 00 000
CRZ	Coastal Regulation Zone
CTH Road	Chennai Tiruvallur High Road
CTP	Chennai Traffic Police
CTS	Comprehensive Transportation Study
CTTS	Comprehensive Traffic & Transportation Study
CUA	Chennai Urban Agglomeration
DCR	Development Control Rules
DDP	Detailed Development Plan
DES	Department of Economics and Statistics
DMRC	Delhi Metro Rail Corporation
DMRH	Director of Medical and Rural Health Services
DoH	Department of Highways
DPHPM	Dept. of Public Health and Preventive Medicine
DR	Development Regulations
ECR	East Coast Road
EIA	Environmental Impact Assessment
ELCOT	Electronics Corporation of Tamil Nadu
EMP	Enviromental Management Plan
ETB	Electric Trolley Bus
EWS	Economically Weaker Section
FMP	First Master Plan
FoB	Foot Over Bridge
FSI	Floor Space Index
GIS	Geographical Information System
GNT Road	Grand Northern Trunk Road
GoI	Government of India
GoTN	Government of Tamil Nadu
GST Road	Grand Southern Trunk Road

GWT Road	Grand Western Trunk Road
H&UD Dept.	Housing & Urban Development Department
ha	hectare
HHI	Household Interview
HIG	High Income Group
HOV	High Occupancy Vehicle
HT Line	High Tension Line
HTL	High Tide Line
ICC	Inner Circular Corriodor
IL&FS	Infrastructre Leasing & Financial Services
IPT	Intermediate Public Transport
IRR	Inner Ring Road
IT	Information Technology
ITES	Information Technology Enabling Services
JNNURM	Jawaharlar Nehru National Urban Renewal Mission
km	kilometre
KWMC	Koyambedu Wholesale Market Complex
LB Road	Lattice Bridge Road
LC	Level Crossing
LIG	Low Income Group
lpcd	litres per capita per day
LPG	Liquified Petroleum Gas
LRT	Light Rail Transit
LT Line	Low Tension Line
m	million
MBI Road	Marmalong- Bridge- Irumbuliyur Road
MEPZ	Madras Export Processing Zone
Mft3	Million cubic foot
MG Road	Mahatma Gandhi Road
mg.	milligram
MICE Tourism	Meetings, Incentives, Convention and Exhibitions Tourism
μg/m3	microgram per cubic metre
MIG	Middle Income Group
Million	10 lakhs =100 000
MINARS	Monitoring of Indian National Aquatic Resources
MLD	Million Litres per Day
MMDA	Madras Metropolitan Development Authority
MPC	Metropolitan Planning Committee
MR	Mixed Residential
MRTS	Mass Rapid Transit System
MTC	Metropolitan Transport Corporation
MUDP	Madras Urban Development Project
MVA	Mega Volt Ampere
MW	Mega Watt
NCTPS	North Chennai Thermal Power Station
NGO	Non-Government Organisation
NH	National Highway
NHAI	National Highways Authority of India
NMT	Non-Motorised Transport
Nos./nos.	numbers

OMR	Old Mahabalipuram Road
ORR	Outer Ring Road
OSR	Open Space Reservation
PCE	Passenger Car Equivalent
PCO	Public Call Office
PCU	Passenger Car Unit
PDA	Pallikaranai Drainage Area
PR	Primary Residential
PTCS	Pallavan Transport Consultancy Services
PWD	Public Works Department
RITES	Rail India Technical & Economic Services
rly.	railway
ROB	Road Over Bridge
RSPM	Respirable Suspended Particulate Matter
RTS	Rapid Transit System
RUB	Road Under Bridge
SCAT	Sydney Co-ordinated Adaptive Traffic System
SCOOT	Split Cycle Offset Optimisation Technique
SEZ	Special Economic Zone
SIDCO	Small Industries Development Corporation
SIPCOT	State Industries Promotion Corporation of Tamil Nadu
SMP	Second Master Plan
SPM	Suspended Particulate Matter
sq.km	square kilometre
sq.m	square metre
T/W	Two Wheeler
TDM	Travel Demand Management
TDR	Transfer of Development Rights
TEU	Twenty Equivalent Unit
TIDCO	Tamil Nadu Industrial Development Corporation
TN	Tamil Nadu
TNEB	Tamil Nadu Electricity Board
TNHB	Tamil Nadu Housing Board
TNHSP	Tamil Nadu Health Systems Project
TNPCB	Tamil Nadu Pollution Control Board
TNSCB	Tamil Nadu Slum Clearance Board
TNUDF	Tamil Nadu Urban Development Fund
TNUDP	Tamil Nadu Urban Development Project
TNUIFSL	Tamil Nadu Urban Infrastructure Financial Services Ltd.
TP	Town Panchayat
TPP Road	Tiruvottiyur- Ponneri- Panchetty Road
TSPM	Total Suspended Particulate Matter
TWAD	Tamil Nadu Water supply And Drainage Board
UGD	Under Ground Drainage
ULB	Urban Local Bodies
UMTA	Unified Metropolitan Transport Authority
UNDP	United Nations Development Programme
V/C	Volume-Capacity Ratio
VP	Village Panchayat
WB	World Bank

Chapter-I

STRUCTURE OF CHENNAI

Study of development trends helps to ascertain where development had occurred in the past and also particularly in recent years and the reasons for the same. It would help us in identifying the potential areas for future development and also plan for the future directions of growth taking into account of all the relevant planning aspects.

History[1]

1.02 The site on which Madras is situated has a long history. Settlements existed in the area of which Mylapore, Triplicane and Santhome were well known. The geographer Ptolemy had recorded it in the second century AD that the port of Mylapore was known to the Greeks and the Romans. The Port had a flourishing trade with the Roman Empire and received considerable quantities of gold in exchange for products like pepper and fine cloth. Mention has been made of the early settlement of Santhome by Arab travelers and merchants of the 9th and 10th centuries. Marcopolo visited this place in the late 13th Centurary and the Portuguese settled around Santhome in the early 16th Century. According to a traditional account it was at Mylapore that Thiruvalluvar, author of the famous Tirukkural-the universal Code of human conduct-lived. Both Mylapore and Triplicane were important Hindu religious centres and inscriptions dating back to the eighth century have been found in the Triplicane Temple.

1.03 The foundation for the development of the present metropolis was however laid in 1639 when the British East India Company acquired the site on which Fort St.George stands. The site was located by Sir Francis Day who obtained a 'firman' from one Damela Venkatadri Naik, the local chieftain of the area for setting up a factory (trading post). Francis Day and Andrew Cogan together with a few writers, 25 European soldiers and a few other Hindu artificers were the first to settle in the site granted by Venkatadri Naik in 1640.

1.04 The name of Madras is said to be derived from *Madraspatnam*, a village that existed here prior to the settlement of the English.

Madras in 1600[2]

1.05 Madras in 1600 was formed of scattered settlements separated by long distances. Each settlement grew around a nucleus of a temple and has its own history. The most important area at that time was Mylapore. In Santhome, the Portuguese, having originally arrived in 1522, constructed a fort and settlement. Mylapore was an

[1] Ref. Master Plan for CMA 1975
[2] Ref. Structure Plan for CMA, 1980.

adjacent settlement with a newly constructed temple. Triplicane on the north was a separate village.

1.06 There were small settlements in Purasawalkam, Thiruvatteeswaranpetta, Egmore, Nungambakkam, and Saidapet. Among the suburbs, Tiruvottiyur, Velachery, Tiruneermalai, Mangadu, Padi, Poonamallee, Kunnathur, Ayanavarum, Vyasarpadi, Villivakkam, Ambattur, Koyambedu etc. already existed. Each of these villages was self-contained and had its own agricultural production and household industries. The changing rulers in the region never interfered with them; with the result that life in the villages continued to exist without much change. Madras being on the coast, had many sand ridges, but during the 16th century the level of the sea rose and inundated lands within the settlements.

1.07 When the sea withdrew, lagoons and ridges were left behind. The lagoons took some time to become filled in and the sandy ridges were places of safety where new temples and settlements were established. There were two ridges - one between Broadway and the Beach of about 12 feet high roughly along Thambuchetty Street and another one along Mint Street. From First Line Beach the land slowly rose to the ridge and then fell to a valley along the present Broadway, where a drainage channel ran. Again it rose to the second ridge in Mint Street and descended gradually to a valley along the present Buckingham Canal. The Mint Street ridge continued along Tiruvottiyur High Road.

1.08 On the southern side, one such sand ridge ran from the mouth of the Cooum to the present site of the Presidency College. On the rear side was a huge depression in which the college grounds have now developed. The ridge is the present Marina. Further south, a ridge in a "U" shape ran along Besant and Lloyds Road enclosing the Ice House. Parthasarathy Temple at Triplicane is just by the northern area of this ridge. There is a wide depression between the southern ridge and Mylapore.

1.09 Another ridge was along the Luz Church Road linking Santhome Cathedral and Luz Church. The developments of Santhome and Adyar have been principally around this ridge. Mount Road ran along a tank bund and was at a high level; to the east of it, the land gradually falling to Triplicane High Road and the Luz. On the western side, the present Vyasarpadi and Perambur areas were low lying lands periodically inundated by floods. They were previously sparsely populated regions. People's Park was in a low-lying area and Purasawalkam High Road was along a ridge. West of the road was a drainage channel called the Otteri channel.

1.10 Important lines of communication linked these settlements. Egmore, Purasawalkam and Aminjikarai lie along Poonamallee High Road. Triplicane and Mylapore lie along a road leading to Tiruvanmiyur. These roads were just earthen cart tracks.

1.11 Within a few years of the founding of the British settlement and the construction of a Fort at the site, the New Town, which had grown up around it, came to be known as Chennapatnam in honour of the father of the local chieftain. This name was later applied to the whole city. The population, which was 19,000 in 1646, expanded to 40,000 in 1669.

Madras in 1700[2]

1.12 The Cooum River and the Elambore River or North River which flows into the Cooum at its mouth were running very close to each other (near the Central Jail area) and during floods they inundated the whole area. The two rivers were linked by a cut, at this point, to equalise the floods in the rivers. A bridge was constructed in 1710 across the cut between the two rivers. The Principal road to Egmore from that time up to 1931 had been the road in front of the present Central Jail. In this era there was a lot of building activity (a redoubt at Egmore, a bridge and churches at the Fort and many private buildings in and around the Fort).

1.13 Due to congestion inside the Fort, the British constructed some garden houses in what is known as Peddanaickenpet. In 1733 there was a lot of congestion in George Town and the weaving Community started settling in Chintadripet area and Collepetta near Tiruvottiyur since abundant open space was available for weaving. The washer men who were in the Mint area then moved towards the west. The Potters from this area moved outside the Fort on the north side and formed a new colony (Kosapet). Because of the construction of a bridge in 1710 to connect Egmore, people moved towards the present Moore Market area and settled.

1.14 During this time, the British found that Triplicane was a good area for settlement and a large number of people moved there. The presence of the Nawab of Arcot increased the economic prosperity of the area and more and more Muslims settled in Triplicane. So from that time onwards, it grew in importance, second only, to George Town. St. Thomas Mount gained religious importance and Mount Road became prominent with the construction of Marmalong Bridge in 1724.

[2] Ref. Structure Plan for CMA, 1980.

1.15 Because the British started living along Cooum River, roads were laid to give access to them and thus Marshalls Road, Halls Road, Montieth Road and Casa Major Road all became thoroughfares even in 1798. Mowbrays Road and Royapettah High Road were formed to give access to these people. The land west of George Town was a low-lying swampy area and remained vacant without development. The surroundings of the Fort area covering nearly 69 kms. and containing within it 16 hamlets were constituted as the City of Madras in 1798.

Madras in 1800[2]

1.16 After the founding of the Corporation, conservancy and improvement of the City were begun. The City was divided into 8 Divisions and the Government selected 4 Commissioners from residents. The broad-gauge line from Royapuram to Arcot was laid in 1864. Central station was formed in 1872 and linked to the main line. By 1861 the British authorities realised the necessity of a harbour. A pier was constructed in 1862 and further development took place from 1868 onwards.

1.17 The formation of Royapuram station in 1862 induced people to move northwards and settle in Royapuram. This gave rise to the establishment of some timber saw mills and depots in Royapuram after a few years. Further the railway line passed through the present Perambur area, which had so far been lying as swampy waste because of its low level. The introduction of the railway line gave development potential to the hitherto uninhabited place. From 1850 onwards, the necessity of providing recreational facilities was perceived. Many parks such as the Peoples Park, Napier Park and Richardson Park were created in this period. A Museum and a Zoo were also established.

1.18 Before 1800, the roads were in a radial pattern, but after 1810 ring roads were developed inside the City. Mount Road was important and access to it was given from Triplicane High Road, Chamiers Road, Edwards Elliot's Road and Royapettah High Road. To the North of Mount Road, Pantheon Road, Halls Road, Marshalls Road, Spur tank Road and Nungambakkam High Road were formed to serve the new residential areas. Later Brick kiln Road and Perambur Barracks Road connected Poonamallee High Road with Konnur Road, which was extended towards the Railway.

1.19 To facilitate trade the harbour was completed in 1896 just to the east of George Town. The principal roads leading out of the area in three principal directions connected the northern, western and southern parts of the presidency and they still remain as the main transportation framework for the City. Railways were also built radiating from the centre in these three principal directions.

1.20 The building of the harbour was responsible for sand accretion to the south of it and the sea which was washing the ramparts of the Fort at one time was then 2.5 km. away with a wide beach between the land and the sea. A number of public buildings were constructed fronting this beach early in the 19th century, which still add dignity to the City.

1.21 The city extended over an area of about 70 sq.kms. and had a population of 5.40 lakhs in 1901. The demographic growth rates during the previous two decades were 5 per cent and 6 per cent. Important buildings and structures which are the land marks of the city such as the Fort, the Marina-the road parallel to the sea-and the fronting public buildings, the High Court, the Moore Market, the Connemara Public Library and other public buildings around it already existed. George Town was the main business centre but substantial parts of it were used for residential purposes also. The main residential areas however were Chintadripet, Triplicane, Egmore, Mylapore, Purasawalkam, Vepery and Royapuram. Most of the areas outside these settlements were covered by gardens and agricultural lands interspersed with bungalows of the elite. Nungambakkam, Chetpet and Kilpauk were some of the areas, which developed in this manner; Saidapet in the southwest was a separate small settlement. Both sides of Mount Road to a distance of 5 to 6 kms. from Fort St.George were occupied by large business houses, clubs and hotels; industries were few and were located in George Town or Perambur area. The West of Nungambakkam was covered by the Nungambakkam Tank and this long tank extending in the form of a crescent for nearly 6 kms covered the west of Nungambakkam. The three railway lines served the city from the north, west and southwest. The city was mainly a commercial, military and administrative centre for the entire South India.

The City in 1941[2]

1.22 The population of the City increased to 8.6 lakhs by 1941. The city occupied an area of about 80 sq.km. and its boundaries having been extended in 1923. The important developments during the period 1901 and 1941 were the commissioning of the electrified suburban metre-gauge railway between Beach and Tambaram in 1931 which gave a fillip for the development of the outlying suburban areas as far as Tambaram, and the development of the area occupied by the long tank at Nungambakkam as a planned residential neighbourhood by the Corporation. There were also considerable in filling within the developed areas. The city had a good water supply system and most of the areas were sewered. It had quick and cheap modes of transport by trams and electric trains. The main roads were wide enough for the road traffic and passed through shady avenues. By 1941 Madras had developed into a provincial

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[2] Ref. Structure Plan for CMA, 1980.

metropolis enjoying the best of both worlds -urban amenity and rural atmosphere. It was still primarily an administrative and commercial centre.

The City in 1971

1.23 The thirty years between 1941 and 1971 saw tremendous growth in population and economic activity in and around the City. The population first passed the million marks around 1943 and then doubled itself in a short span of about twenty years to cross, the two million mark. This happened in spite of two adverse events, viz., the threat of Japanese invasion to Madras in 1944 and the reconstitution of Madras Presidency on a linguistic basis in the fifties, when Andhra, Mysore and Kerala states were formed. In 1950 the boundary of the City was extended to cover 129 sq.kms. by the inclusion of Saidapet and Sembium. This period also saw the growth of new residential as well as industrial suburbs particularly on the west and south.

1.24 The main reasons for this fast growth can be attributed to the forces of economic activity released after the country obtained independence. The five-year plans and the impetus given by the industrial activity in the public sector brought about the transformation of the City from that of a purely administrative and commercial centre into a metropolis of national importance. The major developments in the industrial field during the post independence era have been, the location of a number of public sector undertakings. This was followed by many private sector undertakings. Concurrently this period saw the deterioration in water supply and drainage services and mushrooming of many slum areas all over the city. The setting up of the Tamil Nadu State Housing Board however, helped in the creation of large residential areas like Anna Nagar on the west and Sastri Nagar on the south. The City's boundary no longer remained well defined. The developments extended into the adjoining areas, particularly, on the north up to Ennore, west up to Avadi and south up to Vandalur.

This growth did not take place in a regulated manner nor did it correspond to the available infrastructure facilities. This fact coupled with the rapid growth of population on the one hand and the increase in number of motor vehicles on the other has given rise to the many problems faced by the Metropolis today. The growth of the City since 1633 is depicted in the Sheet No.1.01 and 1.02.

Structure of the Metropolis in 1971

1.25 The City is at the core of the metropolitan area and is the centre for all commercial and social activities as well as a living area for majority of the total population. Its structure approximated to a semi-circle with extensions in five main directions, the North, Northwest, West, Southwest and South. This is due to the fact that early in the development of the city, George Town and Harbour became the commercial centre of the City. Naturally all communication lines led to this centre and these in turn were linked with each other producing a radial and ring pattern of development. When the City grew, lands on the main arteries were much nearer in terms of time needed to reach the centre than areas, which were away from them, and further expansion took place on these arteries. The economics of transportation has led to formation of development corridors the most important of which are on the west and southwest where, physical conditions were more favourable for development.

1.26 The fringe areas in the north, west and Southwest had been the location of large industrial establishments. The Madras Refinery, the Madras Fertilisers, Ennore Thermal Plant, the Ashok Leyland Truck Factory and many accessory industries are located in the north at Manali and Ennore. The T.V.S. Group of companies, Dunlop Rubber Company, T.I. cycles and the larger industrial estates are located on the west at Ambattur with the Heavy Vehicles Factory located further west at Avadi. Almost all cinema studios and a number of medium industries including the first industrial estate of the State - the Guindy Industrial Estate - are located in the southwest between Arcot Road and Annasalai. A large industrial estate at Ambattur was located in the West along MTH Road and the Standard Motor Factory (now closed) was located in the south at Vandalur along GST Road. The main residential areas outside the City are along the Southwest closely following the electrified suburban line and Alandur, Tambaram, Pallavaram are such residential areas. Most of these areas however lack in commercial and public facilities for which they depend on the main City.

1.27 Triplicane, Mylapore, Purasawalkam and also the northern part of George Town and Royapuram are the old residential areas characterised by street housing (i.e. houses constructed in a row without side open spaces) with shopping along main streets. New residential areas of Mylapore and Theagarayanagar had developed with bungalow type houses but densities here were higher than the exclusive older residential areas of Egmore, Nungambakkam and Chetpet, which lie between these areas and the central business district. This intervening area was developed for multi-storeyed apartments, offices and hotels in view of their relative proximity to the central business district.

The rural areas fringing the city were generally fertile then, particularly on the north and west where paddy was raised as an irrigated crop. Casuarina wood for fuel was raised all along the sandy coast in the southern part of the metropolitan area.

1.28 George Town and its extension southwards into Anna Salai together constitute the central business district of the City where most of the wholesale trade, specialised retail trade and banking and financial institutions were located and where commercial activity was intense. More than 2 lakh work places (jobs) constituting 48 per cent of the total work places in the Madras Metropolitan Area in 1971 were located here. Shopping facilities of local significance had developed along almost all major roads.

1.29 The industrial areas within the City were mainly in the northern and western parts, where they were closely mixed up with residential developments but new industrial establishments as mentioned were located outside the City limits and many important industries were on the outskirts. The main industrial areas are in Ennore and Manali on the north, Ambattur and Avadi on the west, along Anna Salai and its extension on the southwest. Nearly 40 percent of the industrial work places were located on the north and northwestern part of the Chennai Metropolitan Area and over 10 per cent along Anna Salai and its extensions.

1.30 Public and semi public uses were dispersed but those, which were established earlier, were concentrated in the older parts of the City such as Egmore and on the Marina. New public offices were located in Nungambakkam and on Anna Salai. Public uses, except for higher educational institutions were few in the suburbs.

1.31 The radial type of development has created green wedges, the edges of which in some cases penetrate into the City boundary itself. The main communication routes were lined with industrial establishments and shopping but the development is diffused and could not be termed "urban" except in uses. The real rural area was not far from these corridors and urbanisation had scarcely touched those areas either in the physical or social sense. Many of the small settlements in fact did not exhibit any sign of being near a large metropolis. Urban development in CMA in 1973 is shown in the map annexed.

Structure of the Metropolis in 2006

1.32 The Master Plan for CMA approved in 1976 proposed the structure of the Metropolis taking into account various factors. The structure proposed was of radial pattern with City as hub. The main elements of the strategy were[2]:

(a) Restriction of density and population growth in the city;
(b) Restriction of industrial and commercial developments within CMA;
(c) Encouragement of growth along the major transport corridors and development of urban nodes at Manali, Minjur, Ambattur, Avadi, Alandur and Tambaram;
(d) Dispersal of certain activities from CBD; and
(e) Development of satellite towns, beyond CMA at Maraimalai Nagar, Gumidipoondi and Thiruvallur.

[2] Ref. Structure Plan for CMA, 1980.

1.33 The Master Plan included spatial plans for CMA with land use zoning and set the urban form during the Plan period. TNHB has played a major role in development of large scale neighbourhood schemes; Anna Nagar (about 5 Sq.km.), Ashok Nagar and K.K. Nagar (about 7 Sq.km.), South Madras Neighbourhood scheme comprising Indra Nagar, Sastri Nagar and Besant Nagar (about 4 Sq.km.), etc. had come up with plots/flats meeting the requirements of all sections of the society from HIG to EWS. Under MUDP-I (1977-92), MUDP-II (1983-87) and TNUDP (1988-95) Neighbourhoods at Arumbakkam, Mogappair, Villivakkam, Velachery, Kodungaiyur, Ambattur, Avadi, and Madhavaram etc. were developed within the City and its suburbs. These large-scale neighbourhood developments of TNHB with full infrastructure acted as catalyst for private developments for residential use around.

1.34 A number of medium and small-scale industrial developments came up in the areas zoned for industrial developments at Madhavaram, Vyasarpadi, Kodungaiyur, Ambattur, Noombal and adjoining areas Pammal and Perungudi. Industrial estates at Villivakkam, Thirumazhisai and Industrial estate for women enterpreneurs were developed at Morai near Avadi. Considering the demand, infrastructure availability, potential for industrial developments, the lands along the Poonamallee bye-pass road near Poonamallee Town in the west and Old Mamallapuram Road (from City limit to Sholinganallur) in the south were comprehensively reclassified for industrial use where a number of medium scale and small scale industries had come up.

1.35 Though Vallalar Nagar (George Town) and Anna Salai area have continued as CBD, Mylapore, Thyagaraya Nagar, Nungambakkam, and Purasawalkam have developed as Regional Commercial Centres and Adyar, Anna Nagar and Perambur have developed as Regional Commercial Sub-Centres. Relocation of wholesale activities in Vegetable, Fruit and Flower from CBD to Koyambedu and City bus stand from CBD to Koyambedu have been carried out; Iron and Steel Market at Sathangadu was developed. Truck terminal at Madhavaram was made operational.

1.36 Major higher educational institutions are located in the central part and southern part of the City and they continued to expand within its premises. After 1980, a number of private engineering and medical institutions have come up dotting the Metropolis.

1.37 In the last three decades, the agricultural activity within this Metropolis had become abysmally minimal for various reasons including non-availability of water for irrigation purposes, labour cost, and cost of agricultural inputs. In the northern wedge between GNT Road and T.P.P. Road, because ayacut rights of Puzhal and Redhills lakes had been cancelled and local water sources are not adequate, the agricultural activity in these areas is very minimal. Similar is the position in the northwestern wedge between CTH Road and GNT Road, which lie in the catchment area of these lakes, which are the main sources of water supply to the City. The Chembarambakkam lake, another large lake in CMA, is being converted as another source for city water supply and very minimal agriculture activity is being carried out in its ayacut area which lie in the south-western wedge. In the southern wedge between Rajiv Gandhi Salai (Old Mamallapuram Road) and GST Road only in few pockets to a limited extent the agriculture activity in the south of Tambaram - Madipakkam Road continue.

1.38 Chennai has become one of the preferred destinations for IT / ITES companies. Tamilnadu is the second largest software exporter in the country, and 90% of the export is from Chennai alone. A large number of IT / ITES developments are located along the Rajiv Gandhi Salai (OMR), the area popularly known as IT Corridor. TIDEL PARK, a self contained IT Park developed with a total floor area of 2.5 million sq.ft. at Taramani houses all the major players in IT Sector. In this corridor, from Taramani to Semmancheri a number of I.T developments are coming up because of locational advantages for the industry such as uninterrupted quality power supply, better transport facilities, accessibility to airport, availability of potable water supply, pollution free environment etc. These developments are spilling over in the areas in the west along the 200ft. wide Pallavaram - Thoraipakkam Road, and also along Velachery – Tambaram Road. In the large I.T Park (layout with all required infrastructures, developed by government agency over an area of 868 acres) at Siruseri, a number of I.T.Parks are being located and will be fully functional by 2007. Bio-tech Park at Taramani was developed by TICEL, and a major bio-tech park came up at Sirucheri. These employment-generating developments resulted in fast development of the areas in the west of the corridor such as Medavakkam, Jalidampet, Perumbakkam, Vengai vasal, Gowrivakkam, Rajakilpakkam etc. as residential areas.

1.39 Chennai is emerging as a major export hub in the South East Asia. International car manufacturers such as Ford, Hyundai, and General Motor etc. have established around Chennai their manufacturing bases to cater to domestic and international markets. New testing and homologation centre for automobile sector with an investment over Rs. 1000 cr. is being established in this region. Saint Gobin glass factory, Nokia cell phone manufacturing industry at Sriperumbudur, and Mahindra Industrial Park developed over 1700 acres, near Maraimalai Nagar new town are some of the major developments around Chennai. All these developments made considerable impact on the structure of this Metropolis. Urban development in CMA in 2006 is shown in the map annexed.

Chapter - II

DEVELOPMENT PLANNING IN CHENNAI METROPOLITAN AREA

Any new plan for the CMA should build on the information of the past. The following are the major plans that had been prepared for Chennai.

(i) General Town Planning Scheme (1957) prepared by Madras Corporation

(ii) The Madras Interim Plan [1967] prepared by D.T.P., Govt. of Tamilnadu

(iii) Madras Metropolitan Plan 1971-91 (1971) prepared by multi- agency group and published by RD&LA Dept., Govt. of Tamilnadu

(iv) Madras Urban Development Project (1974) prepared by MMDA (now CMDA)

(v) Master Plan for MMA (1975) prepared by MMDA (now CMDA)

(vi) Structure Plan for Chennai Metropolitan Area (1980) prepared by CMDA with Alan Turner & Associates as consultants

General Town Planning Scheme (1957)

2.02 The General Town Planning Scheme was prepared by the Madras Corporation and submitted to the then Madras State Government in 1957. The aim of the General Town Planning scheme was to provide for the organised growth of the then Madras City (129 Sq.km.). The Scheme was then remitted to the Directorate of Town Planning, Madras for technical scrutiny. The DTP after careful scrutiny of the scheme had recommended that more detailed and extensive studies in depth would have to be carried out before the scheme could be finalised. Then it was decided to prepare a comprehensive development plan for Madras Metropolitan Area and as a first step to prepare Madras Interim Plan (1967).

Madras Interim Plan (1967)

2.03 The Directorate of Town Planning had prepared the Madras Interim Plan in 1967. In the Plan, the problems of the City were viewed and appraised in the urban context of the urban area being the first of its kind; the Plan devoted itself purely to physical aspects of development and recommended that the fiscal plan should be separately worked out. It was also envisaged that the Master Plan would be in two stages viz. the preparation of an Interim General Plan in the first instance and Comprehensive Development Plan in the second.

Madras Metropolitan Plan 1971-91 (1971)

2.04 To explore avenues for securing financial assistance for the city development from the national and international institutions, it was found then that the plan prepared earlier were inadequate in regard to long range prospective and also the scale of financial effort needed to direct the future development in orderly manner. Then the Government decided to prepare a report which would not only indicate the dimensions of the problems ahead, but also the necessary efforts - physical and financial - needed to make Madras a living city.

2.05 The proposals made in the Madras Metropolitan Plan 1971-1991 mainly contained the following:

(1) Strategy for physical growth of urban areas

(2) Programmes for land acquisition for urban expansion, provision of infrastructure for transport, water supply, sewerage and drainage, slum clearance, provision of facilities for education, health, recreation and refuse collection

(3) Action to bring revenue base in the area in line with the levels of expenditure and

(4) Recommendation on institutional set up required to implement the Plan.

2.06 The Madras Metropolitan Plan achieved its aim of looking to the longer term future and costing of various programmes of developments. It also clearly illustrated the magnitude of the tasks faced in managing the development of the Metropolitan Area.

Madras Urban Development Project (1974)

2.07 MMDA (now CMDA) was set up as an adhoc body in 1973 and became statutory in 1975 (under the Tamil Nadu Town and Country Planning Act, 1971 which superceded the then existed Madras State Town Planning Act, 1920).

2.08 The Madras Urban Development Project report was prepared by MMDA updating the earlier data and presenting a more viable case for exploring additional resources for investments from various financial institutions including World Bank. The report identifyied the overall problems and the needs of the metropolitan area

2.09 It highlighted the then present deficiencies and future requirements in some of the critical sectors and suggested an investment programme for 1974-79. It also stressed the need for land use control.

2.10 The Madras Urban Development Project report represented a further shift towards concentrating on the positive aspects of managing the development of the MMA and it was concerned primarily with presenting a package of realistic projects that could be implemented within a period of 5 years. The Madras Urban Development Project became the basis for appraisal of projects under the World Bank assisted Madras Urban Development Project-I (1977-82) executed at a cost of Rs.56 Crores, MUDP-II (1983-88) executed at a cost of Rs.73.9 Crores and TNUDP-I Chennai Metropolitan Area component executed at a cost of about Rs.450 Crores.

Master Plan For MMA (1975)

2.11 As per the Tamil Nadu Town and Country Planning Act, 1971 one of the statutory functions of CMDA is to prepare Master Plan for the Metropolitan Area. Drawing heavily on the Madras Urban Development Project report, the Master Plan was prepared also updating the land use surveys conducted in 1964. Land use and Development Control Regulations had become part of the Master Plan. On the resource aspects, it echoed the recommendations of the Madras Metropolitan Plan, 1971.

2.12 The first Master Plan for CMA laid down policies and programmes for overall development of CMA taking a long-term view of the requirements. As stated in its preamble, in particular, dealt with the following aspects:

(1) The present trends of the population growth and the future population in the Metropolitan Area and approximate distribution of population in various parts of this Area

(2) The economic level and activities in the Metropolitan Area and policies for future economic growth as well as future location of economic activities

(3) The (then) present physical structure of the Area and policies for its future physical development

(4) The (then) present traffic and transportation and circulation pattern and policies and programmes for efficient functioning of these sectors

(5) The manner in which land and buildings in the area should be used and allocation or reservation of land for residential, commercial, industrial, institutional, recreational, agricultural and other uses taking into consideration the future needs of the population that will live in the Area

(6) The requirements in respect of urban infrastructure viz. water supply and drainage, housing, education, medical facilities and recreation and the policies for their future development

(7) Policies and programmes for sectoral development as well as detailed development of areas for housing, shopping, industries, civic amenities, educational and cultural and other related facilities

(8) Regulation of the use-zones, the location, height, size of buildings, size of yards, and other open spaces to be left in and around the buildings, and also regulation of land sub-division, architectural and advertisement control and preservation of trees, historical and scenic areas and

(9) The areas proposed for acquisition for purposes of urban development.

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2.13 Salient features/main recommendations made in the first Master Plan are:

(1) The proposed population for 1991 and 2001 for Chennai City (128 Sq.m. then) was 3.60 million and 4.0 million respectively and for CMA was 5.80 million and 7.10 million respectively.

(2) Considering the economic characteristics of the metropolitan area (then), the Plan listed the following points relating to economic policy:

(a) The State income and income of the Metropolitan Area are low. It is apparent that the incomes cannot sustain, without outside assistance; massive expenditure on infrastructure in urban areas which in themselves are necessary to promote development. Development projects would need to be oriented towards raising levels of income both at the State level and the Metropolitan level.

(b) The State ranks third (then) in the country in the level of industrialisation but it is far behind Maharashtra and West Bengal. Only large-scale industrialisation can raise income levels since much effort has already been made to increase agricultural incomes and further efforts in this sector are not likely to yield higher results due to constraints like non-availability of cultivable waste lands and inadequate water for irrigation.

(c) Madras is predominantly an administrative and commercial centre. Its industries are of the service and light type with medium range employment and this character needs to be strengthened. The industrial policy for the area should therefore be oriented towards small scale and service oriented industries to satisfy, employment needs of the resident population rather than location of large-scale industries which breed a host of ancillary industries. Such large-scale industrial activity could be more usefully located in the hinter land of Madras

Metropolitan Area. Further utilisation of the capacities of existing industrial units should also be ensured without correspondingly increasing employment in order to raise industrial incomes.

(d) In view of the availability of a modern and major port, export oriented industries with a "high value added" character will need to be considered.

(e) Tertiary sector employment would need to be raised but all employment increase should be within the framework of the State employment policy in order to regulate migration.

(3) The only way to regulate economic activity, optimise the use of existing (then) infrastructure facilities and plan their expansion to meet future needs in an area is to regulate land use and building activity therein.

(4) The strategy of developments of radial corridors linked to satellite towns was found most suitable (then). The form suggested for future development of MMA envisaged creation of six major urban nodes at Manali, Minjur, Ambattur, Avadi, Alandur and Tambaram along 3 corridors apart from limited development along GNT Road, Poonamallee Road and Lattice Bridge Road.

(5) Each major node was designed for a population of 200 to 300 thousands and would be predominantly and self-contained unit providing for a substantial percentage of land for work places, schooling, shopping and other day to day needs.

(6) The nodes would be connected with rapid rail system and expressways to the city on the one hand and to the satellite towns on the other (Satellite towns proposed (then) were at Gummidipoondi, Thiruvallur and Maraimalai Nagar).

(7) The nodes amongst themselves would be connected by both rail and road systems to facilitate intra-urban movement.

(8) The population within City in 1971 was 2.47 million and the city is bound to grow in population at (then) present rates for at least next 5 years (4.28% for the City and 6.3% for the Madras Urban Agglomeration). The main reason for such assumptions are:

(a) The urban nodes will take time to develop and absorb the future urban population;

(b) The areas, currently being developed as residential neighbourhoods within the City would themselves contribute to increase the City's population substantially, and

(c) There are still some vacant and undeveloped lands within the City, particularly on the fringes where development potentialities exist and

(d) The rate of development within City will however taper off once the development of urban nodes gains momentum and the level of population within the City would then get stabilised.

(9) Based on the Traffic and Transportation Plan for MMA, 1974 (prepared by Madras Area Traffic Study Unit of D T & C P.), a modal split of 80:20 was estimated for the year 1991 between the public mass transit modes and private personal transport (against 51:49 existed then bulk of private mode being by cycles and walk). The important projects identified in the Master Plan (then) for implementation were:

Railways:

(a) Construction of a mass rapid transit system along the north-south eastern corridor between Manali and Thiruvanmiyur;

(b) Introduction of electrified suburban train system on Madras-Thiruvallur and Madras-Minjur lines;

(c) Construction of a combined railway terminal; and

(d) Construction of a circular railway.

Roads:

(e) Widening of the arterial roads to carry six lanes of traffic with separate cycle tracks and footpaths:

(i) Anna Salai

(ii) Poonamallee Road including a fly-over opposite Central Station

(iii) Walltax Road and G.N.T. Road

(iv) Arcot Road

(v) Edward Elliots Road and its extension up to Poonamallee High Road.

(vi) Flowers Road and its extension up to G.N.T. Road

(vii) Adams Road and its extension upto Poonamallee High Road and Sydenhams Road up to its junction with G.N.T. Road

(viii) North Beach Road and its extension to Ennore

(f) Replacement of the 26 level crossings over railways with grade separators

(g) Construction of a new western expressway connecting Anna Nagar with Avadi

(h) Construction of the missing links of the inner ring road within the City and construction of intermediate and outer ring roads

Others:

(i) Construction of three terminals for long distance buses and truck terminals on the radial corridors at their junction with the outer ring road.

(10) The standard of water supply adopted for the estimates of Corporation of Madras and Tamil Nadu Water Supply and Drainage Board for the requirements of water for MMA was 227 lpcd. Recognising the severe limitation of the (then) present sources of water supply, it was suggested that number of studies covering all the aspects of water supply system would have to be carried out before a comprehensive programme for water supply system was put through. Similarly before investing on sewerage a full scale investigation was necessary. The success of development of urban nodes for dispersal of population in the MMA so as to restrict the population within (then) present limits to manageable proportions depends upon the speed with which water supply and drainage scheme had to be implemented in those areas.

(11) About 33.5% of City population [7.37 lakhs] lived in 1202 slums in the Chennai city in 1971. Considering (then) current and future requirements, a massive housing programme would have to be initiated and put through; in 10 years 12,000 hectares of land to be acquired (2500 Hectares each in Manali, Avadi, Ambattur and Tambaram, 1500 Hectares in M.M. Nagar, 750 Hectares each in Gummidipoondi and Thiruvallur and developed for housing. Most of the new housing by the public agencies except that intended to rehabilitate slum dwellers on the same site would have to be necessarily provided in the urban nodes as sufficient developable land within City is not available.

(12) Primary education was the responsibility of local bodies, but privately run schools provide school facilities for a large segment of the population then. Recognising the need for improving school facilities it is observed that policy for the provision of future education facilities should aim at organising school buildings and playgrounds in proper locations. Advance action was called for to reserve adequate lands and locations convenient and safe for children. Depending on the need for the community, it was felt necessary to plan future locations for college and higher education institutions.

(13) Observing the over-crowding in wards and other spaces in the hospitals, and ever-increasing demand for hospital facilities (adopting a standard of 500 beds for each hospital on the basis of 4 beds per 1000 population) it was recommended that on an average one hospital will have to be commissioned every year.

(14) Emphasising the need for providing adequate recreational facilities in the shape of parks, play ground and open spaces to serve all levels of population in urban areas, it was recommended as follows:

a) 80 hectares for one lakh population for metropolitan parks, and 400 hectares of land along each of the 3 major corridors, and 800 hectares along the coast in the north and south are required for provision for regional level recreational facilities.

b) Local level recreational facilities have to be provided at the rate of 0.8 hectares per 1000 population for parks and play spaces. The spatial distribution of these facilities would largely depend on the availability of suitable land, particularly in built-up areas and hence had to be taken up at the level of Detailed Development Plans. The total requirement of open space in MMA in 1991 was 9600 hectares (96 Sq.km.)

(15) Development in the Metropolitan area was taking place at the rapid rate especially on the National Highways leading to the City and to a lesser extent on the other roads. One of the major principles underlying the Plan was securing of balanced development by decentralising the places of employment and residential areas. The Plan had allocated land for industries, commerce, housing, play fields and other types of major urban land uses in appropriate locations and interrelated to each other so as to promote orderliness and smooth functioning.

(16) Each land use zone had its special regulations designed to protect residential and recreational areas from harmful invasions of commercial and industrial uses and at the same time promoting business and industry by diverting them to most suitable places. By regulating the spacing of buildings, floor area ratio, set-backs, parking etc. the Plan aimed to ensure adequate light, air, fire protection etc. and to prevent over-crowding in buildings and land and thus facilitating the provision and continued adequacy of water, sewerage, transportation and other facilities.

(17) The proposed land use plan contains 10 land use zones. In each use zone, certain uses would be permitted normally, other uses may be permitted on appeal to the CMDA and all other uses not specified therein would be specifically prohibited. Planning parameters had been prescribed differentially for 3 categories of areas viz. (i) Continuous Building Areas and George Town, (ii) Madras City excluding GT and CBA, and (iii) rest of CMA.

(18) It was observed that the land use plan was in essence a translation into physical form of planning policies and principles. The policies had taken into account the realities of the (then) present situation and were designed more to channelise future development on orderly lines rather than effect wholesale change in (then) existed development.

(19) The proposed land use break-up for various uses as per the Master Plan (1975) is given in table 2.01

Table No.2.01 Proposed Land use under First Master Plan for CMA

Sl.No	Land Use	Chennai City		Rest of CMA	
		Extent in hect.	% to total extent	Extent in hect.	% to total extent
1	Residential	8,081.98	48.57	32,255.78	30.98
2	Commercial	973.28	5.85	895.42	0.86
3	Industrial	1,107.51	6.66	6,361.62	6.11
4	Institutional	2,746.43	16.51	4,935.20	4.74
5	Open space & Recreational	3,254.11	19.55	7,767.21	7.46
6	Agriculture	-	-	50,924.14	48.91
7	Non-Urban	476.11	2.86	978.71	0.94
	Total	16,639.42	100.00	1,04,118.08	100.00

Source: Draft Second Master Plan 2011 (prepared in 1995)

Structure Plan for CMA (1980)

2.14 The structure plan was prepared in CMDA in association with M/s Alan Turner and Associates in 1980 with the assistance of Overseas Development Agency of U.K. Earlier planning in Chennai Metropolitan Area were reviewed and the following main recommendations were made in the plan:

Suggested Strategy

1) The Master Plan strategy which aimed to restrict the size of the City is in need of considerable amendment owing to much higher population growth than previously predicted and the distribution of population and new development, it is recommended that an alternative strategy should be adopted which combines limited intervention with an acceptance of the pressures for growth.

2) Policies for growth should embrace a realistic, acceptance of growth trends and should not attempt to intervene in an impracticable manner. Programmes, which are capable of implementation in the short term, should be devised within the framework of longer-term strategy.

3) In developing the programmes of the various sectors in pursuit of a chosen strategy, it is essential to adopt realistically achievable targets. In this respect, financially self-supporting projects involving maximum cost-recovery should be developed wherever possible. Cross subsidies will help to create viable projects for the lowest income groups.

4) Particular attention should be given to monitoring migration into and out of Madras - not only in terms of numbers, but also in terms of understanding the reasons for migration and the difficulties encountered in being absorbed into city life.

5) Further study should be made of the structure of the population - particularly of changes in the structure - with a view to determining future demand for services and facilities with greater accuracy.

6) Policies aimed at reducing population growth in the MMA over the long term should be pursued rigorously. They should take two forms:

a) Family Planning Programmes
b) Programmes of regional development aimed at reducing the perceived relative attractiveness of metropolitan city life as opposed to rural or small town life.

7) Priority areas should be designated for which Action Plans should be prepared. These should contain social, economic and physical components prepared in greater depth than the current Detailed Development Plans, which are restricted largely to land use planning. Special action planning terms should be set up within MMDA to prepare Action Plans and co-ordinate them with the programmes of other agencies.

8) The efficient use of scarce resources, particularly of water supplies should be considered to be paramount in determining the pattern of future urban growth.

9) The location of new squatter settlements will need very careful monitoring owing to their influence on the pattern of urbanization, particularly with regard to the provision of new infrastructure.

10) In view of progress at Marai Malai Nagar and Manali, MMDA should carry out a review of their present role in the emergent growth strategy for MMA.

11) In addition to the urban nodes, which form a part of the current Master Plan, other local centres should be encouraged to develop in order to spread employment and service facilities throughout the urban area. A policy of multiple centres should help to provide more balance in transport facilities.

12) Industries with water borne polluting effluent should not be allowed to develop south of Madras, where owing to the predominantly south/north drift they could pose a threat to the Marina Beach.

Land

13) Development should be prohibited in some areas where there are natural constraints (such as flood plains of the foreshore area) and restricted in others. Development policies should seek to discourage development in these areas and to encourage development in defined areas, which will tend to be on the periphery of the City. Detailed geologic maps are needed to determine more accurately the location of constraints.

Areas under considerable pressure for urban development (the 'urban fringe') should receive an adequate share of resources, since it is here that many subsequent urban problems are likely to have their origin.

14) A policy of "green wedges" should be adopted to protect agricultural areas between development corridors. These areas would also provide space for recreation.

15) In certain priority areas, MMDA should adopt the role of principal developer in the assembly and disposal of land to meet the various development needs.

16) An open space policy should be formulated which would define areas for both productive uses and recreation and would set standards and guidelines for their use and maintenance.

17) Urban residents should be encouraged to grow their own fruit and vegetables, wherever they have sufficient space. This will help to achieve the best use of private open space.

18) In order to monitor the physical growth of the City, aerial photographs should be taken every five years; the results should be compared with data on population distribution from the Census. In this way the forecast trends in growth can be confirmed or modified.

Rural Development

19) Detailed Studies should be carried out of the rural/urban interface in order to formulate integrated policies. These would help to define in greater detail, which areas should remain rural and which should be developed for urban purposes. Integration will help to minimize waste of scarce land and water resources.

20) A concerted programme of afforestation should be formulated for MMA, in conjunction with the provision of open space and reclamation of derelict land; generally this would be on poor land unsuitable for agriculture.

Financial Resources

21) MMDA should establish a Programme Budgeting Division in order to develop a five-year rolling programme of capital investments, relevant to the selected strategy and to monitor and evaluate results through a series of performance indicators.

22) MMDA should initiate a detailed sectoral analysis with the ultimate aim of linking all the agencies investment programmes in pursuit of a common development strategy consistent with their own objectives

23) MMDA and other public agencies should aim to capture the increment in land value caused by public land development for the benefit of the community as a whole. This will ease the burden of providing relief for the poorest sections of society.

24) Every opportunity should be found to enable the private sector to play a full part in the development process.

Employment

25) Within the framework of national policy a regional strategy should be formulated, which would help MMA through direct investment, in setting up large public enterprises and the extension of financial incentives. The strategy would seek to promote agriculture, fishing and industries catering for the export market.

26) Complementary income generation programmes should be developed in parallel with all sites and services and slum upgrading projects. These should include, the promotion of small business, and informal activities and the provision of short vocational training courses.

27) A series of action programmes for the stimulation of the small business sector should be developed in collaboration with the relevant agencies.

28) A large number of agencies and public corporations are engaged in economic development. In several cases functions overlap and it is recommended that management studies be carried out with a view towards rationalization and amalgamation.

29) Additional investment in electricity generation and distribution should be given priority in the immediate future. Electric power is in short supply and the numerous power cuts are impeding industrial production and growth.

30) Major industrial development should be directed towards the existing industrial areas and zones, largely in the peripheral areas, with good communications and a nearby resident labour force.

31) Retail, service and small office activities should be encouraged to succeed the wholesale merchants relocated from George Town. A policy of upgrading infrastructure and buildings should be followed.

Shelter

32) TNHB should put much more emphasis on low cost housing especially sites and services programmes rather than the current schemes, which favour middle and higher income groups.

33) To encourage greater private sector participation a number of measures must be taken including a review of the Urban Land Ceiling Act, the use of licenses to construct housing on government land, better access to credit facilities and the provision of more housing for workers by major employers.

Transport

34) The various significant transport proposals should be kept under review to determine their viability, timing and role in shaping the overall development strategy. Such proposals include - the inner circular railways, and the intermediate and outer ring roads.

35) Land use planning should attempt to ensure a balance between resident population and facilities over any 3 km, radius, particularly in terms of housing and jobs for the lower income groups.

36) The First Phase of the MRTS (Madras beach to Luz) has reached a point in its processing where it must be regarded as a committed project, it is essential to maximize the advantages to be gained from the investment by planning appropriate developments at the stations along the line. MMDA should carry out a special study in collaboration with MTP(R).

37) A network of arterial roads, which can be developed as the main channels for vehicular movement, needs to be identified. The districts bounded by the arterials each need to be examined to ensure that internal roads will not become mini-arterials.

38) A long-term strategy for roads should aim for better use of existing roads and improvements of conditions for all highway users. The various highway users should be better segregated than at present - particularly local and longer distance travel. The environment for activities along the roads should be improved.

39) Special routes in some areas should be designated for trucks and other streets should be closed to trucks over a certain weight.

40) A parking policy should be developed with restrictions enforced in the more congested areas. Off-street facilities should be provided for certain new developments provided that rigorous enforcement of street parking exists in the area.

41) Apart from the trunk routes along those corridors not served by rail, bus routes should be designed to give a more local service, providing a feeder service to the corridor trains or buses. In addition cross-town routes will be necessary. The attitude of experimentation with new routes, together with continuing market research into route requirements, should be encouraged.

42) A detailed study should be made of the potential for constructing effective bus/rail interchange stations in the first instance on the Tambaram line at Saidapet, Nungambakkam and Chetput.

43) Cyclists deserve special consideration in the Structure Plan. Further detailed study of cycling habits appears to be justified as does construction, if at first on an experimental basis, of some lengths of exclusive cycle-way. It has been suggested that the banks of the Cooum River or the Buckingham Canal might be a suitable location for such an experiment.

44) The improvement of sidewalk facilities for pedestrians is probably the single most important improvement needed to make the best use of road space. Maintenance and improvement programmes should aim to provide continuous smoothly paved sidewalks on all roads. Where possible, routes independent of vehicle routes should be developed.

45) MMDA should formulate integrated policies of street management (including traffic management and highway maintenance) and co-ordinate the activities of the local bodies, the police, the highway authorities, and public transport authorities and works departments. This is essential, in order to make more efficient use of the existing street network.

Education and Health

46) Educational policies should be directed more towards the needs of the lowest income groups, with emphasis on primary education and vocational training.

47) MMDA should co-ordinate the programmes of operating agencies so as to increase the number of hospital bed-spaces; upgrade existing family and child welfare centres; construct new maternity and child welfare Homes; extend the programmes for medicare centres and pre-school care.

Water Management, Water Supply, Drainage and Sewerage

48) It is imperative to identify new sources of water in the short-term, if lack of water is not to pose a serious impediment to growth and development. This is important both in urbanizing areas and in areas identified for long-term agricultural use.

49) An integrated scheme should be prepared for the reuse of wastewater, together with a comprehensive scheme for utilizing existing water resources.

this would reduce the need for imported water from distant sources. Reuse would be related to higher value market garden crops and would enable nutrients to be returned to the soil.

50) The aquifer below the coastal sands south of Madras should be protected from reduction of recharge by urban development.

51) A study should be made to investigate alternative methods of sewage treatment, particularly simple, cheap solutions for use in sites and services or slum upgrading projects.

Detailed Development Plans:

2.15 Detailed development plans are the plan prepared under section 27 of the TN T&CP Act and are more detailed than the Master Plan. It is prepared generally for smaller areas out of about 3 sq.km. The list of approved Detailed Development plans is given in the Table no. 2.02

Table No: 2.02 List of Approved Detailed Development Plans

Sl.No	Name of the D.D.P	Sl.No	Name of the D.D.P
1	Vivekanandapuram Area	29	Todhunder Nagar Area
2	Gangadeswarar Koil Area	30	Nammalwarpet Area
3	Chetpet	31	Thiru-Vi-Ka Nagar Area
4	Chepauk Area	32	Binny Mill Area
5	Guindy Area	33	Ashok Nagar Area
6	Gandhi Nagar Area	34	Perambur North Area
7	Kottur Area	35	Arunachaleswarar Koil Area
8	Periamet Area	36	Theyagaraya College Area
9	Jeeva Nagar Area	37	Rangarajapuram Area
10	Azad Nagar Area	38	Pulianthope Area
11	Krishnampet Area	39	Egmore Station
12	Zam Bazaar Area	40	Perumalpet Area
13	Radhakrishnan Nagar Area	41	Nappier Park Area
14	Kamaraj Nagar Area	42	Govt. Estate Area
15	Avvai Nagar Area	43	Thiruvottiyur TP Scheme No. 8 Area
16	Thiruvottiyur T.P Scheme No.2 Area	44	Thiruvottiyur T.P Scheme No.1 Area
17	Nandanam Area	45	Thiruverkadu Area
18	Urur Area	46	Nakkeerar Nagar Area
19	Killiyur Area	47	Kalaivanar Nagar Area
20	Nungambakkam Area	48	Kanadasan Nagar Area
21	Guindy Park Area	49	Rajaji Nagar Area

22	Mylapore-Santhome Area	50	Anna Salai Area
23	Karaneeswarapuram Area	51	Vallalar Nagar Area West
24	Thiruvatteeswaranpet Area	52	Vallalar Nagar Area South
25	C.I.T.Colony Area	53	Vallalar Nagar Area Central
26	Amir Mahal Area	54	Vallalar Nagar Area East
27	Marina Area	55	Poonamallee High Road TP Scheme Area
28	Azhagiri Nagar Area	56	Saidapet Part II Area

2.16 It is proposed to cover the whole City area for Detailed Development Plans in the next 5 years and also review the earlier Detailed Development Plans.

Chapter III

DEMOGRAPHY

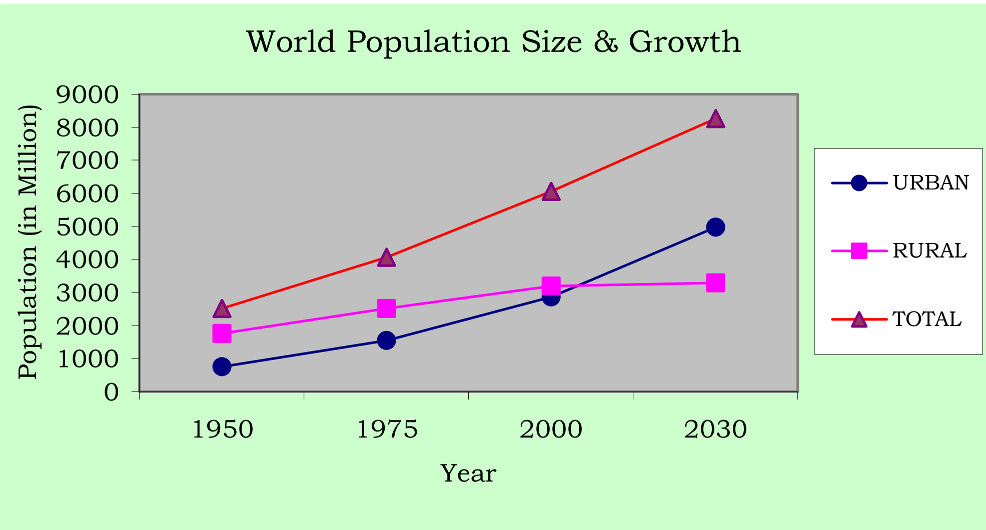
History of the world during the last century has shown that the process of urbanisation and economic growth progress are mutually reinforcing. Cities are the loci and motors of economic and social change. According to United Nations Projections, the World's urban population will grow from 2.86 billion in 2000 to 4.98 billion by 2030 and the World's annual urban growth rate is projected as 1.8 per cent in contrast to the rural growth rate of 0.1 per cent. About 60 per cent of the World's population will live in cities[1] by 2030.

Table No.3.01: World Population Size and Growth, Urban and Rural

	Mid-year population size (million)				Population growth rate (percent)		
	1950	1975	2000	2030	1950-75	1975-00	2000-30
Urban							
World Total	751	1543	2862	4981	2.9	2.4	1.8
High-income countries	359	562	697	825	1.8	0.9	0.6
Middle and low income countries	392	981	2165	4156	3.7	3.2	2.2
Rural							
World Total	1769	2523	3195	3289	1.4	0.9	0.1
High-income countries	219	187	184	139	-0.6	-0.07	-0.9
Middle and low income countries	1550	2336	3011	3151	1.6	1	0.2

Note: High-income countries have gross national income per capita of US$9266 or more based on World Bank estimates.

Source: National Research Council, 2003, P.85.



[1] The state of the World's cities 2004-05 Globalisation and Urban Culture, published by Earth scan in 2004

Urbanisation in India

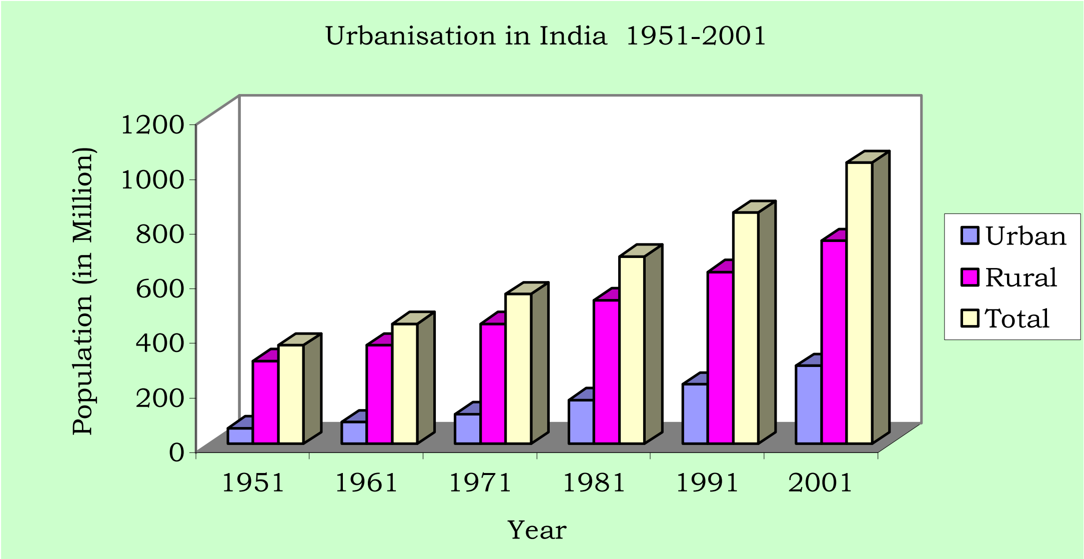
3.02 India, the second largest populous country in the World has one of the longest urban traditions. By the year 2000 BC itself India had an extensive network of towns. Even before the British traders made contact with India (by AD 1600), there was broad regional distribution of towns. These towns were not just of administration, but also of trade and marketing, cultural activities and religious pilgrimage, defence and fortification. New cities - Madras, Calcutta and Bombay were founded by 17th Century. The early British settlements became the means of transforming urban system in India. During the 19th century, the Cities were to be for industrial activities and also became hub of the transport network.

3.03 In a formerly rural economy country like India, because of the need to decrease the number of persons dependent on agriculture and to improve productivity in rural areas, urbanisation is viewed as a prerequisite of growth. The urban population in India has grown from 25.7 million in 1901 to 286.1 million in 2001. After Independence in 1947, the rate of growth of urban population increased from 2.64 percent in 1951 to 3.88 percent in 1981 and thereafter declined to 2.77 in 2001; but the share of urban population to the total population of the country constantly increased from 10.9 percent in 1901 to 15.92 percent in 1951, and thereafter to 27.81 per cent in 2001.

Table No.3.02: Urbanisation in India 1951-2001

	1951	**1961**	**1971**	**1981**	**1991**	**2001**
Urban						
Population (in million)	57.50	78.90	109.10	159.70	217.60	286.10
Annual, Rate of Growth (Percent)	2.64	3.21	3.29	3.88	3.14	2.77
% of urban total population	15.92	17.96	19.90	23.30	25.71	27.81
Rural						
Population (in million)	303.60	360.30	439.10	525.50	628.70	742.60
Annual, Rate of Growth	1.02	1.73	2.00	1.81	1.81	1.68
% of rural to total population	84.08	82.04	80.01	76.70	74.29	72.19
Total						
Population (in million)	361.10	439.20	548.20	685.20	846.30	1028.70
Annual, Rate of Growth	1.26	1.98	2.24	2.26	2.13	1.97

Source: Census of India



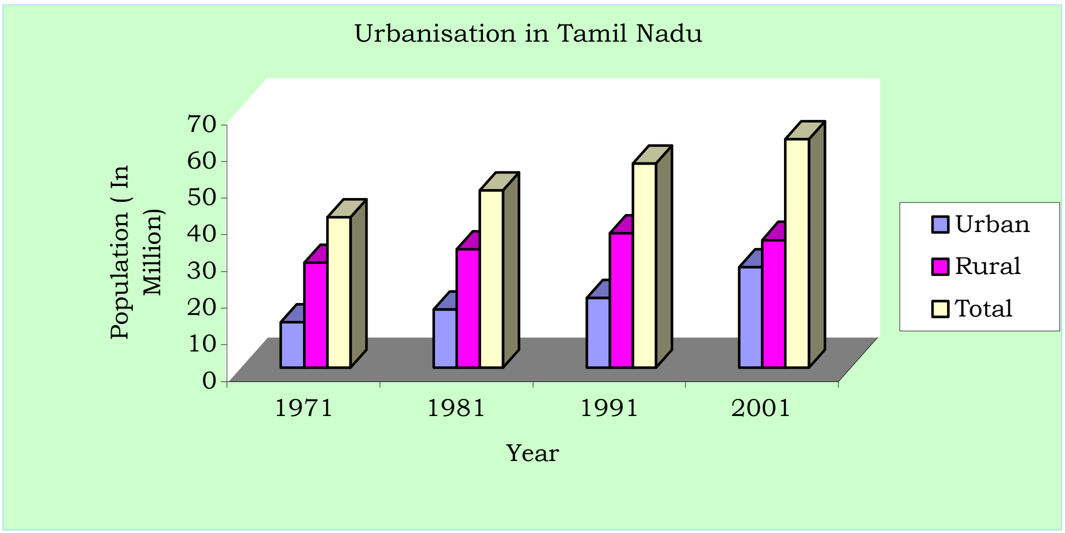
Urbanisation in Tamilnadu

3.04 Tamilnadu has emerged as the third largest economy in India. Cities exist and grow because of economies of urban agglomeration associated with industrial and trade activities. In the recent past, liberalisation, rapidly growing IT sector, an educated, hardworking and disciplined work force etc, accelerating economic development also contributed to the growth of urban areas in Tamilnadu. The extent of the State is 130,058 sq.km. of which the urban area accounts for 12,525 sq.km. Tamilnadu is the most urbanised state in India. It is one of the few states in India with hierarchy of urban areas dispersed fairly uniformly through-out the State. Its urban population has grown from 12.46 million in 1971 to 27.48 million in 2001. Its percentage share of urban population to total population stood always much above the national average.

Table No.3.03: Urbanisation in Tamilnadu (1971-2001)

	1971	1981	1991	2001
Urban				
Population (in million)	12.46	15.95	19.08	27.48
Annual, Rate of Growth (percent)	3.32	2.50	1.91	3.72
% to total	30.26	32.95	34.15	44.04
Rural				
Population (in million)	28.73	32.46	36.78	34.92
Annual, Rate of Growth (percent)	1.53	1.23	1.26	-0.52
% to total	69.74	67.05	65.85	55.94
Total				
Population (in million)	41.19	48.41	55.86	62.40
Annual, Rate of Growth (percent)	2.03	1.63	1.44	1.11
Area in Sq.Km.	130,069	130,050	130,050	130,050

Source: Census of India



Growth of population in Chennai city and CMA

3.05 Chennai (earlier called as Madras) was established in 1639, as one of the East India Company's earliest trading Ports and later became the centre of the company's control over Southern India. By 1700, Madras had become a thriving city with about 3 lakhs inhabitants; most of them lived in the Black Town in the north of the British Fort St. George. By the end of the eighteenth century, according to Dupuis (1968), the north of the city had become profoundly different from the south. The north was densely populated, with Black Town, the heart of the city. To the south were the open spaces and scattered settlements of the Europeans. By the time of the first census in 1871, the city had reached over 4 lakh. The first railway line between Madras and Arcot was opened in 1856 and the Madras Port was improved in 1890, which had attracted industrial developments to the north of the black zones renamed as George Town in 1905[2]. The growth of Chennai City continued in the twentieth century and it has grown to the fourth largest Metro City in India.

3.06 An area about 67 Sq.km. containing 16 hamlets was constituted as the City of Madras in 1798 and subsequently enlarged from time to time. Its enlargement and growth of population since 1901 is given in table No. 3.04.

[2] Urbanisation in India by Robert W. Bradnock, 1984.

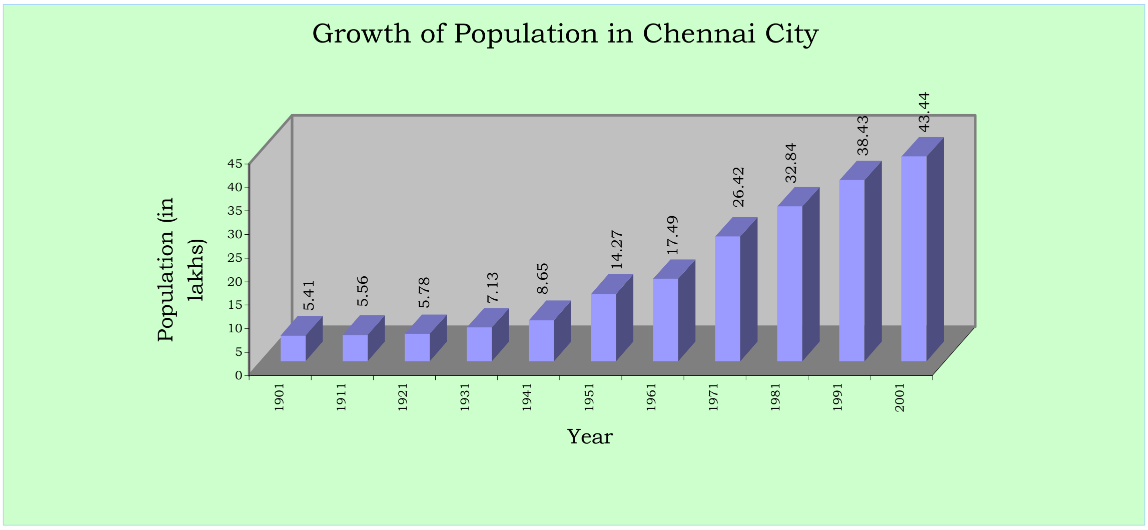
Table No.3.04: Growth of Population in Chennai City, 1901-2001

	1901	1911	1921	1931	1941	1951	1961	1971	1981	1991	2001
City population in lakhs	5.41	5.56	5.78	7.13	8.65	14.27	17.49	24.69	32.85	38.43	43.44
Area in sq.km.	68.17	68.17	68.17	68.17	77.21	128.83	128.83	128.83	176	176	176
Annual Rate of growth of population * (in percent)	--	0.27	0.39	2.12	0.69	-0.11	2.06	3.51	-0.27	1.58	1.23
Population density per Hect.	80	82	85	105	112	111	136	192	187	218	247

*Note: Arrived for a unit area for the purpose of comparison since the city extent varied over time. The figures given are for the decade ended with that year.

Source: Census of India

3.07 The city Corporation area recorded a higher growth of more than 2% per annum during the decades 1951-61 and 1961-71. The reasons for this rapid growth rate can be attributed to industrial developments and increase in economic activities and employment opportunities in the City and its suburbs attracting large migrant population. The negative growth arrived for a unit area during 1971-81 is due to the annexation of lesser dense (then) Panchayat areas around, to the City viz. Velacheri, Taramani, Kanagam, Thiruvanmiyur, Kodambakkam, Saligramam, Koyambedu, Senjery, Thirumangalam, Virugambakkam, Nesapakkam, Kolathur, Villivakkam, Konnur, Erukkanchery, Jambuli, Kodungaiyur, and Selaivoyal in 1978, comprising about 47 Sq.km.



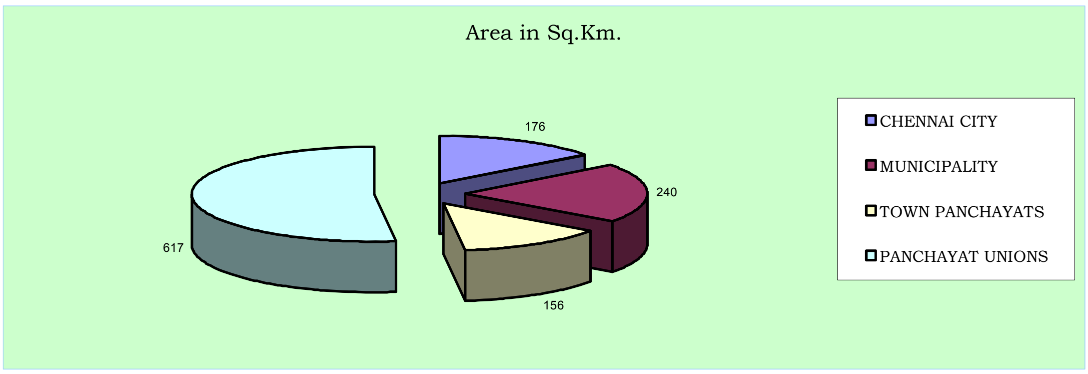
3.08 Chennai City Corporation area consists of 155 divisions within 10 zones (Zone No. I to X) presently. The number of these divisions and its extent varied over time. Hence, for the purpose of comparison, the demographic details have been arrived for the years 1971, 1981 and 1991 keeping the 155 Corporation division boundaries in 2001 as the basis. Spatial distribution of population growth in the City is given in the table no.3.05.

Table No. 3.05: Chennai City - Population Growth in Corporation Zones, 1971-2001

Zone Nos.	Corporation Zone	Area in Sq.km.	Population in Lakhs				Annual rate of growth in percent			Density per Hectare in 2001
			1971	1981	1991	2001	71-81	81-91	91-01	
I.	Tondiarpet	17.30	2.01	2.69	3.72	4.10	2.95	3.28	1.00	243
II	Basin Bridge	11.52	3.59	3.52	3.27	3.76	-0.21	-0.74	1.40	335
III	Pulianthope	13.51	3.34	4.13	4.31	4.60	2.13	0.44	0.64	349
IV	Ayanavaram	19.76	2.22	3.58	4.12	4.97	4.89	1.42	1.89	258
V	Kilpauk	26.38	2.18	3.45	4.94	5.42	4.68	3.66	0.93	211
VI	Ice-House	10.15	3.27	3.49	3.20	3.42	0.63	-0.84	0.65	346
VII	Nungambakkam	12.90	2.91	3.09	3.20	3.48	0.61	0.35	0.83	277
VIII	Kodambakkam	13.00	2.48	3.33	4.39	4.66	2.96	2.81	0.61	368
IX	Saidapet	23.56	1.89	2.61	3.33	4.15	3.25	2.48	2.23	180
X	Mylapore	27.92	2.50	2.97	3.95	4.88	1.70	2.89	2.13	180
	City Total	176.00	26.42	32.85	38.43	43.44	2.20	1.58	1.23	247

Source Census of India

3.09 Population growth in different zones within City Corporation area is found to be not uniform and its rate varied from 0.61% to 2.23%; Saidapet and Mylapore zones have recorded growth rate exceeding 2% during 1991 - 2001.



3.10 Chennai Metropolitan Area comprising City of Chennai and contiguous area around was notified in 1974. It extends over 1189 sq.km. and includes Chennai City Corporation area, 16 Municipalities, 20 Town Panchayats and 214 villages comprised in 10 Panchayat Unions.

Table No. 3.06: CMA - Population Growth in Municipalities - 1971-2001											
Sl. No.	Municipality	Area in Sq.km.	Population in thousands				Annual Rate of growth in percent				Density per Hect.2001
			1971	1981	1991	2001	71-81	81-91	91-01	Avg.	
1	Kathivakkam	4.75	16.14	22.10	27.17	32.59	3.19	2.09	1.84	2.37	69
2	Thiruvottiyur	21.35	82.85	134.01	168.64	212.28	4.93	2.32	2.33	3.19	99
3	Madhavaram	17.41	21.05	29.46	49.26	76.09	3.41	5.28	4.45	4.38	44
4	Ambattur	37.77	45.59	115.90	215.42	310.97	9.78	6.39	3.74	6.64	82
5	Avadi	61.57	77.41	124.70	183.22	229.40	4.88	3.92	2.27	3.69	38
6	Poonamallee	6.55	18.72	23.67	28.83	42.60	2.37	1.99	3.98	2.78	65
7	Thiruverkadu	18.63	13.08	17.23	27.84	32.20	2.79	4.92	1.47	3.06	17
8	Maduravoyal	4.78	6.46	7.45	14.88	43.61	1.44	7.17	11.35	6.65	91
9	Valasarawakkam	2.97	2.41	7.58	21.95	30.98	12.12	11.22	3.50	8.95	104
10	Alandur	8.08	65.04	97.45	125.24	146.29	4.13	2.54	1.57	2.74	181
11	Ullagaram-Puzhithivakkam	3.64	2.38	8.58	16.13	30.42	13.69	6.52	6.55	8.92	84
12	Anakaputhur	2.98	10.88	15.30	24.35	31.92	3.46	4.76	2.75	3.66	107
13	Pammal	5.19	9.05	27.82	36.51	50.00	11.89	2.75	3.20	5.95	96
14	Pallavaram	16.10	51.37	83.90	111.87	144.62	5.03	2.92	2.60	3.52	90
15	Tambaram	20.72	58.81	86.92	113.29	137.93	3.99	2.69	1.99	2.89	66
16	Manali	7.49	3.34	11.96	19.09	28.60	13.61	4.79	4.12	7.51	38
	Total	239.99	484.58	814.02	1183.68	1580.5	5.32	3.82	2.93	4.02	66

Source; Arrived based on Census figures

3.11 The high growth rate of more than 5% was found in Ambattur, Maduravoyal, Valasaravakkam, Ullagaram-Puzhithivakkam, Manali and Pammal Municipalities. Kathivakkam, Poonamallee, Alandur and Tambaram Municipalities recorded low growth rates and the rest fall under moderate growth rate category. Maduravoyal Municipality recorded the highest growth rate of 11.35% in the last decade 1991-2001.

Table No.3.07: CMA - Population Growth in Town Panchayats - 1971-2001											
Sl. No.	Town Panchayat	Area in Sq.Km.	Population (in thousands)				Annual Rate of growth				Density per Hect.2001
			1971	1981	1991	2001	71-81	81-91	91-01	Avge.	
1	Minjur	8.63	7.24	13.62	19.49	23.74	6.52	3.64	2.00	4.05	28
2	Chinnasekkadu	1.67	2.21	2.68	8.42	9.74	1.95	12.13	1.46	5.18	58
3	Puzhal	6.74	6.94	9.26	15.87	20.64	2.93	5.53	2.66	3.71	31
4	Naravarikuppam	20.76	9.96	14.67	17.41	18.33	3.95	1.72	0.52	2.06	9
5	Thirunindravur	14.56	6.94	9.13	16.31	29.33	2.78	7.78	4.27	4.94	20
6	Porur	3.72	3.54	8.63	19.51	28.92	9.33	8.49	4.02	7.28	78
7	Thirumazhisai	7.25	9.56	11.01	14.54	16.29	1.42	2.82	1.14	1.79	22
8	Mangadu	5.63	10.98	11.23	16.06	19.42	0.22	3.64	1.92	1.93	35
9	Nandambakkam	2.61	4.72	7.68	10.56	9.34	4.98	3.24	-1.22	2.33	36
10	Meenambakkam	3.03	2.51	3.09	3.80	3.61	2.13	2.09	-0.52	1.23	12
11	Kundrathur	12.16	14.45	16.42	22.79	25.07	1.29	3.33	0.93	1.85	21
12	Thiruneermalai	5.87	6.31	9.95	17.94	19.23	4.66	6.07	0.70	3.81	33
13	Perungalathur	7.04	3.71	6.29	11.24	19.59	5.40	5.98	5.72	5.70	28
14	Peerkankaranai	1.76	3.58	5.28	10.74	17.51	3.96	7.37	5.01	5.45	99
15	Chitlapakkam	2.90	5.32	11.72	15.90	25.31	8.22	3.10	4.76	5.36	88
16	Sembakkam	6.35	2.60	6.10	13.50	21.50	8.86	8.32	4.75	7.31	34
17	Madambakkam	7.92	2.32	3.49	8.21	17.00	4.14	8.94	7.55	6.88	22
18	Perungudi	4.64	1.74	4.28	9.71	23.58	9.40	8.54	9.28	9.08	51
19	Pallikkaranai	17.43	2.32	3.93	7.82	22.07	5.40	7.13	10.93	7.82	13
20	Sholinganallur	15.35	4.23	5.75	8.53	15.56	3.12	4.02	6.20	4.45	10
	Total	156.02	111.18	164.19	271.35	385.72	4.40	5.13	3.62	4.38	25

Source; Arrived based on Census figures

3.12 From the above, it may be seen that Chinnasekkadu, Porur, Perungalathur, Peerkkankaranai, Chithalapakkam, Sembakkam, Madambakkam, Perungudi and Pallikkaranai Town Panchayats had high growth rates exceeding 5% and Minjur, Puzhal, Thirunindravur, Thiruneermalai and Sholinganallur had recorded moderate growth rates in population; the rest have low growth rates of below 3%.

Table No. 3.08: CMA - Population Growth in Panchayat Unions - 1971-2001

S.No.	Panchayat Union	Area in Sq.Km	Population				Annual Rate of growth in %				Density per Hect.2001
			1971	1981	1991	2001	71-81	81-91	91-01	Ave.	
1	Minjur (4 Villages)	40.02	11015	13900	17032	23368	2.35	2.05	3.21	2.54	6
2	Sholavaram (41 Villages)	131.58	44069	59286	82773	97068	3.01	3.39	1.61	2.67	7
3	Puzhal (28 Villages)	46.87	23739	30493	41661	51081	2.54	3.17	2.06	2.59	11
4	Villivakkam (25 Villages)	84.53	37712	46787	91488	143070	2.18	6.94	4.57	4.56	17
5	Thiruvallur (1 village)	11.39	4409	5505	7789	8719	2.24	3.53	1.13	2.30	8
6	Poonamallee (42 villages)	75.44	56439	60435	66742	71767	0.69	1.00	0.74	0.81	10
7	Kundrathur (30 villages)	80.36	44050	50552	68435	105610	1.39	3.08	4.46	2.97	13
8	Sriperumbudur (4 villages)	20.16	1647	757	0	80	-7.48	-100	--	--	4
9	St.Thomas Mount (33 Villages)	98.39	34773	56118	122435	199235	4.90	8.11	4.99	6.00	20
10	Kattankulathur (6 Villages)	28.26	9490	14905	21891	30695	4.62	3.92	3.44	3.99	11
	Total	617.00	267303	338738	520246	730792		2.67	5.18	4.37	12

Source: Arrived based on Census figures

Population Growth in Villages in Panchayat Unions

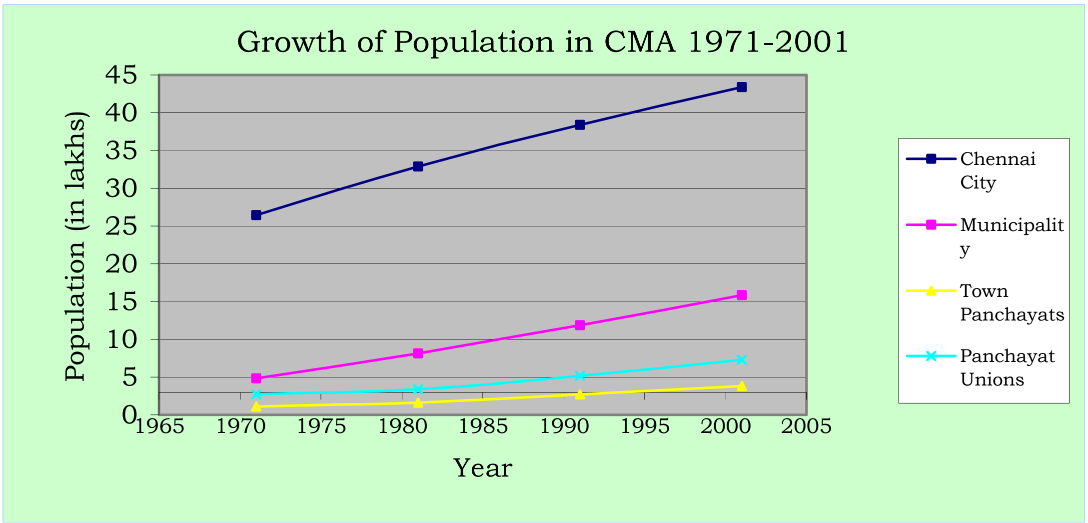
3.13 Attanthangal, Nallur, Sembilivakkam, Vijayavallur in Sholavaram Panchayat Union, Surapattu and Puttagaram in Puzhal Panchayat Union, Keelakondaiyur, Nolambur, Vanagaram, Ayapakkam, Nerkundram and Ramapuram in Villivakkam Panchayat Union, Nemilicheri Nochimedu in Poonamallee Panchayat Union, Chinnapanicheri, Kulathavancheri, Srinivasapuram, Ayyappanthangal, Thelliar Agaram, Mugalivakkam, Manapakkam, Gerugambakkam, Tharapakkam, Kavanur and Nandambakkam in Kundrathur Panchayat Union, Neelankarai, Injambakkam, Okkiam Thuraipakkam, Kovilambakkam, Mudichur, Vengaivasal, Maduraipakkam, Mullaicheri, Arsankalani and Uthandi in St.Thomas Mount Panchayat Union and Mannivakkam in Kattankulathur Panchayat Union have recorded high growth rate during in 1991-2001 exceeding 5%. The villages in St. Thomas Mount Panchayat Union recorded the highest annual growth rate of 6% during 1991 - 2001.

3.14 The overall growth of population in CMA from 1971 to 2001 is given in the table below:

Table No. 3.09: Growth of Population in CMA, 1971 - 2001

Sl. No.		Population (in lakhs)				Annual Rate of growth (in percent)			Area in Sq.km.	Density per Hect. in 2001
		1971	**1981**	**1991**	**2001**	**71-81**	**81-91**	**91-01**		
1.	Chennai City	26.42	32.85	38.43	43.44	2.20	1.58	1.23	176	247
2.	Municipalities	4.84	8.14	11.84	15.81	5.24	3.80	2.91	240	66
3.	Town Panchayats	1.11	1.64	2.71	3.86	4.43	4.94	3.62	156	25
4.	Panchayat Unions	2.67	3.38	5.20	7.31	2.40	4.38	3.58	617	12
5.	CMA Total	35.04	46.01	58.18	70.41	2.76	2.37	1.93	1189	59

Source; Arrived based on Census figures



3.15. The rate of growth in the local bodies within CMA is given in the Sheet No 2.5 & 2.6 from which it may be seen that the proximity to the main city and major urban centres, rail transport availability, ground water availability and residentially developed land availability are the major reasons for faster growth of certain areas within the CMA.

Migration

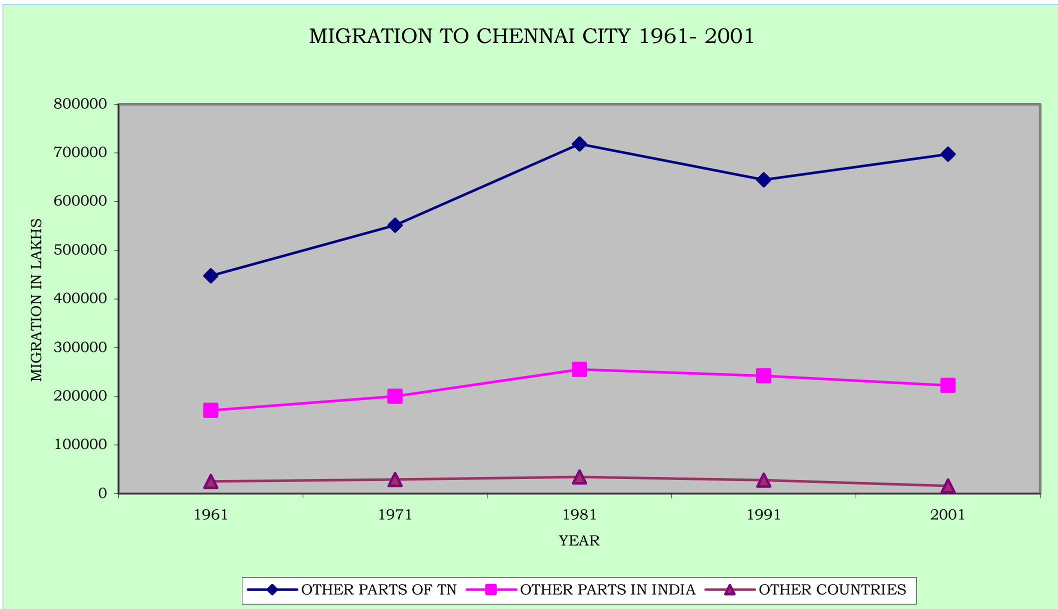
3.16 The cosmopolitan nature of Chennai was a reflection of its attractions to migrant groups from all over India. Migrants came not only predominantly from the surrounding Tamil and Telugu speaking areas but also from southern and northern India. These migrant groups from other states have made their distinctive mark on the patterns of residential and social organisations within this Chennai Metropolis.

3.17 Chennai is a city of migrants like any other metropolitan city in India. According to 2001 Census, migrants to Chennai City from other parts of Tamil Nadu State constitute 74.5 % and the table no. 3.10 shows a downward trend in the migration to the City from 37.24% in 1961 to 21.57% in 2001. Migrants from other parts of India constitute 23.8% and the remaining 1.71 % of the migrants is from other countries.

Table No. 3.10: Migration to Chennai City, 1961-01 (in Lakhs)

Year	Total Population	Total migrants to the city from								% of Total Migrants to the total population
		Other parts of Tamil nadu		Other parts of India (Excluding Tamilnadu)		Other Countries		Un-classi-fiable	Total migrants	
		No.	%	No.	%	No.	%			
1961	17.29	4.47	69.45	1.71	26.6	0.25	3.90	--	6.44	37.24
1971	24.69	5.51	70.61	2.00	25.63	0.29	3.76	--	7.80	31.59
1981	32.84	7.19	71.28	2.55	25.31	0.34	3.41	--	10.08	30.70
1991	38.43	6.44	70.51	2.42	26.47	0.28	3.01	0.04	9.18	23.90
2001	43.44	6.98	74.49	2.23	23.80	0.16	1.71		9.37	21.57

Source: Census of India, 1961, 1971 & 1981, 1991 Social and Cultural Table

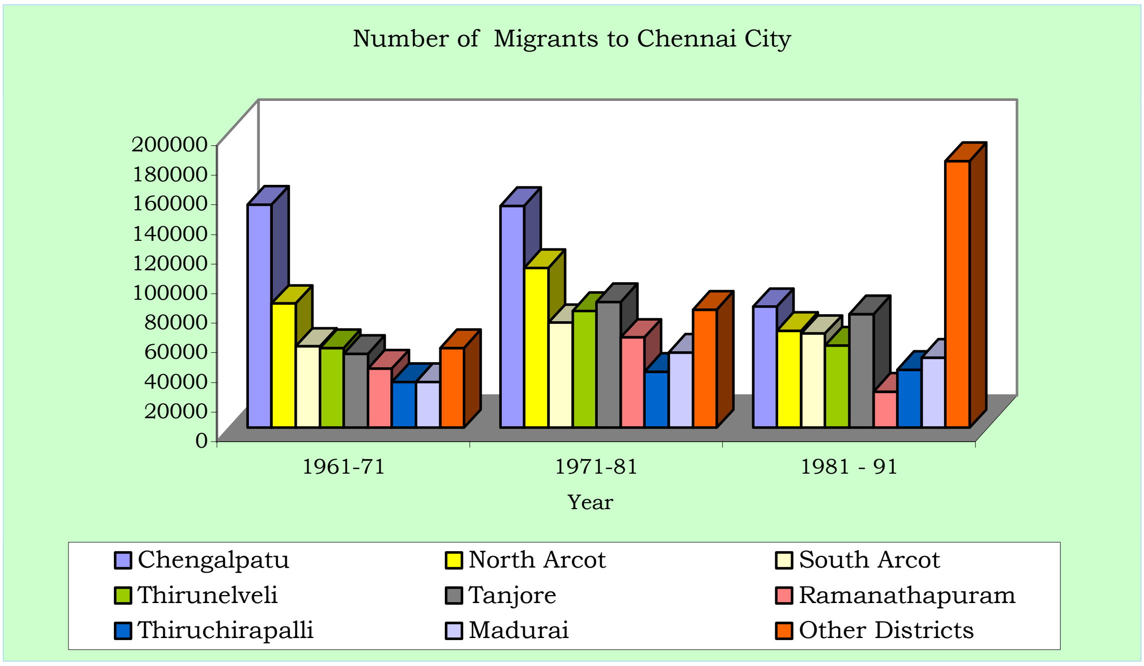


3.18 According to 1991 Census migrants from the nearby districts of Chengalpattu District (now bifurcated as Kancheepuram and Thiruvallur Districts), North Arcot District (now renamed as Vellore District) and South Arcot District (now bifurcated as Villupuram and Thiruvannamalai Districts) alone constitute 33% and Thirunelveli and Thanjavur Districts constitute another 21% of total migrant population to Chennai City. From the table no. 3.11 it may also be seen that the proportion of migration from other districts have almost tripled.

Table No. 3.11: Migration to Chennai city from the Districts of Tamilnadu, 1961-91

District	No. Of Migrants to Chennai city			Percentage to Total Migrants		
	1961-71	1971-81	1981 - 91	1961-71	1971-81	1981 - 91
Chengalpattu	151000	150000	81830	27.5	20.9	12.8
North Arcot	84000	108000	65550	15.3	15.0	10.3
South Arcot	55000	71000	63670	10.0	9.9	10.04
Thirunelveli	54000	79000	55530	9.8	11.0	8.75
Tanjavur	50000	85000	76760	9.1	11.8	12.1
Ramanathapuram	40000	61000	24390	7.3	8.5	3.84
Thiruchirapalli	31000	38000	39170	5.6	5.3	6.2
Madurai	31000	51000	47292	5.6	7.1	7.5
Other Districts	54000	80000	180230	9.8	10.5	28.4
Total	550000	719000	634422	100.0	100.0	100.0

Source: Structure Plan for MMA, and Census of India 1981and 1991 Tamilnadu Migration Tables



3.19 Increase in immigration to the City and also to the CMA is evident from the table Nos. 3.11 and 3.12.

3.20 Study conducted by the Time Research Foundation in 1991 (sponsored by CMDA) showed that:

i) Migrants from other urban areas constitute 63.4% and only 36.3% are from rural areas.

ii) Among migrants from urban areas, female migrants outnumber male migrants.

iii) Among rural migrants, male migrants out-number female migrants.

iv) Among total migrants there is near parity as between male and female migrants.

v) In respect of migrants from southern region of the state, the urban proportion (75%) is higher and the rural proportion being only 25%

vi) Among migrants from other southern states of India, there are more female migrants than male migrants (with migrants from Kerala being an exception) and

vii) The reason on 'family moved' shows the household movement and it constitutes 30%; for marriage reasons another 25% migrated.

Table No. 3.12: Migration into CMA 2001

Sl. No.	Place	Males	Female	Total	Percentage to Total
1.	Total Migrants	855103	753196	1608299	
2.	Elsewhere in the district of enumeration	138235	124844	263079	16.35
3.	Other Districts of the State	549214	471981	1021195	63.50
4.	Outside the state	155431	145307	300738	18.70
5.	Outside India	25360	22360	23287	1.45

Source: Census of India, Tamil nadu Migration Tables-2001

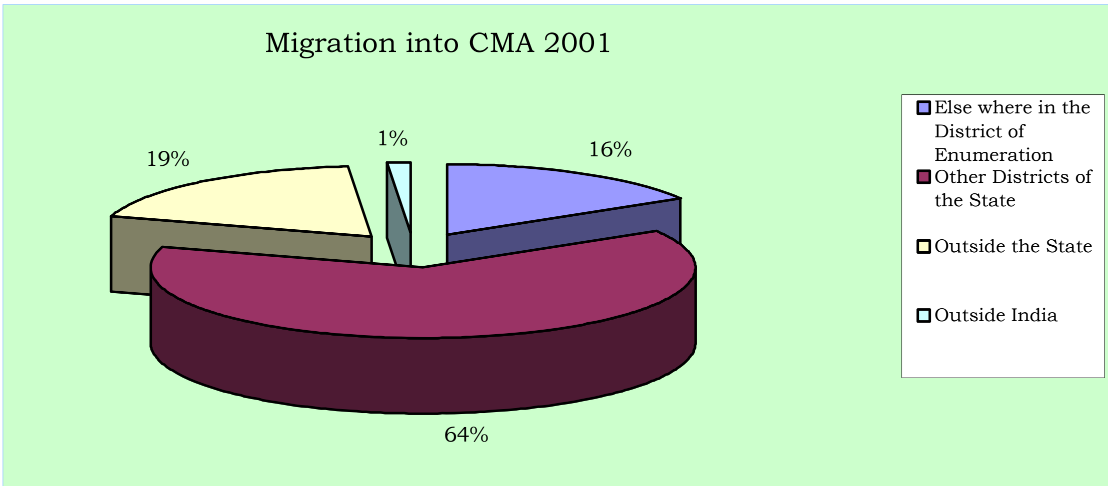
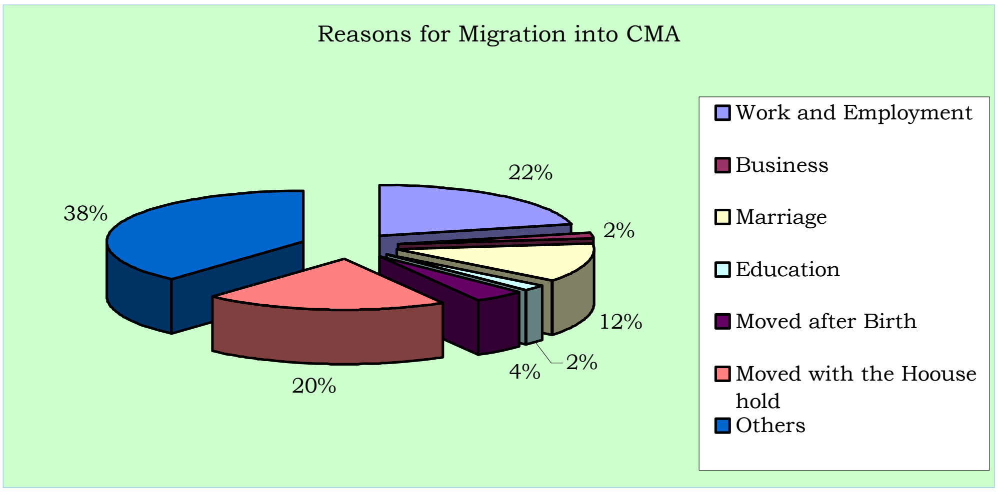


Table No. 3.13: Reasons for Migration into CMA - 2001				
Reason	Migrants			% to total
	Males	Females	Total	
Work & Employment	300939	44761	345700	21.50
Business	20719	4553	25272	1.57
Education	20314	9157	29471	1.83
Marriage	10737	189828	200565	12.47
Moved after birth	38250	30617	68867	4.28
Moved with the Household	144611	175205	319816	19.89
others	319533	299075	618608	38.46
Total	855103	753196	1608299	100

Source: Census of India 2001 Tamil nadu Migration Table



3.21 The origin of the migrant population to the CMA and the reasons for the migration are given in the table Nos. 3.12 and 3.13.

3.22 An interesting and important fact found is the out migration from Chennai City to its suburbs and other areas. The population of the Chennai City in 1991 was 38.43 lakhs which includes 9.18 lakhs migrant population and natural increase of 6.40 lakhs (for 1981-91) population; the net population increase works out to only 5.59 lakhs which shows that there was a net out migration of 10 lakhs (30.4% of 1981 population) from City mostly to the rest of CMA (during 1981-1991). Similarly, an out migration of 10.19 lakhs (26.5% of the 1991 population) is noted during 1991-2001. Though there were large scale building construction activities noted during the above periods, the out migration of resident population from Chennai City proves that considerable conversion of residential premises into non-residential mostly for office, shopping, hotels and other commercial purposes took place; this trend will continue in this metropolis.

Birth rate, death rate and rate of Natural Increase

3.23 Birth rates, death rates and rates of natural increase for the Chennai City are tabulated below:

(Per 1000 population)

Table No. 3.14: Chennai City - Birth, Death and Natural increase rates 1971-1976

Year	Crude Birth Rate	Crude Death Rate	Natural increase rate
1971	38.6	12.3	26.3
1972	37.8	12.7	25.1
1973	36.4	12.4	24.0
1974	34.4	11.5	22.9
1975	34.3	13.1	21.2
1976	32.1	12.0	20.1

Source: Structure Plan for CMA, 1980

(Per 1000 population)

Table No. 3.15: Chennai City - Birth, Death and Natural increase rates 1991-2003

Year	Crude Birth Rate	Crude Death Rate	Natural increase rate
1991	25.89	9.67	16.22
1992	24.01	9.50	14.51
1993	23.82	9.14	14.68
1994	23.39	9.07	14.32
1995	23.75	8.49	15.26
1996	22.68	8.54	14.14
1997	22.50	8.20	14.30
1998	23.81	9.00	14.81
1999	25.68	8.87	16.81
2000	25.53	8.92	16.61
2001	24.50	8.42	16.08
2002	23.72	8.27	15.45
2003	22.62	8.01	14.61

Source: Corporation of Chennai

3.24 From the above two tables, it may be seen that the registered birth rate in Chennai City in 1971 was 38.6 and it was varying from 38.6 to 32.1 during 1971-76. It has now reduced to 22.62 in the year 2003. Similarly the death rate also reduced to a considerable extent from 13.1 in 1975 to 8.01 in 2003. The rate of natural increase declined from 26.3 in 1971 to 14.61 in 2003 as detailed in the tables above.

3.25 The factors which affect birth rate includes age structure of population, the average age of marriage, and the incidence of family planning, among others. Increase in longevity of life due to improved health facilities may also be a reason for reduction in the death rates in Chennai.

3.26 Birth and death rates in 2001 in some of the local bodies within CMA and also the average rates in the Kancheepuram and Thiruvallur districts wherein the rest of CMA falls are given in the Table below which show that these rates vary from local body to local body and that too a large extent.

(Per 1000 population)

Table No. 3.16: CMA - Birth, Death and Natural increase rates 2001

Local Body	Birth Rate	Death Rate	Natural increase rate
Kathivakkam Municipality	24.5	8.42	16.08
Ambattur Municipality	6.52	2.90	3.62
Avadi Municipality	13.05	3.21	9.84
Thiruverkadu Municipality	1.77	6.02	4.25
Maduravoyal Municipality	2.32	2.91	0.59
Valasarawakkam Municipality	16.21	3.91	12.30
Alandur Municipality	18.06	5.59	12.46
Ullagaram-Puzhudhivakkam Municipality	4.96	3.15	1.81
Anakaputhur Municipality	3.04	3.92	0.88
Pammal Municipality	5.86	4.70	1.16
Pallavaram Municipality	26.50	4.80	21.70
Tambaram Municipality	21.88	6.13	15.75
Districts			
Thiruvallur	11.26	4.73	6.53
Kancheepuram	15.58	5.74	9.84

Source: Concerned Local Bodies and Statistical year Book 2001

Age Structure

3.27 Age structure of a population in a city / metropolis plays a major role in urban planning. It gives an idea about dependent population, working population, jobs to be created, the present and future requirements of educational, health and other facilities and amenities etc. It depends on natality (birth rate), mortality (death rate) and also migration. Age structure of population in CMA as per Census 1971 to 2001 is given in the table below:

Table No 3.17: Age Structure in CMA in %					
Age Group	**1961**	**1971**	**1981**	**1991**	**2001**
0-4	13.20	12.51	11.03	8.68	7.31
5-9	12.39	11.74	10.35	9.56	7.97
10-14	10.64	10.97	11.37	10.51	8.95
15-19	8.66	9.97	10.61	10.22	9.55
20-24	10.73	11.05	10.68	11.14	10.47
25-29	10.21	9.29	9.61	10.20	10.33
30-34	7.98	7.15	7.46	8.06	8.46
35-39	6.69	6.99	6.66	7.48	8.04
40-44	5.49	5.14	5.36	5.68	6.19
45-49	4.15	4.33	4.70	4.98	5.50
50-54	3.65	3.51	3.70	3.92	4.37
55-59	2.10	2.46	2.73	2.90	3.10
60-64	2.06	2.30	2.40	2.64	2.83
65-69	1.10	1.15	1.37	1.51	1.96
>70	1.20	1.65	1.97	2.33	3.02
not stated		0.00	0	0.18	1.93
Total	100.00	100.21	100	100.00	100.00

M- Male, F- Female, T- Total

Source: CMDA /TRF Study data Hand book 1991

Census of India, Social and Cultural Tables 1961, 1971, 1981, 1991&2001

3.28 From the above, it may be seen that the proportion of primary school going children percentage has reduced from 12.39% in 1961 to 7.97% in 2001 and the proportion of Secondary school going age group has also reduced from 10.64 in 1961 to 8.95 in 2001. But the proportion of old age group has increased from 4.36 to 7.81% in the said period.

Sex Composition

3.29 Sex ratio is denoted by number of females per 1000 males. In Chennai the ratio has improved over the decades, though it is lower than the Tamil Nadu average. However, it is much above the Indian National average of 900 in urban areas.

Table No. 3.18: Sex ratio in Chennai City, CMA, Tamil Nadu and India						
Year	Chennai City	CMA	Tamil nadu		India	
			Urban	Total	Urban	Total
1961	901	909	963	992	N.A.	N.A.
1971	904	907	951	978	N.A.	N.A.
1981	934	927	956	977	878	933
1991	930	936	951	972	893	929
2001	957	956	981	987	900	932

Source: Census of India, 1961, 1971, 1981, 1991 Social and cultural Tables and 2001 census of Tamil nadu

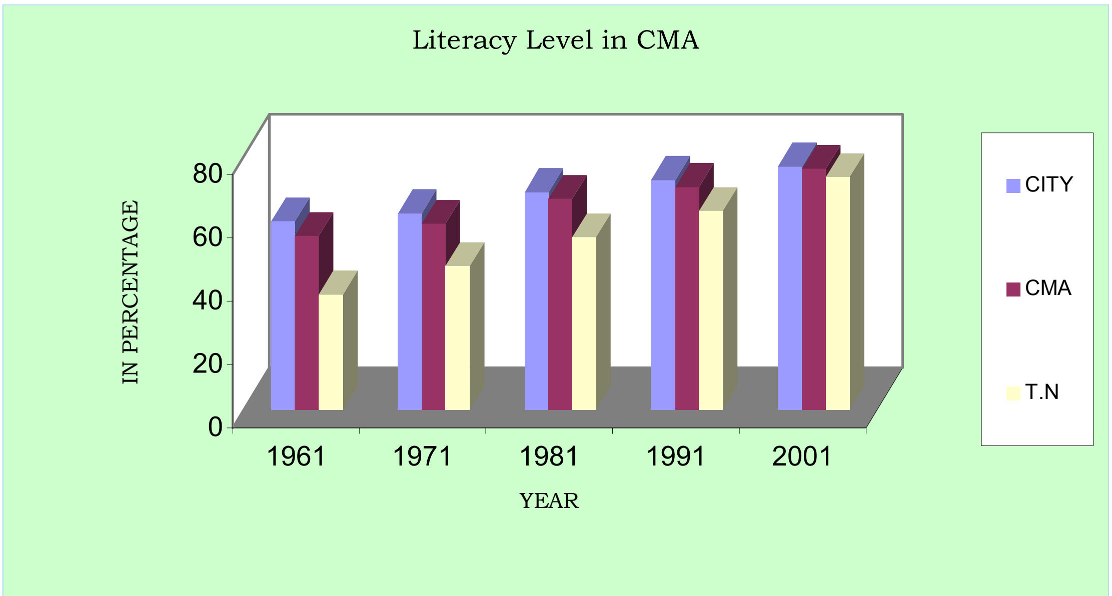
Literacy

3.30 Literacy is defined as the ability to read and write in any single language. The literacy levels in Chennai City and CMA were higher than the state average.

Table No. 3.19: Trends in Literacy levels in Chennai Metropolitan Area - 1961-2001

Units	Literacy rate		(in percentage)		
	1961	1971	1981	1991	2001
City	59.47	62.01	68.68	72.54	76.81
CMA	54.82	58.64	66.56	70.32	76.09
T.N.	36.39	45.40	54.40	62.70	73.50

Source: Census of India, 1961, 1971, 1981, 1991 Social and Cultural Tables and 2001 Census of Tamil Nadu)



Population Projection

3.31 Population projection for Chennai City and CMA had been made earlier on various occasions; the details of the same are given in the Table below summarising the earlier forecasts.

Table No.3.20: Summary of Population Estimates 1991-2011 Population in lakhs)								
Sl. No.	Description	1991		2001		2011		Method Adopted
		CMA	City	CMA	City	CMA	City	
1.	Master Plan [1971] -Projections -Assigned as per the strategy	 58 *58*	 43 *36*	 71 71	 53 40	 --	 --	
2.	Madras Urban Development Project (MUDP I) (World Bank 1974)	59 57	43 43	71 69	53 51	--	--	Exponential Component
3.	MMA Structure Plan (MMDA, 1980)	73	54	--	--	--	--	
4.	NCU: 1988 - Registrar General Census of India - United Nations			74 82				
5.	MMA, Demography, 1981 (MMDA undated)	60 60	43 43	81 77	59 55	-	--	High Medium
6.	Department of Statistics (Government of Tamil Nadu)	61	43	78	54	98	--	
7.	MMDA Projections: 1989	60 59	41 40	79 75	51 49	104 95	58 60	Geometric
	Actual as per Census	58.1	38.4	70.3	43.4	--	--	

Source: Report on Madras 2011, Policy Imperatives-An Agenda for Action by TRF & CMDA1991 (Volume III)

3.32 As part of the Second Master Plan preparation exercise the following population estimate had been made in the 1995 draft Second Master Plan.

Table No. 3.21: Draft Second Master Plan Population Estimate for CMA 1992011 (Population in lakhs)						
Year	Chennai Metropolitan Area			Chennai City		
	Low	Medium	High	Low	Medium	High
1991	54.41	59.17	64.15	37.24	40.34	43.56
1996	58.94	66.76	75.26	39.69	44.69	50.08
2001	63.79	75.22	88.09	42.31	49.47	57.48
2006	69.00	84.64	102.84	45.07	54.72	65.84
2011	74.57	95.09	119.84	48.00	60.46	75.27

3.33. It was stated in the report that the population with medium growth rate could be the more accurate population for the period up to 2011 and concluded that the (then) estimated population were as follows:

(Population in Lakhs)

Year	CMA	Chennai City
1996	66.76	44.69
2001	75.22	49.47
2006	84.64	54.71
2011	95.09	60.46

3.34 As the Tables above illustrate, the population projections are not to be considered exact; rather they reflect the natural growth, migration trends and assumption made at the time of the estimates, including vision then for development in the city/CMA. Review of the population forecast made earlier shows that actually there were reductions in natural increase and migration, when comparing the assumptions made for the projections. However, it is seen that the population projected based on land use assignment and first Master Plan strategy is very close to the actual and it showed that the population increase was as anticipated / planned in the first Master Plan.

3.35 Now, for projecting the population up to 2026, the following methods have been adopted:

(i) Linear method

(ii) Geometric growth method

(iii) Exponential Curve method

(iv) Urban-Urban Growth Difference (UUGD) method

(v) Urban-Urban and Rural Growth Difference (UURGD) method

3.36 Methodology adopted, assumption made, estimate arrived are given in the Annexure to this Chapter. For projection of the population, UUGD method is found suitable for the City and UURGD method is found more appropriate for the CMA as a whole, with the assumption of growth rate based on history and judgment and also the future urbanisation and population policies. For forecast of population of individual municipalities and other local bodies within CMA, the exponential curve method is found suitable (when considering also the trends in growth over the past decades).

3.37 The population projection made for the CMA as a whole, Chennai City, 16 Municipalities, 20 Town Panchayats and for the villages in 10 Panchayat Unions in CMA are given in the Tables below:

Table No. 3.22: Population Projections (Various methods) - Chennai City						
Year	Actual Population	Projected Population				
		Linear	Geometric	Exponential	UUGD	UURGD
1971	2642403					
1981	3284622					
1991	3843195					
2001	4343645					
2006		4706630	4718968	4722228	4627915	4724150
2011		5069614	5126721	5133807	4950145	5144832
2016		5432599	5569707	5581259	5238817	5565962
2021		5795584	6050970	6067710	5540456	6024451
2026		6158569	6573818	6596559	5855804	6524126

Chennail City Population Projection

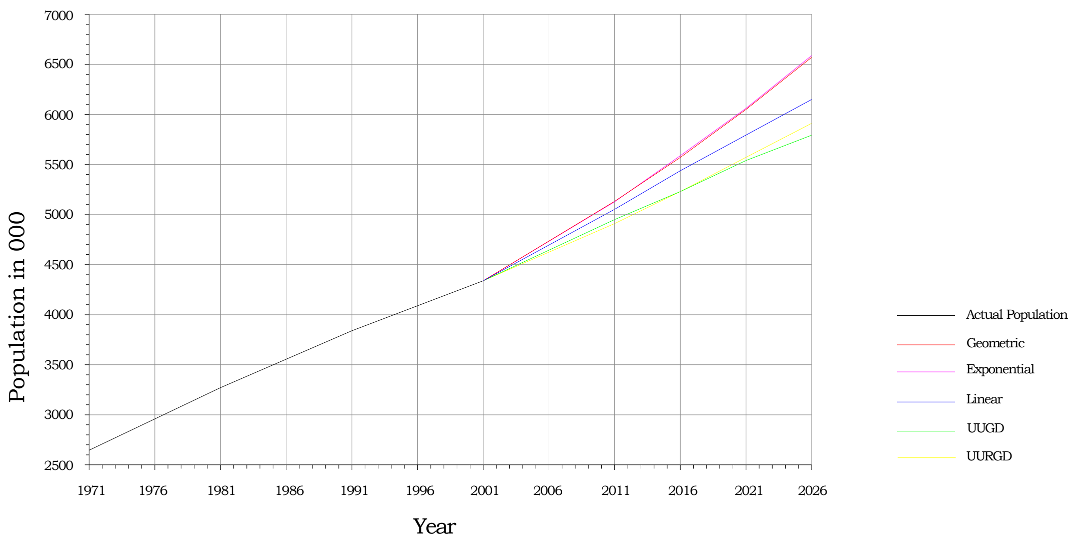


Table No. 3.23: Population Projections (Various methods) - CMA						
Year	Actual Population	Projected Population				
		Linear	Geometric	Exponential	UUGD	UURGD
1971	3505502					
1981	4601566					
1991	5818479					
2001	7040582					
2006		7860735	7927646	7875703	7815056	7896230
2011		8687731	8953451	8702252	8711726	8871228
2016		9514727	10162041	9528802	9710880	9966636
2021		10341723	11600796	10355352	10824396	11197753
2026		11168719	13334095	11181902	12065697	12582137

CMA-Population Projection

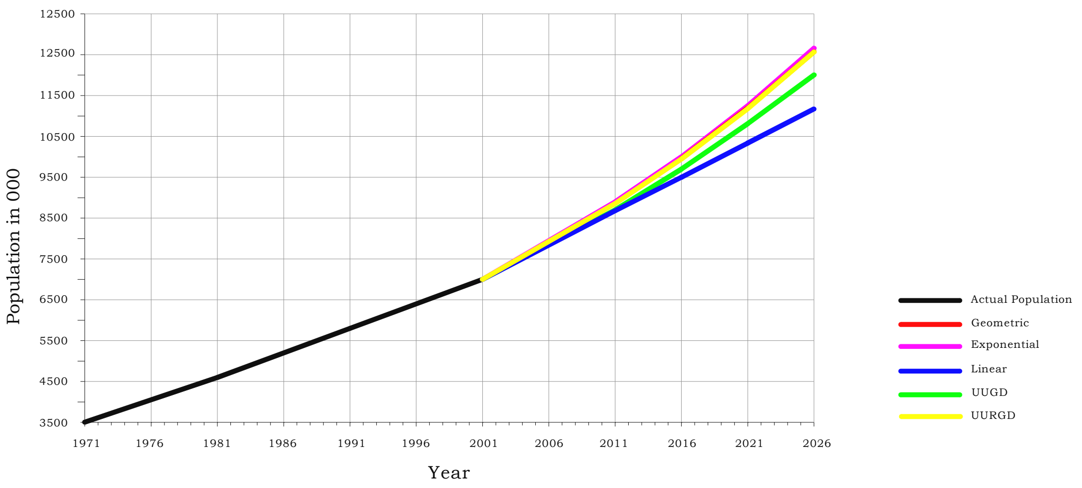


Table No. 3.24: Projected Population for CMA and Chennai City(In Lakhs)							
Sl. No.	Description	Actual	Projection				
		2001	2006	2011	2016	2021	2026
1	CMA	70.41	78.96	88.71	99.66	111.97	125.82
2	Chennai City	43.44	46.28	49.50	52.39	55.40	58.56

Table No. 3.25: Population Projection for Municipalities within CMA (In Lakhs)									
Sl. No.	Description	Actual 2001	Annual rate of growth assumed in %	Projection					Density 2026
				2006	2011	2016	2021	2026	
1	Kathivakkam	0.33	2.0	0.36	0.40	0.43	0.49	0.54	113
2	Thiruvottiyur	2.12	2.5	2.41	2.73	3.09	3.50	3.97	186
3	Madhavaram	0.76	4.0	0.93	1.14	1.39	1.69	2.07	119
4	Ambattur	3.11	4.5	3.89	4.88	6.11	7.65	9.58	254
5	Avadi	2.30	3.0	2.67	3.10	3.60	4.18	4.86	79
6	Poonamallee	0.42	4.0	0.52	0.64	0.78	0.95	1.16	177
7	Thiruverkadu	0.32	3.0	0.37	0.43	0.51	0.59	0.68	37
8	Maduravoyal	0.43	3.5	0.52	0.62	0.74	0.88	1.05	219
9	Valasarawakkam	0.31	3.5	0.37	0.44	0.52	0.62	0.74	250
10	Alandur	1.46	1.5	1.57	1.70	1.83	1.97	2.13	263
11	Ullagaram-Puzhudhivakkam	0.30	4.0	0.37	0.46	0.55	0.68	0.83	227
12	Anakaputhur	0.32	2.7	0.36	0.42	0.48	0.55	0.63	210
13	Pammal	0.50	3.0	0.58	0.67	0.78	0.91	1.05	204
14	Pallavaram	1.45	3.0	1.68	1.95	2.27	2.64	3.06	190
15	Tambaram	1.38	2.5	1.56	1.77	2.01	2.27	2.58	124
16	Manali	0.29	4.0	0.35	0.43	0.52	0.64	0.78	104
	Total	15.81	3.98	18.52	21.75	25.60	30.20	35.69	149

Table No. 3.26: Population Projection to Town Panchayats in CMA (In Lakhs)									
Sl. No.	Description	Actual 2001	Annual rate of growth assumed in %	Projection					Density 2026
				2006	2011	2016	2021	2026	
1	Minjur	0.23	2.5	0.27	0.30	0.35	0.39	0.44	51
2	Chinnasekkadu	0.10	2.5	0.11	0.13	0.14	0.16	0.18	109
3	Puzhal	0.21	3.0	0.24	0.28	0.32	0.38	0.44	65
4	Naravarikuppam	0.18	2.0	0.20	0.22	0.25	0.27	0.30	15
5	Thirunindravur	0.29	5.0	0.38	0.48	0.62	0.80	1.02	70
6	Porur	0.29	4.0	0.35	0.43	0.53	0.64	.79	211
7	Thirumazhisai	0.16	2.0	0.18	0.20	0.22	0.24	0.27	37
8	Mangadu	0.19	3.0	0.23	0.26	0.30	0.35	0.41	73
9	Nandambakkam	0.09	2.0	0.10	0.11	0.13	0.14	0.15	59
10	Meenambakkam	0.04	2.0	0.04	0.04	0.05	0.05	0.06	20
11	Kundrathur	0.25	2.0	0.28	0.31	0.34	0.37	0.41	34
12	Thiruneermalai	0.19	2.0	0.21	0.23	0.26	0.29	0.32	54
13	Perungalathur.	0.20	6.0	0.26	0.36	0.48	0.65	0.88	125
14	Peerkankaranai	0.18	2.0	0.19	0.21	0.24	0.26	0.29	164
15	Chitlapakkam	0.25	2.5	0.29	0.33	0.37	0.42	0.47	163
16	Sembakkam	0.21	7.00	0.30	0.43	0.61	0.87	1.24	195
17	Madambakkam	0.17	8.0	0.25	0.38	0.56	0.84	1.26	159
18	Perungudi.	0.24	5.0	0.30	0.39	0.50	0.64	0.82	177
19	Pallikkaranai	0.22	6.0	0.30	0.40	0.54	0.73	0.99	73
20	Sholinganallur.	0.16	9.0	0.24	0.38	0.60	0.94	1.48	96
Total		3.86		4.73	5.89	7.41	9.45	12.22	78

Table No.3.27: Population Projection for Panchayat Unions within CMA (in Lakhs)									
Sl. No.	Description	Actual 2001	Annual rate of growth assumed in %	Projection					Density 2026
				2006	2011	2016	2021	2026	
1	Minjur (4 Villages)	0.23	1.8	0.28	0.33	0.38	0.45	0.54	13
2	Sholavaram (41 Villages)	0.97	1.6	1.12	1.30	1.51	1.76	2.06	16
3	Puzhal (28 Villages)	0.51	1.4	0.60	0.75	0.95	1.23	1.60	34
4	Villivakkam (25 Villages)	1.43	2.2	1.73	2.12	2.64	3.32	4.27	51
5	Thiruvallur (1 village)	0.09	3.5	0.10	0.12	0.15	0.18	0.21	18
6	Poonamallee (42 villages)	0.72	1.3	0.73	0.82	0.93	1.06	1.21	16
7	Kundrathur (30 villages)	1.06	1.9	1.25	1.49	1.79	2.15	2.59	32
8	Sriperumbudur (4 villages)	--	--	0	0	0	0	0	0
9	St.Thomas Mount (33 Villages)	1.99	2.6	2.50	3.15	3.98	5.03	6.38	65
10	Kattankulathur (6 villages)	0.31	2.7	0.39	0.49	0.63	0.81	1.04	37
Total		7.31	3.45	8.70	10.59	12.96	15.99	19.88	32

Floating Population:

3.38 Arrival of passengers by the 92 inter city trains is estimated to be 1.125 lakhs per day. Similarly arrival of passengers in inter city buses (2028 arrivals) is estimated to be about 83,000 per day. These main arrivals of passengers to CMA accounts to 1.955 lakhs. In addition, there are people who commute every day for work, education, business and others from the adjoining and nearby districts and return home in the evening. These daily commuters estimated to be 20,000. Hence the people arriving in CMA through trains, buses and air is estimated to be about 2.25 lakhs and an equal number of persons may be departing from the metropolitan area. This floating population should also be taken into account in planning for infrastructure development in CMA appropriately.

ANNEXURE - III A

1. The following methods were adopted for projection of population.

(1) Linear Method:

$P_{(t+n)}$ = P_t {l+(n x a)}

where P_t - is population at time 't';

$P_{(t+n)}$ - is population at time 't+n';

n - is no. of time periods;

a - is average change in the past time periods.

(2) Geometric Growth Method:

$P_{(t+n)} = P_t (l+r)^n$

where Pt - is population at time 't';

$P_{(t+n)}$ - is population at time 't+n';

n - is number of time period;

r - is average percentage of change in population over past time period.

Note: Often the rate 'r' is not calculated, but estimated based on historical patterns, and judgment for the future.

(3) Exponential Curve Method:

$P_{(t+n)}$ = P_t x e ^{rt}

where Pt - is population at time 't';

$P_{(t+n)}$ - is population at time 't+n';

n - is number of time period;

r - is average percentage of change in population over past time period.

Note: Often the rate 'r' is not calculated, but estimated based on historical patterns, and judgment for the future.

(4) Urban - Urban Growth Difference (UUGD) Method

$$U^1 = \left[\frac{T^1 + dR}{T}\right] \times U$$

where U - Urban population at time 't';

U^1 - Urban population at time 't+1';

T - Total population at time 't';

T^1 - Total population at time 't+1';

d - u-r, where u - urban growth rate;
r - rural growth rate;

R - Rural population at time 't';

$$C^1 = \frac{U^1 + dU_1 \times C}{U}$$

Where C - City population (Chennai/CMA) at time 't';

C_1- City population (Chennai/CMA) at time (t+1);

U^1 - Urban population (CMA/Tamil Nadu) at time (t+1);

U_1 - Rest of urban population (i.e. U-C);

U - Urban population (CMA/Tamil Nadu) at time 't';

d - r_c - r_u where

r_c - the rate of growth of city population;
r_u - the rate of growth of urban population.

(5) Urban - Urban - Rural Growth Difference (UURGD) Method:

$$C^1 = \left\{ w_1 \left(\frac{U^1 + d_1 U_1}{U}\right) + w_2 \left(\frac{T^1 + d_2 R}{T}\right) \right\} C$$

Where U - Urban population at time 't';

U^1 - Urban population at time 't+1';

U_1 - Rest of urban population (i.e. U-C);

T - Total population at time 't';

T^1 - Total population at time 't+1';

R - Rural population at time 't';

d_1 - $r_c - r_{u;}$

d_2	-	r_c - $r_{r;}$
w_1, w_2 -		Weights of these population components;
C	-	City population at time't';
C_1	-	City population at time't + 1';
r_c	-	rate of growth of city population;
r_u	-	rate of growth of urban population;
r_R	-	rate of growth of rural population.

Note: $w_1 + w_2 = 1.0$. The quantum of weight is determined based on urbanisation level. At low levels of urbanisation contribution from rural areas will be more, and vice versa. The weights adopted based on the U.N. method is as follows:

Level of Urbanisation in percent	w_1	w_2
25	0.2	0.8
30	0.3	0.7
50	0.4	0.6
60	0.6	0.4
75	0.8	0.2

Assumptions made

Tamil Nadu Population

2. The historical pattern of annual growth rate of Tamil Nadu population was as detailed in Table A in Annexure III B. It varied from 0.85% to 3.72% in respect of urban, -0.52% to 1.53% in respect of rural, and 0.34% to 2.03% in respect of total population.

3. The annual rate of growth assumed for Tamil Nadu for population projection as low, medium and high category of growth rate, is 2.0%, 2.5% and 3.0% for urban, and 1.0%, 1.1% and 1.2% for the total population; the rural population growth rate has been arrived out of the total and urban population growth rates assumed.

CMA/City Population:

4. The annual growth rates of population in 1971-81, 1981-91, and 1991-2001 were 2.76%, 2.37% and 1.93% respectively for CMA and 2.2%, 1.58% and 1.23% respectively for the Chennai City. The average rate of annual growth arrived for these three periods was 2.35% for CMA and 1.67% for City and the same was adopted for linear, geometric and exponential growth methods of projection.

5. Projection of population for individual Corporation Division was made based on average annual growth rates. In respect of villages, Town Panchayats and Municipalities, future growth rates were assumed based on the past and judging the future trends. For UUGD and UURGD method, the assumption made based on the historical pattern is given below:

Growth Rates assumed in %

Year	Growth rate category	Tamil Nadu (TN)	TN Urban population	Rest of TN Urban population	Chennai City	CMA
2006	Low	1.00	2.00	2.50	1.00	2.00
	Medium	1.10	2.50	3.00	1.50	2.50
	High	1.20	3.00	4.00	2.00	2.75
2011	Low	1.00	2.00	2.50	1.00	2.00
	Medium	1.10	2.50	3.00	1.50	2.50
	High	1.20	3.00	4.00	2.00	2.75
2016	Low	1.00	2.00	2.50	1.00	2.00
	Medium	1.10	2.50	3.00	1.25	2.50
	High	1.20	3.00	4.00	1.50	2.75
2021	Low	1.00	2.00	2.50	1.00	2.00
	Medium	1.10	2.50	3.00	1.25	2.50
	High	1.20	3.00	4.00	1.50	2.75
2026	Low	1.00	2.00	2.50	1.00	2.00
	Medium	1.10	2.50	3.00	1.25	2.50
	High	1.20	3.00	4.00	1.50	2.75

ANNEXURE IIIB

POPULATION PROJECTION URBAN - URBAN GROWTH DIFFERENCE METHOD AND URBAN RURAL GROWTH DIFFERENCE METHOD

Table No. A Population and Growth Rate - Tamil Nadu

Year	Population			Growth rate		
	Urban	Rural	Total	Urban	Rural	Total
1901	2724781	16527849	19252630			
1911	3149137	17753479	20902616	1.46%	0.72%	0.83%
1921	3428079	18200439	21628518	0.85%	0.25%	0.34%
1931	4230382	19241717	23472099	2.13%	0.56%	0.82%
1941	5173682	21093825	26267507	2.03%	0.92%	1.13%
1951	7333525	22785522	30119047	3.55%	0.77%	1.38%
1961	8990528	24696425	33686953	2.06%	0.81%	1.13%
1971	12464834	28734334	41199168	3.32%	1.53%	2.03%
1981	15951875	32456202	48408077	2.50%	1.23%	1.63%
1991	19077592	36781354	55858946	1.81%	1.26%	1.44%
2001	27483998	34921681	62405679	3.72%	-0.52%	1.11%

Table No. B Population Projection - Tamil Nadu 2006 – 2026 Exponential Method

Year		Projected Population			Annual Growth Rate assumed		
		Urban	Rural	Total	Urban	Rural	Total
2006	Low	30374515	35230771	65605287	2.00%	0.18%	1.00%
	Medium	31143450	34790685	65934134	2.50%	-0.08%	1.10%
	High	31931850	34332781	66264631	3.00%	-0.34%	1.20%
2011	Low	33569031	35399911	68968942	2.00%	0.10%	1.00%
	Medium	35290152	34371939	69662091	2.50%	-0.24%	1.10%
	High	37099517	33262690	70362207	3.00%	-0.63%	1.20%
2016	Low	37099517	35405538	72505055	2.00%	0.00%	1.00%
	Medium	39988981	33611847	73600828	2.50%	-0.45%	1.10%
	High	43103489	31609673	74713162	3.00%	-1.01%	1.20%
2021	Low	41001307	35221161	76222468	2.00%	-0.10%	1.00%
	Medium	45313452	32448812	77762264	2.50%	-0.70%	1.10%
	High	50079109	29254057	79333166	3.00%	-1.54%	1.20%
2026	Low	45313452	34817026	80130478	2.00%	-0.23%	1.00%
	Medium	51346868	30812123	82158991	2.50%	-1.03%	1.10%
	High	58183624	26055231	84238855	3.00%	-2.29%	1.20%

Table No. C		Urban Population Projection - Tamil Nadu 2006 – 2026 Urban -Urban Growth Difference (UUGD) Method				
Year		Growth Rate (GR) Assumed in %		Growth Rate Difference in %		Projected Tamil Nadu Urban Population
		TN Urban	TN Rural	TN Urban GR (-) TN Rural GR	D-TN Urban GR (-) TN Rural GR for 5 years	
2006	Low	2.00	0.18	0.0182	0.0912	30295467
	Medium	2.50	-0.08	0.0258	0.1288	31018215
	High	3.00	-0.34	0.0334	0.1670	31751609
2011	Low	2.00	0.10	0.0190	0.0952	33484836
	Medium	2.50	-0.24	0.0274	0.1371	35157207
	High	3.00	-0.63	0.0363	0.1816	36910278
2016	Low	2.00	0.00	0.0200	0.0998	37010422
	Medium	2.50	-0.45	0.0295	0.1473	39850548
	High	3.00	-1.01	0.0401	0.2007	42913798
2021	Low	2.00	-0.10	0.0210	0.1052	40907826
	Medium	2.50	-0.70	0.0320	0.1601	45173577
	High	3.00	-1.54	0.0454	0.2268	49905719
2026	Low	2.00	-0.23	0.0223	0.1115	45216488
	Medium	2.50	-1.03	0.0353	0.1765	51212639
	High	3.00	-2.29	0.0529	0.2645	58059683

Table No.D		Population Projection – Chennai Metropolitan Area 2006 – 2026 Urban Urban Growth Difference (UUGD) Method					
Year		Growth Rate in %		D-(CMA GR (-) Rest of Urban TN GR} X 5	Projected Population		
		CMA	Rest of Urban TN		TN Urban	Rest of TN Urban	CMA
2001	Actual				27483998	20450259	7040582
2006	Low	2.00	2.50	-0.025	30295467	22673055	7622412
	Medium	2.50	3.00	-0.025	31018215	23210836	7807379
	High	2.75	4.00	-0.063	31751609	23952801	7798808
2011	Low	2.00	2.50	-0.025	33484836	25202586	8282250
	Medium	2.50	3.00	-0.025	35157207	26454087	8703120
	High	2.75	4.00	-0.063	36910278	28212106	8698172
2016	Low	2.00	2.50	-0.025	37010422	28011984	8998438
	Medium	2.50	3.00	-0.025	39850548	30149315	9701233
	High	2.75	4.00	-0.063	42913798	33216377	9697421
2021	Low	2.00	2.50	-0.025	40907826	31132068	9775758
	Medium	2.50	3.00	-0.025	45173577	34359993	10813584
	High	2.75	4.00	-0.063	49905719	39097431	10808288
2026	Low	2.00	2.50	-0.025	45216488	34597079	10619410
	Medium	2.50	3.00	-0.025	51212639	39159059	12053580
	High	2.75	4.00	-0.063	58059683	46014676	12045007

Table No.E Population Projection – Chennai City 2006 – 2026 Urban -Urban Growth Difference (UUGD) Method								
Year		Growth Rate in %		D {Chennai city GR (-) Rest of Urban TN GR} X 5	Projected Population			
		Chennai City	Rest of Urban TN		TN Urban	Rest of TN Urban	City	
2001	Actual				27483998	23140353	4343645	
2006	Low	1.00	2.50	-0.075	30295467	25781777	4513690	
	Medium	1.50	3.00	-0.075	31018215	26390300	4627915	
	High	2.00	4.00	-0.100	31751609	27099216	4652393	
2011	Low	1.00	2.50	-0.075	33484836	28784055	4700781	
	Medium	1.50	3.00	-0.075	35157207	30207062	4950145	
	High	2.00	4.00	-0.100	36910278	31899083	5011195	
2016	Low	1.00	2.50	-0.075	37010422	32117765	4892657	
	Medium	1.25	3.00	-0.088	39850548	34611731	5238817	
	High	1.50	4.00	-0.125	42913798	37628878	5284920	
2021	Low	1.00	2.50	-0.075	40907826	35818385	5089441	
	Medium	1.25	3.00	-0.088	45173577	39633121	5540456	
	High	1.50	4.00	-0.125	49905719	44338989	5566730	
2026	Low	1.00	2.50	-0.075	45216488	39925215	5291273	
	Medium	1.25	3.00	-0.088	51212639	45356835	5855804	
	High	1.50	4.00	-0.125	58059683	52501644	5858039	

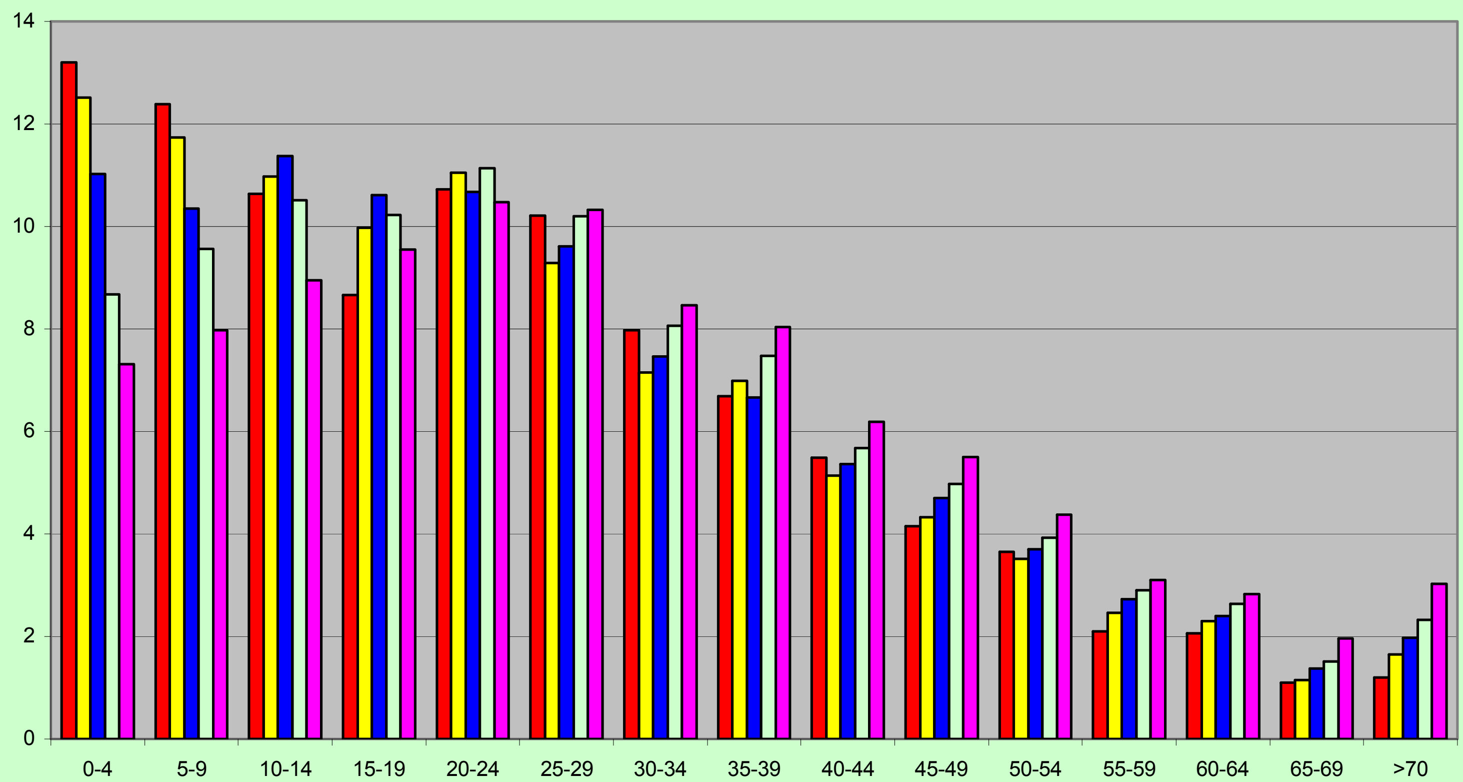
Table No: F Population Projection UURGD Method Weights Assigned					
Urbanisation Level	W_1	W_2	Urbanisation Level	W_1	W_2
35	0.325	0.675	52	0.410	0.590
36	0.330	0.670	53	0.415	0.585
37	0.335	0.665	54	0.420	0.580
38	0.340	0.660	55	0.425	0.575
39	0.345	0.655	56	0.430	0.570
40	0.350	0.650	57	0.435	0.565
41	0.355	0.645	58	0.440	0.560
42	0.360	0.640	59	0.445	0.555
43	0.365	0.635	60	0.450	0.550
44	0.370	0.630	61	0.455	0.545
45	0.375	0.625	62	0.460	0.540
46	0.380	0.620	63	0.465	0.535
47	0.385	0.615	64	0.470	0.530
48	0.390	0.610	65	0.475	0.525
49	0.395	0.605	66	0.480	0.520
50	0.400	0.600	67	0.485	0.515
51	0.405	0.595	68	0.490	0.510
			69	0.495	0.505

Table No G Population Projection Chennai Metropolitan Area, Urban- Urban Rural Growth Difference Method - 2006-2026

Year	Growth Rate in %		D_1 CMA GR (-) Rest of Urban TN GR	D_2 CMA GR (-) TN Rural GR	Projected Population				
	CMA	Rural TN %			TN Total	TN Urban	TN Rest of Urban	TN Rural	CMA
2001									
Actual					62405679	27483998	20450259	34921681	7040582
2006									
Low	2.00	0.18	-0.025	0.0912	65605287	30295467	22591279	35309820	7704188
Medium	2.50	-0.08	-0.025	0.1288	65934134	31018215	23129715	34915920	7888500
High	2.75	-0.34	-0.063	0.1545	66264631	31751609	23781891	34513022	7969718
2011									
Low	2.00	0.10	-0.025	0.0952	68968942	33484836	25038579	35484106	8446257
Medium	2.50	-0.24	-0.025	0.1371	69662091	35157207	26294682	34504884	8862525
High	2.75	-0.63	-0.063	0.1691	70362207	36910278	27855103	33451929	9055175
2016									
Low	2.00	0.00	-0.025	0.0998	72505055	37010422	27752133	35494633	9258289
Medium	2.50	-0.45	-0.025	0.1473	73600828	39850548	29893713	33750280	9956835
High	2.75	-1.01	-0.063	0.1882	74713162	42913798	32623068	31799365	10290730
2021									
Low	2.00	-0.10	-0.025	0.1052	76222468	40907826	30761041	35314643	10146785
Medium	2.50	-0.70	-0.025	0.1601	77762264	45173577	33986861	32588688	11186716
High	2.75	-1.54	-0.063	0.2143	79333166	49905719	38206418	29427447	11699301
2026									
Low	2.00	-0.23	-0.025	0.1115	80130478	45216488	34097620	34913990	11118868
Medium	2.50	-1.03	-0.025	0.1765	82158991	51212639	38642934	30946352	12569705
High	2.75	-2.29	-0.063	0.2520	84238855	58059683	44750479	26179172	13309204

Table No H	Population Projection Chennai City Urban - Urban Rural Growth Difference Method 2006-026								
Year	Growth Rate in %		D_1 Chennai City GR (-) Rest of Urban TN GR	D_2 Chennai City GR (-) TN Rural GR	Projected Population				
	Chennai City	Rural TN			TN Total	TN Urban	TN Rest of Urban	TN Rural	Chennai City
2001									
Actual					62405679	27483998	23140353	34921681	4343645
2006									
Low	1.00	-0.18	-0.08	0.041	65605287	30295467	25685542	35309820	4609925
Medium	1.50	-0.08	-0.08	0.079	65934134	31018215	26294065	34915929	4724150
High	2.00	-0.34	-0.10	0.117	66264631	31751609	26945380	34513022	4806229
2011									
Low	1.00	0.10	-0.08	0.045	68968942	33484836	28587362	35484106	4897474
Medium	1.50	-0.24	-0.08	0.087	69662091	35157207	30012375	34504884	5144832
High	2.00	-0.63	-0.10	0.132	70362207	36910278	31581463	33451929	5328815
2016									
Low	1.00	0.00	-0.08	0.050	72505055	37010422	31806986	35494633	5203436
Medium	1.25	-0.45	-0.09	0.085	73600828	39850548	34284586	33750280	5565962
High	1.50	-1.01	-0.13	0.126	74713162	42913798	37089749	31799365	5824049
2021									
Low	1.00	-0.10	-0.08	0.055	76222468	40907826	35377821	35314642	5530005
Medium	1.25	-0.70	-0.09	0.098	77762264	45173577	39149126	32588687	6024451
High	1.50	-1.54	-0.13	0.152	79333166	49905719	43536872	29427447	6368847
2026									
Low	1.00	-0.23	-0.08	0.062	80130478	45216488	39338796	34913990	5877692
Medium	1.25	-1.03	-0.09	0.104	82158991	51212639	44688513	30946352	6524126
High	1.50	-2.29	-0.13	0.189	84238855	58059683	51092126	26179172	6967558

Age Structure in CMA in %



Chapter IV

ECONOMY

[•]The economic history of the World during the 20th century has dramatically demonstrated that the process of urbanization and economic progress are mutually reinforcing. Urbanization is closely associated with increasing levels of income and improvements in social indicators

such as life expectancy, literacy, infant mortality and access to infrastructure and social services. Data and many supporting studies conclude that cities, especially bigger cites, mean higher productivity and higher per capita incomes. During 21 century, regardless of ideology, cities are loci and motors of economic and social change. Studies have shown that urban-based economic activity account for more than 50 percent of GDP in all countries. Cities and towns are not only the loci of production, but they are also loci of the most important impacts of globalization and hence the places of change and expectation of the future. Undervalueing urban areas can unwittingly place the economic and social future of countries at risk. Improved understanding of the multiple interactors between globalization and city can therefore contribute to identifying new strategies for protecting and maintaining urban economies.

4.02 The transformation of the global economy during the last two decades is perhaps the most important dimension of globalization because it has supported the diffusion of global culture and provoked deep and broad adjustments within countries and cities what began as trade in goods and services is now accompanied by the flow of capital and exchange of currencies in world financial markets. Information technology has allowed the birth of global interest rates and the increasing movement of capital to new opportunities for immediate and short-term financial benefits. Concern about 'footloose industries' have given way to cyber markets for finance and investment. At city level, the rapid entry of new, mostly foreign investors, and new capital led to changes in the composition of economic activity, particularly favouring financial services and those industries able to benefit from connectivity. *

4.03 The need for integrated economic and physical planning has been highlighted by many economists as well as urban planners. As Nigel Harris observes:

> "The city is a framework for the concentration and organisation of power and economic activity for the aggregation and disaggregation of resources. The city's prime advantage is precisely its concentration of activities and skills. Concentration today supports the technical advance which ensures mastery of the economy of the

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[•] The State of the World's Cities 2004 future. To sustain the city's advantages requires much more rapid change within than outside the city. Activities cluster and disperse, are merged and subdivided; some are eliminated, all within a changing modern economy. The future of the urban area likewise is shifted with speed. Buildings rise and are demolished, new streets and railways are forced through.

City planning however often seems to freeze the city's form, to trap it in a physical mode that makes change more difficult. To the planner, the city is too often not an instrument of economic change and growth but an architectural form or a public good for a particular class of consumer or a status symbol of the national government. Where resources are very scarce, the opportunity cost of planning in isolation from the central priorities of economic development is high. (Nigel Harris, 1978)"

4.04 Modern metropolitan planning involves integration of economic, social, environmental, infrastructure and management aspects. Employment and income generation policies must play a key role in urban planning since the main cause of urban poverty is severe limitations of income to the urban poor.

Tamil Nadu over view

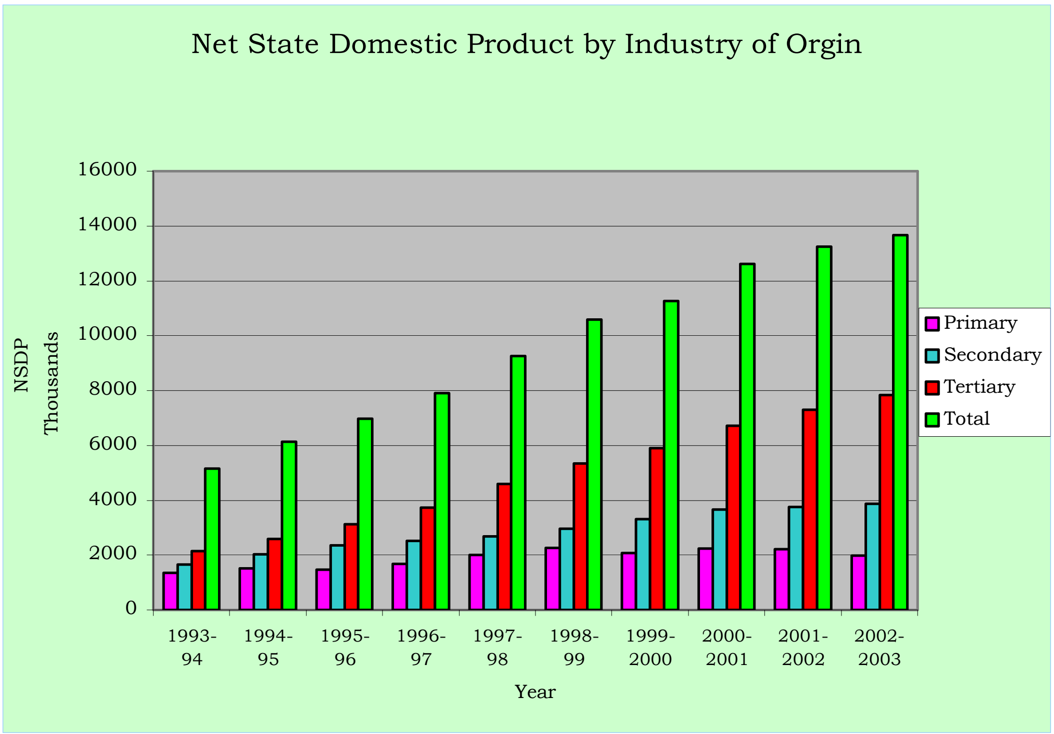
4.05 Tamil Nadu emerged as the third largest economy in India with State Domestic Product of Rs.1530 billion in 2002-2003.

4.06 The table No 4.01 shows the growth of the Tamil Nadu State income in various sectors from 1993 to 2003.

Table No. 4.01: Net State Domestic Product (NSDP) in TN by industry of origin

Sl. No.	Year	Primary	Secondary	Tertiary	Total	Per capita Income (Rs. in lakhs)
1	1993-94	1355029	1660933	2148360	5164322	8955
2	1994-95	1512212	2019035	2596334	6127561	10503
3	1995-96	1472400	2367212	3132304	6971916	11818
4	1996-97	1668087	2507392	3736627	7912106	13270
5	1997-98	2001844	2680994	4586147	9268985	15388
6	1998-99	2256692	2972778	5349845	10579315	17394
7	1999-2000	2065946	3317462	5890654	11274062	18367
8	2000-2001	2231340	3670174	6708443	12609957	20367
9	2001-2002	2212614	3744409	7299696	13256719	21239
10	2002-2003	1980870	3870118	7827099	13678087	21738

Source: Statistical Hand Book of Tamil Nadu, 2003



4.07 The Table No. 4.02 depicts the percentage change in the growth of Gross Domestic Product in Tamil Nadu.

Table No: 4.02. Growth in Gross State Domestic Product (1993-94) Prices - A Trend Analysis

Year	Primary Sector	Secondary Sector	Tertiary Sector	GSDP
1994-95	11.25	15.66	10.82	12.56
1995-96	(-) 12.56	8.83	8.50	3.45
1996-97	(-) 0.84	1.99	10.29	4.96
1997-98	8.51	2.26	11.90	7.82
1998-99	9.04	1.61	6.19	5.21
1999-2000	(-) 4.77	10.00	7.51	5.81
2000-01(RE)	4.56	5.88	9.45	7.35
2001-02(QE)	(-) 2.04	(-) 0.19	6.36	2.68
2002-03(AE)	(-) 12.28	0.16	5.13	0.55
AAGR	0.10	5.13	8.46	5.60

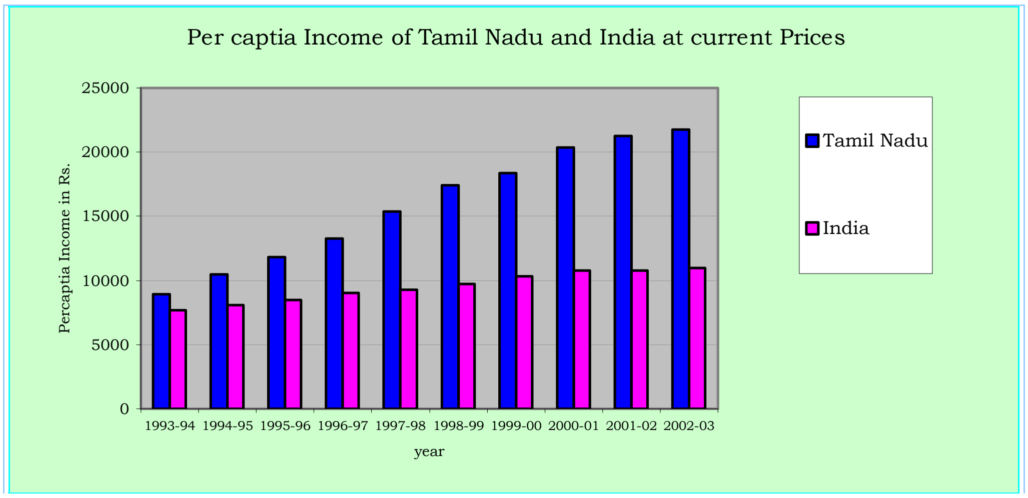
Note: R.E – Revised Estimates, Q.E. – Quick estimates, A.E. – Advanced Estimates:
Source: Department of Economics and Statistics, Chennai-6

4.08 At the National level, the Gross Domestic Product grew by 6 percent via-a-vis 5.6 percent in Tamil Nadu. In per capita terms, it increased by 3.97 percent at the National level whereas it was at 4.29 percent in Tamil Nadu.

Table No.4.03: Per capita Income of State and India at current prices			
Sl.No.	**Year**	**Tamil Nadu Rs.**	**India Rs.**
1	1993-1994	8955	7690
2	1994-1995	10503	8088
3	1995-1996	11818	8499
4	1996-1997	13270	9036
5	1997-1998	15388	9288
6	1998-1999	17394	9733
7	1999-2000	18367	10306
8	2000-2001	20367	10754
9	2001-2002	21239	10774
10	2002-2003	21738	10964

Source: Statistical Hand Book of Tamil Nadu 2003

4.09 The percapita income of Tamil Nadu State has risen from Rs.8, 955 in 1993-94 to Rs.21, 738 in 2002-2003.



Structural Transformation:

4.10 The economic structure of Tamil Nadu has metamorphosised from Primary sector in 1960s to tertiarised activities in 2000s. The percentage share of primary sector has declined from 43.5 percent in 1961 to 16 percent in 2002.

4.11 As regards the secondary sector, though it has increased from 20 percent in 1960 to 35 percent in 1996, it has declined subsequently. The percentage share of secondary sector in 2002-03 is only 29, which is mainly because there is a receding trend in the manufacturing sector.

Table No: 4.04: State Economy - Structural changes (in percentage)				
Year	Primary Sector	Secondary Sector	Tertiary Sector	Total
1960-61	43.51	20.27	36.22	100
1970-71	34.79	26.88	38.33	100
1980-81	25.92	33.49	40.59	100
1990-91	23.42	33.10	43.48	100
1995-96	21.90	34.84	43.27	100
2000-01	19.04	30.49	50.47	100
2002-03	15.97	29.71	54.32	100

4.12 There is considerable increase in the growth of tertiary sector whose share is 54 percent in 2002 and the main reasons attributed to such a rapid growth is the contribution by the Information Technology, Information Technology Enabling Services and Business Process Outsourcing.

4.13 The industrial base of Tamil Nadu is diversified in the manufacturing sectors of automobile, textiles, leather and chemicals. Tamil Nadu is one of the leading States, which attracts more investment in IT. In terms of Foreign Direct Investment (1991-2003) Tamil Nadu stood third in the Nation. The new Industrial Policy of Tamil Nadu 2003 emphasize on optimal use of resources, upgradation of managerial skills and administration, technical improvements and modernization in the filed of manufacturing sector and infrastructure.

4.14 The industrial production in Tamil Nadu, which was passing through a tough phase in the past, has shown a moderate increase in 2002-2003, which is 2.4 percent as against 1.4 percent in 2001-2002. Textile products other than mills' registered a growth of 31.9 percent followed by 'other manufacturing' (26.7percent), 'non-metallic mineral products' (18.0percent) and 'paper and paper products' (12.4percent). The other groups which registered positive growth are 'machinery and equipments' (8.9percent), 'wool, silk and synthetic fibre textiles' (7.7percent) 'metal products and parts' (4.3percent) and 'transport equipment and parts' (4.3percent). In terms of use-based classification except the Basic Goods industries, the other industries i.e. capital goods, intermediate goods and consumer goods have shown moderate growth. The consumer durable goods have recorded the highest growth rate of 16.5percent.

Information Technology Industries:

4.15 The Information Technology (IT) Industry in India is among the fast growing segments of the Indian Industry, compounded with an annual growth rate exceeding 50 percent. Tamil Nadu's contribution to IT industry is significant. The key elements which have made Tamil Nadu an important place in this area are availability of skilled and educated man power, comparatively higher standard of Educational Institutions, sound infrastructure and lower costs of operation.

4.16 As per the Economic Appraisal Report 2002-2003, the Tamil Nadu State is the second largest software exporter in the country next to Karnataka. The number of software companies which was 10 in 1993-94 and exported software to the tune of only Rs.2.4 crores expanded to 934 software units and exported software to the tune of Rs.6315.51 crores in 2002-03 accounting for 17 percent share in all India. It is noted that the global recession did not hinder the software export from Tamil Nadu.

Table No: 4.05: Growth of Software/Hardware Industry in Tamil Nadu

Year	Software Industry		Hardware Industry	
	No. of units	Value of exports (Rs. in Crores)	No. of units	Value of exports (Rs. in Crores)
1993-94	10	2.400	-	-
1999-00	601	1914.00	8	399.82
2000-01	766	3116.00	12	575.65
2001-02	865	5223.00	10	482.43
2002-03	934	6315.51	11	698.25

Source: Software Technology Parks of India, Chennai and SPC Note

Information Technology Enabled Services

4.17 Focusing the immense employment opportunities in Information Technology Enabling Service, Government of Tamil Nadu has taken several steps to leverage the strength of the State to make Tamil Nadu as the prime destination for Information Technology Enabled Services.

Agencies that promote industries in Tamil Nadu

Tamil Nadu Industries Development corporation (TIDCO):

4.18 Tamil Nadu Industries Development Corporation (TIDCO) established in 1985 played a vital role in infrastructure development for industrial growth in Tamil Nadu, including CMA. TIDCO has promoted 75 projects in Tamil Nadu with the employment potential of 1.38 lakh persons. Recently the Government of India has given in principle approval for setting up of Special Economic Zone (SEZ) at Ennore; it is proposed to be implemented by the Ennore SEZ Company formed for this purpose.

Small Industries Development Corporation (SIDCO)

4.19 Small Industries Development Corporation (SIDCO) established in 1970 has developed 74 industrial estates in Tamil Nadu. In CMA, it has developed industrial estates at Ambattur, Guindy, Thirumudivakkam, Thirumazhisai, Thirumullaivoyal, etc. It has constructed 4244 sheds in Tamil Nadu with an employment potential of 41,450.

State Industries Promotion Corporation of Tamil Nadu (SIPCOT) Limited:

4.20 State Industries Promotion Corporation of Tamil Nadu (SIPCOT) limited, a fully owned premier institution, established in the year 1972, has been a catalyst in development of small, medium and large scale industries in Tamil Nadu. The main function of SIPCOT is to establish, develop, maintain and manage industrial complexes, parks and growth centers in Tamil Nadu.

4.21 SIPCOT has created industrial complexes and parks in various strategically located places in Tamil Nadu. SIPCOT has developed the Information Technology Park at Siruseri (3 km south of CMA) extending over an area of 868 acres with all attendant facilities.

Electronic Corporation of Tamil Nadu (ELCOT):

4.22 The Electronic Corporation of Tamil Nadu (ELCOT) is a pioneer in implementing IT related projects in the State. It is the nodal agency for the Government of Tamil Nadu in implementing IT policy and programmes. As on 2002-03, it had promoted 39 projects at the cost of Rs.426 crores in the joint, associate and escort sectors covering a wide range of electronic products.

Distribution of Enterprises and Establishment in CMA:

4.23 Considering the fact that the economy is moving towards liberalization, the need to highlight and monitor its impact has become essential, the Department of Economics and Statistics, Govt. of Tamil Nadu has conducted the Economic Census, in 1998 fourth in the series and published the report in 2000 (when it was conducted, the Kancheepuram District was not bifurcated and hence the Thiruvallur District did not come into existence). The data contained therein are with reference to the following definitions:

(a) Enterprise - is an undertaking engaged in production and or distribution of goods and or services not for the sole purpose of own consumption.
(b) Own Account Enterprises - are enterprises owned and operated with the help of household reasons only.
(c) Establishments – are enterprises operating with at least one hired worker on a regular basis.

Table No.4.06: Distribution of Enterprises, Establishments and Employment in Chennai and Kancheepuram Districts										
		Chennai District			Kancheepuram District			Tamil Nadu		
		Rural	Urban	Total	Rural	Urban	Total	Rural	Urban	Total
1	Distribution of Enterprises									
	OAE	0	38682	38682	31270	26622	57892	918631	506292	1424923
	Establishment	0	109321	109321	20101	25722	45823	489475	599726	1089201
	Total	0	148003	148003	51371	52344	41742	1408106	1106018	2514124
2	No. of Persons working									
	OAE	0	49090	49090	42577	39428	82005	1470096	764967	2235063
	Establishment	0	566747	566747	115941	118769	234710	2113270	2842692	4955962
	Total	0	615837	615837	158518	158197	316715	3583366	3607659	7191025
3	Employment per enterprise									
	OAE	-	1.3	1.3	1.4	1.5	1.4	1.6	1.5	1.6
	Establishment	-	5.2	5.2	5.8	4.6	5.1	4.3	4.7	4.6
	Total	-	4.2	4.2	3.1	3.0	3.1	2.7	3.3	2.9
4	No. of Agriculture									
	OAE	0	476	476	9788	2101	11889	224374	26814	251188
	Establishment	0	603	603	2099	533	2632	40345	11377	51722
	Total	0	1079	1079	11887	2634	14521	264719	38191	302910
5	Employment in Agriculture									
	OAE	0	608	608	14348	2662	17010	339897	39996	379893
	Establishment	0	2175	2175	8379	2141	10520	135275	41824	177099
	Total	0	2783	2783	22727	4803	27537	475172	81820	556992
6	Non-Agriculture									
	OAE	0	38206	38206	21482	24521	46003	694257	479478	1173735
	Establishment	0	108718	108718	18002	25185	43191	44913	588349	1037479
	Total	0	146824	145824	39484	49710	89194	739170	1067827	2211214
7	No. of hired workers in non-agriculture									
	Establishment	0	459753	459753	96466	92731	189197	1654318	2203040	3857358

OAE - Own Account Enterprises

4.24 Agriculture enterprise and establishments in Chennai and Kancheepuram Districts account for minimum and contribute about 5.15 percent to Tamil Nadu. The non-Agriculture enterprise for these two districts works out to 10.58percent of the State. The employment per enterprise is also high when comparing the State average. Number of hired workers in the CMA districts works out to 16.8percent of the State.

4.25 Indian economy is going through structural changes. The share of value added by the primary sector is consistently declining whereas the share of non-primary sector has been increasing Tamil Nadu and CMA are not exceptions to the trend.

Table No.4.07: Non-Agriculture Establishments by size, class of employment

Sl. No.	Employment Size	Chennai District		Kancheepuram District		Tamil Nadu	
		No.	Percent	No.	Percent	No.	Percent
1.	1 – 5	86284	79.37	37132	85.97	871654	84.02
2.	1 – 9	12684	11.67	2949	6.83	92149	8.88
3.	10 – 14	4560	4.19	1238	2.87	37029	3.57
4.	15 – 19	1767	1.63	598	1.38	13346	1.29
5.	20 – 24	1069	0.98	285	0.66	7662	0.74
6.	25 – 49	1652	1.52	552	1.28	10078	0.97
7.	50 – 99	588	0.54	276	0.64	3763	0.36
8.	100 – 199	72	0.07	79	0.18	1025	0.10
9.	200 – 499	25	0.02	62	0.14	539	0.05
10.	500 and above	12	0.01	20	0.05	234	0.02
	Total	108713	100.00	43191	100.00	1037479	100.00

Source: Economics Census of Tamil Nadu, 1998

4.26 From the above, it is seen that the non-agriculture establishments in1998 with 50 and above workers was only 0.64 percent in Chennai and 1.01percent in Kancheepuram Districts against 0.53 percent for the State. The number of non-agriculture establishments in Chennai and Kancheepuram Districts work out to 10.67 percent of the State figures.

Industries in CMA.

4.27 Major industries in CMA are automobile and transport equipment manufacture and their ancillary industries, railway coach building, petro chemicals and fertilizers, automotive tyres, bicycles, electrical and other machinery, and leather products. Some of the large units are located at Ennore, Thiruvottiyur, Manali, Sembiam, Padi, Ambattur, and Porur and along GST Road apart from the Integral Coach Factory at Perambur, and Heavy Vehicles Factory at Avadi. Many small and medium scale industries are located at Vyasarpadi, Ambattur, Villivakkam, Guindy and Thirumazhisai and industrial estates at Madhavaram, Kodungaiyur, Poonamallee, Noombal, Perungudi, Seevaram and Sholinganallur. Simpson, Addison and TVS industries are located in the heart of the City along Anna Salai. MEPZ spreading over an area of 261 acres is functioning at Tambaram. Leather tanneries and leather based industries are located at Pammal and Madhavaram. Thermal Power Plants are located at Basin Bridge and Ennore. Many of the smaller units are scattered in various parts of the Chennai City and the rest of CMA. Industrial estate for leather goods is being developed at Thirumudivakkam.

4.28 Large-Scale automobile engineering, glass and ceramic industries are located within 50 Km. from CMA at Marai Malai Nagar, Irungattukottai, Sriperumbudhur, Thiruvallur and Gummudipoondi; to mention a few are Mahindra Ford factory - manufacturing cars at Marai Malai Nagar, Hyundai Car factory and Saint Gobin Glass factory at Sriperumbudur, Spartek Ceramic tile manufacturing industry and Hindustan Earth Movers and HM Mitsubishi at Thiruvallur. Mahindra Industrial Park developed over an area of 1300 acres located along GST Road (near Chengalpattu) is about 42 Km. from CMA.

4.29 Tamil Nadu accounts for about 21percent passenger cars, 33percent commercial vehicles and 35percent automobile components produced in India. Chennai, the 'Detroit of India' is emerging as a major export hub for cars in South East Asia. In July 2005, the Government of India has decided to establish a new testing and homologation centre near Chennai. It is expected to bring about large savings in the foreign exchange spent on testing exportable vehicles at overseas facilities and also attract foreign exchange inflows by providing a competitive platform for manufacturer abroad to test their vehicles here.

4.30 Tamil Nadu accounts for 70percent of leather tanning companies in India and 38percent of leather footwear and components; most of the footwear industries are located within CMA. A cluster of chemical industries is located at and around Manali in CMA. The following table No.4.08 lists the type of factories in CMA.

Table No.4.08: Number of factories in Chennai, Thiruvallur and Kancheepuram Districts 2002-2003

Sl. No.	Description of Industries	Districts				Tamil Nadu	Percentage to State
		Chennai	Tiruvallur	Kanchee-puram	Total		
1	Food Products and Beverages	85	130	113	328	4166	7.87
2	Tobacco Products	19	3	5	27	237	11.39
3	Textiles	42	44	73	159	6565	2.42
4	Wearing apparel; Dressing and Dyeing	1067	383	258	1708	2105	81.14
5	Tanning & Dressing of Leather products	181	100	279	560	1486	37.69
6	Wood and Products of wood and cork, except furniture; Straw and Plating Materials	11	14	16	41	331	12.39
7	Paper and Paper Products	53	92	56	201	707	28.43
8	Publishing, Printing and Reproduction of Recorded Media	273	31	14	318	927	34.30
9	Coke, Refined petroleum products and Nuclear Fuel	9	18	10	37	67	55.22
10	Chemicals and chemical products	133	182	162	477	2441	19.54
11	Rubber and Plastic Products	121	164	115	400	976	40.98
12	Non-Metallic mineral products	22	132	185	339	1119	30.29
13	Basic Metals	79	238	79	396	921	43.00
14	Fabricated metal products, except Machinery and equipments	233	278	156	667	1263	52.81
15	Machinery and Equipments	170	284	150	604	1536	39.32
16	Office, accounting and computing machinery	6	6	11	23	29	79.31
17	Electrical Machinery and apparatus	96	95	74	265	497	53.32
18	Radio, Television & Communication Equipment and apparatus	63	18	69	150	215	69.77
19	Medical Precision & Optical instruments watches and Clocks	27	22	27	76	108	70.37
20	Motor vehicles, trailers and semi-trailers	96	199	134	429	585	73.33
21	Other Transport Equipments	61	67	41	169	293	57.68
22	Furniture	35	16	45	96	279	34.41
23	Sale, Maintenance & Repair of Motor vehicles and Motor cycles, retail sale of automotive fuel	168	36	40	244	881	27.70
24	Retail trade except of motor vehicles and motor cycles; repair of personal and household goods	25	7	3	35	61	57.38
25	Supporting & Auxiliary transport activities; Activities of travel agencies	6	0	2	8	16	50.00
26	Computer and Relates activities	1	0	0	1	2	50.00
27	Recreational cultural and sporting activities	7	0	2	9	9	100.00
28	Other service activities	10	5	0	15	15	100.00

Source: Annual survey of Industries Frame 2002-2003. Director of Economics and Statistics

4.31 The above table indicates that 70percent of the industries in the sectors of dressing and dyeing, office, accounting and computing machinery, medical precision and optical instruments, watches and clocks, motor vehicles, trailers and semi trailer Industries in the State is located in Chennai, Thiruvallur and Kancheepuram Districts.

Small Scale Industries

4.32 Special emphasis is always given to the development of Small Industrial Sector considering the strength of inherent merits it has on low capital investment, short gestation period, high employment generating potential, capability to induce industrial base etc.

4.33 In the Small-scale industrial sector, there were 56,913 units in Chennai Districts and 37,531 in Kancheepuram and 17,843 in Thiruvallur District as on 31-03-2007, which works out to about 21.16 percent of the units in the State (5,30,552 Units); it was 23percent of the units in the State (1,10,783 units) in 1989. Small-scale industrial sector is high in metals, rubber and plastic products, metal products, electrical machinery, transport equipments, leather and fur products and non-metallic mineral products in Chennai, Kancheepuram and Thiruvallur Districts.

Information Technology:

4.34 Chennai is perhaps the only city in India to have all the top 10 IT Indian multi national companies and the 3 IT majors viz. Infosys, Tata Consultancy Services Ltd and Wipro which have acquired lands in and around Chennai to meet their expansion plans. The Tidal Park I and the IT Park at Siruseri have already been developed in Chennai and its environs. The Tidel Park I is fully functional. The private IT developers have been enthused to build enough IT space and the Government is certain of creating 2.5 million sq.ft of IT space in private as well as in public sectors in the coming years to meet the growing requirements of national and international clients. The first phase of Knowledge Industrial Township is being planned by a Special Purpose Vehicle viz. ELCOT Infrastructure Ltd. in Sholinganallur along the IT Corridor.

4.35 Further, to meet the increasing demand for such parks in the State, steps are being initiated to establish TIDEL Park II with a floor space of 1.5 million sq.ft. in Chennai city. It may be noted that a joint sector IT Park in an area of 1700 acres is being developed with participation of TIDCO, IL&FS and Mahindra near Chengalpattu.

Sales Tax

4.36 Sales tax is the most important head of the tax revenue to the Government. From the table it could be seen that Chennai Metropolitan Area accounts for three-fourth of the total sales tax collection in the State. The share has increased from 69percent in 1994-95 to 76 percent in 2003-04.

Table No. 4.09: Sales Tax collection in Chennai Metropolitan Area and Tamil Nadu (1990 - 2004 year-wise)			
Year	Sales Tax collected (Rs. in Crores)		Share of CMA as percentage of State
	Chennai Metropolitan Area (CMA)	Tamil Nadu	
1994-1995	2886.56	4164.84	69.31
1995-1996	3458.41	5023.37	68.35
1996-1997	4125.50	5751.19	71.73
1997-1998	4462.62	6062.68	73.61
1998-1999	4881.30	6582.60	74.15
1999-2000	5509.94	7380.66	74.65
2000-2001	6575.54	8647.97	76.04
2001-2002	6776.86	8895.75	76.18
2002-2003	7758.86	10191.60	76.13
2003-2004	8897.10	11734.30	75.82

Source: Commissioner for Commercial Taxes

Trade and Commerce:

4.37 The working of economy and the generators of economic momentum are reflected by the volume and value of trade and commerce transacted in the area. Chennai city is the commercial centre of great importance in the southern part of India, having a major seaport and international airport. The direct relevance to economy is the movement of trade and commerce between Chennai and the rest of the world. Increase in port traffic and airport cargo handling and also increase in the sales tax collections and banking transactions indicate the increase in total volume of trade in CMA.

Chennai Port:

4.38 Chennai Port is the largest among all the ports of Tamil Nadu, and is one of the most important ports of India. It is well equipped in terms of shipping facilities (23 berths including 4 exclusive berths for containers), marine services and other associated facilities like warehouses and storages. There are 12 warehouses and the total area of the warehouses is 71,653 sq.m. owned by Chennai Port. Further, 61,222 sq.m. area for warehouse/storage was allotted to private parties. The Port has full-fledged container terminals with road and rail connections which offer all the advantages that containerization could provide such as packaging, landing, pilferage prevention and speedy transportation of cargo. The Port measures a water-spread area of 170 hectares and a land extent of 238 hectares. Table Nos. 3.11 & 3.12 indicate the export and import details of principal commodities at Chennai Port respectively. The principal items of imports are petroleum, oil, lubricants, fertilizers, food grains and fibres. The main items of export are ores (mainly iron ores) granite stones, quartz, Barytes, hides and skins, chemical and cotton goods. The growth of the Chennai Port in terms of cargo handled value of imports and exports are respectively given in Table no. 4.10. Chennai Port handles 60percent of the total cargo handled by the State. The total number of containers handled during 2003 - 2004 was 5, 39,265 showing an increase of 67percent in the last 5 years. The passenger traffic shows an increase of 22percent in the last five years.

Table No.4.10: Export of Principal Commodities - Chennai Port (Tonnes)

Year	Petroleum & oil lubricants	Iron Ore	Other Ores	Food Grains	Containerised Cargo	Other Cargo	Total
1999-2000	1.342.508	6,258,614	299.268		1,958,537	1,146,363	11005290
2000-2001	957,245	7,048,678	535,769		2,880,038	1,063,959	12485689
2001-2002	914,555	7,481,950	613,840		2,757,349	1,258,712	13026406
2002-2003	952,608	7,944,621	443,017		3,536,476	1,204,723	14081445
2003-2004	1,310,489	7,519,527	619,246	19,499	4,106,304	2,832,750	16407815
GRAND TOTAL	5,477,405	36,253,390	2,511,140	19,499	15,238,704	7,506,507	67006645

Note: Other Cargo includes chemicals, Granite stones etc. excludes transshipment Cargo

Source: Port Trust, Chennai, 600 001

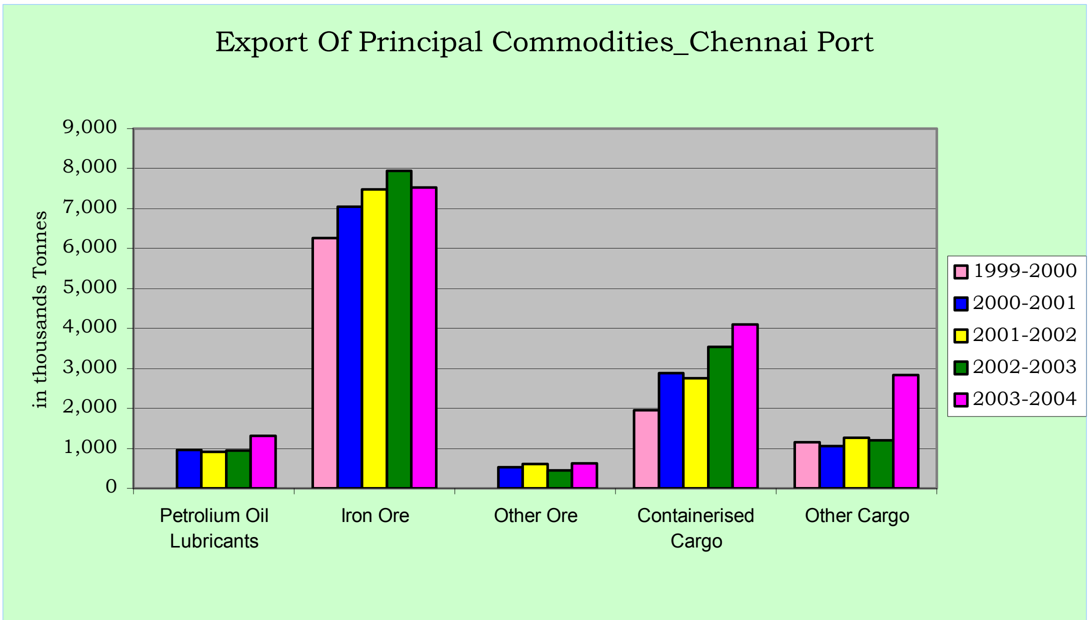


Table No. 4.11: Import of Principal Commodities - Chennai Port (Tonnes)

Year	Petroleum, Oil, Lubricants	Fertilizers (Finished)	Fertilizers (Raw)	Food Grains	Coal & Coke	Containerized Cargo	Other Cargo	Total
1999-2000	8,829,356	729,094	479,435	135,878	12,522,036	2,017,669	1,724,263	26437731
2000-2001	7,938,115	459,578	466,210	0	14,990,972	3,109,181	1,770,370	28734426
2001-2002	7,559,293	459,246	351,657	0	9,655,623	3,099,695	1,963,546	23089060
2002-2003	7,993,042	420,178	287,647	0	5,818,291	3,682,829	1,403,674	19605661
2003-2004	7,890,436	411,972	326,061	0	5,630,922	4,521,936	1,520,908	20302235

Source: Port Trust, Chennai

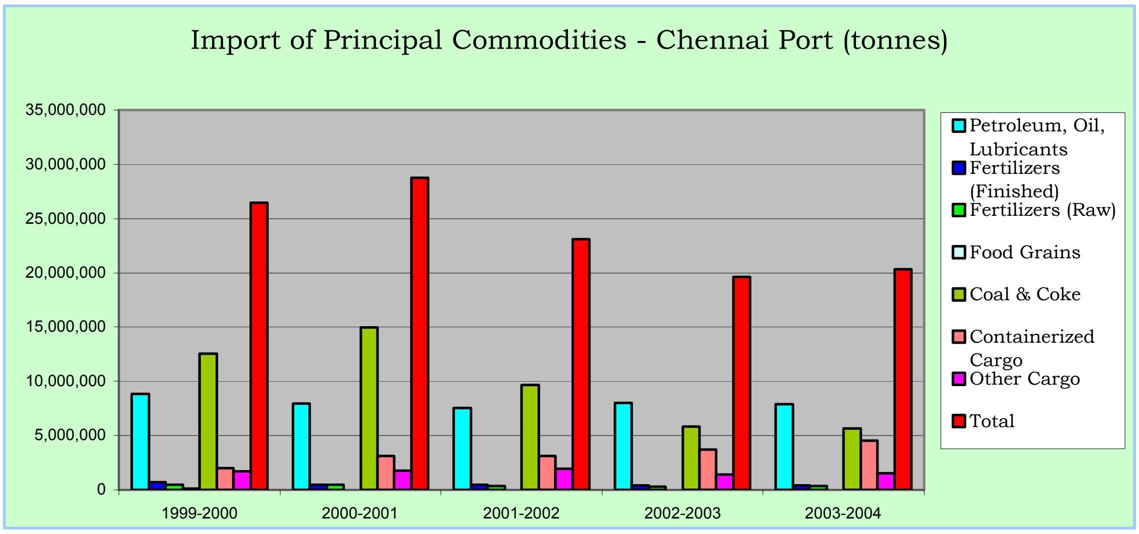
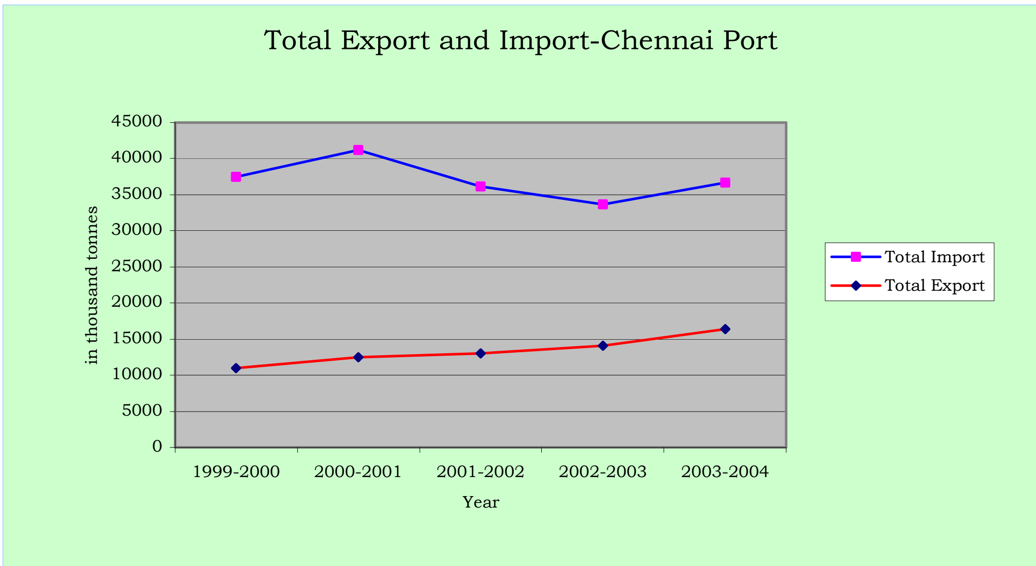


Table No. 4.12: Value of Foreign Imports and Exports - Chennai Port Trust (Rs.in Crores)

	1999-2000	2000-2001	2001-2002	2002-2003	2003-2004
Imports	19277.72	22969.75	22791.28	23903.74	31145.51
Exports	12945.26	15001.17	15194.39	18747.71	24110.03

Source: Port Trust, Chennai-600 001



Ennore Port

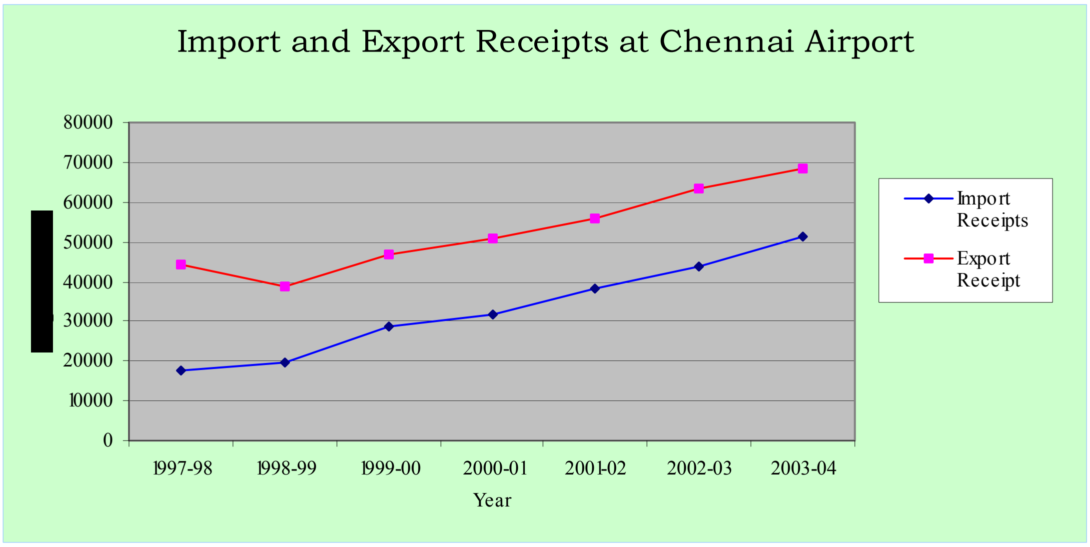
4.39 The Port of Ennore was conceived as a result of Government of India (GOI) policy of locating thermal plants at the coast in order to avail the least cost option of transportation of coal through coastal shipping. The Port was inaugurated on 1st Feb.2001 and the commercial operations commenced on 22nd June 2001. Currently Ennore Port is operating with two coal berths catering to thermal coal requirements of North Chennai Thermal Power Station (NCTPS), Mettur and Ennore Thermal Plants of TNEB. The Port is expanding in a major way by developing Liquid Cargo, LNG, additional coal, iron ore, car and container terminals with additional 10 berths. By 2011, Ennore Port will have capacity to handle 70 million tonnes of cargo including 1.5 million containers and 2 lakhs cars. Development of a port specific SEZ near the port is being planned by TIDCO for enhancing the economic opportunities of the port as well as the region.

Chennai Airport

4.40 The Chennai Airport is one of the major National and International Airports in India. It handled about 20.54 lakh international passengers and 25.01 Lakhs domestic passengers and cargo of about 1.19 lakh tonnes (international) and 0.35 lakh tonnes (domestic) in 2003-2004, and has a significant share in total passengers as well as cargo handled in the four major airports. The international passengers handled by Chennai Air-port have increased 4-fold from 1991 to 2004 whereas the domestic passengers have increased 2-fold in the same years.

Table No.4.13: Import and Export receipt at Chennai Airport							
Import/Export Year	1997- 98	1998-99	1999-00	2000-01	2001-02	2002-03	2003-04
Import Receipts	17766	19780	28721	31739	38305	43570	51120
Export Receipts	44138	38928	46702	50577	55866	63264	68443

Source: Airports Authority of India



4.41 Textile constituting major part of the export receipt i.e. 38.3percent, followed by leather constituting 25.84percent. Electronic goods constitutes major part of the import receipt i.e. 23.05percent followed by personal effects constituting 21.44percent.

4.42 Sixteen international carriers operate from Chennai. Chennai can expect to attract more international carriers after the expansion programme. Government of India has given its consent to expand and modernize the Chennai Airport to handle future air traffic and make it a world-class major airport. Government of Tamil Nadu will make available 1457 acres of land to Air-port Authority of India adjoining the existing airport in the north. The expansion and modernization of Airport will have major impact of the economy of the CMA/the State.

Banking

4.43 The dominant position of Chennai in the State economy is reflected by the trends in banking sector. The banking infrastructure in Tamil Nadu was extensive and the banking network in Chennai, Kancheepuram District and Thiruvallur District is found to be quite strong. The performance in banking could be examined by the indicators like number of branch offices, size of deposits and advances and credit deposit ratio. Bank credit deposit ratio denotes the level of investments in the City being an indicator frequently used by economists. Bank deposit per capita indicates the finance level and disposable income of the population**.** Table No.4.15 indicates the position of banking in Chennai, Kancheepuram and Thiruvallur Districts. Around 16.5 Percent of the total number of bank branches in Tamil Nadu is found to be concentrated in Chennai City. Deposits and advances account for 42 and 48 percent respectively of the State figures. The capacity of Chennai City in credit absorption appears to be very good and it keeps on rising. Further foreign banks also entered in the fray and their operation also gradually expanding. Apart from Banks, large Non-Banking Finance companies also provide financial services in CMA.

4.44 The Table No. 4.14 indicates that there are 780 banks in Chennai City alone (apart from the private financial institutions).

Table No.4.14: Banking Sector in Chennai, Kancheepuram & Thiruvallur Districts

Districts	Bank (nos.)	Deposits (Rs. in lakhs)	Advances (Rs.in Lakhs)	Credit Deposit Ratio (in percentage)
Chennai	780 (16.43)	35654.02 (41.29)	43010.75 (57.66)	120.63
Kancheepuram	158 (3.33)	2896.09 (3.35)	956.88 (1.26)	33.04
Thiruvallur	140 (2.95)	2192.53 (2.54)	849.155 (1.14)	38.73
Tamil Nadu	4746	86343.39	74587.43	86.38

Note: The figures in parenthesis are percentages to State level
Source: Reserve Bank of India

Employment

4.45 As per Census 2001, the total no. of workers in Tamil Nadu is 278 lakhs and the rate of increase in the employment in the last decade is 1.4percent against the population growth of 1.1percent per annum.

4.46 The comprehensive profile of employment in CMA has been made on the basis of secondary sources including Census data. The participation rate i.e. proportion of main workers to the population of CMA was 30.74percent in 1991 and 30.96percent in 2001. The corresponding figures for Chennai City was 30.50percent in 1991 and 31.79percent in 2001. The number of marginal workers both in the Chennai City and in CMA is negligible**.** The table No. 4.15 gives the main workers classified by industrial category for Chennai City and CMA.

Table No: 4.15: Occupational structure CMA _ 1991 and 2001					
1991			2001		
	Chennai City	CMA		Chennai City	CMA
Total Workers	1173062	1675512	Total Workers	1488364	2519278
Main Workers	1171739	1669213	Main Workers	1380757	2284457
Main Cultivators	883	19778	Main Cultivators	15149	33170
Main Agriculture Labourers	199	70085	Main Agriculture	5849	33390
Fishing & Forest	9982	15422	Main House Hold	25836	43394
Mining	1245	3484	Main Others	1333923	2174503
Manufacturing House Hold	7683	20271	Marginal workers	107607	234821
Manufacturing & others	275916	423253	Marginal Cultivators	2026	5728
Main Construction	74856	104913	Marginal Agricultural	1233	22681
Main Trade	300928	372672	Marginal House Hold	5156	10511
Main Transport	125853	166648	Marginal Others	99192	195901
Main Others	374194	472687	Non Workers	2855281	4859201
Marginal Workers	1323	6299			
Non Workers	2668334	3753958			

Source: Census of India

4.47 The workers in primary activity constitute 6.52 percent in CMA and 1.05 percent in City 1991. In 2001, it was 2.91 percent and 1.52 percent respectively in CMA and Chennai City. The percentage of workers in household industry of manufacturing sector more or less remains same in Chennai City and CMA. The percentage of workers in other services has shown a higher increase both in the City and CMA.

4.48 The workers in primary activity are dwindling and it is negligible compared to total, both in Chennai City and in CMA. The occupational structure of Chennai City and CMA reveals that major share of the work force is engaged in tertiary sector.

Non-workers

4.49 The percentage of non- workers was 65.73 percent in City and 69.14 percent in CMA during 2001. Among males, 45.88 percent are non-workers and the corresponding figure for females was 86.48 percent in Chennai City. However, in CMA, 48.83percent and 90.63percent are non-workers, among males and females respectively. The percentage of female non-workers is high in both City and CMA.

Employment in organised sector.

4.50 The following are the distribution of employment in public and private organised sector in Chennai city and Kancheepuram districts.

Table No. 4.16: Employment of organized sector in Chennai, Kancheepuram (including Thiruvallur District)

Year	Chennai Dist.			Kancheepuram & Thiruvallur Districts		
	Public	Private	Total	Public	Private	Total
2000-01	281287	102358	383645	100491	124626	225117
2001-02	316553	94674	411227	108013	136234	244247
2002-03	314743	64914	379657	109678	132298	241976
2003-04	322961	69280	392241	100163	112341	212504
2004-05	323221	68785	392006	81032	118461	199493

Source: Commissionerate of Employment & Training, Chennai - 600 032

4.51 In Chennai District, over these years from 2000, employment in the organized public sector was increasing whereas in private sector it was decreasing with little variance in the total employment figures in this organized sector.

4.52 Chennai and Kancheepuram together contribute about 26percent of employment in organized sector to the State total in 2002-03. Of the total public sector employment, the employment opportunity in Chennai and Kancheepuram Districts are 20percent and 7percent respectively in 2002-03. The percentage of the same in private sector in Chennai and Kancheepuram Districts are 7.8 and 15.9

4.53 Informal sector generally denotes the activities and services, which are readily associated with low wages and low level of skills.

The informal sector comprises broadly in the following groups:

- Self-employed traders, hawkers and family based workers
- Employees and workers in predominantly low wage paying sectors like rickshaw-pulling, repair of bicycles, personal services, etc.

- Employees and workers in comparatively better wage paying occupation such as construction, manufacturing and repair activities.

4.54 When comparing the Census data of 2001 total employees and available figures in Table No. 4.16 on organized sector employment, the employees in informal sector would be in the order of 10.8 lakhs in Chennai City.

Employment Projection:

4.55 Table No.3.17 (chapter III) gives the information about the age structure in CMA from 1961 to 2001.

Table No: 4.17: CMA-Age Structure -2001 & 2011

Age Group	2001		2011
	Male	Female	
0-4	7.92	7.91	6.37
5-9	8.33	8.43	7.21
10-14	9.23	9.17	8.66
15-19	9.72	9.64	9.89
20-24	10.32	10.83	10.52
25-29	10.00	10.95	10.47
30-34	8.69	8.30	8.67
35-39	8.12	8.20	8.51
40-44	6.54	5.89	6.45
45-49	5.79	5.34	5.96
50-54	4.58	4.11	4.61
55-59	3.12	3.06	3.45
60-64	2.73	2.80	3.10
65-69	1.77	1.95	2.29
>70	1.29	1.40	3.84
Not stated	1.87	2.00	0.00

4.56 It is assumed that age structure proportion will stabilise in the ratio projected for 2011.The age group between 15-60 is taken as the eligible age group for employment. It is found that this age group has increased from 59.66 percent in 1961 to 66.02 percent in 2001 and it is assumed that this age group will stabilize at 69.15 percent as projected for 2011 upto 2026.The ratio as found in the last 3 decades is taken as 48% and 52% for male and female for CMA population.

4.57 The percentage of workers as per 2001 Census works out to 54.6 % out of the male population and 18.26 % out of the female population. It is assumed that in future years by 2011, 87.5 % of male and 30% of the female eligible working population will be willing to work and for them jobs would have to be created. Similarly this % for male and female is assumed as 90% and 33% for 2016, 92.5%and 36% for 2021 and 95% and 40% for the year 2026, considering the improvements in the education level, aptitude and willingness to work etc in future years. Based on the above criteria, projection for demand for jobs upto 2026 has been made and tabulated in table No. 4.18.

Table No: 4.18: Employment Projection CMA_2026 (in lakhs)

	2011	2016	2021	2026
Population	88.71	99.66	111.98	125.82
Eligible Workers 15-60 (69.15%)	61.34	68.92	77.43	87.01
Eligible Male Workers (52%)	31.90	35.84	40.26	45.24
Eligible Female Workers (48%)	29.44	33.08	37.17	41.76
Male Willing to Work	27.91	32.25	37.25	42.98
Female Willing to Work	8.37	10.64	13.41	17.19
Total	36.28	42.89	50.65	60.17
Additional Jobs to be created	10.09	16.70	24.47	33.99
Male Willing to Work %(assumed)	87.5 %	90 %	92.5 %	95 %
Female Willing to Work %(assumed)	30 %	33 %	36 %	40 %

Chapter V

TRAFFIC AND TRANSPORTATION

The rapid growth of population in the Chennai Metropolitan Area (CMA) has been causing a strain on the existing urban services and infrastructure, for want of expansion and better management. The transport sector is vital and needs carefully planned expansion to meet the demands of the increasing population.

5.02. The need to take an integrated long term view of transport needs of CMA and to plan road development, public transport services and suburban rail transport, as a part of the urban planning process, have been well recognised.

5.03. A résumé of the changing travel characteristics of the metropolis and the growth of sea and air traffic which have a bearing on the quality of the road and transport infrastructure are indicated in the Annexure I.

5.04. Many studies have been done in the past for development of transportation in CMA. These include Madras Area Transport Study (MATS - 1968), Integrated Transport Plan (1977), Madras Route Rationalisation Study (1986), Traffic and Transportation Study for MMA (Kirloskar 1986), Comprehensive Traffic and Transportation Study (CTTS 1992-95) and other studies done through consultants for specific transportation projects. Based on the recommendations of these studies several major projects such as formation of Jawaharlal Nehru Salai (IRR) , addition of buses, improvements to Metropolitan Transport Corporation (MTC) infrastructure, Mass Rapid Transit system (MRTS) etc have been implemented. But these efforts have not kept pace with the increase in travel demand. A list of major efforts taken is indicated in the Annexure II.

5.05 Major recommendations of some of the recent studies which qualify for implementation on their merits are summarised below. These studies also assisted in the development of a matrix comparing the system characteristics of various alternative transit technologies which is embodied in Annexure III.

CTTS for CMA, RITES Ltd., PTCS & KCL September 1995 & Updating the CTTS (1992-95), RITES Ltd., April 2004

5.06. The comprehensive Traffic and Transportation Study (CTTS) for CMA undertaken through a consortium of consultants, M/s. RITES and M/s. KCL, commenced in May 91 was completed in 1995. The expatriate consultant, W.S. Atkins Planning and Management Consultants Limited, U.K. provided technical assistance for the study. The cost of the study, which was Rs.9.75 m, was shared partly by the GoI and partly under Technical Assistance component of TNUDP I. The schemes from the study report have been included in the Chennai Metropolitan Development Plan (CMDP).

5.07. The Study provided the core inputs for predicting the future travel demand for the CMA. The travel demands in 2004, 2006, 2011, 2016, 2021 & 2026 have been projected on the basis of increase in per capita trips (from 1.32 in 2004 to 1.6 by 2016 and 1.65 by 2026). Table no. 5.01 gives 3 scenarios based on different modal splits between the road and rail system. The 3 scenarios have been worked out gradually increasing the share of the public transport between the public and private transport modes and also increasing the share of the rail transport between the bus and rail modes. Scenario 2 has been selected based on the following assumptions.

i) The modal split between public and private transport will change from 43:57 to 35:65 (2004), 55:45 (2011) and 60:40 (2016), 65:35 (2021) and 70:30 (2026) in line with the trend in share of public transport increasing with City size.

ii) The sub-modal split between bus and rail will have to change from 91:9 to 85:15 (2004), 75:25 (2011), 70.30 (2016), 65:35 (2021) and 60:40 (2026) if the road transport system is not to break down in the context of increased commuter trips.

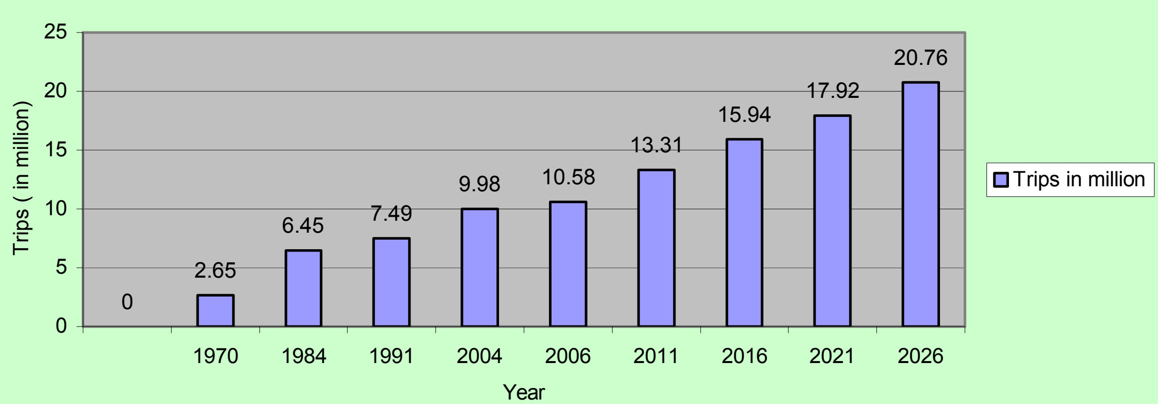
'able No. 5.01: Projected daily trips by public transport

		1991	2004	2006	2011	2016	2021	2026
1. Population in lakh		58.07	75.61	78.96	88.71	99.62	111.98	125.82
2. Daily per capita Trips		1.29	1.32	1.34	1.5	1.6	1.6	1.65
3. Total Daily Person Trips in lakh		74.91	99.81	105.81	133.07	159.39	179.17	207.60
Scenario 1 Modal Split %	Private	57	64.57	60.00	50	45	40	35
	Public	43	35.43	40.00	50	55	60	65
Total Daily Person Trips by Public Transport in lakh		32.21	35.36	42.32	66.53	87.67	107.50	134.94
	By Rail %	9.25	14.54	16.00	20	25	30	25
	By Road %	90.75	85.46	84.00	80	75	70	75
Daily Trips in lakhs								
	By Rail	2.98	5.14	6.77	13.31	21.92	32.25	33.74
	By Road	29.23	30.22	35.55	53.23	65.75	75.25	101.21

Scenario 2 Modal Split %	Private	57	64.57	55.00	45	40	35	30
	Public	43	35.43	45.00	55	60	65	70
Total Daily Person Trips by Public Transport in lakh		32.21	35.36	47.61	73.19	95.64	116.46	145.32
	By Rail %	9.25	14.54	16.00	25	30	35	40
	By Road %	90.75	85.46	84.00	75	70	65	60
Daily Trips in lakh	By Rail	2.98	5.14	7.62	18.30	28.69	40.76	58.13
	By Road	29.23	30.22	39.99	54.89	66.94	75.70	87.19
Scenario 3 Modal Split %	Private	57	64.57	50.00	40	35	30	25
	Public	43	35.43	50.00	60	65	70	75
Total Daily person Trips by Public Transport in lakh		32.21	35.36	52.90	79.84	103.60	125.42	155.70
	By Rail %	9.25	14.54	20.00	30	35	40	45
	By Road %	90.75	85.46	80.00	70	65	60	55
Daily Trips in lakhs	By Rail	2.98	5.14	10.58	23.95	36.26	50.17	70.07
	By Road	29.23	30.22	42.32	55.89	67.34	75.25	85.64

Source : *CTTS(MMDA, RITES, KCL & PTCS, 1992-95) & Short term study to updateCTTS (1992-95)(CMDA, RITES & PTCS, 2004)*

Fig.No. 5.01 Growth of Travel Demand in CMA

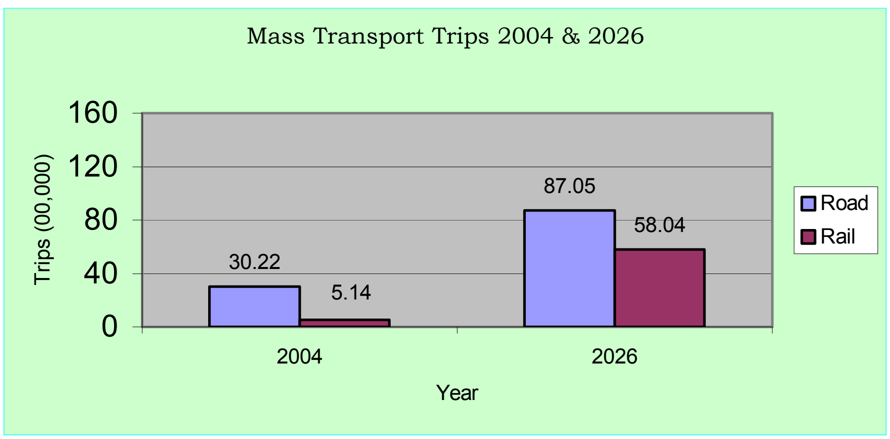


5.08. It will be seen from the Table no.5.01 that the number of trips carried by bus transport in 2004 would become nearly 2.8 times in the year 2026. Similarly the volume of passengers to be carried by rail transport will be nearly 11 times the present volume.

(in lakhs)

Table No. 5.02: Mass Transport Trips 2004 & 2026		
Mass Transport Trips (in lakhs)	2004	2026
Total Mass Transport Trips	35.36	145.09
Increase in 22 years	-	109.73
Total road (bus trips)	30.22	87.05
Increase in 22 years	-	56.83
Total rail trips	5.14	58.04
Increase in 22 years	-	52.90

Source: Short term study to update CTTS (1992-95)(CMDA, RITES & PTCS, 2004)



Detailed Project Report, DMRC & RITES, October 2007

5.09. Following a detailed feasibility study at a cost of Rs.5.4 m undertaken through Delhi Metro Rail Corporation (DMRC) GoTN have retained DMRC for preparing detailed project report at a cost of Rs.33 m for development of metro rail for Chennai for implementation during 2007- 2012. The gist of the DPR is as below:

Corridor-1: NH-45 (Airport) – Guindy – Sardar Patel Road – Kotturpuram High Road – Cenotah Road – Anna Salai – Gemini – Spencers – Tarapore Towers – Along Cooum River upto Rippon building – Central Station – Broadway (Prakasam Road) – Old Jail Road – Tiruvottiyur High Road (upto Tiruvottiyur).

Corridor – 2: Along Poonamallee High Road (Corporation limits) – EVR Periyar Salai – Rajaji Road (North Beach Road) covering Koyambedu – Anna Nagar Arch – Aminjikarai – Kilpauk Medical College – Egmore – Central-Fort-Beach.

Description	Underground		Elevated		Total	
	Length (km)	No. of Stations	Length (km)	No. of Stations	Length (km)	No. of Stations
Corridor 1	14.250	11	8.805	7	23.055	18
Corridor 2	-	-	23.449	19	23.449	19
Corridor 1 & 2	14.250	11	32.254	26	46.504	37

Options	Corridors	Cost including taxes @ 2007 price Rs. in m
Option I *(Recommended)*	Corridor 1 *(Underground from Washermanpet to Saidapet and remaining length up to Chennai Airport elevated)*	56890
	Corridor 2 *(elevated)*	33430
Option II	Corridor 1 *(underground)*	74080
	Corridor 2 *(elevated)*	79780

Detailed Project Report on Hybrid Mono Rail System for Chennai, Metrail India Private Ltd., November 2004

5.10. A detailed project report undertaken by a consortium of Metrail India (p) Ltd. has established the feasibility for development of hybrid monorail system for Chennai for a length of 25 km comprising parts of Periyar EVR salai and Anna salai at a cost Rs.9470 m. It recommended implementation of the project over a period of 27 months. Since the same corridors are being considered subsequently for development of metro rail, the project has not so far been implemented.

Infrastructure Development Plan for Ennore Area, CMDA, May 1999

5.11. At the instance of GTN, CMDA prepared an infrastructure development plan for Ennore area. The inputs for the Plan were collected from the agencies concerned and industries through a Core Group chaired by the Chief Planner. The cost of implementing the Plan was estimated at Rs.51620 m. The draft plan was presented before the High Level Official Committee for Major Infrastructure Projects in Tamil Nadu chaired by Chief Secretary to GoTN in Mar.1999. The schemes under the Plan have since been included in the Infrastructure Investment Plan for the CMA (now termed as CMDP) and the on-going TNUDP III.

MRTS Ph.III (Velachery to St.Thomas Mount), RITES Ltd., May ,2000

5.12. CMDA retained the services of M/s. RITES in February '97 for carrying out a short-term study for establishing the feasibility of the alignment proposed for MRTS from Velachery to Villivakkam for a distance of about 25 km. The cost of the Study was Rs.0.75 m. The consultancy report was approved by the Authority in Mar.2001. As a follow-up, the first stage of the MRTS Phase III (now termed as extension of MRTS phase II) envisaging the elevated MRTS on single pillars along the median of the IRR, projectised at a cost of Rs.4160 m, has since been approved by GoI in Feb.2007 for joint implementation with GoTN. The project has since been commissioned by MTP®.

Densification of MRTS Corridor Development, L&T-Ramboll, October 2001

5.13. The feasibility study on densification of MRTS corridor development undertaken in 2001-03 indicated that the daily transit ridership on the MRTS corridor could be increased to 3.86 lakh trips in 2006, 6.62 lakh trips in 2011 and 11.86 lakh trips in 2021 by allowing a modest density of 400 persons / ha and other minimal intervention. Based on the recommendations necessary changes in the development regulations and land use planning have since been incorporated.

Infrastructure Investment Plan for CMA, CMDA, February 2003

5.14. The investment plan termed Chennai Metropolitan Development Plan (CMDP) to be implemented in the short term (one year), medium term (3years) and long term (5-10 years) comprised urban transportation schemes to be implemented in the short- term at a cost of Rs.6726.3 m, the medium-term at a cost of Rs.34600.2 m and in the long-term at a cost of Rs.61135.6 m. Works on CMDP commenced in 2003- 2004 with budgetary support by GoTN. In tandem with investments by GoTN, a shelf of short and medium-term traffic and transportation subprojects culled out from CMDP (and validated through a quick study to update CTTS (1992-95))has since been included in the on-going TNUDP III at a cost of USD 150 m.

Study on Parking Requirements for CMA, Wilbur Smith Associates Pvt. Ltd., Nov. 2003

5.15. CMDA has undertaken a two-stage parking study for the CMA. The first stage study has principally focused on the problems of parking across the CMA and drawn up a comprehensive parking policy for the CMA as a whole. The upshot of the study is outlined as follows:

(a) Based on field surveys covering 360 critical stretches, the total peak parking demand in the city is in the order of 13,000 PCE against a supply of 5100 PCE. For example the supply in T. Nagar is 794 PCE against a demand of 2151 PCE and the supply in Parrys is 704 PCE against a demand of 4426 PCE. The haphazard parking has led to loss in the road capacity that ranges between15% to 60%.

(b) While the field studies on on-street parking covered 27 critical areas, it covered 15 areas outside City. The field studies also covered parking provided at major industrial areas, various educational institutions, religious places, recreational centres, hospitals, bus terminals and rail terminals.

(c) After taking stock of the entire parking problems and issues in the CMA, the study recommended a parking policy for Chennai on the basis of best practices followed both inside and outside the country. The thrust of the recommended parking policy is as follows:

(i) Short-stay parking is preferably located in proximity to trip destinations and protected from long-stay parkers;

(ii) Institutions (e.g. education institutions), industrial establishments, commercial complexes, cinema theatres, kalyana mandapams, entertainment halls, hotels and restaurants should provide adequate off-street parking facilities for employees, visitors etc;

(iii) Commuter parking should be provided at the railway stations and at the MTC bus terminals by the respective authorities to facilitate the commuters to adopt the park and ride concept;

(iv) Multi level parking (ramp type and mechanical parking) facilities should be planned and developed at suitable locations;

(v) Considering the existing road network and the growth trend in the private vehicle population, it is necessary to bring down the demand on parking spaces, both on-street and off-street;

(vi) Since transportation is a function of land-use, allocation of spaces for various land-uses within the CMA could be done with a view on reducing the use of private motorised vehicles such as high dense developments, exclusive commercial neighbourhoods;

(vii) Parking pricing should be judiciously devised to manage parking problem on the demand side. Till the proposed Unified Metropolitan Transport Authority (UMTA), is formed the agencies / departments which are currently looking after parking related issues should be facilitated to perform their expected roles effectively and in a co-ordinated manner;

An effective institutional structure is necessary to implement the various provisions of the parking policy discussed above.

(d) A critical element of the parking policy is to evolve off-street parking standards for various activities, which would become ultimately part of the Development Regulation of the Master Plan for the CMA. With an objective of promoting transit ridership, a special set of parking standards for the influence area of rail corridors have been suggested where lesser parking standards has been mooted. Separate set of standards has thus been evolved for (i) the areas falling within Corporation limits, Municipalities and IT corridor, (ii) panchayat areas and (iii) transit corridors.

(e) These have since been incorporated in the Development Regulation.The study has also drawn up short-term strategies and solutions for managing the parking problems in the CMA. Recognising the urgency to increase the supply of parking space particularly at critical traffic generating locations, a 6- month long consultancy study has been commissioned in June 2005 through M/s. MEKON to prepare detailed project reports for development of multi-level parking complexes on public-private partnership at 6 locations at a total cost of Rs. 2.1 m. Based on the DPRs, schemes at a cost of Rs.929.8 m have since been included in the the Master Plan II.

Feasibility Study for Development of the Outer Ring Road in CMA, TNRDC / SOWIL, May 2007

5.16. The consultancy study undertaken by CMDA (through TNRDC and SOWiL) at a cost of Rs.6.71 m established the feasibility of implementing the 62.3 km long ORR project as a multi-modal corridor with area development on the western side to a depth of 50 m. The approximate cost of constructing the road with 2-lane divided carriageway with service road and footpath on either side in the first phase is estimated at Rs.9000 m. Based on the Study, action is being pursued for implementing the project with external assistance from JBIC through a SPV.

5.17. For the purposes of reestablishing or validating the need for most or all the major capital-intensive schemes enlisted for implementation in the medium or long term (which have been contained in the volume I of the Master Plan II), it is desirable that appropriate studies are taken up in course of implementation of the Master Plan II. The list of such studies, though not exhaustive shall include the following:

- Comprehensive Transportation Study for Chennai (CTS): The 18-month long study at a cost of Rs.16.2 m has just commenced with World Bank assistance. The broad terms of the study are indicated in the Annexure IV.
- Detailed Engineering Survey for construction of ORR: Once a SPV is floated by the TNRIDC, it would undertake the study through the consultants.

- Detailed feasibility study for the 3rd corridor for the proposed metro rail.
- Preparing a comprehensive road plan for the Outer-CMA as a whole with a grid of say 2 km x 2 km.
- Detailed feasibility study for the proposed elevated highways along City waterways.
- Detailed feasibility study for the elevated highways proposed along other corridors.
- Detailed feasibility study for the freight corridors proposed along selected corridors.
- Detailed feasibility study for the new out-station truck terminals proposed at the intersections of ORR with NHs and SHs.
- Detailed feasibility study for new rail lines in and around CMA.
- Detailed feasibility study for the primary and secondary road network immediately outside CMA but within the region.
- Study establishing the feasibility of the waterways in CMA as inland transport corridors.
- Study establishing the feasibility of the operation of hovercraft along the seacoast.
- Detailed feasibility study establishing the suitability of alternative transit systems customized to the specific corridors.
- Rationalization of bus routes in the context of the corridor(s) of the proposed metro rail.
- Detailed project report for all the corridors identified for full-fledged / partial BRT.
- Detailed project report for the next package of multi-level parking (with or without Government owned land).
- Planning a network of pedestrian-ways / malls / parkways.
- Planning a network of cycle-ways.
- Detailed study to review the adequacy of parking standards.

Annexure I

Travel characteristics of CMA

1. The modal preferences of the commuters in the CMA are best characterised in that in a group of 100, 26 travel by bus, 2 by train, 33 by walk, 13 by cycle, 19 by two wheeler, 4 by car and 3 by other modes (vide Table -1).

Table 1: Daily average person trip distribution by mode in CMA

(Trips in million)

Sl. No.	Mode	No. & percent of total trips by mode									
		1970		1984		1992		2004		2005	
		No.	%	No.	%	No.	%	No.	%	No.	%
1	Bus	1.10	41.5	3.074	45.5	2.84	37.9	2.89	29.0	2.47	25.8
2	Train	0.30	11.5	0.610	9.0	0.31	4.1	0.50	5.0	0.24	2.5
3	Car/Taxi	0.08	3.2	0.103	1.5	0.11	1.5	0.40	4.0	0.36	3.8
4	Fast TW	0.04	1.7	0.219	3.2	0.52	7.0	1.80	18.0	1.83	19.1
5	Auto rickshaw	-	-	0.024	0.4	0.16	2.2	0.20	2.0	0.29	3.0
6	Bicycle	0.57	21.3	0.720	10.7	1.06	14.2	1.30	13.0	1.23	12.8
7	Cycle rickshaw & others	0.00	0.1	0.105	1.6	0.24	3.5	0.10	1.0	0.03	0.3
8	Walk	0.55	20.7	1.895	28.1	2.21	29.5	2.79	28.0	3.14	32.7
	TOTAL	2.65	100.0	6.750	100.0	7.45	100.0	9.98	100.0	9.59	100.0

Source: MATS (1968-69), Short-term Traffic Improvement Programme Report (MMDA & KCL, 1984) & CTTS (MMDA, RITES, KCL & PTCS, 1992-95), & Short term study to updateCTTS (1992-95)(CMDA, RITES & PTCS, 2004), HHI Survey of the DPR for the Chennai Metro Rail Project, DMRC, 2005

2. The growth of sea and air traffic is briefly described as below:

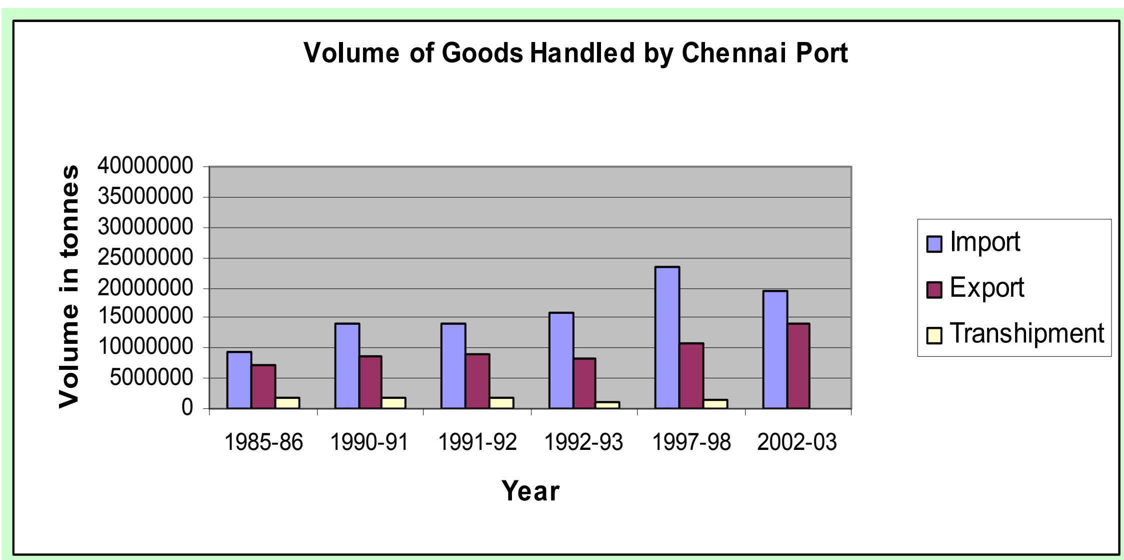
Seaport terminals

3. The Chennai Port Trust (CPT) located in the CBD handled 33.69 million tonnes in 2002-2003. While the imports increased from 1.8 million to 19.61 million tonnes for the period 1951-52 to 2002-2003 registering a growth of 989%, the exports increased from 0.3 million to 14.1 million for the same period registering a growth of 4600%. Of the total import and export the foreign traffic handled accounts for 93% and the coastal traffic 7% for the year 2002-2003. While mineral oils and other POL account for 41% of the imports, iron-ore accounts for 56% of the exports (2002-2003). The imports are predominantly from south-east Asian countries accounting for 41% and the exports are made predominantly to Japan accounting for 14% (vide Table - 2).

Table 2: Growth of Cargo Traffic in Chennai Port

Year	Import	Export	Transshipment	Total (tonnes)
1951-52	1,775,134	279,157	--	2,054,291
1956-57	1,895,703	607,851	--	2,503,554
1961-62	2,268,853	1,198,290	--	3,467,143
1969-70	3,535,771	2,904,372	192,277	6,632,420
1974-75	4,760,511	3,155,479	--	7,915,990
1980-81	6,412,177	3,962,562	72,736	10,44,475
1985-86	9,303,071	7,040,719	1,802,902	18,146,692
1990-91	14,124,933	8,642,713	1,749,944	24,517,590
1991-92	14,182,056	9,100,779	1,763,348	25,046,183
1992-93	15,680,715	8,439,962	1,209,354	25,330,031
1997-98	23,352,921	10,825,638	1,352,450	35,531,009
2002-03	19,605,661	14,081,445	--	33,687,106

Source: Annual Report, Chennai Port Trust, 2004



4. CPT handled container traffic to the tune of 424,665 TEUs in 2002-2003, which is 23.3% more than that handled in the previous year. Some of the salient performance of the CPT includes exporting 8,432 cars in 2002-2003, which is 103% more than that handled in the previous year. The Railways handled 12.3 tonnes of cargo traffic through 5,24,320 wagons from the Port. A total of 1692 merchant vessels and 387 Government vessels entered the Port during the year (vide Table -3).

Table 3 : No. of vessels entered during 2002 - 2003		
S.No.	Classification	No. of ships
1	Foreign	1,111
2	Coastal	581
3	Government	387
	Total	2,079

Source: Annual Report, Chennai Port Trust, 2004

5. The Ennore Port Limited (EPL), developed as a satellite port to CPT, is the 12th major port in India and the first corporate port in India. Designed to develop 22 berths to handle a variety of bulk,liquid and container cargo,EPL has 3.775km channel (250m wide and 16m deep) to capable of handling 65,000-77,000 DWT vessels. The EPL commenced commercial operations on 22.6.2001. It presently handles around 10m tonnes of thermal coal per annum.

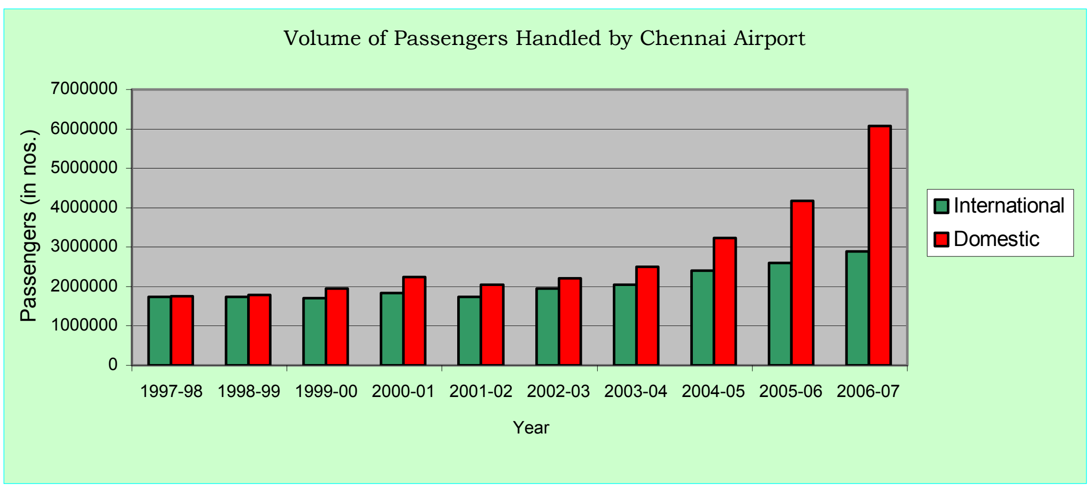
Airport terminals

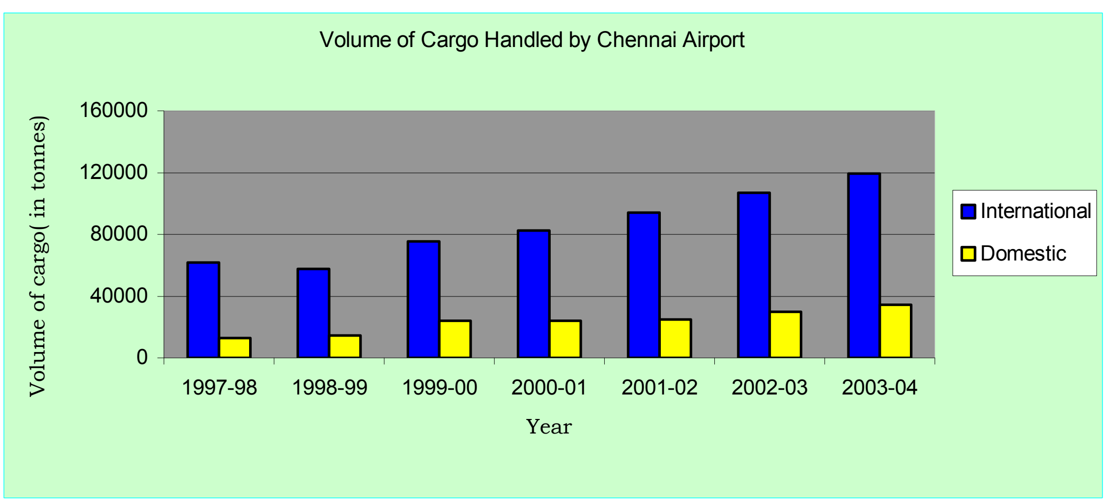
6. Chennai has a national air terminal viz. Kamarajar Domestic Terminal and an international terminal viz. Anna International Terminal located at Meenambakkam. Totally 20 international flights per day are operated from Chennai. While the growth of international traffic is 5%, that of the national air traffic is 7%. The AAI imports 44,000 to 51,000 tonnes of cargo per year and exports 63,000 to 68,000 tonnes per year. While the growth of international passenger movement is 17.8% for the period 1997-98 to 2003-2004, the growth of passenger traffic within the country is 42.5%. Similarly the growth of international cargo movement is 93% for the same period and that within the country is 167% (vide Table -4).

Table 4: Growth of air traffic in Chennai				
	Year	International	Domestic	Total
Aircraft movements (Nos.)	1997-98	10862	20231	31093
	1998-99	11169	19813	30982
	1999-00	11080	23531	34611
	2000-01	12063	25293	37356
	2001-02	12398	25673	38071
	2002-03	14490	29863	44353
	2003-04	14515	36749	51264
	2004-05	18111	43122	61233
	2005-06	21155	47900	69055
	2006-07	23567	76162	99729
Passenger movements (Nos.)	1997-98	1744037	1755854	3499891
	1998-99	1736021	1788005	3524026
	1999-00	1702534	1945587	3648121
	2000-01	1837954	2235011	4072965
	2001-02	1741458	2042784	3784242

	2002-03	1947937	2213409	4161346
	2003-04	2054043	2501778	4555821
	2004-05	2400670	3233256	5633926
	2005-06	2606638	4173345	6779983
	2006-07	2895930	6078196	8974126
Cargo movements (Tonnes)	1997-98	61902	12925	74827
	1998-99	57646	14304	71950
	1999-00	75423	24185	98608
	2000-01	82316	23930	106246
	2001-02	94171	24941	119112
	2002-03	106834	29825	136659
	2003-04	119563	34560	154123

Source: Airport Authority of India, 2004 & Sep. 2007





Annexure II

List of major efforts taken

1. Various measures taken to improve and strengthen the transport supply in CMA in the past include:

a) Procurement of about 305 buses under Madras Urban Development Project MUDP-I, 915 buses under MUDP-II and 1600 buses under Tamil Nadu Urban Development Project (TNUDP) through PTC and DATC (now merged into MTC) was made at a cost of about Rs.63 m, Rs.221 m and Rs.927.50 m respectively.

b) In addition, 3 bus depots and 8 terminals were constructed during MUDP-I and 10 bus depots/terminals under TNUDP.

c) The following rail projects were implemented by Southern Railway.

i) Quadrupling Chennai-Arakkonam B.G. Line (upto Pattabiram);
ii) Gauge conversion of Chennai Beach - Tambaram - Chengelpattu lines including optimisation of the line (by replacing road/rail level crossings by overpasses or underpasses) (under implementation);
iii) Development of Mass Rapid Transit System (MRTS) from Chennai Beach up to Thirumylai for a length of 8.5 km in the I Phase at a cost of Rs.2690 m commissioned in 1997; and
iv) Commissioning of II phase of MRTS from Thirumylai to Velachery (11.165 km) at cost of Rs.7690m on 19, Nov. 2007

2. Critical bottlenecks in the road network have been improved under MUDP I (Rs.15.20 cr), MUDP II (Rs.63 m) and TNUDP (Rs.839 m) through Department of Highways (DoH) and Chennai Corporation (CoC). These included forming an inner ring road for a length of 17.5 km initially and dualling its carriageway subsequently. The balance of it comprising the northern segment for a length of 12.5 km has been formed and the southern segment for a length of 6 km. is being formed. The First phase of Chennai Bypass connecting NH 45 and NH 4 for a length of 19 Km at a cost of Rs.900 m has already been completed and commissioned in November 2002. Development of Chennai Mofussil Bus Terminal (CMBT) at Koyambedu has been completed at a cost of Rs.1030 m and commissioned in Nov. 2002. Chennai City Contract Carriage Bus Terminal (CCCBT) has also been constructed and commissioned. Improvements to radial roads in and around CMA have been carried out for a total length of 250 km at a cost of 2120 m.

3. The major road and rail investments proposed for the future comprise the following:

a) Extension of MRTS Phase II from Velachery to St.Thomas Mount for a length of 5 Km at a cost of Rs.4160 m;
b) Development of II phase of Chennai Bypass from NH4 to NH5 for a distance of 13 km. at a cost of Rs.4450 m;
c) Construction of 4 No. grade-separators, 3 on IRR and 1 on NH-45 opp. airport at a cost of Rs. 2100 m;
d) Gauge Conversion of Chennai Beach-Tambaram rail line at a cost of Rs.2350 m;
e) Development of an Outer Ring Road for a distance of 62 Km at a cost of Rs.9000 m;

f) Developing the 2nd BG coaching terminal at Egmore;
g) Development of 3rd rail line from Beach to Athipattu;
h) Development of 4th rail line from Beach to Athipattu;
i) Quadrupling the rail line between Pattabiram and Thiruvallur;
j) Development of 3rd rail line from Thiruvallur to Arakkonam;

4. Drawing a cue from the document viz. Action plan for IT corridor, 2003 prepared by CMDA, GoTN have commenced implementation of a high quality 6-lane arterial road from Chennai to Mammallapuram for a length of 47 km at a cost of Rs.1217.4 m through IT Expressway Ltd., a special purpose vehicle floated through TNRDC to serve the IT and ITES industries located primarily in the southern outer-CMA.

5. Recognising the fact that the capacity of the urban road network can be appreciably increased by removing the major bottlenecks in the network particularly along such strategic roads such as IRR, it has been proposed to construct well-designed grade separators at all the critical intersections of radial roads with IRR. In the first phase NHAI on its own is developing 3 grade separators at Kathipara, Padi and Koyembedu intersections and one opposite airport on GST Road at a total cost of Rs.2100 crores. These are expected to be completed with a period of 18 to 24 months. The GoTN are proposing to develop the grade-separators at Thirumangalam and Vadapalani intersections of IRR in the second phase.

Non-transport developments

6. The Government in the Transport Department have already taken various initiatives for introducing innovative technologies for motorised vehicles. The Govt. have recently directed for induction of 5000 LPG operated autos in the City. There are also 14 no. ALDS (automatic LPG Dispensing System) in the City. Already electric operated cars manufactured by a company (Reva) are on the roads of Chennai. The strategy to improve the air quality in the metropolis will be principally governed by such conscious measures as to tilt the modal share in favour of public transport modes and the initiatives being taken both by GoI and GoTN to phase out lead in petrol and sulphur dioxide in diesel, making it mandatory on the part of vehicle manufacturers to conform to Bharat II, to introduce pollution-free fuels such as CNG/LPG for vehicle operation.

7. Several non-transport measures were also implemented over the last decade to reduce traffic congestion. These include decentralisation of the CBD, viz., shifting of the whole-sale market to koyambedu, the Iron and Steel Market to Sathangadu, construction of truck terminal at Madhavaram which have relieved the arterials and other City roads considerably from the lorry and bus traffic.

Institutional arrangement

8. The traffic and transportation schemes are presently implemented by several departments and agencies. While long-term planning and coordination is carried out by CMDA, individual schemes are executed by Railways, DoH, CoC, MTC; The traffic enforcement is carried out by Chennai Traffic Police (CTP). There are a number of committees to coordinate the implementation of transport schemes in the CMA. They are:

i) High Level Coordination Committee for MRTS (Chaired by Vice-Chairman, CMDA) for coordinating implementation of MRTS Phase-I&II
ii) Chennai Road Safety Council (Chaired by Commissioner of Police) for traffic enforcement

iii) Coordination Committee (Chaired by Superintending Engineer, Chennai Corporation)

iv) Indian Transport Road Development Association (Chaired by Rane Power Steering Ltd.).

v) Besides these agencies, there are agencies, which are concerned with licencing of vehicles and policy making such as Regional Transport Office etc.

In the absence of financial and administrative powers vested with these committees, the coordination effected by these committees is limited.

9. Under the UNDP-UNCHS supported Sustainable Chennai Project, through the deliberations of the action committee and working groups certain actions for sub-strategies such as

(i) optimising the utility of existing transport infrastructure;

(ii) enhancing the modal share of rail and bus and

(iii) improving the air quality

were discussed and implemented.

Annexure III

Comparison of various urban transit options

Sl.No.	Characteristics	Hybrid Monorail	Sky Bus	Metro	MRTS	Elevated Bus Way
1	Gauge (m)	Standard (1.435)	Standard (1.435)	Standard (1.435)	Broad (1.676)	Standard (1.435) (for LRT later)
2	Speed (KMPH)	37	36 – 150	40	36	30
3	Headway (Seconds)	120	30	180	180	15 -32
4	Passenger / unit or coach	320 (3 car unit)	300 (2 car unit)	2200 (6 car unit)	2844 (9 car unit)	100 (SB) 180 (VB)
5	Capacity (PPHPD)	4500 – 32000	18,000 – 72,000	45,000 – 50,000	60,000 – 90,000	30,000
6	Radius of curvature (m)	20	50	300	250	150
7	Gradient	1 in 6	1 in 25	1 in 50	1 in 70	1 in 25
8	Distance between stations (Km)	0.80 – 1.00	0.70 – 1.35	1.00	1.20	1.10
9	Station size in m (length x width)	50 x 17	18.5 x 3.0	135 x 30 / 180 x 30	300 x 30	60 x 6.5
10	Single column - size	0.80 m x 1.0 m or 1.55 m x 2.0 m	1.0 m dia	1.45 m dia	1.2 m dia	1.5 m dia
11	Power	Battery	Electric	Electric	Electric	Diesel
12	Construction cost/ Km (Rs. in crore) (excluding land cost)	37 – 39	34	70	53	36.78 23.7 (1995 price)
13	O & M Cost Rs. in crore / Km	0.30	6.97	1.20	-	0.02 (maintenance only)
14	Fare Rs. / Passenger / Km.	0.95	0.50	1.00	0.53	0.45

PPHPD – Passengers per hour per direction ; KMPH – Kilometer per Hour ;
SB – Standard Bus ; VB – Vestibule Bus; LRT – Light Rail Transit ; dia –diameter
Source : As given by the corresponding system promoters

Annexure IV

The broad terms of reference of the Comprehensive Transportation Study

SCOPE OF WORK: PRINCIPAL AREAS OF ACTIVITY

The consultant work is to divide roughly into six areas of activity, which are as follows:

- i) Collection / updating of household, land use, and travel demand data
- ii) Development and operation of an urban transport model
- iii) Formulation of transport strategy;
- iv) Identification of a phased program of transport investments and management proposals;
- v) Organising a communication campaign with a view to move the stakeholders, public and govt. agencies to accept and implement the preferred programme of transport investment and management; and
- vi) Training for and knowledge transfer to CMA and other agencies

Activity 1. Collect and Update Household, Land use and Transport Data

The data to be updated and collected are those required for the purpose of metropolitan transport strategic planning. These would include both historical and spatial data. Spatial data are needed for the calibration of the UTP model. A traffic zonal system shall be determined in collaboration with CMDA for the purpose of spatial data collection, traffic forecasts and sub area analysis. Considering the vast area under CMDA, the population and its distribution over the area, the sample size for the survey should be determined based on the size and spread of the geographical sections. A sample size of 2% for household survey for the entire region shall be adopted.

As no recent data on employment exists in Chennai, the consultant should present a methodology to address this lack of the data for the model, in the inception report.

The following tasks shall be conducted under this activity.

- Baseline Data Collection
- Design, Supervision and Implementation of Household Travel Origin-Destination
- Survey including a stated preference survey
- Screenline Traffic Counts
- Suburban Commuter Rail Passenger Survey
- Bus Passenger Survey
- Roadside Motor Vehicle O-D Survey (including Goods Vehicles)
- Survey of large traffic generators (e.g. bus stations)
- Estimating Speed Flow Functions
- Taxi and autorickshaw surveys (IPT Surveys)
- Speed and delay surveys
- Workplace surveys
- Goods focal point survey
- Parking survey (Limited data collection to update the comprehensive parking study carried out in 2002-03)
- NMT survey
- Survey Data Processing and Analysis

Activity 2. Development and Operation of an urban Transport Planning Model

The consultant shall recommend a model package that would be suitable for CMA planning needs. The purpose of the travel demand model is to provide good policy-related and future travel forecasts, and therefore, should be simple. The conventional approach of developing the 4-stage model at aggregate level (except the modal choice at disaggregate level) should be followed.

The model should integrate household activities, land use patterns, traffic flow, and regional demographics. The core of the proposed model system is a household activity simulator that determines the locations and travel patterns of household members daily activities by trip purpose. The model should estimate travel behavior with regard to longer term choices of residential and employment location, and land use and adaptive behavior in response to transportation system changes, including fare and pricing policy.

It should be noted that given the high volume of passengers that travel in the suburban train network, the model should pay particular attention to mode split between bus and rail as well as private car and rail and conducting sensitivity analysis of demand with regard to rail fare. Attention should be paid to the issue of how travel time (walk and wait time), comfort, and mobility or access to transport is treated in the model and how improving these parameters could affect mode split in favour of the urban rail network. Similarly attention should be paid to these issues for travel on the bus network, as well as the impact of traffic congestion on bus speeds.

The consultant should recommend appropriate software package(s) suitable for the development of the requisite model in the Inception Report. The consultant shall undertake an overview of various models such as TransCAD, SATURN, CUBE, etc. with their merits and demerits and recommend a model package that would be suitable for CMA planning needs. The final decision on the selection of software package (s) will be made by CMDA within 10 days of presentation of options along with merits and demerits. The software package (s) should in particular be capable of:

- Modeling mode split, including walk, public transport modes (autorickshaws, taxi, bus and rail) and private transport (motorcycle and car), good's vehicles
- Assigning trips to urban rail and bus networks taking into account the condition in trains, variations in bus speeds and frequency due to changes in overall traffic volume, and fares
- Reflecting the impact of new land use developments and / or control policies, including truck terminals, truck parking lots, interstate bus terminal etc.
- Responding to traffic demand management measures such as parking fees, road user charges and congestion pricing as well as the staggering of working hours, flexi hours and multiple shift work.

Task 2.1 Transport Network Coding

Consultants shall compile a transport network inventory with sufficient details with a view to ensuring traffic assignments to network at link level. The consultants shall compile attribute data for the network (link and node characteristics). The consultant shall prepare input files including network coding based on the inventory collected as a part of Task 1.1 to code the network. In addition, the consultant shall also develop traffic origin and destination matrix by mode and time (day / peak / off peak). The consultant shall review current speed volume function and assess the adequacy of this function for the purpose of planning and update speed / volume if necessary specially the speed / volume for the new road facilities (flyovers, bypass etc.).

The consultants in consultation with the client shall identify the committed schemes, proposed schemes and accordingly prepare do-nothing, do-minimum, do-something networks to assess the impact.

Task 2.2 Model Calibration

The model should reflect the travel behavior of different income and social groups in a disaggregate fashion and should be sufficiently sensitive to test policy measures and physical improvements to the transportation network and services. The model should be calibrated for travel costs, speed and other factors.

Traffic assignment should be done on both peak and off-peak hour. The model result should be sufficiently detailed for the identification of project benefits among different income groups and users by different modes. The consultants, upon calibration, must demonstrate to the client how reliable the model is in replicating the current modal splits and traffic flows at screen lines and cordons. As the model is to be used for evaluating micro-investments including inter alia major landuse changes (commercial development exceeding 10000 sq.m. of built up area), grade separators at junctions, widening of carriageways on a corridor basis, providing bus lanes, introduction of ATC system etc. cost of more than Rs.100 million (US$ 2 million), the consultants should show how reliable the calibrated model is for more detailed flows than at screens / cordons.

The consultant should produce a working paper presenting the main data and assumptions used by the model.

The consultant should present the process that would be put in place to ensure good quality control of the data that are entered in the model and satisfactory performance of the model. Calibration and validation of the model is a well identified step in the process of the study. The consultant should therefore produce a specific report on the same that should be formally accepted by CMDA before the consultant progresses further in the step.

Task 2.3 Establish Economic Evaluation Procedures

Consultants should establish the framework for economic evaluation that would be used in the formulation of a long-term transport strategy and the identification of a phased investment program. The framework should allow the economic feasibility to be expressed in terms of expected Net Present Value (NPV) and Internal Rate of Return (IRR). Special attention should be given to the following major areas:

- Identify major items of economic benefits;
- Establish appropriate vehicle operating costs (VOC) and value of travel time (VTT); and
- Develop appropriate economic evaluation procedures to make full use of UTP model outputs including link traffic volumes and speeds by vehicle type for existing and planned networks

Activity 3. Update the Long-Term Transport Strategy for CMA

Task 3.1 Review of 1992-95 CTTS

The consultant should review the 1992-95 CTTS strategy, assess its relevance to the current and future transport needs of CMA, identify major changes in socioeconomic conditions, land use, and transport since 1995 and suggest strategic areas that need to be considered, improved and strengthened.

Task 3.2 Review of Institutional Arrangement of Transport

The purpose of this task is to clarify current arrangements for formulating, monitoring and implementing transport policy and for identifying, appraising, authorising, sanctioning, financing and implementing transport schemes and operational policies. The study is required to identify the role and responsibilities of the various bodies both within and outside CMA, concerned with transport policy, financing, investment and regulation of operations. This task is intended essentially as a review of earlier work. The main issues to be addressed by the consultant shall include the following:

- the current lack of a clear institutional structure with regard to urban transport planning, management and investment coordination;
- institutional and policy barriers to positive change in the delivery of transport services to the people and businesses;
- the number of staff qualified in transport planning and management;
- the relative roles of public and private passenger transport operations, and the means of regulating them;
- the relative roles of central, state and municipal governments in transport planning, investment and management

The study should advise on the needs and possible scope of (i) institutional and policy reform; and (ii) post-study professional development and training in transport planning and management.

Task 3.3 Review of Transport Financing

The study should pay careful regard to the financial resources available for transport investments, maintenance and operations, both in total and by agency, on the basis of an assessment of the level of funding available annually during the past 5 years, current changes in the fiscal framework and economic trends. The outcomes of this work should be estimates, on the basis of a number of assumptions and scenarios, of resources likely to be available for transport investments, maintenance and operations during the coming years, with distinction between tied resources (such as commuter rail surcharges) and untied resources. Potential sources of private sector finance identified in earlier studies should also be reviewed and updated. Likely candidates for private sector financing / PPP based on feasibility of levying tolls or user fees should be identified.

Task 3.4 Traffic Demand Analysis and Forecast

The consultant shall review the CMA master plan for 2006-2026 and may consider alternate land use scenarios (maximum 3) for future development in consultation with CMDA. CMDA will provide forecasts of population for the years 2019 and 2026. The consultants shall forecast travel demand by mode using the calibrated UTP model for the years 2009, 2014 and 2019 on the existing transport network plus all committed transport investments, under the assumed alternative income, population, and land use growth scenarios. These exercises will give indications on the likely traffic problems in the future years, thus providing a basis for strategy formulation.

Task 3.5 Definite Alternative Long-Term Transport Strategies

The long-term transport development and management strategy should be a combination of policies (e.g. demand management, user charges etc.) and physical improvements. In collaboration with CMDA, the consultant should define the specific objectives, principles, and criteria required to guide the formulation of the long-term strategy.

Having regard to various policy options and transport strategies and systems, alternative feasible scenarios for horizon years should be developed in consultation with CMDA. It is likely that the scope to manage transport demand by control of land use development will be limited but that the implications of transport developments on land use developments will be great. It is likely, for example, that restricted space in the CBD, combined with demand management measures and improvement of rail and road links in the CMA, will lead to faster development of less developed areas in peri-urban areas. The study should seek to identify the scale of such development pressures.

The UTP model linking land use pattern, travel demand and modal split, should be used to test the impact of major modifications of strategy and will be used mainly for the 20-25 year horizon. All options for improving transportation in CMA should be considered and compared. Alternative transport strategies and their evaluation should focus on agreed land use strategy and show whether any significant easing of future transport problems could follow from revision of the land use strategy. The consultant shall recommend an integrated land use transport model. CMDA would confirm the form of land use strategy to be assumed by the consultants for the development of a long-term transport strategy when it is presented to the Steering Committee.

Task 3.6 Evaluation of Alternative Strategies and Selection of Preferred Strategy

A comprehensive evaluation system should be developed in consultation with CMDA, taking into account all relevant factors such as capital and operational costs and environmental, social and political factors, services to the poor and vulnerable. The evaluation should be comprehensive, on the basis of four major criteria: (i) economically viable; (ii) socially acceptable; (iii) environmentally sustainable; and (iv) financially feasible. At the strategic planning level, a preliminary assessment of economic and financial feasibility and social and environmental impacts should be made for each proposed alternative strategy. In defining the alternative strategies, the consultant should pay special attention to the services to poor and the vulnerable.

The comprehensive evaluation should result in the recommendation of a preferred long term transport strategy. The consultant in collaboration with CMDA and other transport agencies should explain the basis for its recommendations.

Task 3.7 Prepare a Draft Transport Strategy Document

The outline strategy should amount to a directional plan indicating the main imperatives of transport policy during the foreseeable future, having regard to need, desirability (with regard to factors such as land use development and environmental and social impacts), affordability and uncertainty. This strategy should take account of economic growth, current initiatives of CMDA, the need to conserve and enhance the urban environment, land use plans and likely land use development patterns.

The strategy document should cover the following patterns:

i) **Current Situation**. Clarification in broad but objective terms of the current situation and trends: the quantity of personal movement by commuter rail, bus, car, autorickshaws, NMT, two-wheelers, taxis, water transport and on foot; the amount of goods vehicle movement; Intercity traffic, long distance Bus / Railway passenger movement; origin-destination patterns; current problems.

ii) **Traffic Growth**. Clarification of current growth rates, potential growth rates (with reference to experience elsewhere), potential problems.

iii) **Current Constraints**. Clarification of current constraints with regard to government policies on vehicle licensing, vehicle and fuel prices and trends, land use policies and the resources available for transport investment and the scope of cost recovery through direct user charges such as tolls

iv) **Long Term Trends and Prospects**. Review of implications for transport (supply and demand) of economic growth and land use developments; and of the implications for transport planning of uncertainty.

v) **Transport Investment Options**. Review of the main means of providing additional transport capacity with regard to their effectiveness, magnitude of orders of cost (including both capital and recurrent costs), economic and financial viability, etc. including;

vi) **Demand Management**. The need for and means of achieving the management of the potentially high growth in the use of motor vehicles (including motorcycles).

vii) **Environmental Measures**. Review the need for measures to ameliorate adverse environmental impacts, either existing or resulting from proposed projects or increased development.

viii) **Land Use Strategy**. Review of the scope and limitations of reducing / controlling traffic demands through land use policy and of the scope for using transport policy to influence land use development.

ix) **Institutional Arrangement**. A diagnostic assessment of the current institutional arrangements for administering and planning transport activities in CMA, including policy formulation, regulation of transport operations, financing, and investments, and recommendations for improving the arrangement.

x) **Conclusions and Recommendations on Transport Strategy**.

Broad conclusions and priorities for development of commuter rail, bus transit, highway construction, goods transport and interregional bus / rail transport, water transport, ferry services, traffic management and demand management in short, medium, and long terms; observations on land use strategy.

Activity 4. Identify a Medium-Term Investment Program

The product required for this Activity is a rolling program of investments and management proposals sufficient and appropriate for the period to 2019. The emphasis of the work will depend upon the conclusions reached on the transport strategy. The Medium-Term Investment Program shall comprise the following parts:

i) **Commuter Rail Development Programme**. Identification of a staged programme of commuter rail development including consideration of the Metro-rail Study and its cost implications, etc.

ii) **Highway Network Development Programme**. Identification of staged programme of highway investments, having regard to traffic demands, economic benefits, system effects, etc.

iii) **Bus System Development Programme**. Identification of proposals for investment in buses and civil works (depots, workshops, terminals, segregated bus lanes, exclusive busways, etc.) and operational management, taking account of bus transport policy recommendations outlined in the transport strategy.

iv) **Traffic Management**. On the basis of the initial batch of traffic management effort funded under the TNUDP III, identification of low-cost physical and regulatory measures to improve the efficiency and safety of traffic circulation. Particular attention should be given to the management of NMT, pedestrians, bus priority measures, etc.

Task 4.1 Identification and Costing of Investment Options

The identification of a phased programme of transport investments proposals shall first involve a sifting of options and pre-feasibility studies. The consultant shall take into consideration of the candidate projects already proposed by the transport agencies in CMA. For each investment project, the consultant shall define a base option and two or more alternative options. For each option, the consultant shall propose preliminary alignment, conceptual design option, technology choice, and timing of construction and start of operations. The consultant should also provide preliminary (or pre-feasibility study level) estimates of the associated capital and operating costs, traffic, revenues if applicable, and operating characteristics.

Task 4.2 Evaluation of Options

The consultant shall evaluate these options using the comprehensive criteria specified in para 3.6 but at a more detailed level. The evaluation should consist of two parts. The first, which may be partially quantitative and largely qualitative, shall consist of categories such as operational feasibility, integration with existing systems and physical environment, ease of response to changing conditions, land use effects, travel generation potential, environmental quality, requirement for involuntary resettlement, and other. The second part shall consist of simplified economic and financial evaluation, based on capital and operating costs and revenues, passenger costs and financial costs of each option.

Evaluation of investment options shall be based upon the network assignment of traffic demand matrices derived from the results of the transport surveys.

Task 4.3 *Prioritise the Investment Projects and Formulate an Investment Programme for Year 2019*

On the basis of comprehensive evaluation of the above options, the consultant shall prioritise the identified investment projects and formulate a Medium-Term Investment Program. The program should focus on identification of capital investments to be made in the 10-year period 2009-2019. The schemes thus identified would be prepared and largely implemented during TNUDP III phase II or as a separate package. The medium-term programme shall include background assumptions, rail, road, road based PT, mass transit system proposals and water transport investment proposals. When proposing the program, the consultants should assess the impact / implication of various scenarios / strategies / fiscal policies on transport network.

The background assumptions shall include a summary of the transport strategy emerging from the outline long-term study, the form of land use distribution assumed to apply in the medium term, assumptions about institutional changes or developments and assumptions about investment levels. Unless good reason emerges to the contrary, one land use disposition should be defined for the medium term.

Activity 5 *Communication Campaign*

The objective of the communication campaign is to

a) to collect suggestions from the stakeholders (public and govt. agencies);
b) to disseminate the results of the study; and
c) to help in determining priorities among the various options that can be envisaged.

Activity 6 *Skill and Knowledge Transfer*

The study should advise on the needs and possible scope of professional development and training in transport planning and management.

The new CTS model should be installed in CMDA and CMDA, CC, MTC, DHRW & S.Rly. personnel should be provided necessary training in its use. A two week workshop shall be conducted for senior officers on the usage of model.

The consultant shall train core staff (7) on model update and calibration so that they could use the model proficiently as a planning tool with only minimal assistance from the consultant on the need basis after the study is completed.

With the objective of making the training more productive, two persons from CMDA would work along the consultants' team. One would work full time on the model. The other would be involved in more strategic aspects. The consultant should define their functions and the results expected from both these trainees in the Inception Report.

The consultant should work with the Transportation Engineering Division, IIT, Chennai. The objective is that the institution would be able to provide support and advisory services to CMDA after the study is completed. While the consultant shares the data with the institution, the institution would provide advice to the consultant in the building of the model and interpretation of the outputs of the model. As the services/support of the institution is made available free of cost, it would be the responsibility of the consultant to ensure that the support of the institution is fully utilised. The above institution would review and validate the data collected by the consultant within 2 weeks of submission.

All data, in the form of both raw data and structured database, should be fully transferred to CMDA. The data collected at various stages of the study should be organised and integrated in a database and provided to both CMDA and the institution mentioned above. The consultant should also provide the final set of data used by the model that are consistent with the results in the final report to CMDA. Partial payment of the consultancy fee would be subject to compliance with this requirement.

Chapter - VI

SHELTER

Shelter is a basic need. When the need for shelter is not satisfied, it becomes almost impossible for an individual to think of satisfying his/her family aspirations and intellectual needs. Primary responsibility of any city is to provide its residents with a decent and habitable shelter. A standard housing does not mean merely land and building, but includes basic service like water supply, sanitation and access roads.

6.02 Demand for housing is a universal phenomenon, which exists in all societies, but it varies from "no shelter" to "better shelter"; consequently it is related to economic level of households. An assessment of housing need, demand and supply becomes necessary to work out a meaningful shelter strategy.

6.03 "Housing Need" is an expression of housing requirements, and it is computed based on (i) available housing stock in the base year, (ii) no. of households at the base year, (iii) rate of demolition of dilapidated/deteriorating structures for reconstruction, and (iv) rate of clearance/conversion of '*Katcha*'/slum structures for better housing.

6.04 'Housing demand' is related to market with reference to purchasing power, affordability, willingness to raise funds and it may be assessed based on the following major factors viz. (i) the economy of doing a house (temporal choices), (ii) affordability, (iii) willingness to pay (including for construction, maintenance, resource mobilization), (iv) availability and accessibility of housing finance, and (v) availability of residential plot/flats at affordable prices.

6.05 Normally, 'housing demand' is less than 'housing need' in a developing economy like ours, and these converge when society's economic level is rich, distributed with less disparity and stable. Hence a housing policy at metropolitan level has to take into account the factors concerned with the housing needs and demand.

Housing Scenario in CMA

6.06 The decadal growth of households and housing units is given in the Table No. 6.01. (It shows that the housing requirement gap is not significant for the period 1971-91 and there is significant gap in the year 2001).

Table No. 6.01: No. of Households and Housing Units in City and CMA								
	(in Lakhs)				Growth rate in %			
	1971	1981	1991	2001	1971	1981	1991	2001
Households in the City	4.44	6.29	7.96	9.62	--	41.7	26.55	20.85
Households in the CMA	6.89	9.04	11.82	16.19		31.2	30.95	36.97
Housing Units in the City	4.80	6.37	7.98	9.57	--	32.7	25.22	20.55
Housing Units in the CMA	6.63	9.15	12.34	15.83	--	38.00	34.90	29.50

Source: Census of India

6.07 The Table No.6.02 shows that the average rate of growth in housing units is declining when comparing the rate of growth of households.

Table No. 6.02: Rate of Change in Population, Households and Housing Units, 1971-2001- CMA			
	Annual rate of growth in population	Annual rate of growth in Households	Annual rate of growth in Housing units
1971-1981	2.76	3.12	3.27
1981-1991	2.36	3.07	3.03
1991-2001	1.93	3.69	2.63

6.08 The structural characteristics of housing stock in 2001 in Chennai City in respect of predominant materials used for wall is given in Table No.6.03 which shows that 92 percent of the houses are with *'pucca'* walls constructed with bricks, stones, concrete and other materials.

Table No. 6.03: Chennai City Distribution of Houses by Predominant Material of Wall, 2001			
Sl.No.	Type of Wall	Number of Houses	% to Total
1.	Grass, Thatch, Bamboo wood, Mud	333959	3.55
2.	Plastic, Polythene	1671	0.18
3.	Mud, Unburnt brick	29438	3.02
4.	Wood	1208	0.17
5.	GI Metal, Asbestos sheets	9061	0.95
6.	Brick	622304	65.03
7.	Stone	49363	5.16
8.	Concrete	208516	21.78
9.	Others	1556	0.16

Source: Census of India

6.09 In Chennai City 75% of the houses are with roof made up of brick, stone, concrete and other materials of pucca nature, about 15% are with semi-pucca roofing materials such as tiles, slate, G.I. metal sheets and asbestos cement sheets, and about 10% are with 'Katcha' materials such as thatched, bamboo etc.

Table No. 6.04: Distribution of Houses by Predominant Nature of Roof – Chennai City

Sl.No.	Type	Number of Houses	% total
1.	Grass Thatch, Bamboo, Wood, Mud, etc.	90,735	9.48
2.	Plastic and Polythene	2,966	0.31
3.	Tiles	71,403	7.46
4.	Slate	1,662	0.17
5	GI metal, Asbestos sheets	65392	6.83
6	Brick	18908	1.98
7	Stone	5246	0.55
8	Concrete	696997	72.83
9	Any Other Material	3767	0.39

Source: Census of India

6.10 Even though the proportion of the housing units with '*Katcha*' roofing materials accounts for only about 10%, in absolute numbers it is large i.e., 93,701 and these are vulnerable to fire accidents, particularly in summer months and every such occurrences of fire accidents burn down the whole area of such thatched roofed slums which is common in Chennai city, some times resulting in casualties. The proportion of dilapidated / deteriorating housing units accounts for only about 0.5% of the total households.

Table No. 6.05: Distribution of Houses by Predominant Nature of Floor – Chennai City

Sl.No.	Type	Number of Houses	% total
1.	Mud	32729	3.42
2.	Wood Bamboo	1003	0.10
3.	Brick	4782	0.50
4.	Stone	5712	0.60
5.	Cement	611892	63.93
6.	Mosaic & Floor tiles	296953	31.03
7.	Any other material	4005	0.42

Souce: Census of India

6.11 According to Census, 2001, about 71% of households live in less than three roomed housing units; proportion of households which live in one roomed, two roomed, three roomed houses etc. is given in the Table No.6.06

Table No.6.06: Distribution of Households by Number of Dwelling Rooms – Chennai City, 2001

Number of Rooms	Households	% Total
No. of Exclusive Rooms	205020	02.47
One Room	318325	38.45
Two Rooms	251659	30.40
Three Rooms	144149	17.41
Four Rooms	57555	06.95
Five Rooms	17938	02.16
Six Rooms and above	17665	02.13

Source: Census of India

6.12 The study on "Shelter Strategy" conducted by the School of Architecture and Planning (Anna University) for PMG, Govt. of Tamil nadu (1995) revealed the following:

(i) The total households in CMA during 1991 were 12.17 lakh consisting of 9 lakh in the non-slum category and 3.17 in the slum category.

(ii) The total number of houses in CMA during 1991 was 12.72 lakh consisting of 9 lakh units in the non-slum and 2.72 lakh units in the slum categories respectively.

(iii) Among the non-slum households 18% belong to EWS, 30% belong to LIG, 22% belong to MIG and 30% belong to HIG.

(iv) Among the slum households 86% belong to EWS, 13% belong to LIG and 1% belongs to MIG and HIG.

(v) 44% of the houses are with one bedroom, 35% with multipurpose room, 17% with two-bed room, and 4% with 3 and more bedrooms.

(vi) 3.47 lakh housing units were delivered in CMA during the decade 1981-91, out of which a little more than half i.e. 1.81 lakh units (52%) were plots, 1.07 lakh units (31%) were houses and only 0.59 lakh units (17%) were flats.

(vii) Out of the total housing units delivered by TNHB, 51% are meant for EWS, 24% are for LIG, 12% are for MIG and 13% are for HIG.

(viii) TNSCB has taken up more than fourteen different schemes out of which ten schemes are meant for physical improvement of slums and four schemes are meant for financial assistance to the slum households for improving their shelter units.

(ix) TNSCB has delivered as much as about 12,000 tenements, about 700 houses and about 5000 serviced plots. Nearly 1.88 lakh households were benefited under different environmental improvement schemes.

(x) The annual delivery rate of all the housing units put together was 16,000 during 1981 which increased to 41000 units during 1991, an increase 1.5 times. The annual delivery rate of plots alone was 10,000 units during 1981, which increased to 18,000 units during 1991, an increase of 80%. The annual delivery of houses has registered a 150% increase from about 7000 units during 1981 to about 18,000 units during 1991. The annual delivery rate of flats increased from 1,000 units during 1983 to about 6,000 units during 1991, a fivefold increase.

(xi) Most of the plots delivered under MUDP I, II and TNUDP were meant for EWS and LIG.

(xii) In the case of MUDP I, II & TNUDP, the above facilities were provided with better quality as the cost per family sanctioned was more. Further under these World Bank aided schemes, patta for the site under enjoyment was also issued to the beneficiary in addition to making available the loan for improvement/construction house.

6.13 Study on "**Effective Demand for Housing in Tamil Nadu**" was conducted in **1995** by the Consultants M/s. STEM for PMG, Govt. of Tamil Nadu. Covering a sample size of 2,255 households in respect of Chennai Urban Agglomeration. The following were the outcome of the study.

(i) Social Profile of Households

Madras Urban Agglomeration is spread over Madras and part of the Chengalpattu-MGR District. Total estimated households are 1.14 million covering 5.4 million population. Nearly 92 % of SC & ST households are from lower income groups i.e., EWS and LIG. The percentage of male population is more than (52 %) female population (48% per cent) in MUA. More than 8% are above 59 years of age group. Two out of every three households have been staying at the same place for more than 20 years. Nearly 45% of the households have migrated to MUA. While more than 75% of migrant households moved in from other towns, thus urban-urban migration is a major flow. About 10% of migrant households are from other states and 51% from the same district. Majority of migrants from other states fall in the higher income groups. About 30% of households migrated only for employment purpose, followed by movement of household or spouse.

(ii) Economic Profile of Households

About 38% of the households have an income less than Rs.1101 per month, while 9% draw less than Rs.501 per month. 3% of the households have an income of less than Rs.250 per month. EWS and LIG groups account for 72% of the households. The city, thus, has more of poor people houses, on the ground than of rich people houses, as a skyline. In the case of EWS, the expenditure is more than their monthly income. Major source of income is wages, salary and pension for 90% of households, other investments account only 7%. Of the earning head of households 14% are above 59 years of age and 6% are females as head of households. Only 29% of the members are earners. And only 28% of the households have assets. About 76% of a household's expenditure go for the major and main item of food and essential.

(iii) Physical Profile of Buildings

Nearly 77% of the buildings are accessible through tar roads, and only 14% are approachable through mud roads. 36% of EWS buildings and 31% of LIG buildings roofs and walls are of temporary materials namely thatch/grass. 35% of EWS buildings are dilapidated. 18% of the buildings comprise flats and 41% are independent buildings. 8% of the buildings are put to residential and commercial uses. 53% of the buildings are within 20 years of age. 14% are between 20 to 40 years of age. Nearly 7% of buildings are of more than 3 floors. On an average 2.81 households occupy each building. Average house cost is thrice the annual income of a household.

(iv) Amenities

On an average only 12% of the households have no power supply, however in the case of EWS category nearly one in four households have no electric connection. Municipal water supply, own well or bore well, public tap or hand pump together cover around 89% of the households. 71% of the households have private bathing facilities. 70% of the households have private toilet facility. 65% of households have facilities to dispose sewage into municipal connections and 33% have septic tank or soak pit and remaining percent have open drain and dry latrines. Availability of amenities increases with the rise in income and EWS has far less amenities than others.

(v) Tenure Status and Mode of Property Acquisition

About 46% of the buildings are rented houses and 54% are own houses. Nearly 95% of tenant households do not have any property. Of the owned households, 23% are ancestral property and 22% are purchased from private party. 32% of the households have made full payment for purchased property and only 7% have paid partial payment. Percentage of houses owned increases with rise in income.

(vi) Perceptions on Major Problems in Housing

Arranging own funds and obtaining a loan were found to be time consuming and difficult, more than 25% of households found it very difficult. Preparation of plans and getting them approved were difficult for more than 30% and very difficult for 18% of the households. Getting water, sewerage and electricity connections were not a problem for nearly 40% of the households. Getting skilled workmen was not a problem in urban areas like Madras.

(vii) Housing Needs and Demand

Housing needs were computed for 3 scenarios. Housing needs vary from 10% to 23% of total housing stock. In the EWS category, it varies from 19% to 40% units. The housing demand is computed for 5 scenarios. It varies from 1.61% to 22%. The effective housing demand should be based on mixed socio-economic variables. EWS households who desire a house, but cannot afford it, do not come under housing demand and need to be addressed separately.

Slums

6.14 The Government of India Slum Areas (Improvement and Clearance) Act of 1954 defines a slum as "any predominantly residential area where the dwellings by reason of dilapidation, overcrowding, faulty arrangement, lack of ventilation, light or sanitary facilities or any combination of these factors are detrimental to safety, health or morals. In 1971, the Tamil Nadu Slum Clearance Board, drafting officials from Survey, Statistical, Revenue and Town Planning Departments then, conducted Socio-Economic survey of Madras Slums. For the purpose of the survey, a slum was taken to mean "hutting areas with huts erected in a haphazard manner without proper access, without protected water supply and drainage arrangements and so congested as to allow a little free flow of air to get in". Some of the observations found made in the report are extracted below:

"Slums generally present the most unhygienic, ugliest, nauseating scene. During rainy season, the whole area gets flooded, the pathways become swampy and the entire colony become as fertile breeding place for mosquitoes, exposing the slum dwellers living in the area to all sorts of diseases. During summer, the thatched huts are prone to fire accidents. Thus, the slum dwellers' life is the most miserable one, devoid of all basic amenities.

6.15 To ease the difficulties of the slum dwellers, attempts were made earlier to clear the slums in Madras City. In North Madras, the Corporation of Madras and in South Madras the City Improvement Trust and subsequently, the Tamil Nadu Housing Board were looking after the slum clearance work. They, apart from constructing flats for the low and middle income groups, allotted open developed plots, measuring 20' x 40' in certain areas to slum families. However, further allotments required vast areas of land, and the scheme was given up owing to scarcity of land within the City.

6.16 The Government of Tamil Nadu hold the view that slums are not acts of God, but of human folly and that they can be banished by wise planning and resolute action.

6.17 The Tamil Nadu Government realized that the feeble, halting, incomplete and disconcerted measures of the past have to give place to a comprehensive, integrated and concerted policy to be put through on an emergency footing. It also realized that unless programmes of slum clearance and settlements of slum dwellers are drawn on a more realistic basis, relating them to economic opportunities and provision of social services and social welfare services, the results may well turn out to be frustrating. It also recognized the fact that the slum dwellers are an essential element in city life, and that they are as necessary as any other section of the population for the life of the city."

6.18 The TNSCB was formed in 1970 and the Tamil Nadu Slum (Improvement and Clearance) Act, was enacted in 1971. The following are the objectives of the Board:

(1) To clear all the slums in Madras city within a targeted period,
(2) To prevent further growth of slums in Madras city,
(3) To give protection to the slum dwellers from eviction and to re-house them in modern tenements and
(4) To provide basic amenities such as drinking water supply, electricity, storm water drainage etc. to certain slum areas until they are finally cleared.

6.19 The Socio-economic Survey 1971 has brought out the characteristics of slums. Main findings are

(1) Causes of slums in Chennai:
   (i) Poverty - Frequent failure of monsoons led to mass influx of agricultural labourers from the adjoining districts to the City. After coming to the City they pick up any manual job (unmindful of the hardships) such as head load carriers, rickshaw-pullers, cart-pullers, domestic servants, petty vendors, carpenters, masons and other manual works. The income they derive from their jobs was very low which was hardly sufficient for a balanced diet, so they were unable to pay for rents for securing decent dwellings and hence squatted on open spaces available near their work spots.
   (ii) Physical conditions - The low-lying areas and waterfronts served as fertile grounds for the growth of slums.
(2) Out of the City extent (then) of 128.83 sq.km. slums covered about 6 percent of the total area. They were situated in government lands, City Corporation lands, and Housing Board lands, lands of religious institutions and private land as detailed in Table No.6.07.

Table No.6.07: Ownership of Lands Occupied by Slums, 1971		
Sl.No.	Ownership of land	Percentage of slum families to the total slum families
1.	Private	31.96
2.	Corporation	08.11
3.	Government (State)	35.69
4.	Housing Board / Slum Clearance Board	13.09
5.	Port Trust	0.03
6.	Hindu Religious Endowment / Wakf Board / other missions	09.01
7.	Others	02.12

6.20 In order to provide a comprehensive data base on the number, location and type of slums and their population for overall planning and implementation of specific programmes, CMDA had drawn the services of the consultants M/s. Economist Group, Chennai to conduct the survey of slums in the CMA; it was conducted between April and December, 1986. In this survey, the number of shelter within slums were collected in respect of the slums which were not covered (then) under MUDP I & MUDP II and slum clearance schemes of TNSCB.

Table No.6.08: Survey of Slums in MMA - Slum Population in the Study Region – Covered under MUDP/Clearance, 1986

Zone	Corporation Divisions	MUDP I		MUDP II		Clearance		All	
		S	F	S	F	S	F	S	F
I Extended Areas	8	3	5276	39	7319	6	2925	48	15520
III South Madras	69	19	3192	70	14060	72	24169	161	41421
IV North Madras	73	51	14102	121	25978	40	17576	212	57656
Madras City	150	73	22570	230	47357	118	44670	421	114597
II Towns in MUA	10 Towns	3	1709	10	3408	-	-	13	5117
MUA	-	76	24279*	240	50765	118	44670	434	119714

Source: Survey of slums in MMA, 1986

S – Slums F – Families Socio-economic part of it was concerned estimation were made based on sample survey.

For the purpose of that study, CMA had been divided into four as follows:

Zone-I - Extended areas of the City covering 10 towns in 8 Corporation Divisions
Zone-II - Peripheral areas comprising 10 towns in the MUA
Zone-III - Madras City : South having 69 Corporation Divisions
Zone-IV - Madras City : North with 73 Corporation Divisions

6.21 Statistics on the growth of slum households and slum population is given in Table below.

Table No.6.09: Growth of Slums in Chennai City				
Sl.No.	Year	No. of slums	No. of the households	Slum Population
1.	1956	306	57,436	2,87,180
2.	1961	548	97,851	4,12,168
3.	1971	1202	1,63,802	7,37,531
4.	1986	996*	1,27,181	6,50,859
5.	2001	1431(CMA)	1,78,000	8,20,000

Pavement Dwellers:

6.22 According to Survey of Pavement-dweller in Chennai City conducted by the consultant SPARC for CMDA in 1989-90, the number of households who were living on pavements was 9491 at 405 clusters at an average of about 23 households at a place; their population was 40,763 (20,811 Male and 19,950 Female) with 40.2% children population. Unlike other old cities in India namely Delhi, Mumbai, Kolkotta the number of pavement dwellers in Chennai is relatively few.

Delivery of Housing

6.23 The delivery agencies in CMA can be broadly classified as public, co-operative and private sector. Under the public sector, the agencies operating mainly are TNHB and TNSCB; the agencies which provided housing to its employees are TNPHC, Railways, P&T and CPWD, Port Trust, etc.

Tamil Nadu Housing Board

6.24 Tamil Nadu Housing Board was established as a statutory body in 1961 under the Tamil Nadu State Housing Board Act, 1961.

6.25 TNHB has been delivering the housing under their different programmes viz. regular programme, MUDP & TNUDP. TNHB was delivering serviced plots and constructed houses and flats under these programmes. TNHB had developed large neighbourhoods with all amenities and facilities within CMA at Arignar Anna Nagar, K.K. Nagar, Ashok Nagar, Bharathi Nagar, South Madras Neighbourhood Scheme (comprising Indira Nagar, Besant Nagar and Sashtri Nagar), Thiruvanmiyur, Tambaram etc. TNHB has also developed Sites and Services Schemes under MUDP-I, MUDP-II and TNUDP at Arumbakkam, Villivakkam, Kodungaiyur, Mogappair (East), Mogappair (West), Maduravoyal, Manali, Madhavaram, Ambattur, Avadi and Velachery. The details of the housing delivered under MUDP & TNUDP is given in the Table No.6.10.

Table No. 6.10: Details of Sites and Services Projects under MUDP & TNUDP				
Project	Scheme	Year of Commencement	Extent (hect.)	No. of Units
MUDP I	Arumbakkam	1977	34.20	2,334
	Villivakkam	1979	71.55	3,751
	Kodungaiyur	1979	84.87	6,094
			190.62	12,179
MUDP II	Mogappair East	1981	74.13	5,062
	Mogappair West	1983	73.00	5,555
	Maduravoyal	1983	26.70	2,048
	Manali I	1986	40.00	2,947
	Manali II	1987	38.00	2,625
			251.83	18,264
TNDUP	Ambattur	1988	141.60	10,806
	Avadi	1988	50.90	4,012
	Velachery	1988	20.90	1,789
	Madhavaram	1988	63.90	4,884
	Madhavaram	1988	60.00	5,000
			337.30	21.441

TNPHC

6.26 Tamil Nadu Police Housing Corporation was established in 1981 to effectively implement schemes for providing adequate housing facilities for the police personnel. It has delivered 4,702 housing units from 1992 to 2002; there are 1,509 housing units under construction by it at various parts of CMA and their proposals in the anvil include construction of 1,378 housing units.

6.27 The Shelter Strategy Study conducted by School of Planning and Architecture, in 1991 revealed that the public agencies had delivered total housing units of 10,555 Nos. from 1981 to 1991 [CPWD (1,854 Nos.), P&T (1,572 Nos.), Railways (2,400 Nos.) Port Trust (500 Nos.), TNPHC (104 Nos.), THADCO (1,125 Nos., and others (3,000 Nos.)].

Co-operative Housing Societies

6.28 In Tamil Nadu, 1,253 Co-operative Societies comprising 196 Taluk Co-operative Housing Societies and 1,057 urban Co-operative Housing Societies, and a State Level Apex institution viz. "The Tamil Nadu Co-operative Housing Federation Ltd" are functioning to cater to the housing needs of its members. The schemes implemented through co-operative housing societies are:

1. Rural Housing Scheme for Economically Weaker Sections
2. Special Housing Scheme for Economically Weaker Sections in Urban areas
3. LIG, MIG Schemes in Rural areas
4. Urban Housing Schemes
5. Valmiki Ambedkar Awas Yojana (VAMBAY) Scheme
6. Repairs and Renewal of existing houses

Though the number of households benefited in the State from these societies upto the end of 2004 was 11.36 lakhs, their role in delivery of housing in CMA is minimal. According to the School of Planning and Architecture's Shelter Strategy Study, the number of plots delivered by the Co-operative Societies from 1981 to 1991 was only 2,489 in CMA.

Private Sector

6.29 It can be broadly divided into two. The first one being the private individuals or owner private who construct the house by himself/herself initially with a smaller floor area and adding incrementally when necessity arises and his fund position improves. The second one being the organized private agencies or real estate developers/promoters who acquire land, develop plots, construct houses/flats and sell. The School of Planning and Architecture's Shelter Strategy Study (1995), the total number of housing units delivered by the private sector during the year 1981-91 was 2.73 lakhs; out of which 53% was plots, 37% was houses and only 10% was flats (Details are given in Table No.6.11).

Table No.6.11: Annual Delivery of Plots, Houses and Flats by Private Sector in CMA During 1981-82 to 1990-91

Year	Approved				Unapproved			Total			
	Plots	Houses	Flats	Total	Plots	Houses	Total	Plots	Houses	Flats	Total
1981- 82	1528	4181	--	5709	8257	2355	10612	9785	6536	--	16321
1982-83	2019	4676	--	6695	8987	2300	11287	11006	6976	--	17982
1983-84	2602	4523	918	8043	8948	2483	11431	11550	7006	918	19474
1984-85	4667	5076	1067	10810	9721	3003	12724	14388	8079	1067	23534
1985-86	3799	5270	2333	11402	10562	2941	13503	14361	8211	2333	24905
1986-87	2755	4883	2681	10319	11008	3120	14128	13763	8003	2681	24447
1987-88	7652	6673	2726	17051	9112	3411	12523	16764	10084	2726	29574
1988-89	6838	7963	6368	21169	8860	3204	12064	15698	11167	6368	33233
1989-90	10447	12806	6874	30127	8487	3653	12140	18934	16459	6874	42267
1990-91	8817	13728	5896	28441	8905	3916	12822	17725	17642	5896	41263
Total	51124	69779	28863	149766	92850	30384	123234	1438974	100163	28863	273000

6.30 The flatted housing units delivered by the real estate developers / promoters serve only the needs of the MIG and HIG households. LIG housing needs are met by the owner private developments.

6.31 Details on the number of Planning Permissions issued by the Municipalities, Town Panchayats and Panchayat Unions for residential buildings are given in Table No.6.12.

Table No.6.12: Planning Permission for Residential Buildings Issued by Municipality, Town Panchayats & Panchayat Unions

Sl. No.	Agency	No. of Planning Permission issued for Residential buildings				
		2000-01	2001-02	2002-03	2003-04	2004-05
1.	Municipalities	5544	6541	8000	8557	8579
2.	Town Panchayat	2749	3689	4544	5719	5421
3.	Panchayat Unions	2629	2921	3160	4154	6319
	Total	10922	13151	15704	18430	20325

6.32 Planning Permissions (Nos.) issued by Chennai Corporation and CMDA for residential buildings are given in the Table No.6.13.

Table No.6.13: Planning Permission issued for Residential Buildings by Chennai Corporation and Chennai Metropolitan Development Authority

Sl. No.	Agency	No. of Planning Permissions issued for Residential buildings				
		2000	2001	2002	2003	2004
1.	Chennai Corporation	4882	4624	4735	6215	5700
2.	CMDA	678	762	688	1032	1323

6.33 Planning Permissions of ordinary residential buildings up to 4 dwelling units and up to G+1 floors are issued by the local bodies concerned and only for the flatted residential developments, group developments and multistoreyed (high rise) developments, Planning Permissions are ordered in CMDA. Assuming the average number of dwellings covered in the Planning Permissions issued by Chennai Corporation as 2, Municipalities as 1.75, Town Panchayats as 1.5 and by Panchayat Union areas as 1.25, and average number of residential Planning Permission issued per annum as 5,500, 8,000, 5,500 and 5,000 respectively, it is estimated that the annual supply of approved housing units will be about 40,000 units permitted by the local bodies. In addition, CMDA issues Planning Permission for annually for about 8,000 housing units. Assuming that the delivery of unauthorisedly constructed small housing units as about 25% of the total supply, then the total delivery of housing units by private will be about 60,000 units per annum if the present trend continues; it excludes the housing plots delivered by public.

6.34 Details on the number of layouts approved by CMDA, residential plots generated etc. are given in the Table No. 6.14. Private delivers annually about 6,000 residential plots in approved plots. In addition, in the past, the private delivered an equal number of plots in unapproved layouts. Hence the supply of housing plots in the CMA is estimated to be about 12,000 plots per annum by the private.

Table No. 6.14: Layout Approvals in CMDA 1991-2004

Year	No. of Approved Layouts	No. of Residential plots	Shop sites	Public Purpose sites	School sites
1991	200	14601	132	429	2
1992	176	9640	120	397	--
1993	160	8063	78	516	9
1994	144	6799	93	360	2
1995	103	4823	110	237	16
1996	63	2648	47	91	--
1997	66	1601	33	39	12
1998	59	1840	49	65	13
1999	79	4894	30	98	--
2000	102	4205	36	157	--
2001	82	4575	63	229	3
2002	84	3769	29	80	--
2003	142	7801	90	121	4
2004	143	6200	84	401	20

6.35 Since 1989 CMDA, ensures that at least 10% of plots excluding roads are provided as EWS plots when according approval in cases of layouts exceeding one hect. By this way at least 10 % of the plotted out area in the layout is generated as EWS plots which can accommodate about one third of the population, which can be accommodated in the layout area.

Projection of Housing Demand

6.36 The housing need for CMA was projected taking into consideration the growth of households, vacancy rate, and demolition rate of old buildings and replacement rate. The housing demand is estimated based on the growth of house-holds, vacancy rate, replacement rate and affordability. The following Table gives the details of projected housing need and demand for 2026.

Table No: 6.15 Projection of Housing Need and Demand						
Housing Need						
	2001	2006	2011	2016	2021	2026
Population	7040582	7896230	8871228	9966636	11197763	12582137
House holds	1619000	1754718	1971384	2214808	2488392	2796030
Number of Houses	1583014					
Shortage w.r to 2001	35986	171704	388370	631794	905378	1213016
Vacancy Rate @.5%	7915	8774	9857	11074	12442	13980
>60 Years old building	2.3	1.5	1.5	1.5	0.75	0.75
Demolition Rate	37237	26321	29571	33222	18663	20970
Replacement Rate	18619	13160	14785	16611	9331	10485
Total Housing Need	**81138**	**206798**	**427798**	**676090**	**936483**	**1247967**
EWS (30%)	24341	62039	128339	202827	280945	374390
LIG (35%)	28398	72379	149729	236632	327769	436788
MIG (20%)	162287	41360	85560	135218	187297	249593
HIG (15%)	12171	31020	64170	101414	140472	187195
Housing Demand						
Shortage	35986	171704	388370	631794	905378	1213016
Vacancy	7915	8774	9857	11074	12442	13980
Replacement of Old Buildings	18619	13160	14785	16611	9331	10485
Total Housing Demand	**62520**	**193638**	**413012**	**659479**	**927151**	**1237482**
EWS (30%)	10796	58091	123904	197844	278145	371245
LIG (35%)	21882	67773	144554	230818	324503	433119
MIG (20%)	12504	38728	82602	131896	185430	247496
HIG (15%)	9378	29046	61952	98922	139073	185622

6.37 TNSCB estimates indicate that the slum families in undeveloped slums work out to 1.10 lakhs; out of which 75,498 families are living in objectionable slums. TNSCB has also estimated that there are about 34,752 families in unobjectionable areas and further there are 6,150 families who live in slum conditions in the encroached parts of the tenement areas which have been reserved as parks, public purpose sites etc.

Slums on unsuitable Locations

6.38 The slums situated on river margins, road margins, seashore and places required for public purposes are categorized as objectionable slums. The areas occupied by them are to be retrieved and handed over to the land owning department to implement programmes like road widening, desilting, strengthening of bunds etc.. Hence the benefits of the various onsite programmes implemented by TNSCB could not be extended to the slums located in objectionable areas. The location of slums is given in Table No. 6.16.

Table No.6.16: Location of Slums in Chennai City, 2000		
Sl. No.	**Description**	**No. of slum families**
1	River Margin	30,922
2	Feeder Canals	5,288
3	Road Margin	22,769
4	Seashore	16,519
	Total	**75,498**

Source: TNSCB

Slum areas on River Margin

6.39 There are three major watercourses in the Chennai City and the banks of which are encroached. The slum families are living there without any basic amenities and subjected to annual flooding, besides polluting the watercourses. The details of the slums located on river margins are as follows:

Sl.No.	Name of the River	No. of Slum families
1	Cooum River	8,432
2	Buckingham Canal	15,733
3	Adyar River	6,757
	Total	30,922

Slums along Feeder Canals

6.40 The feeder Channels like Mambalam - Nandanam Canal, Otteri Nullah, Captain Cotton Canal etc. are encroached on either side preventing the free flow of water and causes stagnation of water during rainy season in the nearby residential areas. It is identified that 5,288 families are living on the margins of these channels.

Slums on Road Margins

6.41 Besides, the slum families are also squatting on road margins affecting free flow of traffic. It is identified that 22,769 slum families are squatting on the road margins and places required for public purposes.

Seashore

6.42 The Chennai District Collector has identified that 16,519 slum families living on seashore have been affected by Tsunami disaster. Of these 2,200 families have already been rehabilitated at Semmancheri. The remaining families are to be resettled in tenements.

Slums on Unobjectionable Locations

6.43 Besides, TNSCB has identified that 34,752 families are living in slum situated on unobjectionable locations as per its survey. These families are living in deplorable conditions and require to be developed through provision of housing and infrastructure. These slums are dense often subjected to fire accidents, flood etc. Based on previous experience, the TNSCB is planning new strategies for solving the problems of slums.

6.44 TNSCB has been unable to clear all slums and provide houses to the EWS. So far the TNSCB has only been able to construct 72,000 houses or tenements (over a period of 35 years), which works out to around 2,000 houses per annum. At this rate it will be difficult to cover all the remaining 1.1 lakh households in Chennai who continue to live in raw slums.

6.45 Given the enormity of the problem it is increasingly becoming difficult for TNSCB to use these traditional modes/strategies (strategy A) of development, reconstruction and resettlement. These traditional strategies of house being built by the government and its agencies may have to be abandoned due to several reasons, including the following:

- Lack of availability of funds and high cost
- Poor recoveries given the economic strata for which the houses are built
- Lack of availability of lands
- Escalation in the cost of lands
- Lack of in-house implementing capacity, specially in respect of newer, faster building technologies
- Lack of manpower and organizational wherewithal to take up huge works
- Delayed execution due to Governmental procedures having to be necessarily followed and
- Indifferent quality of construction

6.46 Keeping this in view there is a need to go in for other strategies (Strategy B) which can supplement the existing efforts even if they do not replace them. Some of these strategies are based on leveraging the high costs of lands that are often government owned.

6.47 The strategies followed for different categories of slums (raw slums, developed slums, dilapidated tenements are as outlined .

(i)Strategy A:

6.48 The traditional strategies can be used here like reconstruction, redevelopment using funds and lands provided by the Government. This can be used in the case of raw slums/developed slums/tenements. In the recent past several new schemes have come up which offer a glimmer of hope since they have the mandate of a slum-free Chennai by 2013 and which offer low cost or zero cost funds. Under these schemes it is proposed to construct house in the next two to five years all of which are for the EWS:

- ETRP and Rajiv Gandhi package (tsunami housing): 13,000 dwelling units (these are both reconstruction/redevelopment and resettlement schemes and are coming up in several places in Chennai including Marina,Thiruvottiyur,Tonidarpet, Okkiyumthorapakkam and Semmencheri)
- JNNURM:10,000 dwelling units (these are resettlement schemes for slum dwellers)
- XII Finance Commission: 5,000 houses (basically in the resettlement mode at several places including Perumbakkam and Semmencheri)

(ii) Strategy B

6.49 TNSCB land (both tenement land and unobjectionable poromboke land) as well as raw slum land can be redeveloped through the BOT route by allowing private developers to reconstruct the existing dilapidated tenements /slum houses on a part of the land and using the rest of the space for commercial exploitation.

Night Shelter

6.50 TNSCB has estimated that annually the people who enter Chennai without provisioning for stay would be around 20,000. A part of them would be finding rental accommodation in LIG areas while some would encroach on public lands. To address their immediate housing needs night shelters need to be constructed.

It is also important that unauthorised encroachments on public lands to be prevented by ensuring that the Government departments protect their lands.

Fishermen Housing

6.51 Chennai is a coastal metropolis and there are 84 fishermen villages along the coast of which 43 are in Chennai City, 30 are in the northern part of City up to Minjur and 11 in the south up to Uthandi. There are 12 landing sites in Chennai (14 and 38 in the northern and southern parts). Housing for fishermen becomes important particularly because the housing has to be close to their working area namely the sea and the restrictions placed by CRZ for several types of development. According to a recent count there are 36,162 fishermen households with an average household size of 3.81. They live in 31,688 pucca as well as kutcha structures of which 16,482 are in Chennai, 8,439 in northern part of CMA and 6,767 in the southern part of CMA. The growth of population among fishermen has increased by 5%between 2000 and 2005 and thus this trend is likely to continue. At present fishermen housing is dealt by TNSCB and Fisheries Department. Any housing or redevelopment projects for fishermen in the coastal areas should take into account the traditional rights and way of life.

Chapter - VII

INFRASTRUCTURE

Water Supply

Introduction

One of the major problems faced in CMA is the inability of the administration to keep pace with the increasing need for utility services particularly water supply and sewerage. This problem is not unique to Chennai alone and almost all the rapidly growing cities in India share the same.

7.02 In Chennai City, Chennai Municipal Corporation was responsible for construction, operation and maintenance of water supply system till August 1978. It was transferred to the (then newly formed) CMWSSB with all assets and liabilities. The major supply sources viz. Poondi reservoir, Cholavaram lake and Redhills lake are under the control of the State PWD (Irrigation); further the PWD (Ground Water Cell) is responsible for the investigation of ground water resources within CMA to augment supplies.

7.03 In the cases of large-scale neighbourhood developments made by TNHB such as Anna Nagar, K.K. Nagar, J.J. Nagar, Ashok Nagar, South Madras Neighbourhood schemes etc., water supply facilities within the area had been provided by TNHB (usually with ground water sources or extending the city water supply) and after installation, maintenance had been transferred to CMWSSB. The same is the case in cases of large-scale TNSCB developments also.

7.04 In the rest of the CMA, construction and water supply schemes were undertaken mostly by TWAD Board at the cost of the local body concerned and after completion, transferred to the local body for future operation and maintenance. CMWSSB is now planning to cover these areas with water supply and sewerage services.

City System

7.05 Chennai City water supply is drawn from the Red hills lake located about six km. from City in the northwest along the GNT Road. This lake mainly receives its supply from the Kosasthalaiyar River across which a dam was constructed to create reservoir at Poondi. The anicut constructed in its downstream at Tamaraipakkam diverts the flow through the upper supply channel to Cholavaram Lake; from there it flows to Redhills Lake. A separate lined channel from Poondi reservoir connects the upper supply channel which prevents any possible loss of transmission through dry beds of the river. The flow into these sources is seasonal.

7.06 A number of small lakes are also connected in the northwest of CMA to the Redhills Lake and the run-off in the catchment areas of these lakes are fed to the Redhills Lake.

7.07 Water tapped from the Redhills Lake is filtered enroute, conveyed through closed conduits to Kilpauk Water Works, treated therein and distributed to various parts of the City.

7.08 Nucleus of the protected surface water supply system now in existence in Chennai City was formed in 1872 while major inputs including filtration and pumping commenced in 1914. The scheme as it exists today consists of the surface storages at Poondi, Cholavaram and Redhills lakes with a total capacity of 183 million cubic metres fed by the Kosatalaiyar river. The maximum water levels of Redhills and Cholavaram lakes were raised in 1972 and the irrigation rights under the lakes had also been acquired by the Government. The lakes are mostly fed by North East monsoon, which is active only for a few months in a year between October and December.

7.09 A statistical analysis of the combined storage of the lakes for a period of thirty years has indicated that the safe potential from the lakes at 95% probability is only 142 mld. As the lakes are shallow, evaporation losses are as high as 43% at present.

Ground Water System

7.10 The ground water sources are ground water from well fields, coastal aquifer; brackish water based Reverse Osmosis Plants, Neyveli Aquifer etc.

7.11 During 1960s & 70s three aquifers at Minjur, Panjetty and Tamaraipakkam located in the north and northwest of the City and the aquifers along the coastal belt from Thiruvanmiyur to Kovalam were identified and tapped. The area North East of the City was taken up for extensive hydro- geological study as part of a UNDP aided project to assess the potential for development of a ground water source. Studies have identified a 'buried channel', which should have been the course of Palar river thousands of years back. In this course, a well field was identified extending to a stretch of about 50 km. length and 5 km. average width hydro geologically suitable for extraction of ground water. The following well fields have been developed.

Table No.7.01 Details of Well Fields				
Well Fields	Year of Commission	No. of wells	Yield MLD during design	Average yield from Dept. wells in 2005
Tamaraipakkam	1969	2 out of 30	50	1.60
Panjetty	1969	1 out of 13	41	0.08
Minjur	1969	5 out of 9	34	3.10
Poondi	1987	4 out of 12	27	1.20
Flood Plains	1987	0 out of 5	14	0
Kannigaiper	1987	0 out of 5	14	0.01
Total		12 out of 74	180	5.99

Source: CMWSSB

7.12 Above table clearly brings out the depletion of ground water source during the last 30 years in that area due to increase in demand resulting in overdrawal of ground water. In addition, due to severe scarcity, CMWSSB has hired private agricultural wells from 2000 to augment water supplies. The average yield from such sources during 2005 is to the tune of 77 MLD.

Details of hiring of Private Agricultural Wells		
Sl.No.	Year	Average Yield in MLD
1	2001	37
2	2002	45
3	2003	55
4	2004	70
5	2005	77

7.13 In the recent past the agriculture activity in the ayacut areas of Chembarambakkam tank has been reduced drastically because of conversion of lands for urban uses and for reasons of uneconomical production costs. Now the Chembarambakkam tank (located about 12 km. from the City in the west) is also used as one of the main sources of water supply to the City. Veeranam tank (located about 230 km. from the City in the south) has been identified as an additional source of water supply to Chennai.

Table No.7.02 Characteristics of Existing Surface Reservoirs						
Reservoir	Maximum Water Elevation (Metre)	Drawl of elevation (Metre)	Maximum usable volume (million)	Full Area surface (million sq.ft.)	Mean Depth (Metre)	Catchment Area (Sq.km.)
Poondi	42.67	34.14	91.44	376.74	8.53	1983
Cholavaram	19.66	14.22	24.93	55.76	5.44	28.49
Redhills	15.30	8.59	87.02	195.15	6.71	60.00
Chembarambakkam	26.03	18.72	103.15	711.18	7.31	357.00
Veeranam	14.48	12.34	37.61		2.14	422.40
Total			306.54			2850.89

Ground Water Regulation

7.14 To regulate and control the extraction, use or transport of ground water and to conserve ground water, the Chennai Metropolitan Area Ground Water (Regulation) Act, 1987 was enacted. The preamble of the Act is extracted below:

> "There is often acute scarcity of water due to consecutive failure of monsoon rains; the available water in the Poondi, Cholavaram and Red Hills reservoirs, which are the main sources of supply of water to the Chennai City, is inadequate to meet the requirements for drinking and other domestic purposes of the people in the Chennai City;
>
> The United Nations Mission, which investigated the possibility of supplementing water supply to Chennai, has recognized that a better economic answer might lie in the development of groundwater potential and had identified the Minjur, Duranallur-Korteliyar basin, the Poondi, Korteliyar Flood Plains and Kannigaipper aquifers and also Poonamallee-Porur aquifer in Cooum-Adayar basin as having groundwater for extraction;
>
> The United Nations Development Programme which conducted pre-investment studies on improving water supply and sewerage systems of chennai concurred with the estimation of the Geological Survey of India that groundwater can be extracted from the twenty kilometre stretch of the coastal zone between South Madras and Kovalam;
>
> The Madras Metropolitan Water Supply and Sewerage Board has reported that all other possibilities of augmenting water supply to the Chennai City have been exhausted and that it is necessary to regulate and control the extraction and use of ground water in any form and to conserve the same in the City of Chennai and the district of Chengalpattu and to regulate and control the transport of groundwater;
>
> Based on the United Nations Development Programme studies, a scheme for artificial re-charge of the Arni-Korteliyar basin with excess flood water flowing into the sea is to be taken up by interlinking Arni and Korteliyar at two or more feasible points and also by constructing sufficient number of check dams at the appropriate places in the Korteliyar river course;
>
> Such re-charge will enable optimum utilisation of ground water and formation of a hydraulic barrier against seawater intrusion;
>
> The Government have, after careful examination of all aspects, decided that it is necessary in the public interest to regulate and control the extraction and use of

groundwater in any form and to conserve the groundwater in the City of Chennai and certain revenue villages in the district of Chengalpattu and to regulate and control the transport of groundwater." The Government of Tamil Nadu has enacted the Act viz. Madras Metropolitan Area Ground Water (Regulation) Act 1987.

7.15 The Act is being enforced by the CMWSSB in the Chennai City and the District Collector in the rest of CMA, and it provides for grant of permit to sink wells in the scheduled areas, registration of existing wells and use of ground water in the area, license for extraction, use or transport of ground water etc.

7.16 The entire area within the Chennai City has potential for restricted drawal through shallow open wells, and tube wells, and these types of wells are predominant in the premises within the City. Though unprotected, it helps to supplement the limited quantity of protected water available from the public system. Assuming a conservative figure of drawal of 50 lpcd of ground water, for the population of Chennai about 45 million in 2005, estimated drawal will be in the order of 225 MLD. Details on TDS level in ground water in Chennai are given in the Annexure VI-A. The level of TDS of sample wells varied from 500 ppm to 2500 ppm.

7.17 Estimated safe yield and average yield at various sources for water supply to Chennai in the year 2004 is given below:

Table No.7.03 Safe Yield from different sources 2004

Sl.No.	Source	Average yield in MLD	Safe yield in MLD
1.	Surface Water including I stage Krishna	17.89	500
2.	New Veeranam Project a) Veeranam Lake b) Neyveli Aquifer	 148.42 50.56	 180 65
3.	Well fields	85.23	100
4.	Distance Source	63.16	100
5.	Porur Wells	2.96	3
6.	Southern Coastal Aquifer	2.12	4
Total		370.34	952

Source: CMWSSB

7.18 The sources of water supply to Chennai in the year 2004 is given in the Table below. The areas served by it are the Chennai City, adjoining urban areas (10 sq.km.) and industries at Manali.

Table No.7.04 Sources of Supply of Water to Chennai, 2004	
Source (in ML) for the whole year 2004	
Veeranam Lake	14,842
Redhills Lake	4,155
Rain water	1,691
Chembarambakkam	133
Erattai Eri	207
Well fields	31,195
Southern Coastal Aquifer	776
R.O. Plants	182
TWAD Source	275
Porur Wells	210
Neyveli Aquifer	5,966
Distance Source	21,357
Total	80,988

Source: CMWSSB

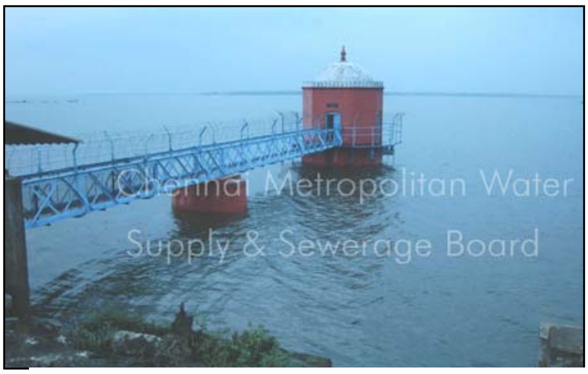
7.19 When there was acute scarcity of water in 2004 to the City due to monsoon failure and it could not be supplied through pipe, CMWSSB had ably managed the crisis by supplying through tanker lorries and the water supply through tankers were 9,930 nos./day (average) from January, 2004 to June, 2004 and 9,003 nos./day (average) from July to December, 2004.

Supply Levels

7.20 The average water supply in Chennai city is 90 lpcd. In slums within the City the level of water supply is 25 lpcd. The current water supply from all the sources is of the order of 550 MLD. However, during the summer season, in times of reduced storage, the supply levels would be as low as 300 MLD. In addition, CMWSSB also provides bulk water to the surrounding municipalities to supplement the other sources.

Treatment Capacities

7.21 The availability of treatment capacities at 750 MLD is much more than the current supply levels. The treatment capacity appears to be high as the current supply is lower than the anticipated levels and would suffice the current requirement if a full supply is envisaged. The location and capacities of treatment plants is presented in Table No.7.05.



View of Redhills Reservoir

Table No. 7.05: Treatment Capacities		
Sl. No	**Location**	**Capacity-MLD**
1.	Kilpauk	270
2	Red hills	300
3	Veeranam	180
	Total	750

Source: CMWSSB

7.22 In addition, a 530 MLD capacity treatment plant at Chembarakkam is currently under construction taking the total capacity to 1280 MLD. This is constructed in anticipation of full realization of Krishna Water to the City, which would augment the existing supply levels.

Distribution System and Storage Capacity

7.23 The distribution network in Chennai City covers a length of 2,582 Km and is divided into 159 Divisions. About 98% of the population of the City is covered through piped supply and the balance is through water tankers. The water is distributed through 27 main and subsidiary water distribution stations comprising Over Head Tanks and Under Ground Tanks. The combined storage capacities are of the order of 322 ML. The per capita supply in Chennai Corporation area, as of July 2006, is 105 lpcd. The Chennai City water supply distribution network has been divided into 16 Zones with independent water supply distribution stations and distribution network with feeder mains. In 12 Water Distribution Stations and 11 distribution network zones, improvement works have been completed and the supply level is almost equitable. For balance 4 Water Distribution Stations and 5 zonal distribution networks, improvement works have been proposed under JNNURM funding, since the earlier attempt to include in the World Bank assisted projects had not been finalised. On completion of the improvement works in the balance 4 Water Distribution Stations and 5 distribution network systems, the supply level will be almost equitable.

7.24 The ULBs manage their own storage capacity, the present supply levels are low and would need augmentation in the future.

Master Plan for Water Supply- Distribution System

7.25 A master plan for the management of the water supply and sewerage for the City, which was prepared in the year 1978, was revised in the year 1991 in order to receive and utilize the water for City supply under Telugu Ganga Project from Andhra Pradesh and later updated in 1997. The master plan contemplated the construction of additional water treatment plant, water distribution stations, laying of additional transmission mains and strengthening of the existing distribution system. The master plan envisages re-organizing the existing distribution system network to the 16 zones and adequate infrastructure to ensure equitable distribution of water supply. The implementation of the master plan has been taken up in stages for water supply management in Chennai city to utilise the Krishna water received from Andhra Pradesh.

7.26 The size of the pipes constituting the distribution system ranges from 100 mm to 1500 mm in diameter with a total length of about 2,582 km. The existing system largely consists of C.I. pipes. In some specific areas PVC mains exist. Of the total length, 85% is estimated to be of diameter 200 mm and less. Also, about 50% of the smaller sized distribution pipelines are estimated to be of age 50 years or more. Ageing of pipeline, incrustation due to intermittent supply and other factors have contributed to the reduction in the capacity of the distribution system resulting in low pressure in the distribution system. More than 50% of the total system is estimated to have zero residual head. Chennai City has been expanding at a fast rate and because of this, it is found difficult to meet the demand, especially at the tail end areas. The 300 MLD capacity Water Treatment Plant was constructed at Red hills with World Bank assistance under the first Chennai Water Supply Project and 3 Nos. of transmission mains were also laid to facilitate the treated water to different water distribution stations located in the City.

Coverage in Slums

7.27 About 3.5 lakhs population of Chennai in slums comprising of 58,631 households are covered through piped supply, tankers and public fountains. CMWSSB supplies water through 8,916 public fountains and 3,542 mini tanks to augment the piped water supply. About 2/3rd of these facilities cater to low income and slum population. Ground water for supplementation purposes is also drawn through 7,726 India Mark II pumps.

Second Chennai Project

7.28 The Second Chennai Water Supply Project was taken up by CMWSSB during February 1996 with the World Bank Assistance at a cost of Rs.778.79 crores and most of the works contemplated under this project has been completed. Some of the major works such as construction of Water Distribution stations (7 nos.), Laying of Clear Water Transmission mains (36 km.), and Strengthening of Water Distribution system in 11 zones (660 km.) out of 16 zones were taken up, including leak detection rectification works covering about 70% of Chennai City Area. Benefits attained due to the implementation of the above project are:

(a) Piped water supply availability has been increased resulting in reduction in defective streets with improved service level and pressure in the mains.

(b) Reduction in the radius of the water distribution station resulted in increased piped water supply availability with reduced quantity of supply.

(c) The level of unaccounted for water has been reduced resulting in additional water availability to the Chennai citizens.

7.29 Due to the construction of additional water treatment plant of 300 MLD capacity at Red Hills in 1996, the work of refurbishment of the existing Kilpauk Water Treatment Plan of capacity 270 MLD was taken up at a cost of Rs.24.57 crores and all the works were completed.

Chennai City Water Supply Augmentation Project-I

7.30 Chennai Water Supply Augmentation Project-I (to add 180 MLD water to Chennai City water requirement) was taken up by CMWSSB in 2004 at a cost of Rs.720 crores. It is to draw 190 MLD of raw water from Veeranam Lake near Sethiathope, situated in Cuddalore District at about 230 km. from Chennai City, pump the raw water to about 20 km. through the pipeline to Vadakuthu for treatment, pump the treated water from Vadakuthu for a distance of about 8 km. to the Break Pressure Tank at Kadampuliyur ridge point and then convey the water from this ridge point by gravity for about 200 km. to the Water Distribution Station at Porur in Chennai and distribute to the public through the distribution network system.

Chennai City Water Supply Augmentation Project-II

7.31 The Government on 27.2.2004 has accorded the revised administrative approval for the Chennai Water Supply Augmentation Project-II (CWSAP-II) at an estimated cost of Rs.124.00 crore. The objective of this project is to augment water supply to the City by intercepting the rainwater runoff into the sea by the construction / rehabilitation of check-dams across Cooum, Adyar and Palar rivers. The water thus collected from the Vayalur check-dam and the available water in Kolavoy lake will be treated in the proposed Water Treatment Plant at Mangalam and will be transmitted to the City. The aim of the project is to tap 20 MLD of water from different sources.

7.32 The following are the components of work to be implemented under the CWSAP-II:

(1) Construction of check-dam at Vayalur across Palar river.

(2) Infrastructure for drawal of 20 MLD from Checkdam at Palar (drawal and conveyance).

(3) Treatment Plant at Vayalur and laying pumping main upto Pallipattu with an intermediate pumping at Tiruporur.

(4) Construction of check-dams across Manapakkam, Nandampakkam, Anakaputhur and Cowl Bazaar.

(5) Improvements to two Check-dams across Cooum at Paruthipattu and Kannapalayam.

This project is scheduled to be completed by end of 2007.

Telugu Ganga Project

7.33 Under this project 12 TMC of water from Krishna River will be received and stored in Poondi, Redhills, Cholavaram and Chembarambakkam lakes.

Construction of 530 MLD capacities Water Treatment Plant at Chembarambakkam

7.34 In order to treat additional water to be drawn under the Telugu Ganga Project additional treatment plant with capacity of 530 MLD is constructed at Chembarambakkam with an estimated cost of Rs134.90 crores with assistance from the French Government and a transmission main at a cost of Rs.90.00 Crores.

Desalination Plant

7.35 Keeping in view, the chronic problem of water scarcity in Chennai and adjoining areas due to frequent failure of the monsoons, Government decided to set up a desalination plant for supply of potable water to the residents of Chennai and adjoining areas. Accordingly CMWSS Board has proposed to construct a 100/200 MLD Sea Water Desalination Plant at Minjur on Design-Build-Own, Operate and Transfer (DBOOT) basis. The required land of 120 acres has also been identified near Minjur. The project is scheduled to be completed in 18 months time and is expected to be commissioned by 2007.

Third Chennai Project

7.36 CMWSSB has proposed to take up further systematic improvement projects in water supply, both for Chennai City and adjacent Urbanised Local Bodies as "Third Chennai Project" with World Bank assistance at a cost of Rs.750 crores. In order to improve the sources, works are proposed for deepening and desilting of Ambattur tank, Korattur tank and Madhavaram tank and for rehabilitation of Porur tank besides formation of check-dams. It is also proposed to install water meters to all the consumers to achieve sustainable revenue. The following are some of the major works envisaged under the proposed Third Chennai Project.

(a) Strengthening of water Distribution system in the left out 5 zones:

Under Second Chennai Project, out of 16 zones, works on strengthening of water distribution system was taken up and completed in only 11 zones. Due to the benefits achieved under Second Chennai Project, it is now proposed to take up the strengthening of the distribution system in the remaining 5 zones also viz. Anna Poonga, Kilpauk, Triplicane, Southern Head works and KK Nagar. The total length of pipelines to be laid is about 305 km. at a total cost of Rs.150 Crores.

(b) Infrastructure facility to draw additional ground water from A.K. Basin:

Under the Second Chennai Project, the consultancy study to reassess the ground water potential in Araniar Kortalayar Basins is under way and will be carried over into the Third Chennai Project after ascertaining the sustainable yield.

Rest of CMA

7.37 Potable water supply system exists almost in all the municipalities with CMA. Alandur, Pallavaram, Tambaram, Anakaputhur and Pammal Municipalities have water from Palar River as source, and other municipalities have CMWSSB bulk supply or the ground water as source. Water supply in Panchayat areas are concerned, it is by local wells and public taps.

Water Demand

7.38 The rate of consumption of water in some Indian cities is given below:

Table No.7.06 Water Consumption in Indian Cities

Town	Consumption litres per capita per day
Bangalore	140
Mumbai	260
Delhi	270
Chennai City	107*
Pune	220

Source: CMWSSB

* 2006-2007

7.39 It was estimated in the Madras Water Supply and Sanitation Project report (1987) that the requirement of water will be 165 lpcd based on need based assessment. Future requirements of water at the rate of 150 lpcd for the City and 100 lpcd for the rest of CMA have been estimated and the estimates are given in the table below:

Table No.7.07 Estimates of water requirements					
I. Chennai City		Year			
		2011	2016	2021	2026
1.	Population in lakhs	49.95	52.39	55.4	58.56
2.	Water requirement in MLD for the resident population				
(a)	@ 150 lpcd	749	786	831	878
(b)	@ 120 lpcd	599	629	665	703
(c)	@ 100 lpcd	500	524	554	586
3.	Water requirement in MLD for the other than residential use such as office, commercial, industrial premises and other places of employment, education, etc.				
(a)	@ 30% of 2(a) above	225	236	249	264
(b)	@ 25% of 2 (b) above	150	157	166	176
(c)	@ 20% of 2(c) above	100	105	111	117
4.	Industrial use				
(a)	@ 10% of the 2(a) above	75	79	83	88
(b)	@ 10% of the 2(b) above	60	63	66	70
(c)	@ 10% of the 2(c) above	50	52	55	59
5.	Total requirement				
	@150 lpcd	1049	1100	1163	1230
	@120 lpcd	809	849	897	949
	@100 lpcd	649	681	720	761
II. Municipalities in CMA					
1.	Population in lakhs	21.75	25.60	30.20	35.69
2.	Water requirement in MLD for the resident population				
(a)	@ 125 lpcd	272	320	378	446
(b)	@ 100 lpcd	218	256	302	357
(c)	@ 75 lpcd	163	192	227	268
3.	Water requirement in MLD for the other than residential use such as office, commercial, industrial premises and other places of employment, education, etc.				
(a)	@ 30% of 2(a) above	82	96	113	134
(b)	@ 25% of 2 (b) above	54	64	76	89
(c)	@ 20% of 2(c) above	33	38	45	54
4.	Industrial use				
(a)	@ 10% of the 2(a) above	27	32	38	45
(b)	@ 10% of the 2(b) above	22	26	30	36
(c)	@ 10% of the 2(c) above	16	19	23	27
5.	Total requirement				
	@125 lpcd	381	448	529	625
	@100 lpcd	294	346	408	482
	@75 lpcd	212	250	294	348

III. Town Panchayats					
1.	Population in lakhs	5.89	7.41	9.45	12.21
2.	Water requirement in MLD for the resident population				
(a)	@ 100 lpcd	59	74	95	122
(b)	@ 80 lpcd	47	59	76	98
(c)	@ 60 lpcd	35	44	57	73
3.	Water requirement in MLD for the other than residential use such as office, commercial, industrial premises and other places of employment, education, etc.				
(a)	@ 30% of 2(a) above	18	22	28	37
(b)	@ 25% of 2 (b) above	12	15	19	24
(c)	@ 20% of 2(c) above	7	9	11	15
4.	Industrial use				
(a)	@ 10% of the 2(a) above	6	7	9	12
(b)	@ 10% of the 2(b) above	5	6	8	10
(c)	@ 10% of the 2(c) above	4	4	6	7
5.	Total requirement				
	@100 lpcd	82	104	132	171
	@80lpcd	64	80	102	132
	@60 lpcd	46	58	74	95
IV. Village Panchayats					
1.	Population in lakhs	10.59	12.96	15.99	19.88
2.	Water requirement in MLD for the resident population				
(a)	@ 80 lpcd	85	104	128	159
(b)	@ 70 lpcd	74	91	112	139
(c)	@ 60 lpcd	64	78	96	119
3.	Water requirement in MLD for the other than residential use such as office, commercial, industrial premises and other places of employment, education, etc.				
(a)	@ 30% of 2(a) above	25	31	38	48
(b)	@ 25% of 2(b) above	19	23	28	35
(c)	@ 20% of 2(c) above	13	16	19	24
4.	Industrial use				
(a)	@ 10% of the 2(a) above	8	10	13	16
(b)	@ 10% of the 2(b) above	7	9	11	14
(c)	@ 10% of the 2(c) above	6	8	10	12
5.	Total requirement				
	@80 lpcd	119	145	179	223
	@70lpcd	100	122	151	188
	@60 lpcd	83	101	125	155

Table No.7.08 Estimate of Water Requirement (CMA)					
Chennai Metropolitan Area		Year			
		2011	2016	2021	2026
1.	Population in lakhs	88	100	112	126
2.	Water Requirement in MLD for the resident population				
a)	Scenario I	1165	1284	1431	1606
b)	Scenario II	938	1035	1154	1296
c)	Scenario III	762	838	933	1046
3.	Water Requirement in MLD for the others from residential use such as office, commercial, industrial premises and other places of employment, education etc.				
	Scenario I	349	385	429	482
	Scenario II	235	259	289	324
	Scenario III	152	168	187	295
4.	Industrial Use				
	Scenario I	116	128	143	161
	Scenario II	94	103	115	130
	Scenario III	76	84	93	105
5.	Total Requirement				
	Scenario I	1631	1797	2003	2248
	Scenario II	1267	1397	1558	1750
	Scenario III	990	1090	1213	1360

7.40 The sources presently available and also to be tapped in immediate future by CMWSSB are given in table No. 7.09.

Table No. 7.09 Assessment Of Abstractable Reliable Quantity of Water from Various Sources

Sl. No.	**Name of Source**	**Safe Yield in MLD**	**Remarks**
1	Poondi - Cholavaram - Red Hills Lake System	227	Based on the assessment during 1997 revision of Master Plan for water supply.
2	Groundwater aquifer from Northern Well Field	68	
3	Other sources like Southern Coastal Aquifer, Rettai Eri, Porur, etc.	5	
4	Receipt of Krishna Water from Telugu Ganga Project (when full agreed quantity of 930 MLD (12 TMC) supplied)	837	10% loss from entry point to Poondi Lake has been considered.
5	Veeranam Lake (CWSAP-I)	180	
6	Desalination Plant	200	a) 100 MLD in 2008 b) 100 MLD in 2009
7	Local sources including Palar River in the CMA area other than City limits.	32	Based on the assessment during 1997 revision of Master Plan for water supply.
8	Abstractable quantity of local groundwater in the city for the use	240	

	of other than drinking and cooking purposes.		
9	Waste water reuse a) Already in use b) Expected in future (SIPCOT use)	 45 120	From 2009
Total		**1954**	

7.41 The source as indicated above will meet the demand up to the year 2011. There is a gap of nearly 300 MLD to meet the CMA demand for the year 2026. This gap will be met by water conservation measures, promoting recycling and by identifying additional sources. The Redhills source has to be protected from possible contamination due to development in the catchment area by not allowing reclassification of land uses or expansion / addition in other zones.

Rain Water Harvesting

7.42 The importance for conservation of water and rainwater harvesting was understood and due consideration and thrust were given from early 90's itself in Chennai. While issuing Planning Permission for construction of major developments such as flats, residential developments, office, shopping and other commercial complexes, the condition to provide rain water-harvesting structures within the premises was imposed and ensured to be provided before issue of Completion Certificates. Provision of rainwater structures in all types of developments, irrespective of size or use was made mandatory by amending DCR and Building Byelaws in the year 2001, not only for the buildings proposed to be constructed but also for all the existing buildings. After implementation of this scheme widely in CMA, a significant increase in the ground water levels and also quality of ground water was noted.

7.43 In 2001, it was also made mandatory that all centrally air-conditioned buildings shall have their own wastewater reclamation plant and shall use reclaimed wastewater for cooling purposes.

Sewerage

7.44 Chennai City Sewerage System was designed in 1910 for an estimated 1961 population of 6.6 lakhs at the rate of 114 lpcd, as a separate system. The system then allowed for admission of storm water from house courtyards and roofs through gullies. The city was divided as north, west and south (independent) drainage areas and sewage from each area was collected by relay pumping and conveyed to the pumping stations at Napier Park, Purasawalkam and Royapuram, and finally discharged into the sea at Kasimedu out-fall. Kodungaiyur sewage farm was developed in 1956 and a portion (27 mld) of the sewage collected at Purasawalkam pumping station was discharged through force mains to Kodungaiyur farm.

7.45 A comprehensive improvements to the city sewerage system was designed in 1958 for an estimated 1976 population of 25.5 lakhs and 1991 population of 27.2 lakhs at a sewage flow rate of 110 lpcd in 1976 and 180 lpcd in 1991; the City was also divided into five zones with proposals for five independent disposal works. It was planned to isolate the system of collection, transmission and disposal of sewage in each zone in order to obviate the difficulties of the relay system.

Coverage

In the present Chennai City Corporation area of 176 sq.km., the sewerage system now covers 99% of the city area. There are 5,15,560 sewer connections as on date to serve the population of Chennai City through a network of 2,663 kms of sewer and 180 sewage pumping stations.

Capacity of the Sewage Treatment Plan

7.46 The sewage generated from Chennai City is treated in 9 Sewage Treatment Plants as detailed below:-

Sl.No.	Location of the Plants	No. of Units	Treatment Capacity in MLD
1.	Kodungaiyur	3 Nos.	270
2.	Koyembedu	2 Nos.	94
3.	Nesapakkam	2 Nos.	63
4.	Perungudi	1 No.	54
5.	Villivakkam	1 No.	5
Total		**9 Nos.**	**486**

7.47 Apart from those zonal systems, other small systems developed by the TNHB in their neighbourhood scheme areas are also functioning and these are connected to the nearest mains of the zone in which they lie.

7.48 As the capacity of sewers was limited, during rainy days they became surcharged due to ingress of storm water. Any surplus of sewage in excess of pumping stations capacity was drained into the nearby natural water courses of the city viz. Cooum river, Adyar river, Buckingham canal and Otteri Nalla.

Chennai City River Conservation Project

7.49 Based on their master plan, CMWSSB had conceived a comprehensive project for sewerage system improvements under the umbrella Chennai City River Conservation Project (CCRCP). In order to cope up with the increased sewage flow resulting from water supply augmentation schemes under implementation, and to prevent the overflow of sewage into the City waterways, the Chennai City River Conservation Project has been taken up at a cost of Rs.720.15 Crores with Government of India grant of Rs.491.52 Crores and the remaining Rs.228.63 Crores borne by CMWSSB. The Project includes investments for providing additional sewage interceptor pipe lines, replacing sewer mains which are worn out, and enlarging the capacities of pumping stations and force mains to cope with the flow in the year 2021. The entire Project contemplated under NRCD share consists of 16 Packages. Packages 1 to 12 consist of interceptor and diversion of works, improvements to pumping station and pumping mains. Package 13 to 16 consists of construction of Sewage Treatment Plants. The works of interceptors, pumping mains, gravity mains, pumping stations and construction of 4 Sewage Treatment Plant units at Perungudi (54 mld capacity), Koyambedu (60 mld capacity), Nesapakkam (40 mld capacity) and Kodungaiyur (110 mld) capacity have been completed.

JBIC Project

7.50 CMWSS Board took up implementation of Chennai Sewage Renovation and Functional Improvement Project under JBIC funds. The following 2 components were taken up for implementation:

(a) **Effluent Conveyance System** - envisages pumping of secondary treated sewage from Koyambedu and conveying it to Kodungaiyur through a 900mm dia DI pipe line to be laid over distance of 16.4 km. to Kodungaiyur where TT/RP Plant was proposed for sewage renovation and supply the permeate to industries at Manali.

(b) **Permeate Conveyance System** - The Permeate from the Sewage Renovation Plant (Tertiary Treatment/Reverse Osmosis) would be conveyed to the 12 industries at Manali for their use for various purposes such as cooling, process, boiler teed and others. Subsequently, due to non-availability of demand for recycled water the project was closed in the year 2003.

Rest of CMA

7.51 For the purpose of planning Sewage System, the Metro Water has divided the CMA into four categories as Chennai City, adjoining urbanised areas classified as Adjacent Urban Areas (AUA), Distant Urbanised Areas (DUA) and Rural Areas. The AUA contiguous to Chennai City covers an area of 165 sq.km. with the 6 Municipalities (viz. (i) Alandur, (ii) Ambattur (iii) Avadi, (iv) Thiruvottiyur and (v) Madhavaram and (vi)

Manali two Municipal Townships (viz. (i) Kathivakkam and (ii) Thiruverkadu) 11 Town Panchayats (viz. (i) Chinnasekkadu, (ii) Maduravoyal, (iii) Nandambakkam, (iv) Pallikkaranai, (v) Perungudi, (vi) Porur, (vii) Puzhuthivakkam, (viii) Valasaravakkam, (ix) Kottivakkam, (x) Nerkundram and (xi) Ramapuram) and the Cantonment area.

7.52 The Distant Urbanised Areas (DUA) comprises 6 Municipalities (viz. (i) Pallavaram, (ii) Tambaram, (iii) Anakaputhur, (iv) Pammal, (v) Thiruverkadu, and (vi) Poonamallee) and five Town Panchayats (viz. (i) Minjur, (ii) Naravarikuppam, (iii) Chitlapakkam, (iv) Kundrathur, and (v) Mangadu).

7.53 Some local bodies like Alandur, Valasaravakkam and Thiruvottiyur have implemented the sewerage system in their municipal areas; others are in the process of providing a full-fledged sewerage system and by engaging consultants they have prepared detailed project reports. While formulating the sewerage proposals for the AUAs/DUAs by the consultants, the approval of the CMWSSB is sought for disposal of the sewage to the nearest STPs of Metro sewerage system since the local bodies could not locate the site for construction of STPs in their area to treat the sewage. Due to various reasons including scarcity of source of water, the STPs of Metro system could not realize the projected quantum of sewage immediately. It was found advantageous to dispose the sewage from the AUAs to the nearest STPs of Metrowater, where further capacity augmentation can be taken up in subsequent years.

7.54 The local bodies like Alandur, and Valasaravakkam have implemented the sewerage system in their areas with the contributions by the consumers and loan from the financial institutions.

7.55 Government in March 2005 has directed that Chennai Metropolitan water Supply and Sewerage Board shall be the Nodal Agency for execution, and maintenance of underground sewerage schemes in AUAs and DUAs in the Chennai Metropolitan Area. The Government have directed as follows:

(a) CMWSSB is required to develop proposals for execution of a comprehensive sewerage system for the entire Metropolitan Area in addition to existing sewerage system to Chennai by CMWSSB.
(b) CMWSSB may to the extent possible use the existing and the planned capacity in its sewerage treatment plants to treat the additional sewage load from AUAs and DUAs; and wherever necessary, additional treatment capacity has to be created.
(c) Funds for the project will be mobilised by the concerned local bodies, which will provide the same to CMWSSB.

(d) CMWSSB will execute the entire scheme including the internal sewering within local bodies and house service connection within these areas. The sewer system will also be maintained by CMWSSB for which the local bodies will be required to pay the fees, which shall be negotiated from local bodies with clear agreement.

(e) An important aspect of the proposed scheme will be the issue of public contribution and tariffs, since it will be difficult to have the different amounts prescribed by AUAs and DUAs with a common sewage management network and it will be desirable to have common tariff structure. The local bodies will collect sewerage charges and taxes internally. CMWSSB should include this aspect in the study proposed to be conducted.

(f) If and when the AUAs and DUAs are brought under the umbrella of a greater Chennai Corporation, the CMWSSB would collect charges directly from the consumers.

(g) Although CMWSSB should aim for the utilisation of existing and enhanced capacity in the 4 STPs, restricting the treatment facility to the existing STPs alone may involve considerable additional expenses on construction and maintenance since some local bodies like Avadi, Poonamallee and Tambaram are located at a distance of more than 15-24 km. Hence, it may be desirable to have one or two additional treatment plants of a similar capacity for these DUAs. This aspect can also be taken up in the preparation of Master Plan.

(h) Keeping in view the high costs, CMWSSB should consider a proper phasing for execution of the scheme.

(i) CMWSSB should quickly organise detailed study of coverage indicated above, principles of revenue generation and management of the facilities proposed.

7.56 The comprehensive sewerage system is in existence in (i) Alandur, (ii) Valasaravakkam, (iii) Thiruvottiyur (50% covered) and (iv) Ambattur (partially covered adjoining City limits). The local bodies have formulated proposals and got detailed project reports prepared by the consultants for provision of sewerage system, are (i) Pallavaram, (ii) Tambaram, (iii) Madhavaram, (iv) Kathivakkam, (v) Porur, (vi) Ullagaram Puzhudhivakkam, (vii) Avadi, (viii) Maduravoyal and (ix) Thiruvotriyur (Rehabilitation and Extension). For Pallavaram the work has been entrusted to CMWSSB and the works are expected to be completed by August 2007.

7.57 The local bodies which are not covered presently with the provision of sewerage system in their area, are (i) Ayanambakkam, (ii) Chinnasekkadu, (iii) Manali, (iv) Nandambakkam, (v) Pallikaranai, (vi) Perungudi, (vii) Kottivakkam, (viii) Nerkundram, (ix) Ramapuram, and (x) St. Thomas Mount Cantonment in the AUA, and (xi) Minjur, (xii) Naravarikuppam, (xiii) Chitlapakkam, (xiv) Anakaputhur, (xv) Pammal, (xvi) Thiruverkadu, (xvii) Kunrathur, (xviii) Poonamallee and (xix) Mangadu in the DUA.

7.58 CMWSSB has taken action to conduct studies with the following main objectives:

(i) to prepare Plans and programmes to link the existing/ongoing sewerage system of the local bodies,

(ii) integrating the system with CMWSSB sewerage system, and

(ii) to provide feasibility studies for providing sewerage system to unsewered local bodies.

7.59 The study will cover provision of sewerage system with treatment and disposal arrangements, including examination of reuse options with reference to the demand for such water, nearness to Sewage Treatment Plants etc. for the Adjacent Urbanised Areas and Distant Urbanised Areas in Chennai Metropolitan Area. Four local bodies are presently served by underground sewerage system and nine local bodies are in the process of providing underground services for which consultants were engaged for preparation of DPRs. The study will cover all these existing and proposed UG system to be integrated. The Phase-I study will cover the evaluation of the DPRs of sewerage system prepared by the consultants for the local bodies like: (i) Pallavaram, (ii) Tambaram, (iii) Madhavaram, (iv) Kathivakkam, (v) Porur, (vi) Ullagaram Phuzhudhivakkam, (vii) Avadi, (viii) Maduravoyal, (ix) Thiruvottiyur. The evaluation will cover the population projection in compliance with the CMDA projections, design criteria, treatment option of sewage, whether in a locally identified site or integrating with the available surplus capacity of the STPs of CMWSSB. The cost of conveyance of sewage to the STP of CMWSSB will be compared with the cost of treatment in a locally identified site. The possibility of reuse of treated wastewater to meet the demand by commercial establishments/factories and feasibility of recovering the costs of such treatment will be examined.

7.60 The Phase-II study will cover preparation of feasibility proposals for providing sewerage system to the remaining AUAs and DUAs viz. 19 local bodies and examining the quantum of sewage generation, updating the population estimates, finalising the design criteria in compliance with the system already provided in the other local bodies, preparation of outline scheme, preliminary cost estimates with alternative proposals, economic feasibility based on the alternate cost proposals for phasing of the investment and financial and administrative arrangement for implementation, operation and maintenance of the system. The study will also address the options on the issue of treating the sewage generation in a locally identified site or to integrate with the STP of CMWSSB and recommend the best available option.

7.61 The observation made in the Tenth Five Year Plan report is extracted below:

"It is infeasible to provide underground sewerage or septic tank latrines in the cities and for all residents. In the first place, highly urbanised, industrialised and densely populated urban centres may be provided with sewerage, with priority being given to installing sewage treatment plants to prevent pollution of water sources. For the majority of the urban centres, low cost sanitation is the appropriate technology.

Low cost sanitation is not a programme solely for the urban poor or slum population. It has to be propagated as the appropriate solution wherever the costly option of underground drainage is not feasible. In this sense, there is a need to offer more options to households that desires sanitation facilities which, while being based on the "twin-pit-pour-flush" model, is in keeping their needs and capacity to invest. Low cost sanitation is best propagated as a part and parcel of the maintenance of environmental health."

"Technologies for treatment of sewage are another area of concern. The conventional technologies require electricity, which has to be paid for. On the other hand, low cost technologies like ponds require large area of land that is not readily available in big towns and cities. There is a need to adopt intermediate technologies, of which there is a choice. Further research and development need to be done in this area."

7.62 As regards rest of CMA, while issuing Planning Permission for constructions of buildings, for sewage disposal from households septic tanks are insisted. In cases of larger developments such as special buildings (flats) and group housing upto 75 dwellings in a site septic tank with upflow filters are insisted with a condition that the upflow water shall be used for gardening purposes within the development site. For very large developments exceeding 75 dwelling units and multi-storeyed buildings, in-situ sewage treatment plants (STP) are required to be provided and the treated water shall be used within the developed site for gardening purposes. This method of sanitation is followed for maintenance of environmental health in the rest of CMA areas where sewerage is not available.

Proposal along I.T. Corridor

7.63 Tamil Nadu Government have declared the area along the Rajiv Gandhi Salai (OMR) as I.T. Corridor. CWSSB has made a study for provision of water supply and sewerage system in the I.T. Corridor and assessed the initial demand for water supply as 15 MLD and the projected future demand as 50 MLD.

7.64 The water supply for the I.T., I.T.E.S. industries and other urban developments in the IT Corridor is proposed to be made in two phases. The First Phase includes tapping 20 MLD of water from Palar river at Vayalur by constructing a collector well at Vayalur and treating the same in a water treatment plant to be constructed nearby with a capacity of 20 MLD. An under-ground tank for storing the treated water with a capacity of 5 million litres, and to pump the water up to Pallipattu with an intermediate pumping at Thiruporur making supplies along the IT Corridor. Water will be supplied to the bulk consumers along the IT corridor and balance water will be conveyed to Pallipattu head works for City water supply. Tentative cost of the project is Rs.46.00 crores.

7.65 In Phase-II, when additional demands come up, it is proposed to construct a desalination plant of 50 MLD capacity at Kelambakkam where 58.75 hectares of land belonging to Salt Corporation has been identified.

Providing Underground Sewerage System

7.66 The sewerage system is proposed to be provided at an estimated cost of Rs.26 crores along IT Corridor. CMWSSB has estimated that sewage generated for the present is 27.12 MLD. The urbanised areas in Okkiam Thoraipakkam, Ekattur, Kazhipathur, Padur, Kelambakkam, Thiruporur, Navalur, Semmenchery, Sholinganallur and Karapakkam are proposed to be provided with a separate collection, conveyance, and treatment and disposal system. From Kottivakkam and Perungudi pumping stations the sewage is to be conveyed to the existing Perungudi STP. It is proposed to provide this sewerage system to the present water supply level and expand the treatment plant capacity in modular form when the supply level increases. The collection and conveyance system will also be expanded as and when supply level increases.

7.67 It is proposed to construct 22 nos. of sewerage pumping stations and 5 nos. of STPs for localised treatment and conveyance of treated effluent. The IT, ITES and educational institutions in the corridor will be encouraged to recycle and reuse waste water generated in their complexes for toilets washing and gardening etc.

Annexure VII A

Water Level & Water Quality in TDS in Chennai City and Chennai Metropolitan Area			
City			
Sl.No.	Name of Village	Water level Range (m)	TDS Range (in ppm)
1	Tondiarpet	4.00-6.00	750-1050
2	Royapuram	4.50-6.50	800-2100
3	Perambur	5.00-12.00	700-1100
4	Villivakkam	4.50-10.50	600-830
5	Anna Nagar	7.00-10.00	500-750
6	T. Nagar	6.60-10.50	800-1200
7	Guindy	4.00-8.50	500-700
8	Velachery	3.00-9.00	900-2500
9	Saidapet	5.00-13.00	600-800
10	Adyar	4.00-9.00	800-1700
11	Kolathur	1.00-7.50	600-1200
12	Kotturpuram	1.50-8.00	1000-2800
13	Kodambakkam	5.00-11.50	500-1700
14	Triplicane	4.00-10.00	800-1600
15	Thiruvanmiyur	4.50-10.00	800-2800
16	Kilpauk	2.50-8.00	600-1200
17	Sembium	4.60-8.50	500-1500
18	Egmore	2.50-6.00	450-600
19	Taramani	1.50-10.00	1200-2800
Ponneri Taluk			
20	Manali	2.00-5.00	400-2000
21	Vichur	3.00-5.50	500-1800
22	Athipattu	3.00-6.50	600-1700
23	Athur	3.50-10.00	500-900
24	Nayar	4.25-9.50	450-1100
25	Seekanjeri	2.50-4.50	300-600
26	Nandhiyambakkam	2.00-4.00	400-1000
27	Minjur	2.50-6.50	700-1500
28	Sholavaram	1.50-7.00	400-1400
29	Ennore	3.00-6.00	500-1700
30	Alamadi	3.50-6.00	400-900

Ambattur Taluk			
31	Pondeswaram	3.00-10.00	400-850
32	Naravarikuppam	2.00-9.00	500-950
33	Athipattu	3.00-8.00	800-1900
34	Padi	2.50-7.50	500-1050
35	Korattur	3.00-9.00	700-1800
36	Mogappair	4.30-8.00	300-695
37	Madavaram	2.00-7.00	200-900
38	Puzhal	1.00-6.50	350-1300
39	Thiruvottiyur	4.25-6.25	600-1100
40	Maduravoyal	4.00-12.00	600-1200
	Mathur	2.00-7.00	200-900
Thiruvallur Taluk			
42	Pakkam	3.50-9.00	400-110
Poonamallee Taluk			
43	Avadi	2.00-11.00	400-800
44	Thirunindravur	1.50-11.50	300-700
45	Thiruverkadu	1.50-10.50	350-800
46	Pattabiram	2.50-9.00	400-1100
47	Ramavaram	1.50-10.00	300-800
48	Chembarambakkam	0.50-10.00	200-500
Sriperumbudur Taluk			
49	Chettipedu	2.00-9.00	300-750
50	Nandambakkam	2.00-10.00	450-1050
51	Meppur	2.50-9.50	400-900
52	Mangadu	3.00-10.00	400-900
53	Kundrathur	2.50-7.50	300-900
Tambaram Taluk			
54	Nanganallur	3.00-8.00	800-1400
55	Thiruneermalai	3.00-8.00	450-1050
56	Pallikaranai	1.50-7.00	700-1600
57	Tambaram	1.50-8.00	300-800
58	Sholinganallur	2.00-7.50	250-650
59	Shithalapakkam	3.00-6.50	400-900
60	Pallavaram	3.50-7.75	450-650
61	Chrompet	3.00-7.00	500-900

62	Meenambakkam	3.00-8.00	800-1400
63	Perungudi	2.50-9.50	1100-2500
64	Nanmangalam	2.00-8.00	300-900
65	Kovilambakkam	1.50-7.00	250-800
66	Chitlapakkam	2.50-8.00	400-900
67	Pammal	2.00-7.50	350-800
68	Nandambakkam	2.50-7.00	300-750
69	Madambakkam	2.00-6.50	250-650
Chengalpattu Taluk			
70	Vandalur	2.50-7.50	400-900
71	Mannivakkam	2.00-8.00	350-800

Chapter-VIII

SOCIAL FACILITIES

Education

Tamil Nadu's human development achievement has been largely a result of its strong educational heritage. Even in the early years, when the State was Madras Presidency, education was actively pursued and promoted. [1]Government enquiry into the state of education in Madras Presidency, initiated by Sir Thomas Monro in 1822, showed that there was one school per thousand population and that the number of boys taught was one fourth of the total school age population. It also showed that the instructions impart in these indigenous institutions was of little practical value and hence a board was appointed to organize a system of public instruction. In 1826, 14 Collectorate and 81 taluk schools with a central school at Madras were opened. In 1836, this scheme was pronounced a failure and the schools were abolished as inefficient. In 1840, a University Board was constituted by Lord Ellenborough's Government to organize to establish a Central School and a few provincial schools. In 1841, the central school was converted into a high school: in 1853, a college department was added to it and later it developed into the presidency college. In 1854, the court of Directors issued its memorable dispatch regarding education. Thereupon the Department of Education, with Directorate of Public Instruction and its inspecting staff was organized; the so-called Madras University was remodeled and designated the Presidency College; Zillah or district schools were opened; and the grant-in-aid scheme was introduced. While in 1853 there were 460 educational institutions with 14,900 pupils, by 1904 this number had risen to 26,771 with 7, 84,000 pupils[1].

8.02 The report of the Elementary Education survey of the Madras Presidency, 1925, points out that there were three agencies managing elementary schools in the province viz. (i) private bodies, (ii) local boards and municipal councils, and (iii) government. Though early initiatives made some headway in education, major breakthrough came with the Madras Elementary Education Act, 1920. Under this Act, local bodies were given the responsibility for elementary education and were also given powers to levy special cess to raise funds for education.

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[1] Source: Madras Gazetteer (M.Francis)

8.03 Tamil Nadu has been a pioneer in the introduction of various schemes to enhance enrolment of children in elementary education. The most important of these schemes is the massive programme viz. Noon Meal Scheme, introduced by the Government in 1982 with the main objective of not only to ensure nutritional support, but also to act as an incentive to achieve universal enrolment and retention in primary schools; about 6.4 million children in the age group of 5 to 14 are covered under this scheme. Provision of free text books and free uniforms by the Government for children in the Government and Government aided schools are aimed at reduction of economic cost of sending a child to school by the parents; these measures have improved attendance and reduced drop out rates.

8.04 In the literacy rate, the Tamil Nadu has attained third position behind Kerala and Maharashtra both in terms of overall and female literacy, as per 2001 Census. Literacy in Tamil Nadu has gone up from 62.7% in 1991 to 73.47% in 2001 (against all India average of 65.38%). Tamil Nadu state Government is committed to the task of providing universal primary (elementary) education for all children upto 14 years.

8.05 Every habitation with a population of 300 and above should have a primary school within a distance of 1 km. is the policy of Tamil Nadu Government and it has been achieved. The levels of basic infrastructure, educational infrastructure and pupil-teacher ratio in primary schools in Tamil Nadu are ranked within first three among the major states in our country. A major legislative effort for universalisation of education has been to introduction of Tamil Nadu Compulsory Education Act, 1994. 85% percent of the habitations in Tamil Nadu have been provided with secondary school facilities within a distance of 5 km and in secondary education also the State is ranked high among the States in our country.

8.06 In the tertiary education during the last decade, Tamil Nadu witnessed a rapid growth in the number of institutions in higher education ranging from industrial training institute (ITI) and polytechnics to arts and science colleges and engineering colleges.

8.07 The Government of Tamil Nadu have ensured 100% schooling access not only at primary level, but also at the middle school level. This has been achieved by opening 1,112 new primary schools during 2001-2004 and by upgrading 2106 primary schools as middle schools during 2001-2005. By these actions of bringing these facilities nearer to the residence of school going children it has helped in bringing down the school trip and ratio in primary schools from 16% in 2001-02 to 8% in 2003-04. In 2005-06, Government have upgraded 90 middle schools into high schools and 60 high schools into higher secondary schools. Government have proposed to improve infrastructure in 549 high schools & higher secondary schools in the State at a cost of Rs.232 crores. A special Literacy programme for women is also implemented to bring down the literacy gap between males & females to achieve a key millennium development goal. The Government are also implementing the technical education quality improvement programme with assistance from the World Bank in 8 engineering colleges and 3 polytechnics with an initial outlay of Rs.63 crores.

I.T. initiative in Education

8.08 Tamil Nadu has been at the foremost in IT and is one of the States to announce a far-reaching, industrial friendly IT Policy. Recognising that the computer education at the school level is essential to enable children coming out of school to be computer literates, and that acquiring basic knowledge in computers would be useful to them either in gaining employment or in pursuing higher studies, it was introduced in school level itself.

8.09 Recognising that the industrial development of the State depends on skilled manpower, the Government set up a vast network of ITIs in the State. There are 53 Government Industrial Training Institutes (ITI), and 590 private ITIs in the State. There are 422 arts and science colleges (67 Government, 161 Govt. aided and 194 Self financing) in the State and the number has almost doubled when comparing the 1991 figure of 224.There are 202 polytechnics and 236 engineering colleges in the State.

8.10 The State spends almost 20% of its revenue expenditure for education. In the Tamil Nadu Human Development Report, it is found stated that though Tamil Nadu's expenditure is not very high, it has managed to sustain its performance due to existing levels of infrastructure as well as strong presence of the private sector, especially in higher education.

Chennai Metropolitan Area

8.11 In Chennai, being the State capital, the educational facilities available are of high and specialized when comparing the rest of the State. CMA comprises of Chennai City (176sq.km.) and parts of urban rural areas in Kancheepuram District (to an extent of 376. sq.km.) and Thiruvallur District (637.sq.km.). Some of the relevant statistics relating to literacy, educational infrastructure are given in the Table No 8.01.

Table No.8.01: Literacy & Educational Infrastructure in the Districts Covered in CMA				
Sl.No.	Description	Chennai City	Kancheepuram District	Thiruvallur District
1	Life expectancy at birth (yrs) (2005)	M-77.14% F-77.56%	N.A	N.A
2	Literacy rate (2001) Male Female	76.81 81.10 72.35	67.84 74.73 60.78	67.73 74.98 60.26
3	Sex ratio (2001)	95.10	96.10	97.10
4	Gross enrolment rate (2005)			
	(a) Primary	93.97	93.88	96.17
	(b) Upper Primary	94.58	97.91	93.81
	Total	93.85	95.29	95.25
5	Gross Dropout rate (2005)			
	(a)Primary	6.75	3.61	7.43
	(b) Upper Primary	6.02	7.04	8.02
6	Pupil-teacher ratio (2005)			
	(a) Primary	47	42	42
	(b) Upper Primary	39	56	55
7	Enrolment of girls in primary schools as % of enrolment of boys (2005)			
	(a) Primary	97.14	96.49	97.00
	(b) Upper Primary	98.60	92.66	94.51

Source : General Education Statistics of Tamil Nadu, Directorate of School Education

8.12 Planning for educational facilities in a Metropolis like Chennai should take into account regional bearings as it cater, not only the requirements within it but also the surrounding districts, and surrounding States in respect of specialized / higher education.

8.13 In CMA, most of the middle schools include primary classes, high schools include middle and primary classes and the higher secondary schools include primary, middle and high school classes. It is provided both by private and public (State and Central Governments, local bodies). A primary school for about 5000 population (370 school going children of that group), a high school for 7000 population (732 school going children of that age group) and a higher secondary school for10,000 population (210 school going children of that age group) is available as per 2001 Census.

8.14 The existing educational institutions, category-wise, in Chennai City and rest of CMA is given in the table No. 8.02:

Table No.8.02 Educational Institutions in CMA								
Sl.No.	Category of institutions	Chennai City			Rest of CMA			
		Govt./ Govt. aided	Private	Total	Govt./ Govt. aided	Private	Total	Grand Total
1	Schools							
	(a) Primary	164	315	479	233	144	377	856
	(b) Middle	125	127	252	65	56	121	373
	(c) High School	55	171	226	37	139	176	402
	(d) Higher Secondary	60	366	426	32	204	236	662
2	Colleges							
	(a) Arts and Science	22	11	33	7	9	16	49
	(b) Training Colleges	5	1	6	-	-	-	-
	(c) Physical Education	1		1	-	-	-	
	(d) Others including Research Institutions	25	13	38	-	-	-	-
3	Technical Education							
	(a) I.T.I's	3	87	90	2	26	28	118
	(b) Polytechnics	11	2	13	2	18	20	33
	(c) Engineering colleges	4	2	6	1	37	38	44
4	Medical Education							
	(a) Medical Education	3	-	3	0	1	1	4
	(b) Dental Colleges	1	0	1	0	6	6	7
	(c) Siddha	1	0	1	0	1	1	2
	(d) Homeo	-	-	-	-	2	2	2
	(e) Unani	1	-	1	-	-	-	1
	(f) Ayurveda	-	1	1	-	1	1	2
	(g) Pharmacy	1	2	3	-	4	4	7
	(h) Nursing	1	-	1	0	5	5	6
5	Veterinary College	1	-	1	-	-	-	1
6	Law College	1	-	1	-	-	-	1

Source: Census of India

8.15 The table No 3.17 [in chapter-III] shows the age group details over the years 1971 to 2001. Because of the family planning and population control measures taken in our country, from 1971 there is large variation in age structure including the school going children age group. It is estimated that in the future years the school going age group would stabilize at 7.5 % for primary school going age group, 5.19% for middle school going age group, 3.71% for high school going age group and 3.96% for the higher secondary going age group. Based on these estimates, the future demand for schools has been worked out and tabulated in table no 8.03:

Table No: 8.03 CMA_ Age Structure for School Going Population-2026						
Age Group	% Assumed @2011	2001	2011	2016	2021	2026
0-5	7.82	549811	692665	778078	874051	981952
6-10	7.50	527593	664675	746636	838732	942273
11-13	5.19	365266	460172	516915	580675	652359
14-15	3.71	260857	328634	369158	414693	465886
16-17	3.96	278203	350488	393706	442269	496866

8.16 For the future population of 2011, 2016, 2021, 2026, the number of primary, high and higher secondary schools to be opened or existing schools to be strengthened to accommodate the growth of school going children is given in the table no 8.04:

Table No: 8.04 Number of Schools Required_2026							
	2001 No. of Schools	Average No of Students_2001	Average Strength assumed	2011	2016	2021	2026
Primary	1427	370	500	1329	1493	1677	1885
Upper Primary	775	471	500	920	1034	1161	1305
High School	998	261	400	822	923	1037	1165
HSC School	662	210	400	438	492	553	621

8.17 Collegiate, technical and other professional higher educational institutions in CMA are concerned, it serves not only the CMA region, but also the state level apart from the national level for certain specialized fields. However periodical reviews of change in demand for this category of educational institutions should be made at least once in 10 years and necessary infrastructures have to be provided. Human resource development for the present and future demands and also Research and Development for economic development depend on investment and improvement on this higher education sector.

Health

8.18 Planning for health becomes an integral part of metropolitan planning and health status of population is an important indicator of human resource development. Investments in health sector have direct relationship with longevity and improvements in physical and mental development of people. Tamil Nadu's health indicators place it near the top among the States of India. Policy of the Government is to provide a healthy and disease free life to the people of Tamil Nadu. Director of Medical & Rural Health Services (DMRH) is in charge of planning and implementation of programmes of Medical Services through a network of 29 district headquarters hospitals, 155 taluk headquarters hospitals, 80 non-taluk hospitals, 12 dispensaries, 11 mobile health care units, 7 women & children hospitals, 2 T.B. hospitals, 2 T.B. clinics and 7 leprosy hospitals. This Directorate provides the health services in the districts except in Chennai City.

8.19 The Dept. of Public Health and Preventive Medicine (DPHPM) is providing primary health care services. Through a network of 1,415 primary health centers (PHC) and 8,682 Health Sub-Centres (HSCs) spread over the State. These PHC's and HSCs provide preventive, promotive, curative and rehabilitative healthcare services and are equipped with basic facilities. Maternal and child healthcare services are most important of the services provided by this Department. School Health Programme provides for comprehensive health care services to the students in Government & Government-aided schools up to higher secondary level; all Thursdays are observed as `school health days' and all Saturdays are observed as `referral days'. Control of communicable diseases, hygiene and health, Malaria control, Fileria control, Japanese & Encephalitis control, National Leprosy Eradication Programme, Dental Health, Integrated Diseases surveillance, Congenital disorders & birth defect registry, Rural Diabetics survey and Mobile Health Service Programme are implemented by this Department. The State has set the norms of 1 PHC for 30,000 population, 1 Community health Centre for 1 lakh population and 1 HSC for 5,000 population.

8.20 Indian Systems of Medicine (Siddha, Ayurveda, Unani, Homeopathy and Yogaand Naturopathy) regained its importance and the Government has attached special importance to the growth and development of Siddha system, which is a part of Tamil culture. Government has proposed to provide one ISM practitioner in each PHC in a phased manner, recognizing that ISM is modern system of medicine can play complementary roles. Directorate of Indian Medicine and Homeopathy deals with teaching as well as providing health care system of Indian Medicine. The National Institute of Siddha established at Tambaram developed at a cost of Rs.47 Crores is a joint venture of GOI and GTN and it has been established and with the objective of imparting post graduate education in Siddha system and to provide medical care through Sidha system of Medicine.

8.21 The Tamil Nadu Health Systems Project (TNHSP), a 5-year project implemented since Jan. 2005, with total outlay of Rs.597 crores. It aims to improve the effectiveness of the health care system, both public and private in the State through increased access to and utilization of health services (particularly by poor and disadvantaged) development of effective interventions to address key health challenges including non-communicable diseases, improved oversight and management of the health care system (both public & private), and increase effectiveness of public sector hospital services.

8.22 In this State there are 11 medical colleges in Government sector, and 2 medical colleges and 9 dental colleges in private sector. There are 42 teaching hospitals in the State under the control of Director of Medical Education.

8.23 Chennai has established itself as the health Capital of the country and is fast becoming the health destination of choice for people all over the world with its excellent facility, competent specialist and good nursing care.

8.24 In Chennai there are 3 major Government Hospitals. The details of no. of bed in the major hospitals are given in Table No.8.05

Table No: 8.05 Number of Beds in Government Hospitals in Chennai

Sl.No.	Name of the Institution	No. of beds
1	Government General Hospital	2029
2	Government Stanley Hospital	1231
3	Kilpauk Medical Care Hospital	512
4	Government Royapettah Hospital	212
5	I.O.C. & H.I. for Women & Children	752
6	I.C.H. & H.I. for Children Hospital	537
7	R.I.O. & Government Opthal hospital	478
8	Institute of Mental health	1800
9	Govt. Kasturiba Gandhi Hospital	695
10	Govt. RSRM Hospital	510
11	G.H.T.M. Tambaram	776
12	Government T.B. Hospital (Otteri)	222
13	Institute of Thoracic Medicine	100
14	Government Peripheral Hospital (AA Nagar)	100
15	Government Peripheral Hospital (KK Nagar)	100
16	Government Peripheral Hospital (Periyar Nagar)	100
17	Government Peripheral Hospital (Tondiarpet)	100
18	Artificial Limb center	60
19	Government Hospital, Saidapet	33
20	Communicable Diseases Hospital	550
21	ESI Hospital (Ayanavarum)	616
22	ESI Hospital (KK Nagar)	400
23	ESI Hospital (Kilpauk)	n.a
24	Government TB Sanatorium	n.a
25	Arignar Anna Government Hospital for Indian Medicine	229
26	Varma Hospital	29
27	Railway hospital	151
28	Port Trust hospital	200
	Total	12522

Source: Corporation of Chennai

8.25 The following table gives the list of major private hospitals with bed strength exceeding 50 numbers in Chennai City.

Table No: 8.06 Number of Beds in Private Hospitals in Chennai City		
S.No	**Name of the Hospital**	**No. of Beds**
1	A.G. Hospital	55
2	Apollo First Med Hospital	80
3	Apollo Hospital	201
4	Apollo Hospital (Greams Road)	600
5	Apollo Hospital (Tondiarpet)	60
6	Apollo Specialty Hospital	200
7	Aysha Hospitals	60
8	Balaji Hospital	75
9	Billroth Hospital	600
10	C.S.I. Rainy Multi Specialty Hospital	250
11	Cancer Institute	156
12	Chennai Kalliappa Hospital	65
13	Childs Trust Hospital	200
14	City Tower Hospital	70
15	Deepam Hospital (P) Ltd.	100
16	Devaki Hospital	100
17	Dr. Agarwal Eye Hospital	112
18	Durgabai Deshmuk Hospital	175
19	Esware Prasad Tottathrya Orthopedic Clinic	111
20	Frontier Lifeline Pvt.Ltd.	120
21	Hande Hospital	50
22	Hindu Mission Hospital	160
23	Kalyani General Hospital	200
24	Kumaran Hospital	100
25	M.V.Diabetics Research Centre	50
26	Madras Medical Mission	218
27	Malar Hospital	160
28	Medical Research Foundation	101
29	MIOT Hospital	296
30	Philips Hospital	75
31	Santhosh Hospital	60
32	Saveetha University	100
33	Sooriya Hospital	110
34	Sri Ramachandra Medical Centre	1650
35	St. Isabel Hospital	250
36	St. Joseph Hospital	50
37	Sundaram Medical Foundation	100
38	Sundaram Medical Foundation	161

39	The Guest Hospital	50
40	Vijaya Hospital	610
41	Vijaya Medical Educational Trust	250
42	Voluntary Health Services	220
	Total	8411

8.26 Government agencies involved in provision of health infrastructure are Directorate of Medical Education, Directorate of Public health and Preventive Medicine, Directorate of Medical and Rural Health Services, Directorate of Family Welfare, Directorate of Drugs Control, Commissionarate of Indian Medicine and Homeopathy, Tamil Nadu State Health Transport Dept.

8.27 A large number of private hospitals deliver health care in CMA, Apollo Hospitals, Sri Ramachandra Medical College Hospital, Malar Hospital, Vijaya Hospital, Devaki hospital, CSI Rainy Hospital, CSI Kalyani Hospital etc. are the major hospitals. According to the approved Government list, there are 130 private hospitals function in the City area itself. Table no 8.07 shows the number of Government & private hospitals zone-wise & bed number wise.

Table No: 8.07 Number of Beds in Hospitals in Chennai City _Corporation Zone Wise

Zone	No. of Govt. Hospitals					No. of approved private hospitals				
	0-50 beds	51 – 100 beds	101-200 beds	201-500 beds	> 500 beds	0-50 beds	51-100 beds	101-200 beds	201-500 beds	> 500 beds
I	5	1	-	-	-	35	-	-	-	-
II	1	-	-	1	-	23	1	1	-	1
III	6	-	-	-	-	26	-	-	-	-
IV	4	1	-	-	2	12	13	-	-	-
V	2	2	-	1	2	41	12	4	-	-
VI	2	1	-	-	-	15	-	-	1	-
VII	7	1	-	-	2	31	-	1	2	-
VIII	2	-	-	-	-	25	1	3	1	1
IX	1	-		-	-	10	-	-	1	-
X	-	-	1	-	-	29	2	-	-	1

Source; Corporation of Chennai

8.28 In Chennai City area, the Corporation of Chennai runs health posts. Zone-wise distribution of health posts is given in the Table No. 8.08:

Table No: 8.08	Number of Hospitals and Dispensaries Run by Chennai's Corporation in Chennai City		
Sl.No.	Corporation zones	Number of Hospitals	Number of Dispensary
1	Zone-I	10	6
2	Zone-II	9	10
3	Zone-III	13	12
4	Zone-IV	11	6
5	Zone-V	9	4
6	Zone-VI	8	13
7	Zone-VII	8	8
8	Zone-VIII	11	6
9	Zone-IX	8	4
10	Zone-X	6	7

Source: Corporation of Chennai

In the rest of CMA, there are 10 primary health centers functioning at Minjur, Naravarikuppam, Avadi, Medavakkam, Porur, Poonamallee, Manali New Town, Madhavaram, Pozhichalur and Pudur.

8.29 The statement recommended in the Urban Development Plans Formulation and Implementation (**UDPFI)** Health care facilities is given in the Table No 8.09:

Table No: 8.09	Urban Development Plans Formulation and Implementation Standards on Health Facilities		
Sl.No.	Name of the Institution	No. of beds	
1	G.H. with 500 beds	500	1 per 2.5 lakh population
2	Intermediate hospitals (category A)	200	1 per 1 lakh population
3	Intermediate hospitals (Category B)	80	1 per 1 lakh population
4	Polyclinics with some observation beds		1 per 1 lakh population
5	Nursing Home, Child Welfare & Maternity Centre	25	1 per 0.45 to 1 lakh population
6	Dispensary		1 per 0.15 lakh population

Source: Urban Development Plan Formulation and Implementation Guidelines

8.30 The no. of beds available in Chennai City and Kancheepuram & Thiruvallur District which pending lie in CMA are given in Table No 8.10:

Table No: 8.10	**Total Number of Beds in Chennai, Kancheepuram and Thiruvallur Districts**			
Sl.No.	District	Population	Total beds	Population per bed available
1	Chennai	4343645	10999	383
2	Kancheepuram	2869920	2735	1049
3	Thiruvallur	2738866	594	4611

Source: Statistical Hand Book of Tamil Nadu - 2003

8.31 From the Census figures, it appears that the total number of beds given relates only to Government hospitals and not private. Specialty and the bed availability in private hospitals may be about 100% more than the Government ones.

8.32 Considering the longevity in life, improved health conditions predicated, it is assumed that the number of beds required in future may be at the rate of one in 500 population. The number of beds required for the projected population, for the year 2006, 2011, 2016, 2021, and 2026 are 15,800, 17,700, 19,900, 22,400 and 25,100 respectively.

8.33 The existing facilities particularly the specialized & higher order ones are concerned, it serves not only the CMA population, but also the rest of Tamil Nadu and the adjoining states population; as regards private sector, it attracts patients from all over India and also some of the foreign countries. It would be difficult to assess the adequacy of these facilities. Because of accessibility of good infrastructure including specialist manpower, technology, private sector investments in health sector high in recent times, the trend is expected to continue. On the government part, with assistance from World Bank, health infrastructure is being improved. A detailed study on the health infrastructure in CMA, delivery to poor, accessibility spatially, future requirements, contribution by private sector, future requirements, modernization requirements in govt. sector etc. have to be made which may be a basis for formulation of Master Plan for Health infrastructure in CMA. The position may be reviewed every 10 years and suitable measures taken on health infrastructure investments.

Recreation

8.34 Recreation becomes an essential part of life in any civilized society. It is an activity people pursue for relaxation, personal enjoyment usually during their leisure time to break from their routine busy work. In an urban environment, such recreation facilities have to provide a variety of opportunities accessible, affordable and attractive to all groups of population.

8.35 Recreation is a broad function being organised and unorganized, indoors and outdoors, daily and intermittent, local and distant. Sometime even sidewalks could be a more important recreational facility than others in a residential area. Television viewing has become a major daily recreational facility within houses apart from music, hobbies & crafts. Indoor recreation activity pursued by people include the ones provided by cinemas, drama halls, music sabhas (halls), clubs, indoor stadium, exhibition and fairs; outdoor recreation facilities includes parks, playgrounds, beaches, zoos etc.

8.36 In order to provide for the preservation and regulation of parks, playfields and open spaces in the State of Tamil Nadu, the 'Tamil Nadu Parks, Playfields and Open Spaces (Preservation and Regulation) Act, 1960' was enacted. Parks, playfields & open spaces are periodically notified under the Act by the local bodies concerned. These spaces shall be maintained for the purposes notified in a clean and proper condition.

8.37 Chennai is endowed with the second longest straight sandy breach in the world, called *Marina*. Elliots Beach, another major beach in Chennai attracts large number of people. Thiruvanmiyur Beach, Kottivakkam Beach, Neelankarai Beach and small beaches at Thiruvottiyur are also being used by people in those areas. These beaches are used by the people throughout the year and the Marina & Elliots Beaches attract thousands of people every day.

8.38 In Chennai City, there are about 195 parks with extent varying from 150 sq.m. to 3.5 hectares and totaling to more than 60 hectares. Playgrounds maintained by the Chennai Municipal Corporation is concerned, it is more than 200 nos., with a total extent exceeding 50 hectares. In the rest of CMA, unlike the City, the parks & playfields are a very few.

8.39 In new layout developments, a proportion of the site (10% of site extent excluding roads in layouts exceeding 3000 m^2) is earmarked and handed over to the local body concerned for maintenance as parks / playgrounds. By this, in the rest of CMA, over these years from 1975, about 300 hectares of parks / playgrounds have been reserved by the developers and taken over by the local bodies. Apart from the layout developments, in cases of residential / commercial developments and multistoreyed developments, where the extent of the site exceeds 3000 sq.m. reservation of parks / playgrounds are made & handed over to local bodies for maintenance. These provisions in D.C.R. generate considerable extent of land required for open spaces and recreational activities for public use.

8.40 CMA also boasts a number of Theme Parks developed commercially in and around CMA, which attracts not only the local population but also tourists.

8.41 Planning for public open spaces has become an important part of metropolitan planning, as a result of the intense demand for outdoor recreation and also the growing need for conservation. Parks are sometimes heavily used, at times breaking down the plant cover, making to disappear its natural character that made them attractive.

8.42 In provision of recreational spaces, one may have to be concerned with the quality of human experience in that space, the release from the intense stimuli of close urban living, the free choice of activity, the chance to become actively engaged to exhibit mastery, opportunity to learn about the non-human world, the ability to meet new people etc. These mostly psychological gains are not achieved by mere conservation of pre-existing state of nature. Ecological balance of the site so that it can renew by itself in spite of the new pressures, which will be put on it.

8.43 Open spaces should be developed not only to provide psychological openness but also for ecological self-renewal. In managing these spaces, carrying capacity to renew itself naturally (for the ground cover to hold, the trees to succeed themselves. or the water to purify itself) should also be taken into account of.

8.44 CMA is dotted with a number of lakes (with minimum water spread in non-monsoon seasons), which may be developed as recreational spaces in a planned way taking into account its environmental aspects also. It will not only help in conserving these water bodies but also preventing encroachments and pollution.

8.45 Sports are an essential requirement for the development of an individual's personality and it teaches one of the values of teamwork and gives the individual the determination to strive, to seek, to find and not to yield. Recognising the importance of sports, the Government of Tamil Nadu have created the Department of Youth Welfare and Sports development in the year 2000. Earlier, Sports was with the Department of School Education and no department was catering to the matters pertaining to the non-student youth community. The main aims of the Department of Youth Welfare & Student Development are

(i) to establish infrastructure facilities in district headquarters, schools and colleges for development of sports & games

(ii) to establish sports-cum-recreation centers in all village panchayat and urban areas to provide facilities for indoor games, small area games and physical exercise to the public

(iii) to identify talented sports persons, to train them and to render suitable financial assistance to enable them to participate successfully in national and international competitions

(iv) to promote among the student as well as non-student youth community a spirit of adventure and love of physical activity

(v) to implement the various schemes of GOI & GoTN aimed at the promotion of sports & youth welfare

(vi) to conduct various sports youth festivals and exhibitions

(vii) A Sports Development Fund was created by GoTN and the fund is used for achieving aims of the department.

8.46 In Chennai City the following facilities are available:

(i) Jawaharlal Nehru Stadium, Periamet, with Football field (grass), Athletic Track (400 m eight lane synthetic track), 3 Volleyball Courts, floodlighting and electronic scoreboards, etc.

(ii) Multipurpose Indoor Stadium, Periamet, with facilities for Volleyball, Basketball, Table Tennis, Badminton, Boxing, Wrestling, Weightlifting, etc.

(iii) Hockey Stadium, Egmore with Astroturf Hockey filed.

(iv) Aquatic Complex with one Racing Pool (50 m x 25 m), one Diving Pool (18 m x 25 m) and one Warm up Pool (20 m x 25 m).

(v) Tennis Stadium at Nungambakkam with 7 courts.

(vi) Nehru Park Sports Complex with Football field, 400 m Athletic Track, 2 Basketball courts, 2 Kabaddi Courts, 2 Tennis Courts, 2 Volleyball Courts and one Squash Court.

(vii) Shenoy Nagar Swimming Pool with 33 m x 21 m swimming pool, 1 Volleyball Court and 1 Shuttle Court.

(viii) Anna Swimming Pool at Marina

8.47 Special coaching camp for sports developments is conducted by other department with GoTN assistance. Sports Development Authority of India has established sports training centers at Jawaharlal Nehru Stadium (for hockey and foot balls for boys) and Nehru Park Sports (for hockey and volley ball for girls). The Department is preparing a perspective plan for development of sports upto the year 2020. A number of national and international sports competition are conducted in Chennai. The Government of Tamil Nadu strive to improve / provide all necessary infrastructure to make it a prefered destination for international sports events.

Chapter IX

SOLID WASTE MANAGEMENT

Solid Waste Management is an obligatory function of municipal corporations, municipalities and other local bodies in India. Due to increase in population, urbanization, change in life style and consumption pattern the problem of solid waste management in urban areas is increasing. Chennai is not an exception to it.

9.02 Chennai Corporation is the responsible agency for solid waste management in the City Corporation area. Chennai Corporation area is divided into 10 zones and each zone is further sub-divided into about 15 Divisions totaling to 155 Divisions. Conservancy responsibility has been delegated to Zonal officials in City Corporation. According to Census 2001, the population of the City was 43.43 lakhs and the average per capita solid waste generated within the City is estimated to be about 585 grams. It has been estimated that 3000 tonnes of solid waste is generated in these 10 zones in the City area daily and in addition Chennai Corporation also handles about 500 tonnes of debris.

9.03 Recognising the importance of solid waste management and its increasing problems in this Metropolis, CMDA had in consultation with the agencies concerned in this matter conducted a study called " Municipal Solid Waste Management Study for Madras Metropolitan Area" with the World Bank assistance during the year 1996 through the consultants M/s Environmental Resources Management, U.K. It was a detailed study covering all aspects of solid waste management (existing situation and also the future requirements). The Study has shown that the solid wastes generated from households, commercial establishments, etc. are at the ratio given in the Table No 9.01.

Table No.9.01 Solid Waste Generated in Chennai City

S.No	Waste generation	% to total
1	Residence	68
2	Commercial	14
3	Restaurants / Hotels/ Kalyanamandapams / Schools and others	11
5	Markets	4
6	Hospitals and Clinics (collected separately)	3
	Total	100

Source: M/S ERM, UK 1996

From the above, it may be seen that the largest generator of solid waste is the household.

9.04. As part of the Study, sample of waste was analysed for composition and characteristics and its results are given in Table No 9.02.

Table No. 9.02: Composition of Waste

Sl. No	Composition	% to total by weight
1	Paper	8.38%
2	Rags	3.11%
3	Organic Matter	51.34%
4	Plastics	7.48%
5	Metals	0.19%
6	Rubber & leather	0.19%
7	Inert	26.01%
8	Glass	0.29%
9	Coconut	2.48%
10	Wood	0.50%
11	Bones	0.01%

Source: NEERI, 2006

9.05 Chemical analysis of Chennai City solid waste samples have shown that it contain as given in Table No. 9.03.

Table No.9.03: Chemical Analysis of Solid Waste

Sl. No	Items	% Value
1	Moisture content	47.00%
2	Ph Value	6.20% to 8.10%
3	Volatile matter at 550^{0}C	42.62%
4	Carbon	24.72%
5	Nitrogen content	0.88%
6	Phosphorous as P_2O_3	0.44%
7	Potassium as K_2O	0.89%
8	C / N Ratio	29.25
9	Calorific Value in Kj/kg	2594

Source: NEERI, 2006

9.06 From the above it is seen that Chennai municipal solid waste contains higher moisture content, small percentage of recyclable materials and more of compostable (organic matter) and inert materials. These characteristics show the low potential for applying refuse derived fuel and waste to energy (i.e. incineration) processing option due to the low combustibles, high moisture and high inert contents of the wastes. However, they indicate high potential for composting of solid wastes.

9.07 NGO's in cooperation with Municipal Corporation are assisting communities to collect solid waste through community based arrangement in some areas of the City. Collection by NGO's from individual houses / establishments using tricycles are deposited in dustbins which are cleared by Chennai Corporation. Municipal Corporation provides street sweepings and scientific collections throughout the City. Municipal Corporation has handed over the solid waste collection and transfer to disposal sites in respect of the zones VI, VIII & X to a private organisation and it handles about 1000 tonnes per day.

9.08 Solid waste from the Chennai Corporation area is taken to the transfer stations and from there it is finally disposed off at two designated disposal sites viz. Kodungaiyur located at northern part of City and Perungudi an adjoining village in the south. Both the sites are located in low lying areas and are adjacent to the Metro Water Sewage Treatment Works. The extent of the Kodungaiyur landfill site is 182 hectares and the Perungudi land site is 142 hectares. About 45% of the total solid waste generated is disposed at Kodungaiyur site and the remaining at Perungudi site. Within the Chennai Corporation there is a well established repair and maintenance system for solid waste management for mechanical vehicles.

9.09 Central Pollution Control Board has estimated that the per capita solid waste generated in small, medium and large cities, towns in India to be 0.1 Kg to 0.2 to 0.4 Kg and 0.5 Kg per capita per day respectively. In the ERM Study conducted in 1996 it was estimated that the per capita waste requiring disposal in respect of Chennai City was 0.585 Kg per capita per day. It has also arrived at the figures of waste generation rate in respect of municipalities as 0.585 Kg, town pancahayts as 0.439 Kg and panchayat Unions as 0.293 Kg per capita per day within the CMA. Applying these arrived norms, the estimation of solid waste (excluding debris) generated in Chennai City, Municipality and other local bodies within CMA in 2026 would be as follows:

Chennai City	3400 Tonnes
Municipalities	2050 Tonnes
Town Panchayats	550 Tonnes
Panchayat Unions	540 Tonnes
Total for CMA	6590 Tonnes

9.10 In Chennai Metropolitan area small, medium and large enterprises (Secondary raw material) are involved in the recycling industry. They get the supplies from dealers who specialize in waste papers, glass, plastic, metals and other reusable material and are in turn supported by vast network of dealers and small traders. Rag pickers, including those at the landfill site, transfer stations and street level, together with itinerant buyers who collect, separate materials from door to door, provide dealers with regular supply of waste. It has been estimated that they recover about 400 tonnes per day of these material.

Rest of Chennai Metropolitan Area

9.11 All solid waste management functions are the responsibility of the executive authorities of the local bodies namely municipalities, town panchayats, and village panchayats.

9.12 In respect of municipalities, most of them do not have any transfer stations and they directly dispose off the waste collected in the land filled sites available within the local bodies. In most of the village panchayat areas the system of solid waste collection and disposal is very limited.

9.13The consultants after conducting a detailed study in 1996 had given the recommendations after considering what sort of waste management system would be appropriate in CMA in another 15 to 20 years, and the various steps that need to be taken to get out from the existing situation to the more ideal situation proposed, based on experience elsewhere and modifying the same as necessary taking into consideration the climatic, locational, cultural differences and situation appropriate to Chennai. Action programme for implementation given by the consultants is given in the Annexure VIII A.

9.14 The consultants observed that the greatest and the most cost-effective improvements in providing a waste management service to the people are likely to come from improvements in the current framework for providing these services rather than from the introduction of new or different technologies. Improvements to existing arrangements, more clearly defined responsibilities, better management and improved training are therefore seen as paramount to the success of the plan recommended by them.

9.15 The study had recommended a new CMA-wide disposal organization should be set up to develop and manage sanitary land fill and transfer/haulage facilities throughout the area and that organization referred to under the working term “Metro Waste” which need to be developed from the existing solid waste management expertise available with Chennai Municipal Corporation whose capabilities in waste disposal should be strengthened. ‘Metro Waste’ should be structured in a manner, which distinguishes the 'client' function (raising finances and controlling operational performance) from the 'contractor' function (carrying out the services on a day-to-day basis). The benefit of structuring Metro Waste in such a manner will allow greater control over the environmental, operational and financial performance of services. It also recommended the phased introduction of private sector in provision of solid waste management services. Whilst public sector must always remain 'responsible' to the public for the performance of solid waste management services, the private sector can be used to efficiently carry out a large proportion of the day-to-day duties. The involvement of community organisations (CO's) in primary collection services is considered to be essential to the sustained success of the strategy.

9.16 Chennai Corporation has taken action to modernize 7 transfer stations and machinery and also to improve basic infrastructure facilities at the landfill sites. It has also taken action to execute the project of making manure from solid waste on Design-Build-Operate- and-Transfer basis. It has employed the consultants M/s National Productivity for preparation of detailed report on modernisation of the Perungudi and Kodungaiyur Solid Waste Disposal sites and they have submitted their report, which is under consideration of the Corporation of Chennai. As a small scale measure at ward level in 115 places, facilities have been created for making manure from community wastes and the Corporation uses the manure for the parks.

9.17 In respect of the municipalities within CMA availability of land for management presently and requirement of land as per the estimates of the local bodies by 2005 are given in the Table No. 9.04.

Table No.9.04: Land Requirement For Land Fill / Compost Sites

Sl. No.	Name of the Municipality	Requirement of land (in acres)	Existing land (in acres)	Expenditure (Rs. in crores)
1	Alandur	20.46	15.00	37.47
2	Pallavapuram	20.16	-	37.47
3	Tambaram	19.27	4.25	117.46
4	Pammal	6.86	1.00	15.00
5	Anakaputhur	4.44	-	10.00
6	Ullagaram Puzhithivakkam	4.26	-	8.00
7	Ambattur	42.35	7.60	20.00

8	Avadi	22.00	7.20	12.50
9	Kathivakkam	4.56	-	-
10	Madhavaram	15.00	3.66	-
11	Thiruvottiyur	29.65	12.00	-
12	Thiruverkadu	4.30	-	-
13	Poonamalle	5.95	-	-
14	Maduravoil	6.18	-	-
15	Valasaravakkam	5.77	-	-
16	Manali	5.30	5.00	-

9.18 All the municipal areas have identified disposal sites for scientific disposal of solid waste. A common land of extent 50 Acres has been purchased for Alandur, Pallavaram and Tambaram Municipalities at a cost of Rs.113.28 crore at Venkatamangalam village for developing the same as a modernized compost yard bringing the segregated wastes for the purpose. In addition, Tambaram Municipality has identified an extent of about 55 Acres at Nallur village in Sriperumbudur Taluk and 25 Acres in Punchai-Pothivakkam village, Chengalpattu Taluk. In respect of Ambattur Municipality 30 acres of land at Vengal village has been identified. For Kathivakkam Municipality, a site of an extent 5.5 Acre has been identified at Manali Village. Thiruvottiyur and Kathivakkam Municipalities are presently using the common disposal site of an extent 12 Acres at Sathangadu and Thiruvottiyur Municipality has taken action to alienate about 10 Acres from the sewage treatment plant site. Madhavaram Municipality has identified a site of an extent 4.70 Acres at Vadaperumbakkam and 4.93 Acres at Manali and taken action to acquire the same. Pammal Municipality has obtained 2.00 Acres of land for this purpose. Thiruverkadu Municipality has taken action to get 10.20 Acres of poromboke land at Koladi Village for this purpose. Valasarawakkam Municipality has also taken action to acquire lands.

9.19 In respect of the panchayat areas are concerned, only in thickly built-up areas, collection is carried out and disposal made in compost yards that are located in close proximity. In these less dense areas the solid waste collection and disposal is very limited, meeting its present requirements.

Hazardous Waste:

9.20 Hazardous waste residue of production process may cause significant damage to environment and human health and / or environment unless handled, stored, transported, treated and disposed of scientifically using environmentally sound technologies. The improper and careless handling of hazardous waste has all too often created problems for human health and environment. Effective management and handling of hazardous waste is of paramount importance for protection of human health and environment.

9.21 Prior to notification of the rules in 1989, most of the industries had stored the wastes in their premises or had disposed the same on land within the premises. These wastes were disposed especially by small-scale units in low-lying areas and on roadsides along with the municipal solid wastes as no infrastructure were available. However, after notification of the Hazardous Waste (Management & Handling) Rules 1989, the TNPCB took efforts to identify the generators and made them store wastes within their own premises in designated areas. Efforts were taken to provide individual secured landfill facilities within the industrial premises or the Chromium bearing sludge and Arsenic bearing sludge.

9.22 Sites for establishing the common treatment storage and disposal facility (TSDF) for the hazardous waste have been identified, detailed EIA have been carried out and the Government has notified the sites. However, there is public opposition for establishment of common hazardous waste TSDF facility due to the apprehension that the nearby land and groundwater will be polluted. Therefore, the Board could not establish TSDFs during the X plan period for scientifically disposing the hazardous wastes despite the best efforts. A TSDF has been notified in the Gummidipoondi Industrial Estate construction of which is in progress.

Bio-Medical Waste:

9.23 Bio-medical wastes are hazardous because of its potential for infection and also for its ingredients including antibiotics, cytotoxic drugs, corrosive chemicals and radioactive substances. The TNPCB has estimated that in-patient hospital services in Chennai generate about 1 to 2 kg of solid wastes per person per day, and Chennai city has 528 hospitals with a bed strength of about 22,180 and the bio-medical waste generate per day is about 12,000 k.g. TNPCB has authorized two sites for location of common treatment and disposal of bio-medical wastes, which are functioning at Thenmelpakkam village, Chengalpattu Taluk, Kancheepuram District and Chennakuppam, Sriperumbudur Taluk, Kancheepuram District to serve Chennai, Thiruvallur and Kancheepuram Districts and adjoining Villupuram and Cuddalore Districts.

E-Waste:

9.24 The rapid proliferation of computers and consumer electronics such as mobile phones, refrigerators, washing machines, radios, tape recorders, microwave oven, calculators, display devices, telecommunication devices and toys have resulted in a global mountain of hi-tech trash (E-Waste) due to its obsolescence in nature. E-waste consists of lead, mercury, arsenic, cadmium, PVC, Brominated Flame Retardants (NFRS) and dozens of other toxic and potentially hazardous compounds. The fruits of our hi-tech revolution are pure poison if not properly managed at the end of their productive life. A personal computer system weighing about 25 kg is estimated to contain about 1.5 kg of lead. The primary source of computer waste in Tamilnadu is from software companies, Government and public companies, PC retailers and manufacturers, secondary market of old PCs dumped from developed countries and domestic uses. There are about 89 large-scale software companies located in Chennai, Thiruvallur and Kancheepuram Districts. Approximately 38,000 computers and laptops are generated as E-waste every year. Other computer peripherals and printers, fax machines, air conditioner, UPS, network accessories to the quantum of 3000 are also generated every year.

9.25 Seven E-waste recycling industries have been authorized by TNPCB within the CMA. They conduct restricted operations of dismantling of computer hardware, manual segregation of scrap after breaking the scrap by using mechanical equipments like jaw crushers and cutters. The scraps are segregated into plastic components, glass, ferrous material and non-ferrous material. The printed circuit boards available in computer are segregated and exported to reprocessing facilities at Belgium, Singapore, Hong Kong, China & Taiwan for metal recovery. Metals recovered are usually copper and gold. Informal recyclers are located at Labbai Street, Periamet at Chennai. Scrap workers are working in the residential buildings. They are segregating the E-waste manually in a crude method by using small tools or with blower arrangement. IC wastes, Printed Wiring Board (PWB) are burnt in outskirts of Chennai to get Aluminium from the burn-out.

Construction Debris :

9.26 In Chennai City about 500 tonnes of construction debris are generated per day. Chennai Corporation has identified few sites within the City wherein the private developers dump this debris. The developers, who require these debris for landfill are collecting them and utilizing the same. Presently, this kind of solid waste does not pose much problem except unauthorized dumping along certain roads. A system of collection and recycling/ disposal of this construction debris should be worked out by the local bodies concerned and implemented.

Bio- Methanation Plant :

9.27 In Koyembedu Wholesale Market Complex (KWMC) perishable wastes of about 100 tonnes per day are generated. With the assistance of Government of India, Ministry of Non Conventional Energy Sources (67%) and from CMDA's share (33%) the project to generate electricity from vegetable waste (30 tonne] has been planned and implemented at KWMC at a total cost of Rs.5.07 Crores. Electricity generated in this project will be about 230 kW per day. The compost generated will be 10 tonnes per day.

Annexure IX A

Municipal Solid Waste Management Master Plan for Chennai - Technical Action Programme				
Description	**Local Body**	**1996-2000**	**2001-2005**	**2005-20011**
Collection				
Coverage	MC Area	90%	100%	100%
	Municipalities	50%	70%	100%
	Town Panchayats	10%	30%	70%
Waste Storage & Primary collection	Municipal Corporation Area	Public scheme for owner container-based collection system operated by MC & NGO's in Zone VI	Complete introduction of new collection system	
		Implement preferred collection system in all zones at the rate of 2 zones per year	Maintain communal containers in some areas as appropriate	
		Communal container for low income, apartment blocks, commercial and institutional premises.	Maintain public awareness campaign as new collection system is introduced	
		Implement bulk collection points where appropriate	Evaluate the potential for using "wheelie bins" in appropriate areas	
		Develop and implement an appropriate local public awareness campaign for the pilot scheme proposed for Zone VI		

		Expand the public awareness campaign to cover each zone where the proposed new collection system is being implemented		
	Municipalities	Pilot scheme for new collection system following and on similar basis to that in MC	Extend coverage of new collection system	
	Town Panchayats		Initiate new collection system	Extend coverage of collection system
Street Sweeping	MC Area	Carry out pilot project trials of new street sweeping system	Complete introduction of trolley system into all Zones	Evaluate potential for street sweeping using mechanical equipment
		Introduce trolley system with 60t containers (x2) into 2 Zones per year	Continue re-evaluation of street sweeping tasks and performance norms	
		Introduce simple task force in each Zone to clear debris and accumulations		
		Introduce central heavy duty task force under direction of Commissioner to work on programmed clearance and emergency response		
		Re-evaluate street sweeping tasks and performance norms waste collection scheme is introduced into each Zone		
	Municipalities		Introduce trolley system for manual street cleaning into 2 municipalities per year	

	Town Panchayats			Evaluate and if appropriate introduce trolley system into Town Panchayats
Secondary Collection	MC Area	Phase out bullock carts	Complete phasing in of standardised collection vehicles	Evaluation and pilot scheme to test the potential of "wheelie bin" type storage and enclosed compactor vehicles
		Phase out 9.5 tonne tippers for collection, use for transporting waste from Transfer Stations	Complete phasing out of non-standard vehicles	
		Phase in standardised collection vehicles: *3 tonne tippers (general) *1.5 tonne tippers (narrow streets) *0.75 tonne 3 wheeler (narrow streets)		
	Municipalities	Phase out bullock carts by year 5	Complete phasing in of standard collection vehicles	
		Phase in standardised collection vehicles as per MC	Evaluate the potential for contracting out the collection and transport of wastes from collection points to the treatment or disposal points	
		Phase out 9.5 tonne tippers for collection, use for transporting waste from Transfer Stations		

	Town Panchayats		Phase out bullock carts by year 10	Complete phasing in of standard collection vehicles
			Phase in standardised range of collection vehicles	Evaluate the potential for contracting out the collection and transport of wastes from collection points to the treatment or disposal points
Transfer System				
Transfer stations	MC Area	Rehabilitate / develop transfer stations one per Zone apart from Zone I, at a rate of 2 per year, all to be completed by year 5		Evaluate the potential for rationalising the number of transfer stations and developing some of them to enable waste to be containerised and conveyed by higher payload vehicle
				If specialised enclosed compactor collection vehicles are considered for introduction, some transfer stations should be closed and collection vehicles diverted either to other transfer stations or directly to disposal

	Municipalities	Develop Transfer Stations in the Alandur, Thiruvottiyur, Ambattur, Avadi, Pallavaram and Tambaram Municipalities		
		Develop transfer points for the transport of waste for disposal in Madhavaram and Kathivakkam Municipality		
	Town Panchayat		Develop transfer points in Town Panchayats to be completed by year 10	
Bulk Transport System	MC Area	Continue to use existing 9.5 tonne tippers for transport from transfer stations to treatment or landfill, replace as necessary		Evaluate potential for container type vehicles in parallel with evaluation of enclosed compaction vehicles and rationalising transfer stations
	Municipalities	9.5 tonne tippers for haulage of waste from transfer stations purchased by new waste body		
		Purchase Roll-on-off vehicles for the transport of containers from the new transfer points		
	Town Panchayats		Purchase Roll-on-off vehicles in line with development of transfer points	

Treatment and Disposal				
Existing landfills		Upgrade existing land fills at Perungudi and Kodungaiyur to controlled landfill status and to accept the co-disposal of Municipal Solid Waste with appropriate industrial and hospital wastes, to include, improved operational practices, improved site drainage and leachate control, improved access roads, relocation of weigh bridge at Kodungaiyur, development of screening bunds around sites, restoration and landscaping of completed areas	Continue with phased development and restoration of existing sites	Continue with phased development and restoration of existing sites, upgrading of Engineered Landfill status
		Purchase new landfill dozer for each site	Develop site extensions on basis of engineered landfill designs	
		Set up small site laboratories for testing and monitoring		
		Carry out detailed environmental assessment of site extension project		
New landfill		Purchase land necessary for access to the proposed landfill	Carry out competitive tendering exercise for selection of Design, Build and Operate Contractor	Continue with phased development and restoration of new site
		Carry out detailed investigation and environment assessment	Develop new landfill site at…… to start operating in year 7	Monitor BOO contractor performance

		Develop Design, Build and Operate contract specifications	Monitor BOO operator performance	
Composting		Evaluate the potential of composting selected high organics waste, including involvement of the private sector	Based on the results of the pilot composting scheme, develop full scale composting facility/ies	
		Develop pilot scheme for composting		
Vehicle Maintenance				
Front line maintenance	MC Area	Retained by Zonal operational depots under direction of new zonal Assistant Engineers (Mech)		Evaluate potential for contracting out front line maintenance to the private sector
	Municipalities	Retain private sector contracting out	Contracted out	Contracted out
	Town Panchayats	Retain limited private sector contracting out	Contracted out	Contracted out
Workshops	MC Area (B&D)	Extend workshops into disused cattle shed areas	Sustain maintenance equipment levels to meet requirements of the vehicle fleet	Evaluate equipment requirements for new collection vehicle types if these are adopted
		Rehabilitate and re-equip workshops to undertake major repairs		
	Municipalities/Town Panchayats	Contracted out	Contracted out	Contracted out

Chapter X

MACRO DRAINAGE SYSTEM IN CMA

Introduction:

CMA lies along the east coast of Southern India and is traversed by three major rivers namely Kosasthalaiyar River, Cooum River and Adyar River. The climate of the region is dominated by the monsoons, which are caused by thermal contrast between land and sea. Monsoon climates are characterised by clearly marked seasons with specific types of wind and weather. The South West monsoon dominates weather patterns in Tamilnadu from July –September and is characterised by periods of sultry wet weather. Rain shadow effects limit rainfall in the east coast in Tamilnadu and it is light or intermittent during this season. This period is followed by North-East Monsoon, which brings cool cloudy weather, relatively free of rain over most of the monsoon-dominated land (India). The exception is South-East-India including Tamilnadu where about 78% of the annual rainfall occurs at this time. The start of the heavy rains usually falls in October lasting up to December. Most of the rainfall is associated with clear synoptic systems of depressions and cyclones with night time rainfall most common. In CMA between October and December most of the rainfall occurs and it is rare between January and April.

10.02 River Nagari which has a large catchment area in the Chittoor District (Andhra Pradesh) region and the Nandi River, which has catchment area in the Vellore District, join near Kanakamma Chattiram and enter Poondi Reservoir. Kosasthalaiyar River, which has its origin near Kaveripakkam and has catchment area in North Arcot District, has a branch near Kesavaram Anicut and flows to the city as Cooum River and the main Kosasthalaiyar river flows to Poondi reservoir. Poondi Regulator was constructed in 1943. From Poondi reservoir, Kosasthalaiyar River flows through the Thiruvallur District, enters CMA, and joins the Sea at Ennore.

10.03 Cooum River from the Kesavaram Anicut flows through the Kancheepuram District enters CMA and finally reaches Sea near Fort St.George.

10.04 Adyar River having its catchment area in the Kancheepuram District and originating from the Pillapakkam Tank Group and Kavanur Tank Group flows through the CMA enters the city and reaches Sea near Adyar area of the city.

10.05 Sholavaram Tank, Red Hills Tank and Chembarambakkam Tank are the major tanks in the CMA. Sholavaram Tank is the secondary storage tank receiving water from the Poondi Reservoir via Poondi Feeder Canal to supply Red Hills Tank. Red Hills Tank is the main source of water supply to the Chennai City and during storm events water is released to Red Hills Surplus Channel, which enters the Kosasthalaiyar River and discharges into the sea. Its maximum storage capacity is 3285 Mft^3 (93 Mm^3).

10.06 Chembarambakkam Tank has recently been developed as one of the sources for water supply to Chennai City and has maximum storage capacity of 103Mft^3.

10.07 Kesavaram Anicut and regulator which is located in the uppermost catchment of the Poondi reservoir controls the discharge from upper catchment entering Poondi reservoir and during storm events the regulator gates can be opened to discharge into Cooum when Poondi reservoir is full.

10.08 Korattur Anicut and regulator control the discharge in the Cooum and direct the flow to the Chembarambakkam reservoir. When there is flow in the Cooum river and the Chembarambakkam Lake is not at full capacity, then the regulator gates are opened to supply water to the reservoir through the link channel.

10.09 Tamarapakkam Anicut located across Kosasthalaiyar River in the downstream of Poondi reservoir controls excess discharge in the Kosasthalaiyar. If Sholavaram is not at its full capacity, then the gates are opened to divert the excess water along the supply channel to Sholavaram. Vallur Anicut is a small check dam constructed near Minjur across the Kosasthalaiyar River to control water levels and feed irrigation channels in the area.

10.10 Like any region in southern India with agricultural lands, CMA also has a network of lakes, canals and channels within its boundary. There are about 320 tanks /lakes that are earlier used as water source for irrigation and now serve as flood accommodators. Apart from these lakes there are a large number of ponds in CMA.

10.11 Buckingham Canal is a man-made canal, which was constructed during the year 1806. It originates at the place called Bedhakanjam in Andhra Pradesh and runs along the area very close to the east coast, enters CMA at Athipattu village, passes through the Chennai City and leaves CMA at Semmencheri village, and it finally connects to Ongur River at Yedayanthittu Kaliveli near Cheyyar. Its total length is 418 km and in CMA its length is 40km. It runs in the north south direction and connects all the major three rivers in CMA. It was dug for the purpose of navigation and transport of goods and also to accommodate flood. But within CMA for various reasons it now serves as flood accommodator only.

10.12 Otteri Nullah is a channel to accommodate flood, which originates from a place called Otteri near Padi, flows through the city at Anna Nagar, Kilpauk, Purasawalkam, and Perambur and joins Buckingham Canal near Basin Bridge.

10.13 Virugambakkam-Arumbakkam Drain originates near Oragadam passes through Virugambakkam-Arumbakkam area of the city and joins into Cooum River.

10.14 Mambalam Drain is also a flood accommodator, which originates from Mambalam area passes through T.Nagar, Nandanam and joins Adyar River.

10.15 Captain Cotton Canal originates from the Vyasarpadi area of the city and joins Buckingham Canal near Tondiarpet.

10.16 Velachery Drain is a flood accommodator originates from Velachery tank and joins Pallikkaranai Swamp.

10.17 The agencies responsible for management of storm water drainage in CMA are presented in Table No.10.01.

Table No. 10.01: Agencies responsible for Management of Storm Water

Agency	Responsibility
Chennai Metropolitan Development Authority (CMDA)	Project packaging and management, monitoring and co-ordination
Public Works Department (PWD)	Plan, design and implementation of macro drainage works
Chennai Municipal Corporation (CMC)	Plan, design and implementation of micro drainage works
Tamil Nadu Slum Clearance Board TNSCB	Formulation and implementation of rehabilitation and resettlement package

Source: Government Records

Waterways in CMA

10.18 The length of the major waterways passing through the City and their total length in CMA is given in Table No. 10.02 and are presented in sheet No.10.05.

Table No.10.02: Length of Major Waterways in CMA *(Length in Km)*

Waterway	City	CMA
River Cooum	18.0	40.0
River Adyar	15.0	24.0
North Buckingham Canal	7.1	17.1
Central Buckingham Canal	7.2	7.2
South Buckingham Canal	4.2	16.1
Otteri Nullah	10.2	10.2
Captain Cotton Canal	2.9	4.0
Kosasthalaiyar	-	16.0
Mambalam Drain	9.4	9.4
Kodungaiyur Drain	6.9	6.9
Virugambakkam-Arumbakkam Drain	6.9	6.9
Total Length	**23.2**	**157.8**

Source: CMDA

In this background this chapter furnishes a gist of various studies made and completed and on going programs to alleviate flooding in the CMA.

Floodings in the past:

10.19 Chennai City and its environs are very flat with contours ranging from 2m to 10m above MSL, with very few isolated hillocks in the south west near St.Thomas Mount, Pallavaram and Tambaram. Cooum and Adyar Rivers play a major role during floods. River Cooum collects surplus from about 75 tanks in its catchment within CMA and Adyar River collects surplus from about 450 tanks in its catchment, apart from overflows from the large Chembarambakkam Tank. The flood discharge of Adyar River is almost 3 times that of the Cooum River.

Mean Annual Rainfall in CMA is about 120 cm. The mean rainy days are about 52 days.

10.20 The siting of the Madras in the 17th Century near the confluence of the river Cooum with the sea had influenced its expansion in the bed and the ayacut lands of tanks in its upstream.

10.21 Mylapore tank bed and nearby tanks viz. Nungambakkam, Kodambakkam, Taramani, Vyasarpadi, etc were converted into settlement areas to meet the housing and other requirements of the expanding city. To make the problem more complicated Govt. in order to provide work to people during the famine in 19th century had constructed the 'B' Canal along the coast. This was mainly to provide waterway for navigation and it had brought about major interference with the drainage courses of the City as all the drainage courses which were otherwise falling directly into the sea, got intercepted by this canal and rendered segments of the drainage courses beyond the Buckingham canal obsolete and obliterated by the urban development. The impact of the problem was not felt very much earlier, as the habitations were thin and the development pace was slow.

10.22 The last century records have shown that there were several catastrophic flooding in Chennai in 1943, 1976, 1985, 2002 and 2005 caused by heavy rain associated with cyclonic activity. These events of catastrophic flooding were found to be attributable to failure of the major rivers and other drainage systems. Flooding of less catastrophic nature occurs regularly in low-lying areas of the city and its suburbs because of inadequacy or inoperativeness of the local drainage infrastructure.

10.23 The floods in 1943 were historic and damaged Cooum river very badly. Based on the Er.A.R.Venkatachary's report the Govt. had improved the Cooum river and provided a sand pump at the river mouth for removal of sand bar.

Er. Sivalingam Committee

10.24 In 1976, there was catastrophic flooding in Chennai and this time it was the turn of the Adyar river. Er.P.Sivalingam Committee had given it's recommendations for prevention of further damages from floods and recommended schemes worth about Rs.12 Crores to be implemented then under priority and schemes worth Rs.10 Crores in the long run (at the 1970's rates).

10.25 CMDA invited Mr.J.H.Kop, a drainage expert from World Bank who had studied the flood problems and given recommendations in his report. He was endorsing the remedies suggested in the above said previous reports and recommended to CMDA to establish a Task Force (Nucleus Cell) to study the schemes of the various departments and to draw up a programme giving the value of the component works and giving priorities between them so that benefits could be realised within a short time.

Nucleus Cell in the CMDA :

10.26 The Govt. of Tamilnadu had ordered for constitution of Nucleus Cell in CMDA in 1979. The Nucleus Cell had submitted its report in 1980. Following are the main recommendations of the cell:

(i) Considering the urban development already occurred, the surplus course of the Vyasarpadi area known as Captain Cotton Canal could be improved to carry only 2,660 cusecs with difficulty due to lack of space and drain into North Buckingham Canal.

(ii) Some of the surplus from the higher tanks in the area should be diverted into Cooum (i.e. Ambattur tank surplus into Cooum River via Mogappair tank; Korattur tank diverted into Madhavaram tank; Madhavaram tank diverted northwards into Redhills surplus course).

(iii) After the above diversions, Captain Cotton Canal designed for 2,660 cusecs would be sufficient to take care of the flood quantity of 2,360 cusecs in all seasons and proved to be permanent advantage for the Vyasarpadi area.

(iv) To remove the difficulty of discharge of floods a second arm across the Manali-Tondiarpet Road through a causeway has to be constructed for the Captain Cotton Canal to discharge 1,500 cusecs into the 'B'canal. Even then, the 'B' Canal could not be in a position to accommodate the flood and discharge into the sea and hence to realize the benefits of the improvements to Captain Cotton Canal a short cut canal to the sea in Thiruvottiyur area has to be constructed.

(v) To prevent flooding of the Manali area and in Kosasthalaiyar river the Kosasasthalaiyar riverbanks have to be strengthened by borrowing the shoals inside the river as far as possible.

(vi) Otteri nullah has its southern watershed starting from Cooum river itself and have all spills from the river enters into the nullah. The main cause for the floods in the nullah was the absence of banks for Cooum river. If this is taken care of, the discharge from nullah could be confined and disposed off after effecting improvements to the nullah.

(vii) In respect of the Cooum river flow, strengthening of the banks at certain stretches had to be made apart from repairing and strengthening of the Bangaru Channel. Thirumangalam causeway, Naduvakkarai causeway and Chetpet causeway, which were all obstructing the flow, should be removed. The old arch bridges, which were causing afflexes of more than 0.37m, had to be replaced in the first phase by a single span bridge (Aminjikarai bridge, Andrews bridge, Law's bridge and Wellington bridge) in the first phase and College road bridge, C-in-C Road bridge in the second phase.

(viii) It was essential to think measures to divert a portion of Cooum flood discharges into neighbouring Kosasthalaiyar basin (Poondi reservoir) as already formulated by Er.A.R.Venkatachary's in 1943. Flood waters from Eagattur area to be taken in the Northwesterly direction reaching Thirupachur Odai and then to the Poondi reservoir by constructing a diversion channel.

(ix) Cooum River collects surplus from 75 tanks under Cooum Tank Group. All these tanks drain ultimately to Cooum tank from which the surplus reaches Cooum River just above Satharai Causeway. Tanks can be made to absorb and mitigate flood by judicious regulation. Flood outflow from large tanks could also be brought and drain in such a manner they not only absorb the flood but also anticipate them and deplete the tank in advance such that the outflow is kept at moderate level over a long period. Flood absorption capacity of the tanks may be improved by lowering the full tank level or by converting the surplus weir into a 'calingulah'.

(x) North Buckingham Canal had been improved and linked between Ennore South lock and Cooum in fall for a distance of 10 miles in 1973-74. In Thiruvottiyur region, it collects the drainages from Sathangadu tanks and Kodungaiyur tank. In Tondiarpet region, it collects drainages from

Vyasarpadi group of tanks. In Vyasarpadi area it is a flood bank of Otteri nullah. The entire city gets inflow through the 'B' Canal and it is a main drain of the city during monsoon. The floodwater in the canal flows southward into a Cooum river and northwards in Kosasthalaiyar river as much as the gradient permits. The stretch of the canal needs to be desilted and the floodgates of the Ennore South lock and Adyar north lock had to be got repaired.

(xi) The vent-ways were very small at Central Station Bridge and all bridges north of Tondiarpet and hence it took nearly 5 to 6 days for the accumulated flood in this region for the canal to drain. From the lagoon area a additional canal on the west of North Buckingham Canal had to be dug and also a short cut canal taking off from the 5/0 mileage from the eastern bank of the North Buckingham Canal and joining the sea near Thiruvottiyur had to be dug.

(xii) Earlier the Virugambakkam drain was flowing over a distance of 6.5 kms. and falling into Nungambakkam tank from which with a surplus entered into Cooum and Adyar rivers. Now, the entire course completely lost because of the urban development that had occurred in the 2 km stretch and the balance 4.5 kms. stretch was available. The course of the Koyambedu drain was also obliterated. The entire surplus was flowing towards north enter into narrow Arumbakkam drain and reaches Cooum river. In 1976, the submersion went up to the rooftop of some of the buildings and breached the causeway across Nelson Manikkam Road. Koyambedu surplus can be diverted northward into the Cooum directly for not only reducing the discharge into the Virugambakkam drain but also improving the chances of damage of the Koyambedu area, which requires the assured means of quick drainage. The Virugambakkam-Arumbakkam drain has to be improved.

(xiii) Surplus from Valasarawakkam to be intercepted by a drain proposed to be excavated from Arcot Road such that the surplus reach Ramapuram tank, from there it drains to Adyar river. Missing links in the drainage system to drain floodwaters from KK Nagar, Ashok Nagar and Jaffarkhanpet area have to be formed by acquiring lands to discharge floodwater to Adyar River.

(xiv) Two new tanks just above the Chembarambakkam tank had to be constructed to capture 1570 Mft^3 of flood.

(xv) Adyar river has two arms. The northern arm comes from Chembarambakkam basin, joins with the southern arm coming from Guduvancheri, and joins at Tiruneermalai. In 1976, floods in the Adyar river were the worst ever experience. For containing the flood and reducing the submersion to the minimum extent, flood banks have to be put up all along the banks of the river. Two causeways at Kathipara and Jaffarkhanpet had to be replaced by all weather bridge as a long-term measure.

(xvi) Masonry weir built long ago at a distance of 300m below the Maraimalai Adigal Bridge obstructing the normal flow and causing a pond for dhobis to wash clothes had to be removed.

(xvii) Mambalam drain is flowing at the heel of the erstwhile Mylapore tank. Kodambakkam High Road and the Mount road between Gemini and Nandanam were the tank bunds of the Mylapore tank. The drain starts from Prakasa Road flows in the T.Nagar and CIT Colony area crosses the Mount road and reaches Adyar near Teacher's college. Improvements to these drains had to be carried out.

(xviii) Pumps must be permanently installed in the Kotturpuram area to drain floodwater since Adyar HFL is higher than the area.

(xix) Veeranam pipe crossing, which had been laid above the riverbed and below the MFL, was a potential source of scour in the Adyar river and it had to be dismantled and re-laid below the riverbed.

(xx) Kapalithottam, a slum, situated on the north of Greenways Road was a low-lying area originally connected the Adyar Backwaters. This area needs to be drained only by means of underground drainage system.

(xxi) It was possible to transfer 10,000-15,000 Cusecs from Adyar River into Covelong Valley by digging diversion channel either from a pond near Perungalathur or from a reach above Tambaram.

(xxii) Though South Buckingham Canal by itself experience high flood levels during monsoon, rain from local precipitation, it is not a flood carrier. The areas south of Adyar basin inside Madras City drain into the South Buckingham Canal and hence the problems in these areas were grouped under this basin. The flood from the developed area of Nangnallur, Velachery and Tiruvanmiyur normally travel southwards and get into the Pallikkaranai Lagoon, which gets drained by the river Covelong.

(xxiii) FTL of the Velachery Tank to be lowered by 1.2m and a new surplus course had to be provided.

(xxiv) Adambakkam drainage scheme had to be implemented reforming the old drains leading to the Adambakkam Tanks and cutting new drains to form the missing links to avoid flooding in the residential area of Nangnallur, Adambakkam, etc.

(xxv) In the Madras city, there were only major drains (Macro level) and hence the city had necessarily got to have internal drains connecting various parts of the City with the macro drains. Regarding the internal drains the city can be divided into the following reaches:

a) Coastal belt draining into sea directly
b) North Madras belt draining mainly into North Buckingham Canal for drainage relief
c) Hinterland between Cooum and Adyar river draining into Cooum and Adyar rivers
d) Southern area draining mainly into South Buckingham Canal

(xxvi) The Macro Drains and internal drains are mutually complementary and both are important. The master plan evolved by the Madras Corporation for internal drainage system has to be implemented fully.

MMFR / SWD Master Plan Study

10.27 Subsequently CMDA had engaged the services of the consultant M/s Mott MacDonald International, UK, to conduct the study titled 'Madras Metro Flood Relief/Storm Water Drainage Master Plan Study ' in 1992-93. The main objective of the study was to bring together the previous studies carried to assess the problem of flooding in Chennai and to identify using modern hydrological and hydraulic modeling techniques/measures to alleviate flooding in the North of the City.

10.28 The study area comprised two parts, (1) 90 sq.km strip north of the river Cooum and (2) 30 sq.km area south of the City named as Pallikkaranai. Both the Macro Drainage Systems (Rivers, Tanks, and Surplus Channels) and the micro Drainage System (Urban Storm Water Drains) had been examined.

10.29 An extensive data collection exercise had been undertaken by the Consultant involving acquiring and processing of data relating to meteorological, hydrological, tidal conditions, weather, topography, land use, existing macro and micro drainage systems, river basin control structures, floods etc. Hydrological catchment models to simulate runoff from rainfall and computational hydraulic river models (HYDRO) to simulate the built-up of flow and its routing to the Macro Drainage System to Sea were developed and calibrated. With regard to urban storm water system, complete drain inventory have been prepared and stored in a database. A review of the capacity of the whole micro drainage system within the study area had been made using MIDUSS (Microcomputer Interactive Design of Urban Storm water Systems) urban storm water drainage analysis program.

10.30 The range of options examined included upstream storage (to be implemented under Krishna Water supply scheme), diversion of flood flows into tanks, canals, channel resection, structural improvements (including outfalls), provision of short cut canal between the Buckingham canal and the sea, formalisation of flood path and provision of flood defences (walls, banks, etc.). For the Macro System 17 principle options had been identified with the number of variants in each of this considering about 40 possible interventions by using a qualitative screening process. Some of these options were eliminated and 20 options examined in depth using the models. The following are the results of the analysis:

(i) The storage and diversion schemes in the upper catchments were generally found to have little impact on flood levels in the downstream coastal reaches of the system where flooding occurs. In the Kosasthalaiyar system, this was because of the major over bank spillage to the north in the middle reaches at Minjur: the effect of storage and diversion was found to lower the amount of spillage at this point, but to have only a small effect on water levels further down the system. In the Cooum system, flows from the upper catchment do not peak at the same time as flows from the middle and lower catchments. Thus, while the diversion of flows in the upper catchment reduces the volume of water, it does not affect the maximum water levels. The exception to this was an option to divert water from the middle reaches of the Cooum to the Red Hills Tank. This was found to have a positive impact on over bank spillage in the city, but is very expensive and will be difficult to implement because of the intense land use along parts of the route. It would also require an upgrading of the Red Hills Surplus Channel.

(ii) With regard to outfalls, a short cut canal from the Buckingham canal to the sea of 100 m^3/sec. capacity was tested using the models but found to have limited impact on water levels: a larger capacity would be required to have a significant effect. This would be very difficult to implement, given the intensity of land use between the Buckingham Canal and the sea. It was thus concluded that it would be better to improve the Buckingham Canal and rely on the existing outfalls. While these outfalls are choked for much of the year due to littoral drift, analysis showed that the bar would wash out quickly during a major flood event and not form a significant obstacle. This has been confirmed by observation. The sand bars are thus principally a problem for public health, rather than flood relief. It is possible that, during minor flooding, the bar may be an obstacle to flow from the urban drains, but this is unlikely to be catastrophic.

(iii) The conclusion reached was that flood defence and re-sectioning options would provide the cheapest and most assured way of dealing with the macro-flooding problem. Schemes have been outlined for the Cooum, the Kosasthalaiyar, the Red Hills Surplus Channel, the Buckingham Canal, the Otteri Nullah and the Captain Cotton Canal, and tested in the model. Several of these schemes are interdependent. On the Kosasthalaiyar, it is not possible to contain the floodwaters with embankments at reasonable cost. Protection has thus been provided on the south bank only, and a controlled floodway is proposed for the north bank upstream of Minjur. This will need to be designated as a flood zone and development restricted in that area. Existing villages will require to be protected, probably using ring banks. The Kosasthalaiyar will also spill near its confluence into the existing wetland areas.

(iv) For the urban storm water (micro) system, existing coverage of drainage provision within the study area boundary was found to be 50%. The design standard currently adopted was found to be equivalent to about a 1 in 1.25 year 60 minute storms. The economic analysis indicated that a design standard of 1 in 2 or 1 in 3 year return period was the most cost-effective for new areas of drainage, but also showed that upgrading the existing system to this standard was not economically worthwhile. However the drainage system was generally found to be in a poor state, with many blockages due to solid waste and services (water pipes, cables etc.) and repairs needed. The principal interventions envisaged are thus repairs/rehabilitation of existing systems and improved maintenance which is seriously under-funded. For Manali, outline schemes and costs were prepared for pumped storm water disposal and for flood protection.

Drainage Master Plan

(v) The Master Plan recommended by the consultant comprised a number of components:

- Structural works for major flood alleviation and for rehabilitation of the urban storm water system;
- Non-structural measures required to support these investments;

- Capacity building, with particular emphasis on system maintenance and master plan implementation;
- Further studies required to progress the plan; and
- Monitoring and evaluation requirements.

(vi) As far as the Master Plan is concerned, the proposed scheme at Pallikkaranai has been treated as committed expenditure, as government has already decided it should be the priority for implementation to aid economic development in the area.

(vii) A phased programme for implementation of the Plan has been developed, with consideration being given to financial and economic constraints, implementation capacity, operation and maintenance requirements, lead times for project preparation and interdependence of projects.

(viii) The structural measures identified were given below:

Table No: 10.03: Structural Measures

Sl. No.	Structural Works	Cost (Rs. in Million)
1	Pallikkaranai Scheme	160
2	Flood defences and channel improvement on the Cooum	348
3	Flood defences and channel improvement on the Buckingham Canal	96
4	Flood defences and channel improvement on the Otteri Nullah	125
5	Flood defences and channel improvement on the Captain Cotton Canal	20
6	Flood defences and channel improvement on the Madhavaram Surplus Channel	10
7	Flood defences and channel improvement on the Red Hills Surplus Channel	215
8	Flood defences on the south side of the Kosasthalaiyar	50
9	Manali Township drainage and flood protection	40
10	Urban storm water rehabilitation and repairs	35

Note: All cost estimates are preliminary (year 1993)

(ix) Non-structural measures recommended were:

a) Designated floodways on the north side of the Kosasthalaiyar with associated planning controls and flood warning/evacuation procedures
b) Design guidelines for drainage systems
c) Planning and regulatory controls to prevent development in old tank beds unless adequate flood defence measures are in place
d) Planning and regulatory controls to prevent encroachment of squatter settlements in old tank beds and watercourses

e) Provision of good facilities (vehicles, communications) for flood emergency management
f) Public education (e.g. to prevent solid waste dumping in urban drains)
g) Flood risk mapping

(x) A unitary authority should be responsible for the drainage system within the City area and it should be Chennai Municipal Corporation. The lead agencies PWD, and Chennai Corporation are responsible for the general day-to-day management of the system. The authority responsible for the drainage and flood control should be required to meet a number of management objectives, principally.

(xi) In respect of operation and maintenance regular inspection has to be carried out and routine and periodic maintenance involving desilting, embankment regarding, weed/bush clearing, vermin control, clearance of rubbish/debris, etc., had to be carried out according to a regular programme. The study also identified in outline a number of studies including environmental impact assessment, flood risk mapping, etc.,

(xii) The drainage study for Pallikkaranai was included as part of the MMFR/SWD master plan study. The aim of the study is to identify ways of providing protection to an area about 30 sq.km. lying in and around Pallikkaranai. The area was earmarked for development and the development was to be promoted by number of government and private bodies then. For the purpose of the study the area was referred as Pallikkaranai Drainage Area (PDA). The aim of the project was to protect an area of approximately 30 sq.km. from flooding. It could be achieved by a diversion of substantial portion of run off from upstream catchments along a cut off drain linking the existing surplus channel close to a village called Karanai with the Kovalam Backwaters. Northern boundaries of Pallikkaranai Drainage Area cuts off the centre of the existing swamp area at Pallikkaranai. The area to the north will continue to be subject to inundation as run-off enters the area from north and west. East west flood protection bund protects this side of PDA. In order to provide internal drainage to the PDA two pumping stations have to be located in the East-west flood protection bund to lift the water over the bund. A balancing pond is also to be located near the pumping station. However, protection to the area is to be provided by three interceptor drains, which carry overland flows from local catchment around the boundary of the PDA. An arterial drain has to be constructed along the centre of the PDA, which will pick up drainage flows within the area. Details of the proposal are given in the sheets annexed.

Others

(xiii) The consultant M/s Mott MacDonald International had submitted the Storm Water Drainage Master Plan for the Madras city and Pre feasibility Study for the Madras Metropolitan Area in October 1994. With regard to the urban storm water system the detailed technical and economical analysis were carried out in a sample area in Pulianthope representing about 10% of the urban area within the MMFR study boundary. It was used to examine possible interventions to improve the functioning of the micro drainage system in Chennai City. Based on this work, guidelines for the design of future drainage provisions were prepared.

(xiv) A study was also made for the flooding and internal drainage problem at Manali New Town. A detailed inventory of the storm water drainage was prepared and stored in a database. The conclusion reached, was the flood defence and resection options would provide the cheapest and most assured way of dealing with the macro-flooding problem.

(xv) For the Central Buckingham Canal isolation from the Adyar during flood flows offers significant advantages. The Canal will then serve as a collector of local urban run-off, rather than a flood path within the macro system. For the urban storm water (micro system) existing system of drainage coverage within the study area is found to be 50% with additional areas draining overland into the urban drainage network. The drainage system was generally found to be in a poor state, with many blockages due to solid waste and municipal waste and services (water pipes, cables, etc.) and repairs needed. The principal intervention thus requires repairs /resection of the existing system and improved maintenance which is seriously under funded. Capital works include new works including area of drainage in particular at Arumbakkam-Virugambakkam along with some investment in new drain or drain enlargement within the existing system and investment of new outfall facilities.

(xvi) Outside the City, there are currently few drainage facilities beyond the drainage provided by the primary and secondary network of rivers and surplus channels. Improvements to this drainage network will be essential if development is to proceed within the CMA at the pace envisaged. A number of strategic initiatives had been identified for the development of CMA including flood banks on the south side of the Kosasthalaiyar, improvements to the Ambattur/Korattur/Madhavaram surplus channel system, drainage works to allow the Pallikkaranai area to be drained.

(xvii) In addition to this specific proposal there is a need to develop a strategic planning policy within CMA, which would focus on some of the important requirements for the continued functioning of the drainage system as development advances. These might include retention of tanks as open space, provision of river corridor, prevention of encroachment in tanks, the definition of secondary drainage system and development of tertiary and quaternary system and links between these and secondary/primary system.

(xviii) Planning the system includes preparation of contour mapping for the CMA and identification of low-lying areas for discouraging developments.

(xix) Clear policies must be developed for the CMA to ensure the objectives of the flood studies are met. Such policies should embrace the interlinked issues of water supply, wastewater disposal, solid and hazardous waste disposal and flood alleviation.

Environment Impact Assessment of the Drainage Plan:

10.31 In March 1995, CMDA engaged M/s KBN Engineering and Applied Sciences Inc. supported by M/s Mukesh & Associates to conduct an environment impact evaluation of the preferred drainage scheme and redevelopment in this area. The study component comprised of an environmental impact assessment (EIA) of the drainage plan for the study area and a land use compatibility analysis for the proposed Pallikkaranai Drainage Area.

10.32 This EIA was conducted in co-ordination with CMDA and other agencies concerned viz. PWD, TNPCB, CMWSSB, Central Ground Water Board, State Government Board, Geological Survey of India, Anna University, TNSCB, Tamil Nadu Guidance Bureau, Archeological Survey of India (Madras Circle), Central Institute of Brackish Water Aquaculture, Chennai. The major findings of the study are given in the Annexure IX A.

Review of Pallikaranai Development Plan :

10.33 Subsequently in 1997, government had directed CMDA to arrange for a review of the above consultants report in respect of the Pallikkaranai Development Area (PDA) engaging M/s NEERI. Accordingly, NEERI was engaged in June 1998, which had studied these recommendations of the consultants and suggested two alternatives for development of the PDA. On the preferred alternatives, PWD had stated that formation of reservoir in Pallikkaranai in swamp as suggested by M/s NEERI (Alternative-B) either for drinking water supply or for recreational purpose is not desirable due to following facts:

i. The catchment areas of the swamp are fully habited. Hence, the runoff from the catchment that flows into Pallikkaranai swamp is highly contaminated. The effluent from the Perungudi STP and leachate from the solid waste dump also flow into the swamp. So, treating the raw water with this contamination into drinking water to the BIS standard is difficult and highly expensive.

ii. Maintaining the water quality for recreational purpose is also difficult in a location surrounded by habitation.

10.34 In consultation with departments concerned viz. Environment & Forests, PWD, TNPCB, Housing and Urban Development Department it was finally recommended to Government as follows: -

i. Environment and Forest Department was processing the proposal for protecting the swamp lying between the 200 ft. MMRD Scheme Road in the north and Sholinganallur – Perumbakkam Road in the south. The swamp area in the north of MMRD scheme road was mostly occupied by Chennai Corporation, Chennai Metropolitan Water Supply and Sewerage Board & MTP (Railways).

ii. It was reported that the study revealed that in the Pallikkaranai swamp, the contiguous low lying areas received run-off from a catchment area of 235 sq.km. During monsoon large pools of water in the valley confluence in the central portion of the study area. Run-off enters the valley from Velachery in the north and also from Arsankalani and other villages in the south. Due to the presence of network of lakes and surplus channels there is significant degree of regulation that reduces peak flows into the valley (except in the north-western built up areas in Velachery, Madipakkam, etc. which gets flooded during monsoon). The storm water eventually passes thro' the Okkiyam maduvu to reach the south 'B' Canal which flows south and enters the Kovalam estuary. The 'B' Canal is not capable to drain storm water effectively, mainly because of its inadequate size and slope. The peak flow reaching the study area was estimated to be about 350 m^3/sec. (according to PWD, the discharge from the catchment area into the valley is 11542 cubic feet). The members present agreed that the water holding capacity of the valley should be retained to avoid flooding in the catchment areas particularly in the northwest built up areas.

iii. The Government lands lying in between MMRD Scheme Road in the north and Sholinganallur-Perumbakkam Road in the south lying in revenue villages of Pallikkaranai, Sholinganallur, Perumbakkam, Karapakkam and Jalladampettai may be declared as protected marshy land prohibited for any urban development, on the lines of the Adyar Estuary if necessary.

iv. The area on the south of the Sholinganallur-Perumbakkam Road may be put to urban development/use by reclaiming these low-lying areas as recommended in the NEERI Report with proper central drainage arrangements and access roads. By this, about 15 sq.Km of land would be available for development according to NEERI Report (Government lands in Sholinganallur and Perumbakkam account to 800 acres in this area).

v. In order to ensure that this reclaimed area does not pollute swamp and other water-bodies, it may be developed for I.T, ITES and electronic industries. Considering its location, accessibility and proximity to airport, road infrastructure to be provided by the IT Corridor, it would be suitable for such urban development.

vi. Regarding the area north of the MMRD, it could be preserved as it is considering its necessity for flood accommodation, its eco-sensitivity.

Outline Project Report on Flood Alleviation:

10.35 CMDA in consultation with the Line agencies viz. PWD, Chennai Municipal Corporation and TNSCB had prepared an outline project report on flood alleviation and improvement of storm water drainage system in Chennai Metropolitan Area with a total project outlay of Rs.300 Crores to be implemented over a span of 5 years and submitted to government. The project cost abstract and the responsibilities of the agencies involved in the implementation are given in the Annexure IX B**.**

10.36 The Government in G.O.Ms.No.321, H&UD Department dated 12/08/1998 had given their administrative sanction for the project. The project has been taken up in phases for execution and up to mid 2005 it has been executed including an expenditure of about Rs.106 Cores. Under R&R Component implemented by TNSCB 3000 tenements had been constructed at Okkiam Thoraipakkam to resettle the slums in the flood alleviation project along 'B' Canal and Adyar river. Desilting of South Buckingham Canal, construction of retaining wall, formation of jeep track along the banks and construction of 10 vents in North Buckingham Canal, repairs to its linings had been completed. Construction of flood defences and resection of the rivers Kosasthalaiyar had been completed. In respect of Adyar river, construction of flood defences and resection works are nearing completion except for the desilting work east of the Thiru Vi Ka Bridge. Works in Ambattur Tank Surplus Course, Madhavaram Tank Right Flank Surplus Course and Pallikkaranai Drainage Works were taken up. About 70% of the works in respect of the Red Hills Surplus Course works were completed; remaining works in this Course and the works in Madhavaram Tank, Chembarambakkam Tank, Korattur Tank Surplus Courses are to be completed after required lands are acquired. Improvements to Otteri Nallah, Virugambakkam – Arumbakkam drain had been completed. Drainage relief works to Velachery area is nearing completion except for the court stayed short stretches. Improvement to the Cooum river from sea mouth to Periyar bridge has been completed and from Periyar Bridge to Koyambedu will be taken up after completion of R&R works.

Integration of Macro and Micro Drainage :

10.37 The Micro Drainage works to the tune of Rs.43 Crores have been implemented by the Chennai Corporation for improvement of the drainage system in Chennai Corporation area integrating with the Macro Drainage System.

Study on Waste Water Outfall into Waterways :

10.38 The study conducted by the consultant M/s Wardrop Engineering Inc. in 1995 revealed that the waterways in Chennai convey treated and untreated sewage and receive debris and solid waste also though they were originally natural flood discharge channels. The addition of untreated liquid waste had led to a very high level of pollutants and the disposal of the solid and the encroachment of slums had severely reduced flows particularly during monsoon periods. A summary of wastewater outlets existed in 1995 is given below:

Table No. 10.04: Waste water Outfall Details for Inner Chennai Waterways

Waterway	Nature of Outfall				
	Sewage	Storm water	Industrial	Others	Total
River Cooum	109	6	1	-	116
Adyar River	58	23	-	-	81
Otteri Nullah	42	4	1	-	47
South 'B' Canal	26	1	-	-	27
Central 'B' Canal	30	-	-	1	31
North 'B' Canal	58	5	3	1	67
Redhills Surplus Channel	-	-	4	-	4
Mambalam Drain	14	8	-	1	23
Captain Cotton Canal	13	-	-	-	13
Kodungaiyur New Drain	2	-	-	-	2
Ambattur Surplus Tank	5	-	5	2	12
TOTAL	**357**	**47**	**14**	**5**	**423**

Source: M/s Wardrop Engineering Inc, 1995

Environmental Improvement of Watercourses

10.39 The consultant M/s Severn Trent International conducted the study on environmental improvement of watercourses in Greater Madras in 1991. They have recommended for extension of sewerage system to unsewered areas and use of low cost sanitation wherever appropriate, purchase of jetting equipment and replacement of smaller pumps with submersible pumps, extension of storm water drainage system, resolution of interconnections between sewers and storm water drains, short term improvements at sewage works, extension and long-term improvements at sewage works, equipment to clean storm water drains, equipment for street cleaning, sanitary waste disposal, dredging and grading of river Cooum, completion of river Adyar flood protection scheme, desludging of Buckingham Canal, North Ennore Lock to the river Cooum, filling up of central portion of the Buckingham Canal between the rivers Cooum and Adyar, desludging of Buckingham Canal between the River Adyar and the City limit, pumping of Otteri Nullah for treatment, dredging and regrading of Otteri Nullah and construction of groyne to improve the Cooum outfall to the sea, if needed.

Chennai Waterway Conservation Programme

10.40 The sludge disposal consultancy study conducted in 1994 by the consultant M/s MMI has revealed that contamination of water-ways and anaerobic digestion of waste water flowing in the water-ways had led to the accumulation of sludge causing hindrance to the hydraulic functioning of the water-ways and also causing contamination of water-ways in the eco system. The following are the estimates of the sludge accumulation in Chennai waterways:

Table No. 10.05 Estimates of Sludge Accumulated in Inner Chennai Waterways

Waterways	Length in Km	Typical width in dry season (m)		Characteristics		Sludge volume in 'ooo m		
		Water	Total	Durability	Access	Hydraulic nuisance	Envtl.* Nuisance	Both
Cooum	18.00	23-40	45-120	D	O	1210	350-750	1280
Adyar	15.00	15-200	90-500	E,P	O,O*	1880	340-200	1960
North B' Canal	7.10	15	20	D/E,P	O/N	150	40-100	200
Central 'B' Canal	7.20	5	20	E	O	200	30-50	200
South 'B' Canal	4.40	9	15	D,P	C	10	20-50	20
Otteri Nullah	10.20	4-5	7-20	E/D	O/C	100	30-60	110
Captain Cotton canal	2.90	5-40	25-45	E,P	C	20	40-60	50
Total	64.80					3570	840-250	3820

Source: M/s MMI, Sludge Disposal Consultancy, 1994

Note: Abbreviations for Durability
D-Dewatering possible difficult and expensive
I-Dewatering impracticable, floatation of pontoons, barges and small dredges possible after dredging
E-Easy to dewater with bunds and bypasses, with some pumping
Abbreviations for Accessibility
O-No continuous access
O*-occasional access, but from private roads
N-No access
C-Continuous access

10.41 Earlier the consultants had estimated that the number of slum families to be resettled and rehabilitated from the objectionable areas of waterways to do the improvement is 22,800. But the PWD had estimated that at least 10,000 slum families have to be resettled and rehabilitated to do the minimum required improvements to the waterways.

10.42 The length of the major waterways passing through the City and its total length in CMA is given below:

Table No: 10.06 Length of Major Waterways in CMA

Waterway	City	CMA
River Cooum	18.0	40.0
River Adyar	15.0	24.0
North Buckingham Canal	7.1	17.1
Central Buckingham Canal	7.2	7.2
South Buckingham Canal	4.2	16.1
Otteri Nullah	10.2	10.2
Captain Cotton Canal	2.9	4.0
Kosasthalaiyar	-	16.0
Mambalam Drain	9.4	9.4
Kodungaiyur Drain	6.9	6.9
Virugambakkam-Arumbakkam Drain	6.9	6.9

Source : PWD

10.43 The details of the catchment areas of the waterways within Chennai City are given in the sheet annexed. Also the areas, which were affected by the flood events during 1976 and 1985 are given in the sheets annexed.

Suggestion by the National Institute of Ocean Technology :

10.44 To open the mouth of Cooum River, National Institute of Ocean Technology (NIOT) had suggested that groynes at stages may be constructed and the suggestion is under active consideration for implementation by PWD.

Chennai City River Conservation Project

10.45 As part of the Chennai City River Conservation Project (CCRCP) CMWSSB had proposed project for prevention of sewage flow into waterways, treatment of sewage, construction of interceptor sewers and enhancement of pumping station capacity with a total project cost of Rs.720 Crores. But the National River Conservation Directorate (NRCD), Government of India, had finally accepted to fund the project to the tune of Rs491.82 Crores in the year 2000. Out of the project costing Rs.382 Crores approved by the NRCD, CMWSSB had executed works to the tune of Rs.325 Crores.

Chennai Metropolitan Development Plan

10.46 Under Chennai Metropolitan Development Plan (CMDP), projects for macro and micro drainage system to the tune of Rs.39 Crores has been executed during 2003-2004, Rs.41 Crores during 2004-2005, Rs. 103 crores during 2005-06 by PWD, Chennai Corporation, Municipalities and Other Local Bodies within CMA and also by the Highways Department. An outlay of Rs.98.99 Crores has been proposed for the year 2006-07 under this component.

Conclusion

10.47 Flooding in the CMA has become a recurring feature. During dry season, the City experiences acute water scarcity. Abundance of data are available on the macro drainage system. Thanks to numerous studies conducted, Chennai City River Conservation Project has added new dimension to the system. With the co-ordinated efforts of Government agencies, involvement of stakeholders and with the application of modern technology for map making and networking, it is earnestly hoped that flooding in the CMA will become a thing of the past.

Annexure X A

Major findings in the Study on Environmental Impact Assessment of the drainage and redevelopment proposal for Pallikaranai Area (consultants: KBN Engineering and Applied Sciences, Inc., Florida and Mukesh & Associates, Salem):

1. No significant adverse impacts will be caused by the preferred drainage scheme.
2. The preferred drainage scheme will allow for limited development in the study area. Severe flooding including flooding from the 50 and 100-year storm events, which now precludes development, will be reduced to acceptable levels to allow for development. Further reduction in the effects of flooding can be achieved by raising plinth levels of buildings onsite to approximately 2.0 m. above Survey of India datum (m-ASoI) and the incorporation of onsite stromwater retention and detention ponds.
3. The proposed PDA will have a significant positive benefit to the people and economy of the region. Given proper site planning, no significant adverse effects will occur from development of the PDA.
4. Development of the PDA in the study area is most desirable because of:
   a) Protection from flooding,
   b) Compatibility with existing land uses,
   c) Proximity to existing transportation infrastructure,
   d) Available labour force and
   e) Negligible environment impacts.
5. The most compatible land uses for the PDA are a combination of medium, light and service industries. The development of heavy industry in the PDA is not recommended given the generally good environmental quality in the area; the requirements for stringent pollution control, monitoring and enforcement, and its incompatibility with existing land use.
6. The three most important future impacts to the preferred drainage scheme and PDA development once the preferred drainage scheme is implemented are:
   a) Uncontrolled flow of offsite contaminated water to the site.
   b) Effects of offsite development affecting onsite water elevations and
   c) Uncontrolled induced development in the PDA which may result in relocation issues for future development.
7. Relocation to the proposed reservoir is recommended for several reasons including

a) Potential water quality impacts from existing and proposed offsite development
b) Conflicts with proposed TNHB residential development
c) The existence of better alternative locations, i.e. use of tanks adjacent to the study area.

8. Relocation of the proposed MMRD road to the PDA is recommended because of the potential water quality impacts it will cause north of the East-West Flood Protection Bund. It can be located in the PDA where it can better serve the traffic needs of future development within the PDA as well as meet the Area's traffic needs.
9. A relocation of the proposed TNHB development in the northern part of the study area needs to be considered since it is planned to occur in areas identified for portions of the drainage infrastructure. This proposed development could be incorporated in the PDA.
10. There are major offsite developments including the Perungudi landfill and the Pallikkaranai sewage treatment plant which significantly affect onsite water quality including water flowing to the Oggiam Maduvu. Future expansion of these and other development needs to be evaluated in light of impacts to the preferred drainage scheme and the PDA.
11. A basin-wide storm water management plan is necessary to assure the proposed drainage scheme will operate effectively as future development occurs adjacent to the study area and PDA.
12. Restoration and enhancement of the Velacheri Marsh should be undertaken to assure protection of surface water quality entering the study area.

Annexure - X B

Flood Alleviation Measures and Improvements to Storm Water Drainage Systems in Chennai Metropolitan Area

Project Cost - Abstract

Packages	Cost Rs.in Million *
Macro Drainage Network	
Cooum Improvements	557.20
North Buckingham Canal Area	467.70
Adyar Improvements	460.60
North Chennai	334.60
Pallikkaranai Works	252.00
Additional works in Chennai Metropolitan Area	448.00
COMPONENT COST - MACRO DRAINAGE	***2520.00***
Micro Drainage Network	
Cooum (includes Arumbakkam)	75.60
North Buckingham Canal	39.20
South Buckingham Canal	16.80
Central Buckingham Canal	23.80
Adyar	50.40
Captain Cotton Canal	46.20
Otteri Nullah	134.40
Mambalam/Nandanam System	44.80
COMPONENT COST - MICRO DRAINAGE	***431.20***
TOTAL PROJECT COST	2951.20 OR **Rs. 3000 m** (roundly)

(*Cost does not include technical assistance, management and contingencies)

PROJECT IMPLEMENTATION RESPONSIBILITIES	
Agency	**Responsibility**
Chennai Metropolitan Development Authority (CMDA)	Project Packaging and Management, Monitoring and co-ordination
Public Works Department (PWD)	Plan, Design and Implementation of Macro Drainage Works
Chennai Municipal Corporation (CMC)	Plan, Design and Implementation of Micro Drainage Works
Tamil Nadu Slum Clearance Board (TNSCB)	Formulation and Implementation of Rehabilitation and Resettlement Package

Chapter XI
DISASTER MANAGEMENT

Natural disasters will occur. It can neither be predicted nor prevented. The problem before us is how to cope with them, minimizing their impact. Tamil Nadu has witnessed havoc caused by cyclones and storm surge in the coastal regions, earthquakes, monsoon floods, landslides, and recently the Tsunami hit. Increase in urban population coupled with the construction of man-made structures often poorly built and maintained subject cities to greater levels of risk to life and property in event earthquakes and other natural hazards. One of the main objectives is to reduce the risk of loss of human life and property and to reduce costs to the society. We have to recognize that in such cases of natural disasters, we deal with phenomena of enormous magnitude that cannot be controlled by any direct means of human intervention. But what we try to do is to reduce the impact on the human beings and property.

11.02 People have continued to live/settle in the disaster-prone areas, inspite of knowing about the risk and occurrence in the past may be due to certain cultural and historical reasons coupled with advantages of living in these areas. The risk gets amplified when the population increases, the area gets densified and activities increase thereby aggrevating the situation and putting a large number of lives at risk. To cope with the disasters, preparedness and planning are the only ways. There is an urgent need and ample justification for bringing in the required regulations for construction of buildings in the disaster/hazard-prone areas. Apart from improving building safety through better design and construction, it is very important to improve the quality of urban planning including provision of well-designed road networks and open spaces to facilitate disaster management. Finance for disaster management/mitigation should be treated as an investment and not expenditure. Local governments should see disaster management as a key issue and should be well-prepared to handle disasters when they occur and they should strengthen their technical and financial capacity for adequate pre-disaster planning and mitigation. By virtue of being closest to the events the local bodies are most likely to be given the responsibilities for coping with disasters.

11.03 Natural hazard means the probability of occurrence, within a specific period of time in a given area, of a potentially damaging natural phenomenon. Natural hazard prone areas mean the areas likely to have (i) moderate to very high damage risk zone of earth quakes, OR (ii) moderate to very high damage risk of cyclones OR (iii) significant flood flow or inundation, OR (iv) Tsunami proneness (v) landslide proneness or potential, OR (vi) one or more of these hazards. Natural disaster means a serious disruption of the functioning of a society, causing widespread human, material or environmental losses caused due to earthquake, cyclone, flood, Tsunami or landslide which exceeds the ability of the affected society to cope using only its own resources.

11.04 Earthquake-prone areas mean the areas under seismic zones III, IV & V (as specified in IS: 1893), which are likely to have moderate to high damage, risk due to earthquake. "Cyclone-prone areas" mean the areas close to the coast where the cyclonic wind velocities exceed 39 m/sec. (specified in the wind velocity map given in I.S. 875) and prone to cyclonic storms. In this State of Tamil Nadu, it normally extends to a distance of 20 km. from the coast in all the costal districts. "Flood-prone areas," mean areas likely to have significant flood flow/ inundation. "Tsunami-prone areas" means the areas affected in the past and the areas likely to be affected by Tsunami, as identified and notified by City Corporation of Chennai as Tsunami-prone areas.

11.05 Hazard-prone Areas in Chennai Metropolitan Area may be classified as follows:

(i) Earth quake-prone Areas:

Chennai Metropolitan Area falls under Seismic Zone – III. The whole of Chennai Metropolitan Area falls in this zone.

(ii) Cyclone-prone Areas:

In this Chennai Metropolitan Area, it extends to a distance of 20 km. from the coast in all the coastal districts. In these areas, the risk is due to (a) cyclonic wind velocities combined with heavy storm, (b) flooding by seawater due to high waves and (c) flooding due to heavy storm. The map showing the cyclone-prone areas in Chennai Metropolitan Area is annexed.

(iii) Landslide-prone Areas

Unstable geological conditions, indiscriminate construction activity, heavy rainfall and flash floods coupled with poor drainage due to urbanisation are the main factors causing landslides in hilly regions; earthquakes also trigger landslides. No area in Chennai Metropolitan Area qualifies for zoning as landslide-prone area.

(iv) Flood-prone Areas:

From the flood hazard map of India (mapped by meteorological department, New Delhi), it is seen that no area in Tamil Nadu falls in the risk zone. But within a local body area, particularly with reference to an area's proximity to a major drainage system like rivers, canals and also water bodies like lakes, and further with reference to contour levels/low-lying areas, flood-prone area mapping has to be done.

In Chennai Metropolitan Area, there are a few areas along the rivers and canals and low-lying areas which are susceptible to flooding/inundation during heavy storms. Map showing the floodable areas [macro level] identified in the Madras Metro Flood Relief / Storm Water Drainage Master plan is annexed. Existence of macro and micro drainage networks in Chennai Metropolitan Area facilitate draining of these areas within a reasonable time. Developments in such low-lying areas are allowed only when a proposed development conforms to standards and after getting clearance from PWD on the measures to be taken to make it free from inundation.

(vi) Tsunami-affected Areas:

Mapping has to be done on the areas wherein Tsunami had directly hit and flooded the coastal areas in Chennai Metropolitan Area. These areas may have to be zoned as Tsunami-affected areas. However this area within Chennai Metropolitan Area is concerned will fall within the CRZ area 500 m from HTL along the coast.

11.06 Development Control Rules for CMA provide for regulating the constructions with reference to zone, location, height, number of floors, size of buildings, set back spaces to be left around and the use of the building and land. Building Rules under the Local Bodies Acts are concerned they provide for regulation of location of buildings, foundations, plinths, superstructures-walls, floors and rooms, licensing of surveyors and inspection of municipal engineers at various stages of constructions, regulations on dead and superimposed loads, wind load/pressure, reinforced cement concrete and framed structures, construction materials, etc. Structural safety and soundness are regulated under the Building Rules under the Local Body Acts. Hence early action should be taken to include Special Rules for Hazard Prone Areas in the Building Rules of the Local Bodies.

Mitigation

11.07 Mitigation means the measures taken in advance of a disaster aimed at decreasing or eliminating its impact on society and on environment including preparedness and prevention. The Government of India-UNDP Disaster Risk Management Programme (2002-2007) is a national initiative to reduce vulnerability of communities in some of the most hazard-prone area in our country; 38 cities urban earthquakes vulnerability reduction programme is a sub-programme. The main objectives of the programme are:

(i). Awareness Generation - at national level by conducting meetings of municipal authorities and placement of National UN Volunteers, at state level consideration and approval of nodal department's plans. Its agenda includes arranging for regular consultations of city-specific strategies, awareness generation campaign, use of mass media in awareness and sensitization, adoption and dissemination of information, education and communication materials and manuals, dissemination of appropriate construction technologies including retrofitting and seismic strengthening techniques.

(ii). Development of techno-legal regime - including identification of apex body, nodal agency in the state for co-coordinating development of techno-legal regime, need for sensitization of policy makers, orientation of officials in urban local bodies, review of existing bye-laws, other technological aspects and recommendations, orientation of engineers, architects, planners and city managers on the salient features of bye-laws and codal provisions, compulsory certification of engineers and architects, developing a city-specific safety audit of building practices and institution of a state ombudsman for safe building practices, developing of partnerships with financial institutions and insurance agencies for ensuing seismic safety in new/old constructions through loans and legislation for certification of seismic safety of existing buildings within a time frame.

(iii). Earthquake preparedness and response plans - including sensitisation meetings with communities, residential welfare associations, identification and training of volunteer trainers, hazard mapping and vulnerability mapping, preparation of ward/zone preparedness and response plan, formation of Disaster Management Committees (DMC) and task force training (Disaster Management Teams), training DMCs and DMTs, compilation of ward plans to city plans, review of city preparedness and response plans, conversion of city fire services to emergency rescue and fire services, and training of the trainers.

(iv). Training capacity building - including the development and implementation of the training calendar and models with training/response institution, training of practicing engineers and architects from government departments having programme modules of retrofitting and capacity building of government functionaries and technical institutions in the formulation and review of earthquake preparedness and response plans.

11.08 In the state of Tamil Nadu, the Revenue Administration, Disaster Management and Mitigation Department is the nodal agency for the UNDP-Government of India sponsored Disaster Risk Management programme and it is effectively implemented in

Tamil Nadu including Chennai Metropolitan Area. A note on the programme is given in Annexure-I.

11.09 A subcomponent of the DRM Programme is the Urban Earth Quake Vulnerability Reduction Programme (Annexure-II). Most of the components of the GoI-UNDP Urban Earthquake Vulnerability Reduction Programme are also applicable to other natural disasters viz. cyclones, landslides, floods and Tsunami. Hence the awareness generation, development of techno-legal regime, earthquake preparedness and response plans, training and capacity building should be done covering these natural hazards also and the State Nodal Agency may take appropriate action on these.

11.10 Even after the GoI-UNDP DRM programme period, the State Nodal Agency should continue these measures. Pre-disaster preparedness and disaster management plans should be periodically reviewed and up-dated.

11.11 Early action to amend the Building Bye-Laws of local bodies to include special provisions for hazard-prone areas and enforce the same since the whole of Chennai Metropolitan Area falls in Seismic Zone-III now and it also includes cyclone-prone areas to majority of its extent.

Mechanisms

11.12 Disaster management mechanisms exist at district level in the State. The structure includes District Disaster Management Committee, Disaster Management Teams, Crisis Management Groups, Emergency Operation Centre, Site Operation Centres, Modalities of involvement of army and other defence forces, NGOs and other institutions. District Response Plans focus on operational direction and coordination, emergency warning and dissemination, rapid damage assessment and reporting, search and rescue, medical response, logistic arrangements, communication, temporary shelter management (including free food/kitchen management), law and order, missing persons search/media management, animal care, involvement of NGOs and voluntary organisations. Depending on the disaster and damage caused Relief Management Plan need to be prepared identifying the relief needs, mobilization points, transportation and co-ordination with relief teams.

11.13 To restore normalcy to lives and livelihoods of the affected population, recovery and reconstruction plan has to be prepared, both for short term and long term, taking into account restoration of basic infrastructure, reconstruction / repair of life-line buildings / social infrastructure / damaged buildings, medical rehabilitation (both physiological and psycho-social interventions) and restoration of livelihoods through assistance /aid / grants.

11.14 Every year a pre-monsoon preparedness review meeting under the Chairmanship of Chief Secretary to Government is conducted to make assessment of the probabilities of monsoon floods and cyclone well in advance and to co-ordinate and prepare detailed plans to tackle real life situations. The Disaster Management Plan is also prepared. The collectors are instructed to prepare a hand-book containing information about the warning system given by the Indian Meteorological Department in flood and cyclone to the line departments/agencies, dissemination of information to line departments/agencies and general public, the level of preparedness and planning in the district administration, identification of vulnerable areas prone to flood and cyclone, contact telephone numbers of departments concerned / voluntary organisations/ NGOs/ CBOs, SHGs etc. Copies of handbook are distributed to all the line departments, educational institutions, and offices of the elected representatives.

11.15 District Co-ordination Committee conducts meetings with various organisations, local bodies etc. to review the preparedness and to take effective action.

11.16 Mock drills by the Police, Fire Department, NCC, and Home guards are conducted to create awareness for search and rescue purposes in the event of any emergency.

11.18 A control room has been established in the districts to monitor the activities during the monsoon period. Two-way communication system of VHF / HF system has been installed in all the 13 coastal districts and in the office of the State Relief Commissioner. Toll-free public utility service telephone is put into service at the district head quarters. A permanent control room is functioning round the clock at the Ezhiligam in the Office of the State Relief Commission.

11.18 In order to avoid flooding on account of torrential rains, the following steps are taken before the monsoon begins:

a. The PWD is directed to keep a constant watch over all the dams and major anaicuts and take necessary measure to plug the breaches and to ensure advance intimation to public before release of excess water. Cyclone shelters and buildings identified for accommodating the displaced persons are inspected for keeping them in all readiness.
b. The Highways Department is directed to keep all machineries like bulldozers, power-saws etc. in adequate quantity and in good condition for the purpose of clearing obstructions / road blocks caused by uprooted trees, electric posts etc. during the time of cyclone flood etc.

c. The collectors are directed to check the stock position of essential commodities like rice, kerosene in the godown and make available adequate number of lorries in good running condition to move the commodities like rice, kerosene to the affected people.
d. The Public Health Department ensures the availability of disinfectants and essential measures.
e. In times of emergencies, the assistance of navy and coast guard authorities, Army and Air force is taken to tackle the problem of evacuation, dropping of food to the affected people etc.
f. All oil corporation are required to keep adequate stock of fuel at the vulnerable points in the State and to provide refueling centres for helicopters that are pressed into service.
g. The medium of TV and Radio are utilised to caution the public to take precautionary measures and to move to safer places and
h. All the line departments including EB are instructed to have an effective coordination at the district level under the leadership of collectors. Hill Districts are instructed to be alert to deal with landslides.

11.19 Chennai City and its environs are very flat, and traversed by three river viz. Adyar, Cooum and Kosasthalaiyar. Mean annual rainfall is about 120 cm, and mean rainy days are 52 days. Records have shown that there were several catastrophic floods in Chennai in the last century in 1943, 1976, 1985, 2002 and 2005 caused by heavy rain associated with cyclonic activity. Flooding of less catastrophic nature occurs regularly in low-lying areas of the City and its suburbs because of inadequacy or in operativeness of local drainage infrastructure. In the year 2005 total rainfall was 240.8 cm and in the month of October it was 107.7 cm causing flooding of many parts of the City and to suburbs.

11.20 A disaster management cell functions in the City Corporation of Chennai to tackle flooding problem. Before the monsoon in October-December every year as a precautionary measure, drains are desilted and obstructions if any, are removed. Nodal officer and zonal officers are identified and vested with adequate powers to tackle such emergencies. Relief centres are identified and notified and responsibilities fixed for ensuring shifting of affected people to the relief centres, supply of food and other relief measures. Equipments to take out flood water from low-lying / submerged areas are kept ready. Arrangements are made for flood relief and details about whom should be contacted in such emergencies notified, published in news papers and announced over radio and television. All the agencies required to be involved such as Police, Fire and Rescue services, Metropolitan Transport Corporation, District administration, Health, Army and Navy are alerted / kept informed to be ready. Readiness to tackle the flooding situation is reviewed at the Government level and ensured before monsoon every year.

Cyclone Risk Mitigation Project

11.21 The Ministry of Home Affairs in consultation with the States prone to cyclone risks has drawn up a National Cyclone Risk Mitigation Project to be implemented with the assistance of World Bank. Under this project Cyclone Risk Mitigation investments will be undertaken in the States and Union Territories and it involves construction of cyclone shelters, shelter belt plantations, mangrove regeneration and construction of embankments to stop sea water inundation, construction of missing road links and commissioning of technical assistance / studies to sustain these initiatives in the States.

11.22 The World Bank has offered to support this project covering all 13 cyclone prone States and the State of Tamil Nadu is one among them. The State of Tamil Nadu which was placed in the Category II (Low Vulnerability) with the Project outlay of Rs. 26 crores has subsequently been placed under Category I (Higher Vulnerability) with a project outlay of Rs. 176 crores. The project cost has to be shared by the Centre and State Government as grant to State and 25% matching resources will have to be provided by the State in the Budget.

Annexure – I

Disaster Risk Management Programme (2002-2007)

Government of India (GoI) and United Nations Development Programme (UNDP) have signed an agreement in August 2002, for implementation of Disaster Risk Management programme to reduce the vulnerability of the communities to natural disasters in identified multi-hazard prone areas.

Tamil Nadu State is one of the States selected for this project and a Memorandum of Understanding has been signed between the UNDP and the Government of Tamil Nadu on 30.10.2003 in the presence of the Hon'ble Chief Minister of Tamil Nadu. This project will be implemented in the districts of Tiruvallur, Kancheepuram, Cuddalore, Nagapattinam, Kanniyakumari, and The Nilgiris and in the cities of Chennai and Coimbatore.

Goal:

"Sustainable Reduction in Natural Disaster Risk" in some of the most hazard prone districts in selected states of India.

Objectives:

1. National capacity building support to the Ministry of Home Affairs.
2. Environment building, education, awareness programmes and strengthening the capacity at all levels in natural disaster risk management and sustainable recovery.
3. Multi-hazard preparedness, response and mitigation plans for the programmes at state, district, block, village panchayat and ward level in select programme states and districts.
4. Networking knowledge on effective approaches, methods and tools for natural disaster risk management, developing and promoting policy frameworks.

Urban Earthquake Vulnerability Risk:

Reduction Programme:

It is proposed to take up urban earthquake vulnerability risk reduction measures in 166 districts under this programme. In the first phase 38 earthquake prone cities all over the country having a population of more than 5 lakhs including Chennai and Coimbatore Cities in Tamil Nadu falling in zones with moderate to very high risk to earthquake have been identified. The goal of the programme is sustainable reduction in earthquake risk in the most earthquake-prone urban areas across the country. Chennai and Coimbatore cities have been identified as earthquake-prone cities in Tamil Nadu.

Institutional Arrangements

Village	Village Disaster Management Committee
Panchayat	V P Disaster Management Committee
Block	Block Disaster Management Committee
District	District Disaster Management Committee
State	State Steering Committee (State Nodal Agency)
National	Project Management Board (MHA-DM Division)

Implementation Process:

The disaster risk management plan would start at the village level and would be consolidated through similar planning at the Panchayat, Block, District and Urban Local Body levels in the selected districts. At village level volunteers will prepare the plan and at other levels concerned Disaster Management Committees (DMCs) will prepare it. The plans would focus on disaster risk prevention and early recovery through community based preparedness and response plans, skill development for construction of hazard resistant housing and enhanced access to information as per the need of the community. Disaster Management Teams (DMTs) would be formed at various levels to implement the plans.

Awareness Campaign Strategy:

An effective disaster, risk management campaign strategy will be developed by the State Nodal Agency and will include:

- Rallies
- Mass meeting
- Wall paintings
- Posters

Competitions like essay, debate and painting competitions among school and college students.

Training / Capacity Building:

- Training of Trainers at the State, district and block levels to enhance the capacity of the DMC
- Specialised training to DMTs
- Exposure visit of Government officials, PRIs and DMTs
- Studies, research and workshops to be conducted at the national and state level
- Training manuals, standard operating procedures and documentation on best pictures.

Development of Disaster Risk Management Plan:

Multi-hazard risk management plans to be prepared by trained volunteers at the Village Panchayat, Block, District, State and National levels. **Mock Drills** to be carried out before the disaster seasons, by the key players, to find out the feasibility of the plan and to ensure greater role clarity.

Other activities to be taken up under the programme to reduce vulnerability:

- Construction of disaster resistant and cost effective demonstration units and retrofitting of non-engineered buildings
- Training of masons and engineers for wider dissemination and adoption of technology
- Advocacy for standard building codes and bye-laws
- Emergency rescue kits to be provided to the vulnerable districts
- Strengthening of the State and District Disaster Management Information Centres
- Resource Inventory database
- Urban earthquake vulnerability preparedness programme
- Strengthening training institutes
- Vulnerability and risk indexing

Sustainability of the programme:

- All disaster preparedness and mitigation plans to be approved by Village Panchayat, Block and Urban Local Bodies
- Disaster preparedness and mitigation plans to be an integral part of all developmental planning process
- DMCs and DMTs to conduct mock drill regularly to enhance preparedness
- Well-equipped and functional disaster management information system
- Adequate human resources capacity building
- Manuals and guidelines will be available for all emergency operations
- Availability of local trained masons in appropriate disaster-resistant housing technology.

Outcomes:

- Administrative and institutional framework at State level
- Enhanced capacity of all stakeholders
- Aware and informed community
- Comprehensive disaster risk management and mitigation plans in the programme states based on vulnerability and risk assessment of women and children towards natural disasters

- Enhanced capacity of DMTs in first aid, shelter management, water and sanitation and rescue evacuation
- Capacity building in earthquake risk management at national, state and city (Ward / Community) level, including strengthening of key resources institutions and establishing of linkages
- Earthquake preparedness and response plan for all the 38 cities across the country
- Well-equipped disaster risk management information centers at State and District Headquarters
- Manuals, training modules and awareness strategies
- Emergency kits at all selected districts
- Enhanced capacity of the training institutions
- Trained, skilled masons engineers for hazard-resistant housing
- Technology demonstration units for dissemination
- Knowledge network for enhanced involvement of stakeholders
- National and State database on natural disaster risk management
- Vulnerability and risk reduction reports.

Annexure – II

A step towards Urban Earthquake Vulnerability Reduction

National Disaster Risk Management Programme

The GOI – UNDP National Disaster Risk Management Programme (NDRM) is a national initiative aimed to reduce vulnerabilities of communities in some of the most hazard-prone districts (169 districts in 17 states) of India. The programme (2002-2007) aims to contribute to the social and economic development goals of the National & State Governments, enable them to minimize losses of development gains by reducing their vulnerability to natural disasters.

Six districts namely Cuddalore, Tiruvallur, Nagapattinam, Kanniyakumari, Kancheepuram, The Nilgiris, and two cities of Chennai and Coimbatore in Tamil Nadu have been taken under this NDRM initiative. Revenue Administration, Disaster Management & Mitigation Department will implement the Programme in Tamil Nadu.

UEVR Project – Background

The Urban Earthquake Vulnerability Reduction Project is a sub-component of the NDRM. Earthquake is a natural event which may cause tremendous loss of life and property damage. One of the major challenges facing our country is to reduce the vulnerability to this uncontrollable and unpredictable hazard by having a greater understanding about its causes and effects and also by adopting suitable preparedness and mitigation measures.

Seismic Map of Tamil Nadu (Zone II & III)

As per the latest seismic zoning map of India brought out by the Bureau of Indian Standards, 59% of land area of the country is prone to seismic intensity of MSK VII or more during earthquakes. Some of the most intense earthquakes of the world have occurred in India, but fortunately, none of these have occurred in the vicinity of the major cities. India has highly populous cities including the national capital of New Delhi, located in zones of high seismic risk. Majority of the constructions in these cities are not earthquake-resistant. Thus any earthquake striking of one of these cities would turn into a major disaster.

It is most important in the medium and long term to formulate strategies to reduce the vulnerability to and losses arising from a possible earthquake striking any of these cities. Six significant earthquakes have struck different parts of India over a span of the last 15 years. Five of them occurred in rural or semi-urban areas and hence the damage in terms of human lives and property were relatively small. On the other hand, the 2001 Bhuj earthquake struck both rural and urban areas and reiterated the scale of vulnerability. If any of these earthquakes strike the populous urban centres, the damages would be colossal.

The Programme

The programme envisages strengthening capacities of communities, urban local bodies & the administration in mitigation, preparedness and response across 38 cities in the country having population above half a million and falling under seismic zone III, IV and V. Chennai & Coimbatore Cities (Seismic Zone III) are among the 38 cities taken under the UEVRP initiative. City Disaster Management Committee has been formed in these two cities with Commissioner, Corporation of Chennai & The District Collector, Coimbatore as the Chairman of the CDMC. The programme would demonstrate a suitable model for mainstreaming of earthquake risk management initiatives at all levels and help reduce seismic risk in the most earthquake-prone urban areas in India.

Past Seismic Events in Tamil Nadu

Date	Intensity	Location
28.01.1679	IV	Felt at Chennai
16.09.1816	IV	Felt at Chennai
29.01.1822	V-VII	Felt at Chennai, Vellore area
03.01.1859	VI	Felt at North Arcot Coimbatore District
03.07.1867	V	Felt at Villupuram
08.02.1900	VI-VII	Coimbatore, felt all over South India
07.01.1916	VI	Felt at Nilgris area
26.06.1941	VIII	Felt almost Coromandal Coast
29.02.1944	VI	Felt at Madurai, Coonoor area
07.02.1962	VI-VII	Coonoor area
29.07.1972	VI	Coimbatore area
07.07.1988	VII	Kambam, Madurai area
26.09.2001	VI	Chennai area

Goal

Sustainable reduction in earthquake risk in the most earthquake-prone urban areas across the country

Objectives

- Create awareness among government functionaries, technical institutions, NGOs, CBO's and communities about earthquake vulnerability and possible preventive actions.

- Development and institutionalizing of earthquake-preparedness and response plans and practicing these through mock drills.
- Development of regulatory framework (techno-legal regime) to promote safe construction and systems to ensure compliance.
- Capacity building for practicing engineers, architect, builders, contractors & other professionals dealing with emergency response.
- Networking knowledge on best practices and tools for effective earthquake risk management, including creation of information systems containing inventory of resources for emergency operations.

Outcomes

- Capacity building in earthquake risk management at national, state and city (ward / community) level, including strengthening of key resource institutions and establishing of linkages.
- Disaster management team formed at the city level along with sectoral preparedness plan for all nodal agencies in the Urban Local Body.
- An aware and informed community.
- Integration of seismic risk management into development programmes.
- Enhanced capacity of the practicing engineers / architects & the training / academic / resource institutions.
- Review of enforcement mechanisms for the byelaws etc.
- Replication of the programme to other urban centres across the state.

Earthquakes don't kill people. Unsafe buildings do.

Three points to remember

- If you are building a house you can build for safety
- If you are living in a house / flat you can improve its safety
- If you are looking for a place to stay you can look for safety

Before

- Insist upon earthquake resistant features while constructing / buying a house / flat. Ensure construction complies with building byelaws and BIS Codes.
- Consult an engineer / architect for retrofitting your house to make it earthquake resistant.
- A common meeting point inside the city and a contact outside the city should be identified and known to all members of the family.
- Keep a list of important telephone numbers, torch, transistor, first-aid kit, water and non-perishable food at a designated place as a family emergency kit always ready.

- Train yourself in basic first aid. Form teams for First-aid, Search & Rescue etc. in your area and conduct preparedness drills for what to do in case of an earthquake.

During

- Keep calm and help others to keep calm. Do not panic.
- If you are at home or inside a building:
  - don't try to run out of the building, protect yourself by ducking under a sturdy table or a bed and stay there until the shaking stops.
  - Turn off electricity and gas.
- If you are on the road in a built up area:
  - immediately move away from buildings, slopes, streetlights, power lines, hoardings, fly-over etc. towards open spaces.
  - Do not run or wander haphazardly.
  - Keep the roads free for movement.
- If you are driving:
  - stop the vehicle away from the slopes, buildings and electric cables; come out of the vehicle, hold it and stay by its side.

After

- Keep calm and expect aftershocks. Check if you or anyone else is hurt.
- Use first aid and wait for medical help.
- Do not move seriously injured people.
- Do not turn-on electrical appliances and gas.
- Do not spread rumors and don't panic.
- Do not waste water and do not block telephone lines.
- Keep the streets clear for emergency services.
- Volunteer to help.

Bureau of Indian Standards (BIS) has the following seismic codes:

IS 456, 2000, Indian Standard Code of Practice for Plain and Reinforced Concrete (4^{th} Revision)

IS 800, 1984, Indian Standard Code of Practice for General Construction in Street (2^{nd} Revision)

IS 875, Code of Practice for Design Loads (Other than earthquake) for Buildings and Structures.

IS 1893 (Part I), 2002, Indian Standard Criteria for Earthquake Resistant Design of Structures (5^{th} Revision)

IS 4326, 1993, Indian Standard Code of Practice for Earthquake Resistant Design and Construction of Buildings (2^{nd} Revision)

IS 13827, 1993, Indian Standard Guidelines for Improving Earthquake Resistance of Earthen Buildings.

IS 13828, 1993, Indian Standard Guidelines for Improving Earthquake Resistance of Low Strength Masonry Buildings.

IS 13920, 1993, Indian Standard Code of Practice for Ductile Detailing of Reinforced Concrete Structures Subjected to Seismic Forces.

IS 13935, 1993, Indian Standard Guidelines for Repair and Seismic Strengthening of Buildings.

Chapter XII

ENVIRONMENT

Sustainable cities are fundamental to social and economic development. As stated in the tenth plan document of the National Planning Commission, sustainability is not an option but imperative. For a better world to live in, we need good air, pure water, nutritious food, healthy environment and greenery around us. Without sustainability, environmental deterioration and economic decline will be feeding on each other leading to poverty, pollution, poor health, political upheaval and unrest. The environment is not to be seen as a stand-alone concern. It cuts across all sectors of development. We have to improve our economic growth rate, provide basic minimum life support services to large section of our population and deal with the problems of poverty and unemployment. At the same time, we have to pay attention to conserving our natural resources and also improving the status of our environment.

12.02 Environmental deterioration is not a necessary or inescapable result of urbanization; what needs to be done is striking a right balance - in making development in such a way that they are more effectively attuned to environmental opportunities and constraints.

12.03 The metropolitan environment can be looked comprising of mainly two components viz. (i) environment per se, and (ii) the habitat. The environment per se relates to natural features and resources including the air, noise, water and land (open spaces, forests etc.). The habitat is related to built environment and infrastructures such as water supply, sewerage and solid waste disposal.

12.04 The conservation of natural resources includes management of air, noise, water & land.

Air:

12.05 Air pollution is a matter of concern in metropolitan cities. Increasing urban activities mainly the industrial and transportation have resulted in increased emission into the air and the Tamil Nadu Pollution Control Board (TNPCB) has identified the major source of air pollution in Chennai is the emissions from vehicles.

12.06 Under the Air (Preservation & Control of Pollution) Act, 1981 and in the Environment (Protection) Act 1986, the National Standards for ambient air quality have been notified.

Table No: 12.01 National Ambient Air Quality Standards				
Pollutant (micrograms per m^3)	Time-Weighted Average	Concentration in Ambient Air		
		Industrial	Residential	Sensitive
Sulphur Dioxide (SO_2)	Annual Average 24 Hours	80 120	60 80	15 30
Oxides of Nitrogen as NO_2	Annual Average 24 Hours	80 120	60 80	15 30
Suspended Particulate Matter (SPM)	Annual Average 24 Hours	360 500	140 200	70 100
Respirable particulate matter (Size less than 10 micron)	Annual Average 24 Hours	120 150	60 100	50 75
Lead	Annual Average 24 Hours	1 1.5	0.75 1	0.5 0.75
Carbon Monoxide (in mg/m^3)	8 Hours 1 Hour	5 10	2 4	1 2

Source: TNPCB

12.07 The monitoring of air quality in Chennai is undertaken by TNPCB in their own Ambient Air Quality Monitoring Programme (CAAQM) and also under the National Ambient Air Quality Monitoring Programme.

12.08 TNPCB in its Environment Management plan for Chennai city, 2003 has identified that the major pollutant generated in the City are the particulate matter, sulphur dioxide, oxides of nitrogen, carbon monoxide, hydrogen sulphide, and ammonia gas. The major sources of air pollution are domestic (fuels for cooking), commercial (fuel consumed by commercial establishments, trade, industry, hotels etc.), industrial (due to wood, coke, furnace oil LPG, kerosene etc.) vehicular (petrol & diesel fuels), generator sets (diesel and kerosene fuels), natural sources (odour pollution due to gases emanated from polluted stretches, waterways – 'B' Canal, Adyar, Cooum). The TNPCB estimation of the pollution load in different sectors in Chennai City is given in Table No.12.02.

Table No: 12.02 Calculated Pollution Load in Different Sectors in Chennai City							
Sources	Pollutant (T/day)						
	SPM	SO_2	NO_2	HC	CO	Total	Percentage
Domestic	0.032	0.170	1.049	0.101	0.243	1.865	2.65%
Commercial	0.875	1.466	0.731	0.120	0.087	3.279	4.66%
Gen Sets	0.296	0.509	0.268	0.039	0.026	1.138	1.61%
Industrial	2.510	4.565	6.085	0.3119	0.4320	13.9039	19.78%
Vehicular	9.300	0.200	10.250	10.240	20.100	50.090	71.28%
Total (T/d)	13.280	6.910	18.380	10.810	20.880	70.260	100

12.09 The major contribution to Chennai air pollution load is vehicular sector (71.28%) followed by industrial sector (19.70%). TNPCB has mapped the air pollution impact areas and listed the impact areas as follows:

Impact Areas	Areas
Impact Area I (high)	T. Nagar, Part of Saidapet, Choolaimedu, Vadapalani, Royapettah, Egmore, Part of Adyar, Tenampet, some part of Tondiarpet, Koyambedu, Purasawalkam, Mandaveli, Mylapore, George Town
Impact Area II (medium)	Korattur, Anna Nagar, Velachery
Impact Area III (low)	Other than above areas

The following are the observations of TNPCB:

(i) Out of the 8 monitoring points, 4 points recorded low, 2 points medium and 2 points high rating.

(ii) The vehicular sector is the major source of air pollution in Chennai City (followed by the industrial sector).

(iii) In addition to the pollution caused by the vehicles, the untidy roads and gathering of sand on the sides of the roads causes rise of dust.

Water pollution:

12.10 Chennai is traversed by three rivers (viz. Kosasthalaiyar, Cooum and Adyar), B' canal and other natural and man-made canals & drains. The waterways of Chennai are not perennial in nature and receive flood discharge only during monsoon season; in the rest of the year it acts as a carrier of wastewater from sewage treatment plants and others.

12.11 The TNPCB monitors the discharge of sewage and trade effluents generated by local bodies and industries into the water bodies / waterways. The basin-wise sewage generation during the year 2000 is given in the table 12.03.

Table No: 12.03 Waterways Basin-wise Sewage Generation (Year 2000)

Sl.No.	Waterways Basin	Drainage Area (sq.km)	Sewerage Generation (MLD)
1	Adyar River	12	87
2	Cooum River	20	92
3	Buckingham Canal		
	North Buckingham Canal	10	83
	Central Buckingham Canal	5	46
	South Buckingham Canal	2	29
4	Captain Cotton Canal	10	47
5	Otteri Nullah	24	129
6	Mambalam Drain	6	19
	Total	89	532

Source: TNPCB

12.12 The outfalls identified by the TNPCB in waterways are given in table no.12.04.

Table No: 12.04 Source-wise Details of Wastewater Outfalls in Chennai City Water ways During year 1994 and 1999

Sl.No.	Outfalls source	No. of Outfalls in Waterways							
		River Adyar		River Coovum		Buckingham Canal		Otteri Nullah	
		1994	1999	1994	1999	1994	1999	1994	1999
1	Industries	20	11	18	1	14	13	13	4
2	Commercial Institutions	38	38	18	11	21	21	3	3
3	Sewage Treatment Plants	1	1	1	1	1	2	-	1
4	Sewage Pumping Stations	4	1	2	-	9	4	2	1
5	Sewer/Storm Water Drain Overflows	148	147	281	276	63	64	43	43
6	Discharges from Slum settlements	17	17	24	24	20	19	5	5

Source: TNPCB

12.13 However, the recent data collected by TNPCB during 2003 has shown that certain industries have plugged their outfalls and provided effluent treatment plants (ETP) and were using the treated trade effluent inside their premises itself. Pollution Control Board observed that the institutions such as slaughter houses (2 nos.), Central Railway station, water treatment plant (at Kilpauk) and S.T.P's contribute a major quantity of pollution level to the waterways.

12.14 TNPCB under the MINARS programme periodically monitors the water quality of the city waterways. Water samples are collected and analysed by TNPCB every month at 'B'canal (at north, central and south stretches), Otteri Nallah, Adyar River and Cooum River. According to TNPCB, all these water bodies in the City are polluted and not suitable for any designated uses (viz. drinking, bathing, propagation of wild life like animal husbandry & fisheries, industrial, cooking and washing and agriculture); level of contamination is relatively high in 'B' canal followed by Otteri Nullah and Cooum River.

Flood Alleviation Project, 1998

12.15 Government of Tamilnadu have sanctioned a project with an outlay of Rs.300 Crores for Macro and Micro drainage improvements to alleviate flood problems in Chennai Metropolitan Area which was implemented by PWD, Chennai Corporation and Tamilnadu Slum Clearance Board.

Chennai City River Conservation Project (CCRCP), 2000

12.16 In order to improve the conditions of waterways in Chennai a comprehensive package of projects with an estimated outlay of Rs.1700 Crores was prepared. The projects proposed included sludge removal and disposal from waterways, relocation of slums and encroachments, structural works and strengthening of waterway banks, improvement of macro drainage network in the catchments, improvements of micro drainage network in the City, improving the water quality of rivers and waterway, strengthening urban drainage network in the City, construction of sewage flow interceptors and treatment facilities. It was posed to Government of India for funding under National River Conservation Programme (NRCP). Out of the proposed package of projects the Government of India had approved in 2000 the schemes for interception, diversion and treatment in Chennai City at a cost about Rs.491.52 Crores under NRCP and it was taken up for implementation by CMWSSB. In addition, the remaining works to the cost of Rs.228.63 Crores was also taken up by CMWSSB to improve environment.

Ground Water:

12.17 Chennai is underlain by various geological formations from ancient Archaeans to recent Alluviums. It can be grouped into three viz. (i) Archaean Crystalline Metamorphic rocks (ii) Upper Gondwanas comprised of sandstones, siltstones and shoals, tertiary (Eocene to Pliocene) sandstones and (iii) coastal and river Alluvium.

12.18 Central Ground Water Board has taken up the task of long term monitoring of ground water levels and quality. The State PWD and CMWSSB are also monitoring the ground water level fluctuations.

12.19 The agencies have observed that the chemical quality of ground water in Chennai City is generally brackish and not suitable for drinking purposes. In general it is alkaline with pH value from 7.8 to 9.0 and many pockets have high chloride and sulphate; very few selected pockets have potable quality at Besant Nagar, Greenways Road, Nungambakkam, Kilpauk etc. and also good fresh water aquifer is found in the stretch between Thiruvanmiyur and Uthandi along the coast. In areas like K.K. Nagar, Ashok Nagar, Sastri Nagar, Mylapore, Anna Nagar etc. excess iron has been found resulting in reddish colour of water, chocking pipes with yellowish - brown precipitate and also disagreeable taste.

12.20 The quality changes due to seawater intrusion in the past are evident in Triplicane, Mandaveli and other areas along the coast. Mandatory provision of rainwater structures within the City has improved the recharging potential for the ground water and also the water quality and ground water table in the recent past.

Noise Pollution:

12.21Under the Air (Prevention & Control of Pollution) Act, 1981, noise is a pollutant. Noise is described as an unwanted sound that produces deleterious effect on health and affects the physical and psychological well-being of the people. In the recent times, public concern about noise pollution also increased.

12.22 Ambient air quality standards in respect of noise for different use zones notified by TNPCB under the Environment Act is given in the Table No 12.05.

Table No.12.05 Ambient Noise Standards

Sl.No.	Zone	Limits in decibels	
		Day	Night
1	Industrial area	75	70
2	Commercial area	65	55
3	Residential area	55	45
4	Silence zone	50	40

Source: TNPCB

12.23 The noise level survey conducted by the TNPCB reveals that noise level exceeded the limits mostly in commercial areas, mainly due to vehicular movement.

12.24 During festive seasons in Chennai, the noise levels were noted high and particularly during Diwali it exceeded 120 dB (A).

Natural features:

12.25 Sea coast and beaches are a gift of nature to mankind. The coastal line of Bay of Bengal in the east throughout its length bound Chennai Metropolis. We are bound to maintain the aesthetic qualities and recreational utility of the same. Conservation of this gifted coastline may require application of precautionary principles and also the principles of sustainable development and inter-generational equity. The precautionary principle requires the government agencies to plan and prevent the environmental degradations particularly by non-conforming developments in this area. Prudent use of the natural resources sustaining economic growth and preserving the cultural and national heritage inherited from the previous generation for the next generation is also required to be done.

12.26 Earlier, the developments in the coastal stretches were regulated under the building rules of the local bodies concerned and the development control rules of planning authorities. The first Central Government initiative came when the then Prime Minister Tmt. Indira Gandhi wrote to the state Chief Ministers as follows:

> " I have received a number of reports about the degradation and mis-utilisation of beaches in our coastal states by building and other activity. This is worrying as the beaches have aesthetic and environmental value as well as other uses. They have to be kept clear of all activities at least upto 500 metres from the water at the maximum high tide. If the area is vulnerable to erosion, suitable trees and plants have to be planted on the beach sands without marring their beauty.
>
> Beaches must be kept free from all kinds of artificial development. Pollution from industrial and town wastes must be also avoided totally.
>
> Please give thought to this matter and insure that our lively coastline and its beaches remain unsullied."

12.27 The Government of India in their notification dt.19.11.1991 under the Environment (Protection) Act, 1986 declared the costal stretches of seas, bays, estuaries, creeks, rivers and back-waters which are influenced by tidal action (in the landward side) upto 500 m. from the High Tide Line (HTL) and the land between the Low Tide Line (LTL) and HTL, as Coastal Regulation Zone; and imposed restrictions (stated therein) on setting up and expansion of industries, operations or process etc. in the said CRZ. It was also stated that the distance from the HTL shall apply both side in the case of rivers, creeks and backwaters and may be notified on a case by case basis for reasons to be recorded while preparing the CRZ management plans and however, this distance shall not be less than 100 m. or the width of the creek, river or backwater whichever is less.

12.28 In CMA, developments in the coastal stretches are regulated as per the CRZ management plan approved by the Government of India on 27.9.1996 and the CRZ regulations notified by the Government of India under the Environment (Protection)) Act.1986.

Rivers:

12.29 Rivers which traverse through the CMA, are Adyar, Cooum and Kosasthalaiyar and these rivers are monsoon-fed; as stated (in earlier parts of this chapter), in the remaining period it accommodates only the wastewater flows from the STPs and urban areas. Pollution levels in these waterbodies needs to be monitored and needs to be reduced to the level that it could be used for recreational purposes, in the longer term. Measures need to be taken include, eviction & rehabilitation of encroachments, strengthening of banks and other structured measures for flood alleviation, desilting, introducing green cover on its banks, etc.

12.30 Chennai Metropolitan Area is also dotted with a large number of lakes; some of them encroached. They mainly act as flood accommodators during monsoon. These also require desilting, improvements and conservation.

12.31 A detailed study on conservation and exploring the feasibility of developing the waterbodies (including rivers, estuaries, creeks, swamps, lakes & large ponds) as recreational areas may be done, planned and implemented.

Green Areas:

12.32 Chennai City has only about 2% of the area as declared parks. In Chennai Metropolitan Area, the declared forest cover is about 24 sq.km, which is about 2 percent of the CMA area. However, satellite imageries show that green cover over the City due to trees along roadside and within the sites is of considerable extent. There is ample scope for further development of this green cover within the City and also in the rest of CMA, particularly along roads drains, riverbanks etc. Increase in green cover in urban habitats becomes necessity not only to alleviate the problems of pollution, but also to ensure ecological stability.

Chapter-XIII

INFRASTRUCUTRE INVESTMENTS FOR CMA

A. Chennai Metropolitan Development Plan (CMDP)

Cities play a vital role in social and economic development. Efficient and productive cities are essential for national economic growth and equally, strong urban economies are essential for generating the resources needed for public and private investments in infrastructure, social facilities, improved living conditions and poverty alleviation.

13.02 Environmental degradation holds back economic development. The urban challenge put forth before us is to improve the environmental quality and make the city livable. The level of services have to be improved by better planning, management and maintenance of the services, which could be achieved by channalising the investments in the right directions.

13.03 Every citizen has his/her own perception about city where one lives. The benefits to the citizens of CMA because of the proposed Infrastructure Investment Plan named as CMDP are:

- Improvement in the quality of life
- Higher standard of living
- Improvements in service level
- Eco-friendly environment
- Reduced pollution levels
- Less congested roads and better parking facilities
- Hassle free travel by public and private transport and
- Better employment opportunities by attracting more industries/ commercial establishments.

Benefits to The Investors

13.04 In the recent past Chennai has attracted many industries including the Information Technology Industries. Chennai has many natural advantages including availability of a Sea Port and other infrastructure attracting the large-scale industries. The strength of Chennai City is mainly the availability of good infrastructure facilities, skilled labour and availability of land for developments.

13.05 CMDA considering the need for making investments for infrastructure developments in a planned and co-ordinated manner, has proposed infrastructure investment plan called as “CMDP” in consultation with the department agencies and local bodies concerned.

13.06 The major objectives of the Plan are:

i. to identify the infrastructure needs in each sector and work out the projects accordingly;
ii. to identify the quantum of finance for execution of these projects;
iii. to identify the source of finance;
iv. to prioritize the projects by working out the annual programme, medium term programmes and long term programmes;

13.07 In Chennai Metropolitan Area the development and maintenance of urban services are vested with the specialised agencies. CMWSSB, TWAD and concerned local bodies provide water supply electricity by TNEB, solid waste management by the concerned local bodies, roads by Highways and local bodies, storm water drains by local bodies and PWD. These agencies depend on Central Government, State Government and financial institutions for funding their projects. The cost of the projects to be executed in various sectors estimated to be of several crores and these projects cannot be implemented with the State Govt. budgetary support alone; additional resources have to be mobilized.

13.08 The infrastructure requirements for the Chennai City, 16 Municipal areas, 20 TP, Panchayats and 214 village Panchayats and 10 Panchayat unions which lie within Chennai Metropolitan Area have been collected from the concerned agencies and the Investment Plan for them has been prepared.

13.09 The Investment Program has been categorized into 3 categories viz. annual, medium and long-term projects. The projects that need immediate investment have been included in the annual program. The projects which have to be executed to fulfill the infrastructure requirements in the medium term have been included in the medium term programmes. Projects which require heavy investments with long gestation periods have been included in the long term programmes.

Table No.13.01 Investment Plan			Rs. in Crores
Programme/Agency	**Annual Plan (2003-2004)**	**Medium Term Plan (2004-2007)**	**Long Term Plan**
Traffic and Transportation			
Urban Rail Projects	165.00	519.24	3528.00
Road/River Bridge	3.00	7.00	-
Traffic Management	11.94	3.00	-
Resurfacing of Major Roads	43.11	-	-
Strengthening of peri urban Roads	4.54	13.62	4.56
ORR	-	250.00	250.00
Widening & Strengthening of city roads	57.80	-	-
New Road formation	21.50	-	350.00
Relaying of City Roads	99.00	297.00	-
Study for Augmenting the capacity of Major Arterial Roads	0.25	3.55	-
Widening & Strengthening	-	1595.58	-
Cement Conc. - Anna Salai	-	101.32	-
Road Over/Under Bridge	-	832.71	-
Bus Terminal	-	6.00	-
Multilevel Parking	-	48.00	-
Urban Road Projects	-	-	1126.00
Urban Transit System	-	-	855.00
Truck Terminal	-	80.00	-
Sub Total	**406.14**	**3757.02**	**6113.56**
Housing			
TNHB	5.00	570.00	-
TNSCB	27.70	330.41	325.00
Sub Total	**32.70**	**900.41**	**325.00**
Water Supply			
CMWSSB	**730.00**	**946.00**	**649.18**
Sewerage			
CMWSSB	295.00	607.00	897.78
CMA (Thiruvottiyur Mpty)	28.00	-	-
Sub Total	**323.00**	**607.00**	**897.78**

Storm Water Drain			
Chennai Corporation	40.00	47.00	-
PWD	44.65	3.65	-
Highways	3.68	-	-
SIDCO	0.68	-	-
CMA	13.77	-	-
DTP	8.94	10.59	-
Sub Total	**111.72**	**61.24**	-
Solid Waste Management			
Chennai Corporation	16.00	35.00	-
CMA	11.19	16.25	-
Sub Total	**27.19**	**51.25**	-
Electricity			
TNEB	**150.00**	**1910.81**	-
Total	**1780.75**	**8233.73**	**7985.52**
Grand Total			**18000.00**

13.10 The main sources of finance identified are loans from financial institutions, budgetary support/grant from government. Public-private participation is also proposed for the housing, water supply and sewerage projects.

13.11 Based on this overall plan, annual plans are formulated in consultation with the sectoral agencies / departments, Government budgetary allocations made and implemented as Annual Plans since 2003-04. CMDA monitors the progress of implementation and report to the Government. The investment made since 2003 sector-wise under CMDP is given below.

Annual Programme 2003-04, 2004-05, 2005-06 & 2006-07				**(Rs. in Crores)**
Programme	Investment in 2003-04	Investment in 2004-05	Investment in 2005-06	Investment in 2006-07
Traffic & Transportation	412.16	237.28	179.48	199.28
Housing	40.77	65.17	65.27	68.91
Water Supply	581.60	309.03	99.61	92.67
Sewerage	83.16	122.88	66.23	49.58
Storm Water Drain	38.97	40.39	103.10	17.71
Solid Waste Management	-	3.695	6.06	9.94
Electricity	164.88	179.21	211.32	188.52
Others	-	-	3.36	0.44
Total	1321.54	957.66	795.58	670.23

B. Jawaharlal Nehru Urban Renewal Mission (JN-NURM)

13.12 The Government of India in 2005 has introduced the scheme called Jawaharlal Nehru Urban Renewal Mission (JN-NURM) to encourage reforms driven, fast tract planned development of identified cities with a focus on efficiency in urban infrastructure / service delivery mechanisms, community participation and accountability of urban local bodies towards citizens.

13.13 The Mission objectives are to ensure the following in the urban sector including the basic services to urban poor:

a. Integrated development of infrastructure services in cities covered under the Mission.
b. Establish linkages between asset creations and asset management for long term project sustainability.
c. Ensure adequate funds to fulfill deficiencies in the urban infrastructural services
d. Planned development of identified cities including peri urban areas, out growths and urban corridors leading to dispersed urbanisation.
e. Scale up delivery of civic amenities and provision of utilities with emphasis on universal access to urban poor.
f. Special focus on urban renewal program for the old city areas to reduce congestion.
g. Provision of basic services to urban poor including security of tenure at affordable prices, improved housing, water supply and sanitation and ensuring delivery of other already existing universal services of the Government for education, health and social security.

13.14 The Mission strategy includes preparing perspective plan called as City Development Plan (CDP), preparing projects, leveraging of funds and incorporating private sector efficiencies. The duration of the Mission is 7 years beginning from the year 2005-06. The nodal agency for the State of Tamilnadu is TUFIDCO and the concerned department at Government of Tamilnadu is Municipal Administration and Water Supply. Chennai is one of the identified cities for Government of India assistance under JN-NURM. To comply with the prerequisite of preparation of CDP, the City development plan for Chennai as called as 'Development Plan for Chennai Metropolitan Area' was prepared and Government of India's approval obtained.

13.15 Strengths-weakness-opportunities and threats in respect of Chennai are given below:

Strengths	**Opportunities**
Strong Commercial and Industrial Base	High telecom penetration
Skilled and educated man power	Growth oriented reforms
High standard educational and health institutions	Public Private Participation
Good urban land market and availability of developable lands.	
Uninterrupted and quality power supply	
Weaknesses	**Threats**
Traffic congestion	Automobile pollution
Inadequate infrastructure	Overcrowding in certain pockets
Water shortage	Decrease in manufacturing industry

Institutional Arrangements in Development Planning

13.16 The rapid urban population growth and consequent pressure on the developments in major urban centres have led the Government of Tamil Nadu to set up specialised agencies to tackle the urban development issues in various sectors such as housing, water supply and sewerage etc. The principal ones are CMWSSB, TNSCB, and TNHB. The Chennai Metropolitan Development Authority was set up in 1973 with the main objective of being in charge of long term planning and co-coordinating the various urban development activities in the Metropolis.

13.17 The agencies involved in the infrastructure planning and development in CMA are presented below. The details of agencies and its responsibilities are presented in the table no.13.02 below.

<table>
<tr><th colspan="4">Table No.13.02 The Details of Agencies and its Responsibilities</th></tr>
<tr><th>Sl.No.</th><th>Agency</th><th>Responsibility</th><th>Jurisdiction</th></tr>
<tr><td colspan="2">Local Government</td><td></td><td></td></tr>
<tr><td>1</td><td>Chennai Corporation-</td><td>Provision of roads, construction of ROBs, RUBs, pedestrian subways etc., streetlights, solid waste collection and management, micro-drainage, parks and play grounds in their area of jurisdiction</td><td rowspan="4">Within the local body area</td></tr>
<tr><td>2</td><td>Municipalities-</td><td rowspan="3">Provision of roads, construction of pedestrian subways etc., streetlights, solid waste collection and management, micro-drainage, parks and play grounds in their area of jurisdiction</td></tr>
<tr><td>3</td><td>Town Panchayats</td></tr>
<tr><td>4</td><td>Village Panchayats</td></tr>
</table>

Parastatal Agency			
5	TNHB	Neighbourhood development including provision of plots and ready built houses, Sites and Services schemes.	Tamil Nadu State but focus is more on CMA
6	MTC	Bus transport	CMA
7	Traffic Police (Greater Chennai)	Traffic management schemes	Greater Chennai
8	TNEB	Electricity generation and supply	Tamil Nadu State
9	PWD	Implementation & maintenance of macro drainage system	Tamil Nadu State
10	CMDA	Urban planning, co -ordination of project implementation	CMA
11	CMWSSB	Water supply & sewerage facilities for CMA	CMA*
12	TNSCB	Provision of housing, infrastructure and livelihood programs in slum areas	Tamil nadu State but focus is more on CMA
13	Highways Department	Major roads within Chennai City, all bus route roads and major district roads, construction of ROBs, RUBs, pedestrian subways etc.	Tamil nadu State

* Though CMWSSB has jurisdiction over the CMA as per their Act, their area of operation are limited presently to Chennai City Corporation area and a few adjoining areas such as Mogappair, I.T.Corridor etc. However they have proposed to expand their area of operation covering the entire CMA.

13.18 Though the geographical areas of jurisdiction of these agencies overlap, their operational jurisdictions and functions are well defined. For example in the Chennai City Municipal Corporation Area, CMWSSB is solely in charge of water supply and sewerage; in respect of National Highways and roads except for the major roads handed over to State Highways for maintenance, others are maintained by the Chennai Municipal Corporation. In respect macro drains such as rivers and canals the PWD is responsible of maintenance; the micro drains are provided and maintained by the City Corporation of Chennai and other local bodies within CMA.

13.19 Capital investment plan for the various sector proposed under JN-NURM are given below.

Water Supply

13.20 The proposed investments for water supply projects for CMA is of the order of Rs. 6,321.00 Crores and the break up is presented in Table 13.03.

Table No. 13.03 Proposed Water Supply Investments in CMA			*Rs.in Crores*
Component	**Activity**	**Institution**	**Total**
Planning, Reforms and Institutional Strengthening	Comprehensive Water Sector Development Master Plan	CMWSSB	4.00
	Leak Detection Studies		9.00
	Energy Audit Studies		10.00
	Water Quality Studies and Monitoring		4.50
	Design and Implementation of Communication Strategy		7.00
	Baseline Survey		2.00
	Human Resources Development		7.00
	Establishment of Regulatory Authority		1.00
	GIS Mapping of Water and Sewerage Utility Mapping		9.00
	Modernizing Financial Management and MIS		7.00
	Community Initiatives Support		3.50
	Sub-total		**64.00**
Service Delivery	Source Augmentation		3,600.00
	Distribution Network Extension		800.00
	Rehabilitation and Modernisation of Existing Network		1,150.00
	Energy Conservation Measures		70.00
	Metering		210.00
	Piloting 24x7 zone by zone		350.00
	GIS Mapping & Refurbishment		70.00
	Sub-total		**6,250.00**
Citizens Relations Management	Citizen feedback mechanisms established		7.00
	Sub-total		**7.00**
	Grand Total		**6,321.00**

Sewerage

13.21 The proposed investments for sewerage projects for entire CMA is of the order of Rs. 2,299.00 Crores and the break up is presented in Table 13.04.

Table No. 13.04 Proposed Sewerage Investments in CMA			*Rs in Crores*
Component	**Activity**	**Institution**	**Total**
Design & Implementation of Communication Strategy	Comprehensive Sewerage Master Plan	CMWSSB	4.00
	Energy Audit		7.00
	Sewerage Quality Studies and Monitoring		7.00
	Design & Communication Strategy		7.00
	Human Resources Development		7.00
	Community initiative Support		7.00
	Sub-Total		**39.00**

Service Delivery (Sewerage)	Network coverage	CMWSSB	950.00
	Remodeling of old sewers		550.00
	Construction of STPs		550.00
	Recycling Plant & Reuse		175.00
	Energy Conservation Measures		35.00
	Sub-Total		**2,260.00**
	Grand Total		**2,299.00**

Storm Water Drainage

13.22 The proposed investments for storm water drainage, macro drainage for CMA and renovation of lakes is estimated Rs. 1,423.85 Crores and the break up is presented in Table 13.05.

Table No. 13.05 Proposed Storm Water Drainage Investments in CMA

Item	Rs. In Crores
Macro Drainage	865.00
Micro Drainage	
Chennai Municipal Corporation	252.00
Surrounding Municipalities	72.34
Town Panchayats	38.05
Village Panchayats	146.80
Renovation of Lakes	
Chennai Municipal Corporation	-
Surrounding Municipalities	9.91
Town Panchayats	3.36
Village Panchayats	1.57
Culvers	34.82
Total	**1,423.85**

Solid Waste Management

13.23 The proposed investments towards solid waste management sector for CMA is of the order of Rs. 847.81 Crores. The break up is presented in Table 13.06.

Table No. 13.06 Proposed Solid Waste Management Investments in CMA

Sl. No.	Solid Waste Management	Rs. in Crores
1	Chennai Municipal Corporation	808.70
2	Surrounding Municipalities	10.20
3	Town Panchayats	3.18
4	Village Panchayats	5.66
5	Slaughter Houses	20.07
	Total	**847.81**

Traffic and Transportation

13.24 The proposed investments for traffic and transportation for CMA is of the order of Rs. 24,854.08 Crores. The break up is presented in Table 13.07.

Table No.13.07 Proposed Transportation Investments in CMA		
Sl. No.	**Traffic & Transportation**	**Rs. in Crore**
1	Urban Transport Components	
2	Widening Strengthening and resurfacing of arterial, sub arterial and collector roads to at least 4 lane width	1,700.00
3	Elevated highways and Flyovers	4,655.00
4	Road/Rail crossings - ROB/RUB	150.00
5	River bridges	45.00
6	Link Roads	1,650.00
7	Concreting of Major roads	278.00
8	Junction improvements with white topping and Landscaping	75.00
9	Pedestrian sub-ways	102.00
10	Bus lay-byes & shelters	58.35
11	Utility duct and storm water drains along major roads	200.00
12	Peripheral road improvements	197.00
13	Construction of mini flyovers	48.00
14	Concreting of City-roads	95.00
15	Multi level car parking	145.00
16	Road/Rail Crossings - ROB/RUB	396.00
17	Nesapakkam Road widening	14.00
18	Road works including bridges/culverts	675.15
19	Road works	480.32
20	Metropolitan public transport (Bus)	1,300.00
21	Traffic management basic infrastructure needs	4,982.04
22	Street Lighting	8.22
23	Rapid Transit System	600.00
24	Metro rail (about 45 km)	7,000.00
	Total	**24,854.08**

Heritage and Recreation

13.25 The proposed investment towards Heritage and recreation projects is estimated Rs. 103.08 Crores and the break up is presented in Table 13.08.

Table No.13.08 Proposed Heritage and Recreation Investments in CMA		
Sl. No.	**Heritage**	**Rs. Crores**
1	Renovation/Development of Ripon buildings	18.00
2	Renovation/Development of V.P. Hall Building	5.00
3	Recreation Developments in City area	60.00
4	Recreation Developments in Municipalities	8.46
5	Recreation Developments in TP	3.24
6	Recreation Developments in Village Panchayats	8.37
	Total	**103.08**

Urban Basic Services for Poor

13.26 The proposed investments for urban poor population for entire CMA are estimated Rs. 3,887.23 crore for the next seven years and break up is illustrated in Table 13.09.

Table No.13.09 Proposed Investment for Urban Poor in CMA

Sl. No.	Basic Services for Poor	Rs. in Crore
1	Chennai Municipal Corporation	1,319.96
2	CMA (Except Chennai City)	536.27
3	Housing for LIG/EWS groups	2031.00
	Total	**3,887.23**

Parking Lots and Spaces

13.27 The proposed investment for civic amenities in CMA is estimated Rs. 43.85 Crores and the break up is presented in Table 13.10.

Table No.13.10 Proposed Civic Amenities Investments in CMA

Sl. No.	Parking lots / Spaces	Rs. In Crores
Parking Lots / Spaces		
1	Parking Lots / Spaces for TP	15.38
2	Parking Lots / Spaces for VP	28.47
	Total	**43.85**

Satellite Towns / Cities around CMA

13.28 The proposed investments for development of satellite towns/ cities around CMA as a decongestion measure during the next six years estimated is Rs.5000 crores.

13.29 The total estimated cost of the projects for proposed is Rs. 44,779.92 Crores. The summary of investments is given in the table below.

Table No. 13.11: Proposed Summary of investments in CMA *Rs. Crores*

Sl. No.	Component	Total Cost	JNNURM		
			GoI	GoTN	IR/IF
1	Water Supply	6,321.00	2,212.35	948.15	3,160.50
2	Sewerage	2,299.00	804.65	344.85	1,149.50
3	Solid Waste Management	847.80	296.73	127.17	423.90
4	Storm Water Drainage	1,423.88	498.36	213.58	711.94
5	Transportation	17,254.08	6,038.93	2,588.11	8,627.04
6	Mass Rapid Transit System	600.00	210.00	90.00	300.00
7	Metro Rail (45 km.)	7000.00	2450.00	1050.00	3500.00
8	Parking Lots and Spaces	43.85	15.35	6.58	21.92
9	Heritage and Recreation	103.08	36.08	15.46	51.54
10	Satellite Town development	5000.00	1750.00	750.00	2500.00
	Total	40892.69	14312.45	6133.9	20446.34
11	Urban Basic Services for Poor	3,887.23	1943.61		1943.62
	Grand Total	**44779.92**	16256.06		28523.86

Chapter XIV

LAND USE AND PLANNING STRATEGY

With the increase and concentration of population in urban areas, urban problems have increased. It requires and continues to require restriction in respect of the use and occupation of land in urban areas. In order to regulate the growth of the metropolitan area in an orderly manner and also to ensure its economic viability, social stability and sound management for the present and the foreseeable future, the Master Plan with zoning and development regulation is necessary.

14.02 The idea of zoning is that the segregation of certain uses from others reduces the effect of negative externalities, which some uses have on others. Zoning provides spatial segregation of conflicting uses. It also has the benefit of increasing positive externalities because many uses find an advantage in being grouped with other similar uses. These external effects include air and water pollution, excessive noise levels, traffic congestion, and aesthetic disamenities. Because of its predominant role, modern zoning encompasses expanded objectives for supplying certain public goods such as preservation of open space, prime agriculture land and ecologically sensitive areas also. Zoning is also desired on reduction of costs of providing certain public services.

Land use regulation in CMA prior to 1975:

14.03 Land use control in CMA prior to 1975 was not significant. Only about 20% of the Chennai City area, and less than 1% of the remaining CMA area (outside the city) were covered by the Detailed Town Planning Schemes prepared under the (now repealed) Madras State Town Planning Act, 1920. Under Section 89, of the Tamil Nadu Public Health Act, 1939 City Corporation of Madras and some of the Municipalities in CMA had declared residential areas (covering very small part of the local bodies concerned) and in such areas industries other than cottage industries were not permitted; shopping (except offensive trades) were not prohibited provided they did not use power. Industrial areas were declared under the Municipal Acts where industries existed then, and on individual application further industrial areas notified/expanded. These minimal land use regulations existed then orienting more towards preserving the then existed areas rather than channelsing the developments optimising the land and other resources in a planned way.

14.04 Land sub-division regulations prior to 1975 in CMA were limited to the control of making public and private streets under the local body Acts and they were not framed either to discourage such developments in such areas where it is not suitable or divert them to the areas where suitable; further they did not provide for necessary standards statutorily for roads, open spaces and other essential infrastructure.

Land use regulation under Master Plan for CMA, 1975:

14.05 The Master Plan for Chennai Metropolitan Area come into force from 5.8.75 from the date of notification of the Government consent for the plan (in G.O. Ms. No.1313, RD&LA Dept. dated 1.8.75); it was finally approved by the Government in G.O. Ms. No.2325, RD&LA Department, dated 4.12.1976. The Master Plan laid down policies and programmes for the overall development of the CMA. The land use plan designated the use to which every parcel of land in CMA could be put to. The land use plan was enforced through a set of regulations under Development Control Rules, which formed part of the master plan. Any person intending to make any development is required to apply under Section 49 of the Tamil Nadu Town and Country Planning Act, 1971, and obtain Planning Permission.

14.06 The break up of the then existed Land Use in 1973 is given in the Table below:

CMA - Existed Land Use - 1973

Land use	Chennai City		Rest of CMA		Total	
	Extent in Hect.	%	Extent in Hect.	%	Extent in Hect.	%
Residential	7788	44.46	9144	8.67	16932	13.77
Commercial	820	4.68	68	0.06	888	0.72
Industrial	893	5.10	2976	2.82	3869	3.15
Institutional	3045	17.38	2260	2.14	5305	4.31
Open space and Recreational	920	5.25	4822	4.57	5742	4.67
Agriculture			73689	69.88	73689	59.92
Non Urban			1633	1.55	1633	1.33
Others	4052	23.13	10865	10.3	14917	12.17
Total	17518	100	105456	100.00	122974	100.00

14.07 The land use zoning classifications as per the First Master Plan for CMA are (i) Primary Residential Use zone, (ii) Mixed Residential Use zone, (iii) Commercial use Zone, (iv) Light Industrial use zone, (v) General Industrial use zone, (vi) Special and Hazardous

Industrial use zone, (vii) Institutional use zone, (viii) Open Space and Recreational Use zone, (ix) Agriculture use zone and (x) Non-urban use zone. Further considering the character of (then) existed developments, the CMA was divided into three areas viz. (i) George Town and Continuous Building Area, (ii) Chennai City, Municipal and Township areas (excluding the areas mentioned in (i)) and (iii) rest of Metropolitan area.

14.08 The break up of the land use proposed in the First Master Plan (1975) is given in the table below:

Land use proposed as per First Master Plan

Land use	Chennai City		Rest of CMA		Total	
	Extent in Hectares	%	Extent in Hectares	%	Extent in Hectares	%
Primary Residential	10888.05	61.80	22554.94	22.27	33442.98	28.12
Mixed Residential	127.95	0.73	8095.69	7.99	8223.64	6.92
Commercial	521.95	2.96	970.81	0.96	1492.76	1.26
Institutional	3080.67	17.49	3395.39	3.35	6476.07	5.45
Light Industrial	263.11	1.49	285.73	0.28	548.84	0.46
General Industrial	934.26	5.30	3970.34	3.92	4904.60	4.12
Special & Hazardous Industrial	39.97	0.23	1440.39	1.42	1480.36	1.24
Open space & Recreational	1448.99	8.22	6848.67	6.76	8297.66	6.98
Agricultural	94.47	0.54	37084.34	36.61	37178.82	31.26
Non Urban	113.01	0.64	979.48	0.97	1092.49	0.92
Others (Forest, water bodies, Roads etc.,)	105.26	0.60	15672.65	15.47	15777.91	13.27
Total	17617.70	100.00	101298.42	100.00	118916.12	100.00

14.09 A few problems were faced in implementing the first Master Plan zoning and DCR; the major ones are given below:

(1) In certain areas where a site has been zoned for two or more uses (with reference to the abutting road and adjoining uses) an optimal development of single character could not be made taking into account the total extent of the site;

(2) In certain cases, though the existing activities and their performances are not hazardous or objectionable, they could not be allowed to expand even marginally because the zoning made in the Plan is not conforming to accommodate such activities;

(3) Sites zoned for public purposes such as open space and recreational use zone, institutional use zone, stand designated throughout the Plan period and the activities permissible in these zones are public/communal facilities mostly for which the Government/or Government agencies have to acquire and provide. Neither the planning authorities nor the Government or its agencies have acquired these lands and provided these facilities. (The reasons may be paucity of funds. Such zoning could have been restricted to mainly on Government lands, and to the private lands where this activity exist, or essentially required to be acquired for providing the same);

(4) Grouping of permissible activities made in zones such as institutions, open space and recreation are such that these appear so exclusionary, not even permitting residential activity, which is a lower use which a land owner/developer may like to put to at least. Such zoning could have been minimised in the midst of contiguous developed areas;

(5) According to the existing procedure, rezoning (land use variation) consumes a lot of time, with the result discouraged people to apply and get the site reclassified;

(6) Same way, Urban Land Ceiling Act (now repealed) had contributed to unauthorised sub divisions/layouts;

(7) The present zoning and DCR is criticised by a section of developers, for being negative rather than positive; it is more oriented towards the provision of safety, convenience and tranquility in the area rather than encouragement of housing activity, industries and employment;

(8) The standards set for plot extent; frontage etc., for residential developments was high when comparing the affordability of people even for middle-income group of the society.

Reclassification:

14.10 Reclassification of land uses on request from the land owners are received, examined and decided on individual merits of the cases under Section 32(4) of the Tamil Nadu Town and Country Planning Act. From 1976 to 1981, there were about 100 land use reclassifications within CMA, mostly in the areas of sanctioned Town Planning Schemes approved prior to 1975. The number of reclassifications made year wise within City and rest of CMA are tabulated below:

Table No. 14.01: Reclassifications within City and Rest of CMA, 1977-2004						
Sl.No	Year	City	Municipality	Town Panchayat	Panchayat Union	Total
1.	1977	2	4	--	--	6
2.	1978	1	1	--	--	2
3.	1979	--	--	1	--	1
4.	1980	--	6	4	3	13
5.	1981	--	--	--	--	--
6.	1982	17	13	11	20	61
7.	1983	11	9	5	10	35
8.	1984	23	8	55	36	122
9.	1985	44	5	12	29	90
10.	1986	18	5	16	19	58
11.	1987	17	4	8	14	43
12.	1988	14	5	14	64	97
13.	1989	2	1	2	13	18
14.	1990	27	3	19	56	105
15.	1991	30	11	11	59	111
16.	1992	17	11	9	42	79
17.	1993	22	7	8	38	75
18.	1994	22	6	9	22	59
19.	1995	18	14	20	26	78
20.	1996	22	4	5	11	42
21.	1997	17	1	7	25	50
22.	1998	34	9	13	45	101
23.	1999	18	2	4	29	53
24.	2000	19	1	7	10	37
25.	2001	4	--	6	11	21
26.	2002	12	3	8	19	42
27.	2003	10	5	17	25	57
28.	2004	14	2	16	55	87
Total		435	140	287	681	1543

14.11 The extent of sites reclassified ranged from single plot of about 150 sq.m. to more than 25 hectares. From the above table, it may be seen that in spite of sincere attempt of planners to assess the demand and estimate the directions of growth at the areas for development, the reclassification had to be resorted to, because of changing demands of public; of course within the objectives of zoning.

14.12 From the reclassification details given in Table No14.02 it may be seen that 47% of the cases reclassified from Agricultural use zone which indicates the expansion for urban developments and 30% reclassified from Primary Residential use zone which indicates the conversion into higher order use zones for commercial, industrial and institutional activities. Of the total number of reclassifications made, reclassification for residential accounts to 41%, for commercial activities 25%, industrial activities 19% and institutional 8%. As regards Chennai City, out of 435 reclassifications made 65%, 17%, 12% and 5% of the numbers were for commercial, residential, Institutional and industrial activities respectively. In the case of Village Panchayat areas, 53%, 29%, 7% and 6% of the reclassification were made for residential, industrial, institutional and commercial activities respectively.

Table No.14.02: Reclassifications in CMA from January 1977 to February 2005

Land use from \ Land use to	Primary Residential	Mixed Residential	Commercial	Light Industrial	General Industrial	Special & Hazardous	Institutional	Open Space & Recreational	Agricultural	Non-Urban	Total
PR	(EWS) 3	203	89	46	27	12	65	3	14	--	462
MR	7	--	11	18	7	2	3	--	6	--	54
COMM	4	--	--	1	8	--	1	--	2	1	17
LI	4	--	1	--	4	1	1	--	1	--	12
GI	17	2	1	--	--	--	4	1	5	--	30
S&H	5	1	--	--	--	--	--	--	--	--	6
INS	74	21	10	1	8	--	--	4	2	--	120
O&R	49	11	4	11	17	--	4	--	6	--	102
AGRI	471	30	10	95	47	11	36	8	--	17	725
NU	6	--	3	2	--	--	3	--	--	--	14
Total	640	268	129	174	118	26	117	16	36	18	1543

14.13 The Government in G.O. Ms. No.419, H&UD dated 1.06.1984 has delegated the power for reclassification of land uses in individual cases to the Authority in order to reduce delay.

Unauthorised Sub-divisions/Layout:

14.14 The Tamil Nadu Urban Land Ceiling Act came into force from 3rd August 1976. Fearing acquisition under the Act, people had resorted to large-scale unauthorised sub-divisions/layouts in 1975 and 1976. Unapproved layouts/subdivisions were taking place in the Chennai Urban agglomeration, even after the Act came into force, and the unauthorised layout developers were selling these plots as agricultural land evading the provisions of the Act. Lack of knowledge among the villagers in the rest of CMA about the regulations, profiteering motive of layout developers and also the delays in approval of layouts for various reasons also added to the growth of unauthorised layouts/sub-divisions.

14.15 Those unapproved sub-divisions/layouts are perennial problems faced by the public authorities. These unauthorised layouts contain substandard roads, which have not been properly laid out, and have no pavements, drains, culverts or streetlights provisions, and some of them made even on low-lying areas which are not suitable for residential developments. Further they are at scattered locations, to which the local bodies could not provide amenities economically. In many of these unauthorised layouts, purchase of plots mostly for speculation purposes had resulted in no or a few constructions adding security problems to those few households, who have occupied the scattered constructed houses. The policy of CMDA and the procedures adopted in CMA for regularisation of plots in such old unauthorised layouts, which have to be merged with the urban fabric at some point of time considering the public demand, are given in detail in Annexure XIV - A

14.16 The Tamil Nadu Urban Land (Ceiling and Regulation) Act was repealed in June 1999. It didn't have any marked effect on the proliferation of unauthroised layouts. Tamil Nadu Government reviewed the situation. As per Section 22A of the Registration Act, 1908, the State Government may by notification in Tamil Nadu Gazette declare registration of any document or class of documents as opposed to public policy. It also envisages that the registering Officer shall refuse to register any document, which is opposed to public policy. The Government have notified in G.O. Ms. No.150/CT Dept. dated 22.9.2000 certain categories of documents, registration of which is opposed to public policy; one among them is registration of documents relating to house sites in unapproved layouts. Unless a 'No Objection Certificate' issued by the authorities concerned viz. Corporation or Municipality or Town Panchayat or Panchayat Union or village Panchayat or CMDA, is produced before the registering officer, the document will not be registered. It was held by the High Court of Madras that the G.O. will have prospective effect and applicable only for the transfer of house sites by the landowner or promoter of the unapproved layout for the first time.

14.17 The above action of the Government curbed the new unauthorised layout developments after the year 2000, relieving the people and the public authorities from the problems of such new unauthorised layouts.

Development Control Rules:

14.18 The main purpose of the development control rules or the development regulation was to ensure that buildings are built.

(1) Achieving optimum utilisation of land within the infrastructure levels in an area (either within existing or proposed/ planned infrastructure levels) within the plan period;
(2) With adequate natural lighting and ventilation for healthy living;
(3) Within the desired/planned density of population (preventing negative social, economic and environmental impacts of high densities);
(4) With open spaces on ground to improve urban environment;
(5) With adequate space for parking within the plot itself;
(6) With aesthetics and taking into the requirements of urban design aspects such as solids and voids, views and vistas, and also breaking monotony;
(7) With aspects of safety, convenience and economy (within reasonable limits);
(8) With no or minimal negative externalities on the adjoining buildings/uses.

14.19 The development regulations also used as a tool to encourage amalgamation of smaller sites into larger ones for achieving higher FSI with provision of adequate space around the buildings and space for recreational purposes within the site, and also to discourage over-utilisation of road and other infrastructure, resulting in congestion, traffic delay, and choking and also breaking down of water supply or sewerage or electricity infrastructures.

14.20 Development standards prescribed for building constructions vary from country to country, region to region and city to city mainly for the reasons of local climatic conditions, public infrastructure availability/ planned, terrain, culture and way of life of the people, affordability of the people, and also the land availability for development.

14.21 In CMA, prior to 1975 (i.e. the Master Plan for CMA came into force), under Chennai City Corporation Building Rules, Tamil Nadu District Municipalities Building Rules, and Tamil Nadu Panchayat Building Rules, the building constructions were regulated. In respect of planning parameters, it contained a very few provisions viz. requirements of a site for construction, minimum extent for a residential plot, front set back requirement and height requirement. It dealt mainly about requirements of plinth level, foundation and other structural aspects of buildings, sizes of windows and openings for natural lighting and ventilation, corridor and passage width, head room height requirements, drains and sanitary convenience, and chimneys and flues. In 1974, the Multistoried and Public Buildings Rules under the Chennai City Corporation

Act and Tamil Nadu District Municipalities Act were notified which contained the provisions regarding the areas where multistoried buildings (i.e. buildings exceeding 4 storey or 15 mtr. in height) are permissible, setback requirements with reference to height of buildings, minimum plot extent, minimum frontage, minimum road width requirements. FAR, plot coverage and parking requirements.

14.22 In cases of Detailed Town Planning Schemes sanctioned under the Madras State Town Planning Act, 1920, the regulations for building constructions stated in the schedule of such sanctioned schemes were in force prior to 1975, and it contained regulations on minimum plot extent, street alignment and building line, side setback, rear setback, minimum road width and betterment levy requirements. It varied from scheme to scheme and covered only a dismal percent of the metropolitan area.

14.23 Development Control Rules for CMA formed part of the Master Plan came into force on 5.8.75. It contained detailed regulation on zoning related activity control, site requirements, plot extent, frontage, height, front, side and rear setbacks, minimum road width, plot coverage, FSI, parking, open space for recreational purpose reservation, public purpose sites reservation. These planning parameter requirements were prescribed differentially for various activities such as residential, commercial, institutional, cottage industrial, light industrial, general industrial, special and hazardous industrial activities, and also differed for the Chennai City area, rest of the CMA and the George Town and Continuous Building Areas; it further differed for Multistoried developments.

14.24 Planning and development control is a dynamic process. In 1975, for the first time such a comprehensive development control rule was brought in. It was reviewed within few years of starting implementation of the same and when found necessary it was amended. In 1979, amendments to DCR were made in respect of extent of open space recreation to be made for layouts, declaration of MSB areas, permissibility of projection in setback spaces, permissibility of cooling gas storage, public utility buildings, Government and semi-Government offices serving local needs, banks, cottage industries with 5 HP, craft centres, temporary touring cinemas in Primary Residential use zone, permissibility of certain activities in Mixed Residential use zone, height of commercial buildings, permissibility of certain categories of industries in Light Industrial and General Industrial use zones, horse power restriction in Light Industrial use zone, permissibility of religious buildings in Institutional use zones, in Non-urban and Agricultural use zones, certain specific provisions for sites and services and slum Improvement projects, economically weaker section housing exemption of certain floor areas from calculation of floor area ratio and plot coverage, additional conditions in Form-A and Form-B, parking standards in respect of hotels, etc.

14.25 In 1980, the Development Control Rules provisions were comprehensively reviewed and amendments made wherever necessary particularly in respect of plot extent for residential and commercial developments, plot frontages etc. In 1982, need for separate regulation for flatted residential and commercial developments felt and the DCR was amended including provisions for special buildings and group developments, including reservation of OSR space for such large developments; two wheeler-parking requirements were included in the DCR. In 1983, road width standards for industrial subdivisions were brought in and modifications in planning parameters for certain non-residential activities permissible in Primary Residential use zone made. In 1982 to 1983, the Government had clamped temporary ban for construction of multi-storeyed buildings in Chennai, which was lifted in December 1983; the special rules for multi-storeyed buildings were reviewed, a total revision made after consulting various departments/agencies concerned and the amended rule was given effect to from September 1984. In that year, the multi-storeyed building permissible area was also extended to cover the entire City (from the limited inner part bounded by Cathedral Road, Nungambakkam High Road, Brick Kiln Road and Railway line to Thiruvallur). In 1985, assemble of electrical and electronic parts for manufacture of radios, computers etc. was made permissible in Primary Residential use zone and setback requirements for special building were modified. In 1986, certain further modification and special building and group development rules were made in respect of OSR, and provisions made not to insist SSB in CBA.

14.26 After reviewing the DCR in 1989/90, comprehensive amendments to DCR proposing reduction of minimum plot extent, frontage, passage width, redefining special building, prescribing an undertaking for builders/promoters, prescribing uniform FSI for all the areas within CMA, etc. were sent to Government in 1990 which was approved and notified in 1993. The security deposit rate was enhanced in 1995 for amending DCR. In 1997, further amendments to DCR defining road width, prescribing minimum extent for OSR space, etc. were made. In 1998, a separate set of rules for regulation of I.T. parks, and associated hardware/software technology buildings were notified giving certain concessions to I.T. developments. Minimum road width prescriptions for layouts, industrial and institutional developments were brought in, permissible height for multi-storeyed buildings increased, parking standards modified by amending DCR in 1999. Rainwater harvesting provision was made mandatory in 2001 and provision for solar water heating in certain type buildings was made mandatory in 2002. Provision of certain further facilities for physically handicapped in public buildings was made mandatory in 2003, and also the multi-storeyed buildings permissible area was modified in that year. MSB Panel was reconstituted by amending DCR in 2004.

Density and FSI:

14.27 Chennai is one of the high-density cities in India. Its density varies from 180 persons per hec. in Saidapet and Mylapore Corporation zones and 368 persons per hec. in Kodambakkam zone within the Corporation limits and the gross density for Chennai City is 247 persons per hec. FSI is the main tool used in urban planning to regulate the densities of population with reference to infrastructure provision. Density of population needs to be regulated for various reasons including carrying capacity of infrastructure (existing as well as proposed), sociological reasons such as crime rate etc and other physical factors.

14.28 The existing developments in Chennai can be categorized as high dense medium raised developments mostly of buildings up to 15 m. heights. FSI allowed for such development up to 15 m. height presently is maximum 1.5. Multi stroeyed developments (high rise development) within CMA are very few; in order to encourage amalgamation of smaller plots into larger size and construction of buildings with large open space around, a higher FSI of 2.5/2.75 is allowed in multistoreyed developments.

14.29 The prevalence of high density in Chennai is attributed to the following:

(a) Smaller lot sizes
(b) Smaller dwelling unit sizes
(c) Large family size (on an average 4.5) against about 2.51 in USA and other western countries.

14.30 Comparative statement of densities in various cities in India and select cities in the World is given in Annexure - B. Mumbai has reversed the trend by fixing 1.33 as maximum FSI anywhere in the Metropolis in the early 80's itself. Delhi followed suite. Other cities have clamped it down to around 1.5.

14.31 The practice of increasing FSI under the disguise of high land cost needs to be discouraged. At the same time allowing higher FSI in the suburbs and lower FSI in central areas of the City also needs very careful consideration considering the carrying capacity of infrastructure, impact on environment including ground water and traffic volumes.

14.32 Considering availability of land for development and infrastructure (now and also in the Plan period), land use, travel and traffic aspects, optimum density of population, onslaught of private real estate developers on the small house owners/tenants in the central parts of the City, affordability etc., any experimentation in changing already fixed FSI without proper detailed study on this may have to be discouraged. In the present condition, any negative impact triggered due to increase in intensity of population/FSI cannot be reversed.

14.33 In many of the major cities in USA and Europe, even though it look high dense because of it's physical mass of construction, they are low in density when comparing the Indian conditions.

14.34 For the above reasons, after examing the issue in detail, it is proposed to retain FSI almost as existing and follow it.

Existing Land use 2006				*Extent in Hectares.*
Land use	**Chennai City**		**Rest of CMA**	
	Extent	%	Extent	%
Residential	9523	54.25	22877	21.87
Commercial	1245	7.09	390	0.37
Industrial	908	5.17	6563	6.28
Institutional	3243	18.48	3144	3.01
Open space & Recreation	366	2.09	200	0.19
Agricultural	99	0.57	12470	11.92
Non Urban	82	0.47	2433	2.33
Others (Vacant, Forest, Hills, Low lying, Water bodies etc.,)	2087	11.89	56507	54.03

Proposed Land use 2026				*Extent in Hectares.*
Land use	**Chennai City**		**Rest of CMA**	
	Extent	%	Extent	%
Primary Residential use zone	5916.35	33.58%	31090.68	31.68%
Mixed Residential use zone	2426.90	13.78%	13503.10	13.34%
Commercial use zone	714.24	4.05%	880.35	0.86%
Institutional use zone	2868.97	16.28%	3888.85	3.83%
Industrial use zone	691.83	3.93%	7274.33	7.18%
Special and hazardous Industrial use zone	130.67	0.74%	3416.08	3.38%
Open space & Recreational use zone	1000.65	5.68%	392.86	0.38%
Agriculture use zone	------	-------	7295.81	7.20%
Non Urban	113.31	0.64%	2332.92	2.3%
Urbanisable			2075.89	2.05%
Others (Roads, water bodies, hills, Redhills catchments area, forests etc.,)	3754.79	21.31%	28147.55	27.79%
Total	17617.70	100.00%	101298.42	100.00%

Annexure - A

Regularisation Of Unapproved Layouts In Chennai Metropolitan Area

1. In Chennai Metropolitan Area, a layout is defined as sub division of land into plots exceeding 8 in numbers. Sub-division of lands into plots up to 8 in nos. is called as sub- division.

2. Any layout laid out without approval of the competent authority viz. Directorate of Town &Country Planning before 5.8.75 the date of coming into force of Master Plan for Chennai Metropolitan Area or Chennai Metropolitan Development Authority after 5.08.1975 is called as unauthorized or unapproved layout.

3. The Development Control Rules for CMA, forming part of the Master Plan came into force from 5.8.75. Any unauthorized layout laid out on private land [not encroachments on public land] prior to 5.8.75 is recognized as it is. No equivalent land cost in lieu of Open Space Reservation is collectable for these pre- 5.08.75 unauthorised layouts.

4. It was found that before coming into force of the Tamil Nadu Urban Land Ceiling Act on 3.8.76 a number of unauthorized layouts had come up. For considering regularization of individual plots in those pre- 3.8.'76 layouts, the following guidelines were issued by CMDA in A.P.Ms.No. 110, dated 7.10.1976.

1. Site comprised in an unauthorized/unapproved, unsanctioned layout or sub divisions, but abutting on roads taken over by local body	The site will be considered for approval, subject to the plot owner agreeing to pay proportionate cost of the open space reservation needed for the area as per Development Control Rules 19(a)(iv)
2. i) Site comprised in an unauthorized unapproved/unsanctioned layout where developments have been carried out but the roads have not been handed over to the local body. ii) Site comprised in an unauthorized unapproved/unsanctioned layout where the developments have not been carried out.	The site may be considered approval, subject to the following: - i) The road in question can be taken over by the local body ultimately. Note: For the purpose of the above a road will be deemed to be a road that can be taken over if. a) the width of the road is not less than 16 feet and b) the road is connected at least at one end to a public street. ii) The plot owner agreeing to pay the proportionate cost of development as estimated by the local body for the full work/ for any unfinished works. iii) The plot owner agreeing to pay proportionate cost of the open space reservation needed for the area as per Development Control Rules 19(a)

5. The powers to regularize such pre 3.8.76 layouts were delegated to the executive authorities of the local body concerned within CMA.

6. In the Authority proceeding No. Rc.S5/43464/01/CPP/13/82, dated 7.12.82 the following guidelines were issued for regularization of individual plots in unapproved layouts made prior to 3.8.76: -

Planning Permission applications relating to construction in unauthorized layouts made prior to 3.8.76 shall be forwarded to CMDA for consideration with evidence of registration by the Registration Department to prove that sale of one or more plots in the said unauthorized layout had occurred prior to 3.8.76, and proposed sub division also formed part of the registered document, by a sketch or by narration apart from those particulars.

7. Many unauthorized layouts had come up after 3.08.1976. In 1985, a subject was placed before the Authority on the basis of a representation from one of the members of the Authority for extension of the cut- off date for regularization of plots in unapproved layouts. After detailed examination it was decided as follows in respect of the unapproved layouts made after 3.8.1976[a circular in Lr.M1/9838/90, dated 30.4.90].

(1) Executive officers of the local bodies in Madras Metropolitan Area are forwarding planning permission applications for construction of buildings in unapproved layouts/unauthorized sub divisions made after 3.8.1976, which are in violating of ULC guidelines. Government after consultation with MMDA and Commissioner of Land Reforms have directed that such unauthorized sub divisions made after 3.8.1976 should be regularized subject to the condition that the road in front of the applicant's site is public/under the maintenance of the local body concerned.

(2). The status of the road furnished by the local bodies in most of the cases is not clear. Roads of the local bodies are to be taken over by passing a resolution to that effect by the council. Roads should be taken over for maintenance only after obtaining approval of the council for incurring expenditure in providing amenities to the road in the area.

(3). The Executive Officers of the local bodies are therefore requested to forward planning permission applications for constructions in unauthorized layouts/sub divisions made after 3.8.1976 violating ULC guidelines with the following certificates:

i) the road for entire length from the applicant’s plot to its connection to nearest public road is vested with the local body. (This should be supported by adequate proof in the form of an extract from the street inventory register with council’s resolution with date).

ii) the road for entire length from the applicant’s plot to its connection with nearest public road is under the maintenance of the local body (supported with council resolution with date for having authorized incurring the expenditure)

iii) In the event, such a certificate is issued by the President of the local body, it should be countersigned by the Executive Authority of the respective local body i.e. Block Development Officer of Panchayat Union.

(4). The certificate wording in (i) and (ii) shall be as given and any altered or ambiguous wordings shall not be accepted.

(5). It was also decided to regularize only those unauthorized sub-divisions made prior to 31.12.1989. The actual date of registration would be taken into consideration for deciding whether the unauthorized sub divisions have taken place before or after 31.12.1989. The Executive Officers are therefore requested not to forward planning permission applications for regularization of unauthorized sub-division made after 31.12.1989.

8. In 1992, the powers to regularize pre- 31.12.’89 unauthorised subdivision were delegated to the officials of local bodies within Chennai Metropolitan Area subject to satisfying Development Control Rules and the following guidelines: -

(i) a) The abutting road should be a public road (Public road means a road taken over or declared public by the local body / Department of Highways and Rural works).

b) If the abutting road is not declared a public road, then it should be a road where the local body had incurred expenditure for providing infrastructures like streetlights, pavements, storm water drainage etc.

(ii) The unauthorized layout wherein the plot lies should have been made prior to 31.12.1989 (i.e.) the plot for planning permission applied for or any one of the plots in the unauthorized layout should have been registered prior to 31.12.1989 evidenced by the layout plan registered with the sale deed.

(Apart from Development charge under Town Country Planning Act, OSR Charges, local body's road improvement charges etc., additionally regularisation charge at the rate of Rs. 5 to Rs. 15 per sq.m. plot area were collectable when the plot was considered for regularisation)

9. Regularisation scheme for unauthorized development was introduced for Chennai Metropolitan Area in 1999 and extended in 2000, 2001 and 2002, the cut-off dates were also extended and as per the latest schemes which was closed on 8.07.2002, the following are the rule provisions in respect of unauthorised layouts/ subdivisions.

(i) the unauthorized plot should have been registered on or before 31.03.2002.

(ii) Irrespective of whether any plot or land is vacant, partially or fully built, if there is violation in terms of unauthorized sub-division or layout, or plot extent or frontage, the plot or land alone shall first be considered for regularization as one category and regularization fee shall be assessed as specified subject to the following conditions: -

a) In the case of a plot in an unauthorized layout, the plot shall abut a road of width not less than 4.8 metres and the unauthorized layout roads shall be connected to a public road or street maintained by the local body by their resolution.

b) In case of a plot in an unauthorized sub-division, the plot shall abut on a public road or street or gain access through an exclusive or a common passage of width not less than one metre and connected to a public road or street maintained by the local body evidenced by their resolution.

c) No part or whole of any unauthorized layout as such is regularisable. Only individual plot shall be considered for regularization, provided it has been divided and registered on or before 31st March 2002.

(iii) It shall be in conformity with the following: -

a) The Coastal Zone Regularisation of the Ministry of Environment and Forest under the Environment (Protection) Act, 1986 (Central Act 17 of 1986);

b) The Civil Aviation Regulations of the Ministry of Tourism and Civil Aviation under the Aircraft Act, 1934 (Central Act XXII of 1934);

c) The Ministry of Defence Regulations for developments in the vicinity of the Air force Stations within 900 metres around;

d) Regulations of the Chennai Regional Advisory Committee constituted by the Ministry of Communications;

e) Any development for regularisation shall abut on a public road or gain access from a road or passage over which the applicant has right of access;

f) In case of multi-storied developments, No Objection Certificate from the Directorate of Fire Services with regard to the fire protection measures shall be furnished by the applicant;

g) No development shall be regularized in the lands affected by the alignments of proposed Inner Circular (Rail) Corridor, Inner Ring Road, National Highways By-pass Road, Outer Ring Road and Mass Rapid Transit System (Rail) Projects;

h) No special and hazardous industry or and industry categorized as "RED" by Tamil Nadu Pollution Control Board shall be regularized in a non-conforming zone, it shall be regularized only with the clearance form the Tamil Nadu Pollution Control Board;

i) No industry categorized as "ORANGE" by the Tamil Nadu Pollution Control Board shall be regularized in any use zone without the clearance from the Tamil Nadu Pollution Control Board; and

j) Any development for regularisation shall conform to the rule 79 and 80 of Indian Electricity Rules, 1956 in respect of clearance form high tension and extra high voltage lines.

(iv) Regularisation fee collected for the plot area at the following rates

Serial Number	Location of the plot	Rate of Levy (Rupees per sq.m.)
1.	Chennai City Corporation and Banned area	20.00
2.	Municipalities and Municipal Townships.	12.50
3.	Rest of Chennai Metropolitan Area not included in Sl.No.1 &2	7.50

Note: These rates shall apply irrespective of whether the plot is vacant or constructed partly or fully.

(This Regularization fee is in addition to the normally payable changes such as Development Charges, Open Space Reservation Charges, Local body's Road Improvement Charge and License fee etc.)

10. As a large number of applications were received seeking planning permission for constructions in the post- 31.12.1989 unapproved layouts, and the cut-off date prescribed dates back to more than 15 years, CMDA has proposed to Government separately, for extension of the cut-off date for regularizing individual plots in these unauthorised layouts made after 31.12.1989.

Annexure - B

Details of Population in select cities in the World

Sl.No.	City	Area (Sq.km)	Population-2001	Density per hect.
	India			
1	Mumbai			
	a) Mumbai City	157	3326877	212
	b) Mumbai suburban	445	8587561	193
	c) Thane	9032	8128833	9
	d) Wardha	6153	1230640	2
	e) Raigarh	7353	2205972	3
2	Delhi			
	a) Central Delhi	25	644005	258
	b) East Delhi	64	1448770	226
	c) New Delhi	35	171806	49
	d) North Delhi	60	779778	130
	e) North East Delhi	60	1763712	294
	f) North West Delhi	438	2847395	65
	g) South Delhi	251	2258367	90
	h) South West Delhi	417	1749492	42
	I) West Delhi	129	2119641	164
3	Kolkata			
	a) City	185	4580544	248
	b) Hoogli	3150	5040047	16
	c) Howhra	1474	4274010	29
	d) North 24 parganas	4059	8930235	22
	e) South 24 parganas	9870	6909015	7
4	*Chennai*			
	a) City	*176*	*4343645*	*247*
	b) Rest of Metropolitan area	*1013*	*2690094*	*27*
5	Hydrabad City	217	3686460	170
6	Bangalore UA	2174	6523110	30
	Rest of the World			
7	Washington			
	City	159	606900	38
8	NewYork City	800	7322564	92
9	Los Angeles	1216	3485298	29
10	London	1572		53
	City			95
	Outer London			36
11	Singapore	640	43,00,000	67
12	Hong Kong			
	a) Island	80	1335469	166
	b) Kowlon	47	2023979	432
	c) New Territories	974	3343046	34

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Software used: Arc Info, ENVI, Auto Cad, Auto Cad Map 3D.

Aerial Data used: Ikonos 2001,2003 &2004 (1 metre resolution)
Cartosat 2005 (2.5 metre resolution)

Supported by

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Support for certain data collection and Tamil translation was provided by various officials and staff working in different Divisions in CMDA.

Committees which examined objections / suggestions and made recommendations

1. Committee on Land Use

1. Thiru. N.V. Rakhunath	- Chief Planner, CMDA
2. Thiru. P. Thyagarajan	- Addl. Director, DTCP
3. Thiru. K. Rajamanickam	- Dy. Director, Commssionerate of Municipal Administration
4. Thiru. Durganand Balsovar	- Architect
5. Thiru. S. Ramaraj	- President, Institute of Architects (TN Chapter)
6. Tmt. S. Chithra	- Senior Planner, CMDA

2. Committee on Transport

1. Thiru. N. Dharmalingam	- Retd. Chief Planner,CMDA
2. Thiru. T.K. Shanmugasundaram	- Chief Engineer (Genl.), Highways Dept.
3. Dr. A.M. Thirumurthy	- Professor Anna University
4. Thiru. R. Balasubramanian	- Managing Director, MTC
5. Thiru. P.T. Krishnan	- Architect rep. Citizen Consumer and Civic Action Group, Chennai
6. Thiru. K. Kumar	- Senior Planner, CMDA

3. Committee on Development Regulations

1. Thiru. Md. Nasimuddin I.A.S.	- Member Secretary, CMDA
2. Tmt. Tara Murali	- Architect
3. Prof. Suresh Kuppusamy	- School of Architecture & Planning, Anna University
4. Thiru. P.T.G. Sundaram	- Representing Institution of Engineers (TN Chapter)
5. Thiru. M.K. Sundaram	- Vice Present, Builder's Association of India (TN Chapter)
6. Thiru. C. Palanivelu	- Chief Planner, CMDA

4. Committee on Water supply and Drainage

1. Thiru. M. Dheenadhayalan	- Adviser to Govt.(Schemes), PWD
2. Thiru. D. Madavamoorthy	- Engineering Director, CMWSSB
3. Thiru. K. Balasundaram	- Chief Engineer, Corporation of Chennai
4 Thiru. R. Arul	- Secretary, Pasumai Thayagam.
5. Thiru. S. Santhanam	- Chief Planner, CMDA

5. Committee on Solid Waste Management

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3. Thiru. T. Chandrasekaran	- SE, (SW) Chennai Corporation
4. Thiru. R. Raghunathan	– SE, Commssionerate of Municipal Adminstration

5. Thiru. M. Madhivanan	- JD, Directorate of Town Panchayats
6. Thiru. M.B. Nirmal	- EXNORA
7. Thiru. S.Balaji	- Jt. Chief Engineer, TNPCB
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2. Thiru. R. Jayaraman	- CE, TNSCB
3. Thiru. D Ganesan	- CE, TNHB
4. Thiru. Dr. A. Srivatsan	- Architect / Journalist
5. Tmt. K. Radhai	- All India Democratic Women's Association, Chennai
6. Thiru. M.Sivashanmugam	- Deputy Planner, CMDA

7. Committee on Environment

1. Dr. K.S. Neelakandan IFS	- Director of Environment
2. Thiru. M.G.Devasagayam IAS(Retd)	- SUSTAIN
3. Dr. S. Mohan	- Head, Dept. of Civil Engg. IIT-Madras
4. Dr. T. Sekar, IFS	- Member-Secretary, TNPCB
5. Thiru. Rajesh Rangarajan	- rep. Citizen Consumer and Civic Action Group, Chennai.
6. Thiru. A. Subash	- Better Environment for Long Life Foundation, Chennai
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